Modern Intelligent Systems: A Graduate Companion

Neural Networks, Fuzzy Logic, and Evolutionary Optimization

Haitham Amar

First edition, 2026

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Preface

I have taught courses that engage with artificial intelligence both directly—through machine learning and neural networks—and indirectly, at the level of data collection and system design. Over time, I noticed a recurring gap. While the literature offers many excellent treatments of individual techniques, it rarely provides a coherent path that guides students from foundational notions of intelligence to the advanced models used today. What is often missing is a unified narrative that connects intuition, mathematical formulation, and practical deployment across the diverse tools that constitute intelligent systems.

Courses in machine learning tend to emphasize neural networks and deep learning; others focus on optimization and operations research, from classical search strategies to genetic algorithms. Each approach is valuable, yet time constraints and disciplinary boundaries often force a narrowing of scope, sacrificing breadth for depth or vice versa. In one graduate course in particular, a broader framework emerged—one that treated these methods not as isolated topics, but as complementary responses to recurring modeling challenges. That lens, refined over years of teaching and debugging, proved invaluable: it allowed students to situate linear regression, neural networks, transformers, fuzzy inference systems, and evolutionary algorithms within a single, coherent perspective on intelligent system design. This book attempts to make that perspective explicit and durable.

The manuscript evolved into a concise graduate companion that blends classroom-tested explanations with laboratory-style checklists and reflective prompts. The chapters move from supervised learning foundations to fuzzy logic and evolutionary computing, mirroring the trajectory of the underlying lecture material while adding connective tissue so that a reader can revisit the material years later without searching for missing context.

Origins in the classroom

This book began as lecture notes for a graduate course. Early versions leaned heavily toward soft computing, reflecting the emphasis of prior offerings. Prof. Karray, who originally shaped that course arc, was generous in letting me broaden the scope and reshape the narrative around a more general view of intelligent system design.

The material has since been rewritten to stand alone as a book. Any offering-specific details (schedules, grading, local policies) are intentionally kept out of the published edition; if you are using this text in a course offering, consult the accompanying course handout (or a course-edition build) for those logistics.

Over the following years, I came to view fuzzy inference systems as one important piece in a larger mosaic rather than the organizing principle. I iterated on the syllabus—moving topics, adding or removing chapters, and tightening mathematical through-lines—toward a narrative that is coherent, broad in coverage, and implementable in engineering practice.

At the time of this first edition (2026), the field is in the era of large language models, and this book covers them (see Chapters 13 and 14). But it also emphasizes other ideas and toolkits that may underwrite future breakthroughs in intelligent systems: careful probabilistic thinking and diagnostics, principled optimization, sequence modeling beyond any single architecture, and hybrid reasoning approaches that have repeatedly re-emerged in new forms.

Perspective

This book treats intelligence as engineered self-correction: choose a representation, define an objective, pick an update mechanism, and audit failure modes under realistic constraints. Paradigms differ mainly in how they update (rules, gradients, energy descent, population search), not in whether they must be evaluated honestly.

The intent is deliberately anti-mystical: every method earns its place by making its assumptions and breakdown points explicit, not by fashion or rhetoric.

For a practical reader's guide—roadmap, prerequisites, and suggested reading paths—see the final sections of Chapter 1. Notation and reading conventions are collected near the front of the book in *Notation and Conventions*.

The goal is a self-contained reference for researchers and engineers who want a rigorous but narrative-friendly treatment of neural networks, soft computing, and hybrid reasoning systems.

Acknowledgments

This book grew out of teaching and of revising lecture notes across multiple offerings until they could stand on their own as a coherent reference. I am especially grateful to Prof. Fakhri Karray, who originally shaped that course arc and generously invited me to broaden its scope—an act of intellectual generosity that made this book possible.

I thank the students who kept me honest. Their questions (often asked at exactly the wrong time in a derivation) exposed where notation was overloaded, where an argument quietly skipped a step, and where an example needed numbers rather than slogans. Many of the "audit" habits emphasized throughout the book—sanity checks, edge-case testing, and careful reporting—reflect the kinds of failures students encountered when implementations did not match the math.

I also thank colleagues in the Department of Electrical and Computer Engineering who shared perspective on how these topics fit together as an engineering discipline rather than as isolated techniques, and who encouraged a book-length treatment. The open-source ecosystem around LaTeX, Python, and scientific computing made it possible to cross-check derivations and reproduce figures with minimal friction. If you have used this text in a course offering, the teaching assistants and staff who supported the logistics deserve credit as well.

Finally, I am grateful to readers who provided detailed feedback on clarity, typography, and scientific consistency. Any remaining errors and omissions are my responsibility.

Contents

Preface **3**

Acknowledgments **5**

Notation and Conventions **27**

Part I: Foundations and the ERM toolbox **31**

1 About This Book **31**
1.1 Historical Foundations of Intelligent Systems 32
1.2 Defining Artificial Intelligence and Intelligent Systems . . . 34
1.3 Intelligent Systems 36
1.4 Case Study: AI-Enabled Camera as an Intelligent System . 38
1.5 Levels and Architectures of Intelligent Systems 40
1.6 Intelligent Systems and Intelligent Machines 44
1.7 Levels, Meta-cognition, and Safety 45
1.8 Audience, Prerequisites, and Scope 46
1.9 Roadmap and Reading Paths 47
1.10 Using and Navigating This Book 48

2 Symbolic Integration and Problem-Solving Strategies **51**
2.1 Context and Motivation 52
2.2 Problem Decomposition and Transformation 52
2.3 Heuristic Transformations 54
2.4 Summary of the Approach 55
2.5 Heuristic Transformations: Revisiting the Integral with $1-x^2$ 56
2.6 Transformation Trees and Search Strategies 60
2.7 Algorithmic Outline for Symbolic Problem Solving 61
2.8 Discussion: What this example illustrates 65

3 Supervised Learning Foundations **68**
3.1 Problem Setup and Notation 71
3.2 Fitting, Overfitting, and Underfitting 71

3.3 Empirical Risk Minimization and Regularization 72
3.4 Elastic-net paths and cross-validation 75
3.5 Common Loss Functions 75
3.6 Model Selection, Splits, and Learning Curves 80
3.7 Linear regression: a first full case study 84

4 Classification and Logistic Regression 89
4.1 From regression to classification 90
4.2 Classification problem statement 91
4.3 Bayes Optimal Classifier 91
4.4 Logistic Regression: A Probabilistic Discriminative Model . 94
4.5 Probabilistic Interpretation: MLE and MAP 99
4.6 Confusion Matrices and Derived Metrics 99

Part II: Neural networks, sequence modeling, and NLP 105

5 Introduction to Neural Networks 106
5.1 Biological Inspiration 107
5.2 From Biological to Artificial Neural Networks 108
5.3 Neural Network Architectures 110
5.4 Activation Functions 112
5.5 Learning Paradigms in Neural Networks 114
5.6 Fundamentals of Artificial Neural Networks 114
5.7 Mathematical Formulation of the Neuron Output 116
5.8 The McCulloch–Pitts Neuron: Logical Computation and Limits 117
5.9 From MP Neuron to Perceptron and Beyond 117

6 Multi-Layer Perceptrons: Challenges and Foundations 126
6.1 From a single unit to the smallest network 127
6.2 Performance: what are we trying to improve? 130
6.3 Gradient descent: how do weights move? 131
6.4 Why hard thresholds block learning 132
6.5 Differentiable activations and the sigmoid trick 133
6.6 Deriving weight updates for the two-neuron network 134

6.7 From two neurons to multi-layer networks 136
6.8 Summary 137

7 Backpropagation Learning in Multi-Layer Perceptrons 140
7.1 Context and Motivation 140
7.2 Problem Setup 141
7.3 Loss and Objective 142
7.4 Notation for Layers and Neurons 143
7.5 Forward Pass Recap 144
7.6 Backpropagation: Recursive Computation of Error Terms . 146
7.7 Batch and Stochastic Gradient Descent 149
7.8 Backpropagation Algorithm: Brief Numerical Check 153
7.9 Training Procedure and Epochs 156
7.10 Role and Design of Hidden Layers 157
7.11 Case Study: Learning the Function $y = x \sin x$ 160
7.12 Applications of Backpropagation-Trained Networks 160
7.13 Limitations and Known Failure Modes of MLPs 161
7.14 Summary 161

8 Radial Basis Function Networks 166
8.1 Overview and Motivation 167
8.2 Architecture of RBFNs 168
8.3 Radial Basis Functions 172
8.4 Key Properties and Advantages 172
8.5 Transforming Nonlinearly Separable Data into Linearly Separable Space 173
8.6 Finding the Optimal Weight Vector $\mathbf{w}$ 174
8.7 The Role of the Transformation Function $g(\cdot)$ 175
8.8 Examples of Kernel Functions 175
8.9 Interpretation of the Width Parameter σ 176
8.10 Effect of σ on Classification Boundaries 177
8.11 Radial Basis Function Networks: Parameter Estimation and Training 178
8.12 Remarks on Radial Basis Function Networks 183
8.13 Preview: Unsupervised and Localized Learning 184

9 Introduction to Self-Organizing Networks and Unsupervised Learning 186

9.1 Overview of Self-Organizing Networks 187
9.2 Clustering: Identifying Similarities and Dissimilarities . . . 189
9.3 Dimensionality Reduction: Simplifying High-Dimensional Data 191
9.4 Feature Mapping: From Prototype Vectors to Grid Coordinates 192
9.5 Self-Organizing Maps (SOMs): Introduction 193
9.6 Conceptual Description of SOM Operation 195
9.7 Mathematical Formulation of SOM 196
9.8 Kohonen Self-Organizing Maps (SOMs): Network Architecture and Operation 197
9.9 Example: SOM with a 3×3 Output Map and 4-Dimensional Input 198
9.10 Key Properties of Kohonen SOMs 199
9.11 Winner-Takes-All Learning and Weight Update Rules . . . 200
9.12 Numerical Example of Competitive Learning 201
9.13 Winner-Takes-All Learning: Consolidation After the Numerical Example 202
9.14 Regularization and Monitoring During SOM Training . . . 203
9.15 Limitations of Winner-Takes-All and Motivation for Cooperation 206
9.16 Cooperation in Competitive Learning 208
9.17 Example: Neighborhood Update Illustration 211
9.18 Summary of Cooperative Competitive Learning Algorithm . 211
9.19 Synthesis: SOM Training Rules, Annealing Schedules, and Design Choices 212
9.20 Applications of Kohonen Self-Organizing Maps 214

10 Hopfield Networks: Introduction and Context 221

10.1 From Feedforward to Recurrent Neural Networks 222
10.2 Hopfield's breakthrough 223
10.3 Network Architecture and Dynamics 224
10.4 Encoding conventions 225

10.5 Energy Function and Stability 226
10.6 Energy Minimization and Stable States 226
10.7 Example: Energy Calculation and State Updates 228
10.8 Energy Function and Convergence of Hopfield Networks . . 230
10.9 Asynchronous vs. Synchronous Updates in Hopfield Networks 232
10.10 Storage Capacity of Hopfield Networks 233
10.11 Improving Storage Capacity via Weight Updates 234
10.12 Example: Weight Calculation for a Single Pattern 234
10.13 Finalizing the Hopfield Network Derivation and Discussion 236

11 Convolutional Neural Networks and Deep Training Tools 242
11.1 Motivation and map 243
11.2 Why fully connected layers break on images 244
11.3 Sparse connectivity and parameter sharing 246
11.4 Convolution and pooling mechanics 247
11.5 Pooling as nonparametric downsampling 249
11.6 Channels and feature maps 250
11.7 Training Neural Networks: Gradient-Based Optimization . 252
11.8 Vanishing and Exploding Gradients in Deep Networks . . . 253
11.9 Strategies to Mitigate Vanishing and Exploding Gradients . 254

12 Recurrent Neural Networks: Dynamics, Memory, and Gating 263
12.1 Motivation for Recurrent Neural Networks 264
12.2 Key Idea: State and Memory in RNNs 265
12.3 Comparison with Feedforward Networks 266
12.4 Recap: Feedforward Building Blocks 267
12.5 Input–Output Configurations and Mathematical Formulation 270
12.6 Mathematical Formulation of a Simple RNN Cell 272
12.7 Recurrent Neural Network (RNN) Architectures and Loss Computation 272
12.8 Stabilizing Recurrent Training 276
12.9 From Recurrent State to Attention: Bridging Gated RNNs and Transformers 281

12.10 Synthesis: RNN Derivations, Gating Decisions, and Sequence Modelling Limits 282

13 Neural Network Applications in Natural Language Processing 289

13.1 Context and Motivation 290
13.2 Warm-up: from symbols to vectors 291
13.3 Key Insight: Distributional Hypothesis 293
13.4 Contextual Meaning and Feature Extraction 294
13.5 Word2Vec at a glance 294
13.6 From lookup to objective: compact derivation path 296
13.7 Word2Vec objectives in detail 297
13.8 Efficient Training of Word Embeddings: Hierarchical Softmax and Negative Sampling 300
13.9 Local Context vs. Global Matrix Factorization Approaches 302
13.10 Global Word Vector Representations via Co-occurrence Statistics 303
13.11 Finalizing the Word Embedding Derivations 306
13.12 Bias in Natural Language Processing 309
13.13 Responsible deployment checklist 311

14 Transformers: Attention-Based Sequence Modeling 316

14.1 From encoder–decoder bottlenecks to attention 317
14.2 Scaled Dot-Product Attention 319
14.3 Self-attention vs. cross-attention 322
14.4 Multi-Head Attention (MHA) 322
14.5 Positional encodings (order without recurrence) 325
14.6 Objectives and masks 326
14.7 Transformer blocks and training stabilizers 328
14.8 BERT vs. GPT vs. encoder–decoder 331
14.9 Long Contexts and Efficient Attention 334
14.10 Fine-Tuning and Parameter-Efficient Adaptation 335
14.11 Decoding and Evaluation 335
14.12 Audit and failure modes (short list) 337
14.13 Alignment (Brief) 337

14.14 Advanced notes (optional) 338
14.15 RNNs vs. Transformers: When and Why 339

Part III: Soft computing and fuzzy reasoning 342

15 Introduction to Soft Computing 343
15.1 Hard Computing: The Classical Paradigm 345
15.2 Soft Computing: Motivation and Definition 345
15.3 Why Soft Computing? 346
15.4 Relationship Between Hard and Soft Computing 347
15.5 Overview of Soft Computing Constituents 347
15.6 Distinguishing Imprecision, Uncertainty, and Fuzziness . . . 348
15.7 Soft Computing: Motivation and Overview 349
15.8 Fuzzy Logic: Capturing Human Knowledge Linguistically . 350
15.9 Comparison with Other Soft Computing Paradigms 351
15.10 Zadeh's Insight and the Birth of Fuzzy Logic 352
15.11 Challenges in Fuzzy Logic Systems 352
15.12 Mathematical Languages as Foundations for Fuzzy Logic . 352
15.13 Fuzzy Logic: Motivation and Intuition 355
15.14 From Crisp Sets to Fuzzy Sets 356
15.15 Wrapping up soft computing and previewing the fuzzy trilogy 358

16 Fuzzy Sets and Membership Functions: Foundations and Representations 361
16.1 Motivating example: designing a membership function from measurements 362
16.2 Fuzzy sets and the universe of discourse 363
16.3 Membership Functions: Definition and Interpretation . . . 364
16.4 Discrete vs. Continuous Universes of Discourse 364
16.5 Crisp Sets versus Fuzzy Sets 365
16.6 Membership Functions in Fuzzy Sets 366
16.7 Comparison of Membership Functions 368
16.8 Example: Overlapping weight labels 369
16.9 Fuzzy Sets: Core Concepts and Terminology 371
16.10 Probability vs. Possibility 373

16.11 Fuzzy Set Operations 375
16.12 Additional Fuzzy Set Operations 377
16.13 Example: Union and Intersection of Fuzzy Sets 378
16.14 Cartesian Product of Fuzzy Sets 378
16.15 Properties of Fuzzy Set Operations 379
16.16 Fuzzy Set Operators 380
16.17 Complement Operators in Fuzzy Logic 381
16.18 Triangular norms (t-norms) 382
16.19 Triangular conorms (t-conorms / s-norms) 384
16.20 T-Norms and S-Norms: Complementarity and Properties . 385
16.21 Examples of common t-norm / s-norm pairs 385
16.22 Fuzzy Set Inclusion and Subset Relations 386
16.23 Degree of Inclusion 386
16.24 Set Operations and Inclusion Properties 386
16.25 Grades of Inclusion and Equality in Fuzzy Sets 387
16.26 Dilation and Contraction of Fuzzy Sets 389
16.27 Closure of Membership Function Derivations 390
16.28 Implications for Fuzzy Inference Systems 392
16.29 Worked Example: Mamdani Fuzzy Inference (End-to-End) 393

17 Fuzzy Relations: Composition, Projection, and the Extension Principle 398
17.1 Context and Motivation 399
17.2 Problem Statement 399
17.3 Intuition and Challenges 400
17.4 Formal Definition of the Transformed Membership Function 400
17.5 Interpretation 401
17.6 Transformation of Fuzzy Sets Between Universes 401
17.7 Extension Principle Recap and Projection Operations . . . 405
17.8 Projection of Fuzzy Relations 405
17.9 Dimensional Extension and Projection in Fuzzy Set Operations 409
17.10 Fuzzy Inference via Composition of Relations 410
17.11 Properties of Fuzzy Relation Composition 412
17.12 Alternative Composition Operators 412

18 Fuzzy Inference Systems: Rule Composition and Output Calculation 415
18.1 Context and Motivation 416
18.2 Rule Antecedent Composition 417
18.3 Rule Consequent and Output Fuzzy Set 418
18.4 Aggregation of Multiple Rules 419
18.5 Summary of the Fuzzy Inference Process 419
18.6 Worked example: thermostat inference with numbers 421
18.7 Mamdani vs. Sugeno/Takagi–Sugeno systems 424
18.8 Neuro-Fuzzy Hybrid: ANFIS 425

Part IV: Evolutionary optimization 429

19 Evolutionary Computing: Search, Selection, and Adaptation 430
19.1 Context and Motivation 431
19.2 Philosophical and Historical Background 436
19.3 Problem Setting: Optimization 437
19.4 Illustrative Example 438
19.5 Why Not Brute Force? 438
19.6 Summary 439
19.7 Challenges in Continuous Optimization and Motivation for Evolutionary Computing 439
19.8 Evolutionary Computing at a Glance 440
19.9 Biological Inspiration: Evolutionary Concepts 440
19.10 Implications for Genetic Algorithms 441
19.11 Summary of Biological Mechanisms Modeled in GAs 442
19.12 Genetic Algorithms: Modeling Chromosomes 443
19.13 Mapping Genetic Algorithms to Optimization Problems . . 445
19.14 Encoding in Genetic Algorithms 446
19.15 Population Initialization and Size 448
19.16 Genetic Operators 449
19.17 Selection in Genetic Algorithms 449
19.18 Crossover Operator 452
19.19 Mutation Operator 453

19.20 Summary of Genetic Operators and Their Probabilities . . 454
19.21 Known Issues in Genetic Algorithms 455
19.22 Convergence Criteria 456
19.23 Summary of Genetic Algorithm Workflow 457
19.24 Pseudocode Representation 459
19.25 Example: GA for a Constrained Optimization Problem . . 459
19.26 Genetic Algorithms: Iterative Process and Convergence . . 462
19.27 Beyond canonical GAs: real-coded strategies 463
19.28 Particle Swarm Optimisation (PSO) 464
19.29 Genetic Programming (GP) 465
19.30 Wrapping Up Genetic Algorithms and Genetic Programming 467
19.31 Multi-objective search and NSGA-II 469

Back matter **474**

Key Takeaways **475**

A Notation collision index **493**

B Reproducibility and Reporting Standards **495**

C Kernel Methods and Support Vector Machines **496**

D Linear Systems Primer **498**

Index **502**

List of Figures

1 Roadmap of the book strands (core supervised path through attention models; SOM/fuzzy; optimization/evolutionary). It also serves as a quick prerequisite map when you want to jump ahead and later backfill foundations. 47
2 Transformation tree for the running example integral. Badges [**S**]/[**H**] mark safe vs. heuristic moves; the left branch shows the sequential safe path, while the right branch terminates in explicit backtracking. The tree view makes solved and dead-end leaves visible. 77
3 Schematic underfitting and overfitting as a function of model complexity. Training error typically decreases with complexity, while validation error often has a U-shape; regularization and model selection aim to operate near the minimum of the validation curve. 78
4 Why L1 promotes sparsity (geometry). Minimizing loss subject to an L2 constraint tends to hit a smooth boundary; an L1 constraint has corners aligned with coordinate axes, so tangency often occurs where some coordinates are exactly zero. . 78
5 Illustrative lasso path as the regularization strength increases. Coefficients shrink, and some become exactly zero, yielding sparse models; the shrinkage order depends on feature correlation and scaling. 79
6 Classification losses as functions of the signed margin z. The curves highlight how different losses treat confident mistakes and near-boundary points. 79
7 Regression losses versus prediction error. The Huber loss transitions from quadratic to linear, reducing sensitivity to outliers while staying differentiable near zero. 79
8 Dataset partitioning into training, validation, and test segments. Any resampling scheme should preserve disjoint evaluation data; when classes are imbalanced, shuffle within strata so each split reflects the overall class mix. 81

9 Mini ERM pipeline (split once, iterate train/validate, then test only the best model on the held-out set). This keeps tuning decisions separate from final reporting. 82
10 Illustrative learning curves reveal under/overfitting: the validation curve flattens while additional data continue to decrease training error only marginally. A shaded patience window marks when early stopping would halt if no validation improvement occurs. 82
11 Calibration and capacity diagnostics (reliability and double descent) 84
12 Ridge regularization shrinks parameter norms as the penalty strength increases. This controls variance without forcing sparsity. 85
13 Synthetic binary dataset. 92
14 Bayes-optimal boundary for two Gaussian classes. Equal covariances and similar priors (LDA setting) yield a linear separator; unequal covariances yield a quadratic boundary. The boundary is near the equal-posterior line (vertical, pink); left-/right regions map to predicted classes R0 and R1. 93
15 MAP estimates interpolate between the prior mean and the data-driven MLE. As the sample size grows, the MAP curve approaches the true mean; with little data, the prior dominates. 93
16 The sigmoid maps logits to probabilities (left). The binary cross-entropy (negative log-likelihood) penalizes confident wrong predictions sharply (middle). Regularization typically shrinks parameter norms as the penalty strength increases (right). 96
17 Gradient-descent iterates contracting toward the minimizer of a convex quadratic cost (schematic). Ellipses are level sets; arrows show the “steepest descent along contours” direction. Poor conditioning stretches the ellipses and slows convergence. 97
18 Illustrative logistic-regression boundary 98
19 ROC and PR curves with an explicit operating point 101

20 Confusion matrix for a three-class classifier; diagonals dominate, indicating strong accuracy with modest confusion between classes B and C. 101
21 Perceptron geometry. Points on opposite sides of the separating hyperplane receive different labels, and signed distance to the boundary controls both prediction and update magnitude. Compare with Figure 18 in Chapter 4: both are linear separators, but logistic smooths the boundary into calibrated probabilities. 122
22 The minimal neural network used in this chapter is a two-neuron chain. The first unit produces an intermediate signal, and the second unit maps that signal to the final output. . . 129
23 Think of performance as a surface over the weights. Gradient descent moves in one vector step (blue), whereas coordinate-wise updates can zig-zag (orange). 132
24 Hard thresholds block gradient-based learning because the derivative is zero almost everywhere. A smooth activation like the sigmoid provides informative derivatives across a wide range of inputs. 133
25 Computational graph for backpropagation (reverse-mode automatic differentiation) 152
26 Forward (blue) and backward (orange) flows for a two-layer MLP. Cached activations and layerwise deltas travel along these arrows; backward signals use next-layer weights and activation derivatives. 152
27 Canonical activation functions on a common axis. Solid curves show the activation; dashed curves show its derivative. 159
28 RBFN architecture. Inputs feed radial units parameterized by centers and widths; a linear readout with weights and bias is trained by a regression or classification loss. In the classical pipeline only the output weights are learned, while centers and widths come from clustering or spacing heuristics; variants also fine-tune centers and widths end-to-end. 169

29 Localized Gaussian basis functions (dashed) and their learned output-weighted combination $\sum_j w_j \varphi_j(x)$ (solid). Overlapping bumps allow RBF networks to interpolate complex signals smoothly. 170
30 Center placement and overlap (schematic). K-means prototypes tend to tile the data manifold more evenly than random picks, giving more uniform receptive-field overlap. Random centers can leave gaps or create excessive overlap, which affects the width (sigma) choice and the conditioning of the later linear solve. 171
31 XOR before and after an RBF feature map. Left: in (x_1, x_2), no single line separates the labels. Right: in (g_1, g_2), the transformed points are linearly separable; one valid separating border is $g_1 + g_2 = 1.3$ (equivalently $w = [1, 1]$, $b = -1.3$). 174
32 Schematic illustration of how the width parameter σ influences decision boundaries: too-large σ underfits (bases overlap heavily and wash out locality), intermediate σ captures the boundary, and too-small σ can produce fragmented regions. For a computed XOR example, see Figure 34. 177
33 Primal (finite basis) vs. dual (kernel ridge) viewpoints. Using as many centers as data points recovers the dual form; using fewer centers corresponds to a Nyström approximation. The same trade-off appears in kernel methods through the choice of kernel and effective rank. 181
34 Effect of σ on an RBFN XOR boundary (4 centers at the data corners, ridge $\lambda = 10^{-3}$, threshold 0.5). Too-large σ makes bases overlap heavily, producing a very smooth, low-contrast boundary; intermediate σ yields a cleaner separation; too-small σ makes the model extremely local, producing small "islands" around prototypes. 183
35 Learning-rate scheduling intuition (schematic). On a smooth objective (left), large initial steps quickly cover ground and roughly align prototypes, while a decaying step-size refines the solution near convergence. Right: common exponential and multiplicative decays used in SOM training. 189

36 Classical MDS intuition (schematic). Projecting a cube onto a plane via an orthogonal map yields a square (left), whereas an oblique projection along a body diagonal produces a hexagon (right). The local adjacency of vertices is preserved even though metric structure is distorted. 191
37 Gaussian neighborhood weights in SOM training (schematic). Early iterations use a broad kernel so many neighbors adapt; later iterations shrink the neighborhood width $\sigma(t)$ so only units near the BMU update. 197
38 Illustrative bias–variance trade-off when sweeping SOM capacity (number of units or kernel width). The optimum appears near the knee where bias and variance intersect. 203
39 Regularization smooths an objective surface (schematic). Coupling neighboring prototypes (right) yields wider, flatter basins than the narrow, ill-conditioned unregularized basin (left). 204
40 Illustrative objective surface combining quantization error and an entropy-style regularizer (a modern SOM variant; for example, a negative sum of $p \log p$ over unit usage). Valleys arise when prototypes cover the space evenly; ridges highlight collapse or poor topological preservation. 205
41 Illustrative validation curves used to identify an early-stopping knee. When both quantization and topographic errors flatten (shaded band), further training risks map drift. 207
42 Voronoi-like regions induced by fixed prototypes in input space (left) and the corresponding soft assignments after sharpening the neighborhood kernel (right). Softer updates blur the decision frontiers and reduce jagged mappings between adjacent units (schematic illustration). 208

43 Left: a 5-by-5 SOM grid with the best matching unit (blue) and neighbors inside the Gaussian-kernel radius (green). Right: a toy U-Matrix (grayscale-safe colormap) showing average distances between neighboring codebook vectors; larger distances suggest likely cluster boundaries. Treat a U-Matrix as a qualitative boundary hint unless preprocessing and scaling are fixed. 209
44 Component planes for three features on a trained SOM (toy data). Each plane maps one feature across the grid; aligned bright/dark regions reveal correlated features and complement the U-Matrix in Figure 43. Interpret brightness comparatively within a plane rather than as an absolute scale. 210
45 SOM grid with the best-matching unit (BMU) highlighted in blue and a dashed neighborhood radius indicating which prototype vectors receive cooperative updates (schematic). . . . 211
46 Hopfield energy decreases monotonically under asynchronous updates. Starting from a noisy probe $\mathbf{s}^{(0)}$, single-neuron flips move downhill until a stable memory is recovered. 229
47 Energy-landscape intuition (schematic). Hopfield recall is a descent process toward a nearby basin (a stored memory), but other minima can exist and act as spurious attractors when the network is heavily loaded. 237
48 Receptive field growth across depth. Even with small 3×3 kernels, stacking layers expands the spatial context seen by deeper units, which is why depth can replace very large kernels. 252
49 Dropout effect on training/validation curves (validation flattening) 259
50 Batch normalization first maps each channel to a standard normal shape and then applies a learned scale and shift (γ, β); the normalized distribution shown here is the pre-affine step that stabilizes gradients. 260
51 Schematic loss curves illustrating typical behavior: Adam often drops loss quickly early, while SGD+momentum can be steadier later. 261

52 Decision boundaries for logistic regression (left) versus a shallow MLP (right). Linear models carve a single hyperplane, whereas hidden units can warp the boundary to follow nonconvex manifolds such as the moons dataset. 268
53 Binary cross-entropy geometry (left), effect of learning-rate schedules on loss (middle), and the typical training/validation divergence that motivates early stopping (right). 269
54 Unrolled RNN over three time steps. The brace marks that the same parameters $\mathbf{W}_{xh}$, $\mathbf{W}_{hh}$, and $\mathbf{W}_{hy}$ are reused at every step; backpropagation through time (BPTT) accumulates gradient contributions through all copies before updating those shared weights. 272
55 Backpropagation through time (BPTT): unrolled forward pass (black) and backward gradients (pink) through time. 275
56 Vanishing and exploding gradients 276
57 Gradient norms (left) explode without clipping (orange) but remain bounded when the global norm is clipped at tau (green). Training loss (right) stabilizes as a result. 277
58 Teacher forcing vs. inference in a sequence-to-sequence decoder. Gold arrows show supervised targets; orange arrows highlight autoregressive feedback that motivates scheduled sampling. 278
59 Long Short-Term Memory (LSTM) cell (Hochreiter and Schmidhuber, 1997; Gers et al., 2000). 279
60 Gated Recurrent Unit (GRU) cell (Cho et al., 2014). 279
61 Attention heatmap for a translation model. Rows are target tokens (decoder steps) and columns are source tokens (encoder positions). Each cell is an attention weight; the dot in each row marks the source position receiving the most attention. . 282
62 Analogy geometry in embedding space 305

63 Reference schematic for the Transformer. Left: scaled dot-product attention. Center: multi-head concatenation with an output projection. Right: pre-LN encoder block combining attention, FFN, and residual connections; a post-LN variant simply moves each LayerNorm after its residual add (dotted alternative, not shown). 324
64 Transformer micro-views. Left: positional encodings (sinusoidal/rotary) add order information. Center: KV cache stores past keys/values so decoding a new token reuses prior context. Right: LoRA freezes a base weight matrix $\mathbf{W}$ and trains a low-rank update $\Delta\mathbf{W} = \mathbf{BA}$ for parameter-efficient tuning. 325
65 Attention masks as heatmaps (queries on rows, keys on columns). Left: padding mask zeroes out attention to padded *key columns* of a shorter sequence in a packed batch; padded-query rows are typically ignored downstream by output/loss masking. Right: causal mask enforces autoregressive flow by blocking attention to future tokens. 327
66 Trapezoidal membership functions for grades C and B with the overlapping region shaded. Scores near 78–82 partially satisfy both grade definitions. This overlap design keeps boundary cases smooth instead of forcing abrupt label switches. 369
67 Overlapping membership functions for Small/Medium/Large labels 370
68 Fuzzy AND surfaces for common t-norms 384
69 End-to-end fuzzy inference example. (A) Consequent membership functions with clipping levels from firing strengths at T = 27 deg C. (B) Aggregated output set (max of truncated consequents) and a centroid marker near s* approx 0.58. This trace provides an end-to-end sanity check for clipping, aggregation, and centroid defuzzification. 395

70 Mapping a fuzzy set through the function “y = x-squared”. The membership at an output value y is the supremum over all preimages x that map to y; shared images such as $x = \pm 1$ map to y = 1 using the maximum membership. Helpful when applying the extension principle to a non-invertible function. 403
71 Alpha-cuts under the non-monotone map “y = x-squared”. A symmetric triangular fuzzy set on X maps to a right-skewed fuzzy set on Y. Each alpha-cut on A splits into two intervals whose images union to the output alpha-cut. Use this to propagate a fuzzy set through a non-monotone map via alpha-cuts. 406
72 Illustrative fuzzy relation table (left) together with its projections onto the error universe (middle) and the rate-of-change universe (right). These are the exact quantities used in the running thermostat example before composing rules. Use it to build rule antecedents from a relation and verify which universe each projection inhabits. 408
73 Evolutionary micro-operators. Left: fitter individuals get sampled more often (roulette/tournament). Middle: crossover splices parents by a mask (one-point shown). Right: constraint handling routes offspring through repair/penalty/feasibility before evaluation. Use this to map an implementation to the canonical operator loop. 444
74 Illustrative GA run showing the best and mean normalized fitness over 50 generations. Flat regions motivate “no improvement” stopping rules, while steady separation between best and mean indicates ongoing selection pressure. Helpful for diagnosing premature convergence versus ongoing exploration. 457
75 GA flowchart showing the iterative process: initialization leads to fitness evaluation and a termination check. If not terminated, the algorithm proceeds through selection, crossover, mutation, and replacement, which then feeds the next generation’s fitness evaluation. Reference when verifying that your implementation preserves the intended control flow. 458
76 Sample Pareto front for two objectives 470
77 Map of model families 479

List of Tables

1 Common losses and typical use (reference for Chapters 3 to 5). Match the loss to your modeling assumptions (noise, margins) and to the downstream decision metric you care about. 76

2 Handling class imbalance for logistic models (Chapter 4 reference table). A compact menu of resampling, weighting, and thresholding strategies for rare positives. 103

3 Single-neuron flips from $(1, 1, -1)$; all raise the energy, so the state is a local minimum. 229

4 Feature-based word vectorization example. Each word is mapped to a vector of graded semantic features; fractional entries (e.g., 0.5) indicate mixed usage across contexts. Treat this as an intuition pump: real embedding coordinates are learned from data rather than hand-assigned. 292

5 Fuzzy vs. probabilistic reasoning. Distinguishes randomness (probability) from graded concepts (fuzziness) when interpreting uncertainty. 349

6 Boolean vs. fuzzy operators. Mapping from crisp logic rules to graded operators for fuzzy inference. 350

7 Typical operator choices in fuzzy inference and their qualitative effects. Here the t-norm implements fuzzy AND, the s-norm implements fuzzy OR, and the implication shapes consequents. The table supports default-operator selection for a Mamdani-style pipeline and anticipates qualitative output behavior. 393

8 Popular t-norms and their typical roles. Reference when choosing a default conjunction operator and understanding its qualitative behavior. 407

9 Toy GA generation on a bounded interval. One crossover and mutation illustrate how the fitness function guides selection before the next generation. Use this to explain how variation operators interact with selection pressure. 458

10 Big-picture view of model families across the taxonomy and learning paradigms. Each entry represents a family introduced in the book; supervision labels indicate the dominant training signal rather than strict exclusivity. The table serves as a compact lookup for model-family placement across levels and learning signals. 480

Notation and Conventions

Symbol overloads. A small number of symbols are intentionally reused across chapters (for example, $\sigma(\cdot)$ as the sigmoid nonlinearity versus σ as a width/scale parameter). For a one-page index of the most common collisions and the disambiguation rule used in this book, see Appendix A.

This section collects book-wide notation, conventions, and a few reading aids. Symbols may be locally redefined within a chapter when explicitly stated; when a symbol is reused, the local meaning is made explicit in context.

Conventions

Symbol	Description
$\mathbf{x} \in \mathbb{R}^n$	Input vector (features)
$y_{\text{reg}} \in \mathbb{R}$	Regression target (continuous)
$y_{\text{bin}} \in \{0, 1\}$	Binary class label (Bernoulli outcome)
$\hat{y}$	Model prediction
$\mathcal{D}$	Dataset or feasible domain
$\mathcal{L}$	Loss function (objective)
$\sigma(z)$	Sigmoid function $1/(1 + e^{-z})$
$\tanh(z)$	Hyperbolic tangent activation
$\text{ReLU}(z)$	$\max(0, z)$ activation
$\mathbf{W}, \mathbf{b}$	Weights and biases (parameters)
h_t, c_t	Hidden and cell states (RNN/LSTM)
n	Sequence length (tokens)
d_{model}	Model (embedding) width
h	Number of attention heads
d_k, d_v	Per-head key/query and value widths
$\mathbf{Q}, \mathbf{K}, \mathbf{V}$	Query, key, value matrices
$\mathbf{W}_i^Q, \mathbf{W}_i^K, \mathbf{W}_i^V$	Per-head projection matrices
$\mathbf{W}^O$	Output projection after concatenating heads
KV cache	Stored past keys/values for decoding
$\mu_A(x)$	Membership of x in fuzzy set A
T, S	t-norm (AND) and s-norm (OR)
$\text{softmax}(z)$	Normalized exponential mapping
$\lVert \cdot \rVert_2$	Euclidean norm
∇	Gradient operator
η	Learning rate
λ	Regularization strength
k	Number of clusters/classes/neighbors (context-dependent)
$\mathbb{E}[\cdot]$	Expectation
$\text{Var}[\cdot]$	Variance
$\text{diag}(\cdot)$	Diagonal matrix formed from a vector
$\odot$	Hadamard (elementwise) product
$\phi(\cdot)$	Feature map; in kernels, $k(\mathbf{x}, \mathbf{z}) = \phi(\mathbf{x})^\top \phi(\mathbf{z})$
$\mathbf{1}$, $\mathbf{I}$	$\mathbf{1}$: all-ones vector; $\mathbf{I}$: identity matrix

Throughout the book we follow a consistent notational style:

- Bold lowercase ($\mathbf{x}, \mathbf{w}$) denote vectors; bold uppercase ($\mathbf{W}, \mathbf{X}$) denote matrices; plain symbols (e.g., x, y, σ) denote scalars.
- Random variables are written in uppercase (X, Y) when needed, with low-

ercase (x, y) for their realizations.

- Transpose is always indicated by the superscript $\cdot^\top$, as in $\mathbf{W}^\top\mathbf{x}$; we avoid using a plain superscript T to reduce ambiguity.
- The logistic sigmoid is written as $\sigma(z) = 1/(1 + e^{-z})$; the same letter σ without an argument (e.g., σ^2) denotes a standard deviation. The intended meaning is clear from whether an argument is present. In fuzzy chapters, $\mu_A(x)$ denotes membership rather than a mean; context and arguments disambiguate overloaded symbols.
- For activation notation, $f(\cdot)$ is the generic placeholder used when the chapter does not commit to a specific nonlinearity, $\sigma(\cdot)$ is the logistic sigmoid, and $\varphi(\cdot)$ denotes radial-basis responses; $\phi(\cdot)$ remains reserved for generic feature maps.
- Embedding matrices are written in bold $(\mathbf{E}, \mathbf{W}, \mathbf{U})$ and should not be confused with the expectation operator $\mathbb{E}[\cdot]$; when expectations appear, they are always typeset with the blackboard bold $\mathbb{E}$. We use row embeddings by convention (matching the row-major dataset convention used throughout the book): a one-hot row vector $\mathbf{x} \in \{0, 1\}^{1\times|V|}$ selects a row via $\mathbf{xE} \in \mathbb{R}^{1\times d}$.
- Feature maps in kernel methods are written as $\phi(\cdot)$; we reserve $\varphi(\cdot)$ for radial-basis kernels. When probability density functions are needed, we write them as $p(\cdot)$ (or $p_X(\cdot)$ when the variable must be explicit). The corresponding design matrix Φ collects feature-map evaluations on data.

These conventions are occasionally restated in local "Notation note" boxes where multiple meanings could collide (e.g., in chapters on statistics or recurrent networks). As a practical guide: row vectors are written as $1 \times n$ objects, column vectors as $n \times 1$; a *design matrix* collects one data point per row (features in columns), so data matrices are row-major $N \times d$.

Reading Aids

Reading aids: how to read the visuals and what to ask

- **How to read the visuals:** captions state what each color/line style represents (training vs. validation, safe vs. heuristic transformations, etc.), and table captions mention the chapter(s) they support so you can jump back quickly when a later chapter references them.
- **Four recurring questions:**
  1. What is the core scientific idea, and how does it relate to earlier material?
  2. Which methodological cautions should a practitioner keep close at hand?
  3. How do the accompanying figures or derivations anchor those ideas visually?
  4. Where does the topic sit within the broader landscape of intelligent systems?

Author's note: intuition before algebra

The math that follows is intentionally tight, but the spirit of this book is to keep the *reason* for each tool front and center. When I introduce a method, I start from the question an engineer would actually be wrestling with and then introduce the equations needed to make it precise. Read each chapter with that heuristic in mind: first ask what story the technique lets you tell, then check that the derivations honor that story.

Part I: Foundations and the ERM toolbox

1 About This Book

Intelligent systems are engineered artifacts that perceive, reason, and act under constraints. This chapter sets the shared vocabulary (historical context, core definitions, recurring design themes), and Figure 1 shows how the strands connect and where to enter.

Learning Outcomes

- Define what we mean by an *intelligent system* and distinguish it from an *intelligent machine*.
- Place the main strands of modern AI (logic, computation, learning) in a short historical arc.
- Use the book's organizing lenses (components, levels, and the representation→objective→update→audit loop) to situate a method and anticipate its failure modes.

Use this chapter as a standing reference: later tools reuse the same system vocabulary and the same habit of checking assumptions against constraints.

Design motif

The unified contract. Across this book, every intelligent system instantiates the same engineering loop:
(i) Representation: choose what the system stores about the world/problem.
(ii) Update/action: choose how that representation changes (rules, gradients, energy descent, population operators).
(iii) Objective: define what "better" means (loss, likelihood, energy, constraint satisfaction, fitness).
(iv) Audit: define checks that catch silent failure (leakage, miscalibration, instability, constraint violations, metric gaming).
Paradigms differ mainly in the *space being searched* (parameters, rules, programs, prototypes, populations) and the *feedback signal* available (labels, self-supervision, rules, fitness). The rest of the book is a tour of design points under this contract; the back-matter synthesis in Figure 77 and Table 10 summarizes the map once the chapter tour is complete.

What we mean by a model. In this book, a *model* is any system that maps observed information to a prediction about an outcome. Sometimes the prediction is a single value (regression), and often it is a probability distribution over possible outcomes (classification). The details change from chapter to chapter, but the contract stays the same: choose a representation, define what it means to be wrong, and then tune the parameters so the predictions improve under an honest evaluation.

1.1 Historical Foundations of Intelligent Systems

A brief historical sketch helps place intelligent systems within a longer tradition that runs from early mechanical devices, through symbolic logic and computation, to modern machine learning.

Mechanical Automata and Scholastic Logic In the 12th–13th centuries, engineers such as Al-Jazari designed programmable water clocks and mechanical automata whose gears, cams, and valves executed fixed sequences of actions. Although these devices lacked learning or internal models, they embodied the idea

that artifacts could sense (via floats and levers), transform signals mechanically, and act on their environment. In parallel, medieval scholars such as Ibn Sīnā and Thomas Aquinas refined Aristotelian syllogistic logic, systematizing patterns of valid inference even though a fully symbolic notation did not yet exist.

The Mechanical Computer and Early Programming In the 19th century, Charles Babbage designed the mechanical computer now known as the *Analytical Engine*. Ada Lovelace is often cited as one of the first programmers; her notes on the Analytical Engine include an algorithm for computing Bernoulli numbers and helped establish programming as a discipline.

An important (and still practical) lesson from this era is the "garbage in, garbage out" principle: if incorrect input is provided to a computational system, the output will also be incorrect. In modern terms, this is a reminder that data quality and validation are part of the intelligence pipeline, not an afterthought.

Mathematical Logic and Formal Reasoning The symbolic formalism used in modern AI emerged in the 19th and early 20th centuries. Works by George Boole (1847), Gottlob Frege (1879), Giuseppe Peano (1889), and later Bertrand Russell and Alfred North Whitehead (1910–1913) introduced algebraic and predicate-calculus notations that underpin automated reasoning. Formal inference rules such as:

$$\text{If } A = B \text{ and } B = C, \text{ then } A = C.$$

This exemplifies the transitivity of equality—an example of a valid inference rule operating on equality relations—and provides a basis for reasoning systems that manipulate symbols according to formal rules.

The Turing Test and the Birth of AI The mid-20th century marked a pivotal moment with Alan Turing's proposal of the *Turing Test* in 1950. This test was designed to assess a machine's ability to exhibit intelligent behavior indistinguishable from that of a human. The Turing Test shifted the focus from mechanical computation to the broader question of machine intelligence.

Early Machine Learning and Symbolic AI Following the Turing Test, research into machine learning and symbolic AI accelerated. In the 1950s, the perceptron model was introduced as an early neural network capable of binary classification. Around the same time, James Slagle developed an early influential AI program: a symbolic integration system capable of performing calculus operations symbolically rather than numerically (Slagle, 1963). This line of work anticipated themes later formalized in decision procedures for elementary integration (Risch, 1969) and demonstrated that machines could manipulate abstract symbols to solve problems, a core idea in symbolic AI.

Summary of Key Historical Milestones

- **12th–13th Centuries:** Mechanical automata (e.g., Al-Jazari) and scholastic refinements of syllogistic logic.
- **19th Century:** Charles Babbage's Analytical Engine and Ada Lovelace's pioneering programming notes; Boole and contemporaries formalize symbolic logic.
- **Early 20th Century:** Frege, Peano, Russell, and Whitehead develop predicate calculus and logicist foundations.
- **1950:** Alan Turing's Turing Test frames the question of machine intelligence.
- **1950s:** Development of early machine learning models (perceptrons) and symbolic AI programs (e.g., Slagle's integration system).

This historical arc sets the stage for contemporary intelligent systems: programmable artifacts whose behavior is grounded in formal models, implemented on digital hardware, and increasingly trained or tuned from data. The sections that follow make the working definitions and modeling assumptions explicit.

1.2 Defining Artificial Intelligence and Intelligent Systems

Artificial Intelligence (AI) is often misunderstood as merely a collection of popular applications such as image recognition or voice detection. However, these are just subsets of a much broader field. Instead of defining AI by its famous applications, it is more accurate to view AI as a body of collective algorithms,

research, and engineering practice aimed at enabling machines to perceive their environment, perform inference, and take purposeful actions.

Core Definition of AI Following the agent-centric view of Russell and Norvig (2021), artificial intelligence studies computational agents that map percepts to actions through algorithms operating over explicit representations (state graphs, feature vectors, logical predicates, or probabilistic models) subject to domain constraints (physical limits, safety rules, resource budgets). See also Poole and Mackworth (2017) for a complementary treatment focused on agent architectures. Each model we study is evaluated on whether its assumptions support competent *perception* (information acquisition), *reasoning and decision-making* (information processing), and *action* (environment intervention), where a *percept* denotes the data received at a decision epoch (a discrete sensing-and-decision instant; e.g., sensor readings, feature vectors, linguistic tokens) and an *action* denotes the command issued to the environment or downstream system.

Many model-based systems generate hypotheses and test them, yet the field also includes purely reactive controllers (e.g., subsumption architectures in behavior-based robotics or PID loops) that optimize behavior without explicit hypothesis testing. Classic behavior-based robotics research (Brooks, 1986; Arkin, 1998) treats such controllers as intelligent agents. They satisfy the perception–action cycle even in the absence of symbolic reasoning. We flag them as boundary cases: they remain control-theoretic constructs, yet they highlight the continuum between classical control and adaptive AI systems. Throughout this book we discuss both deliberative reasoning (planning, inference, search) and reflexive intelligence (engineered feedback loops that achieve goals without symbolic reasoning), and we try to make clear which lens is being used in a given chapter.

For now, if we adopt a value-centric view of AI, we can characterize intelligent systems by the kinds of questions they help us answer. In practice, three capabilities dominate:

- Explaining the past,
- Understanding the present, and
- Predicting the future.

Framed this way, the parallel with human intelligence becomes explicit: both artificial systems and humans are judged by how well they can reconstruct what has happened, make sense of what is happening, and anticipate what is likely to happen next. For example, humans use memory and narrative to explain past events, situational awareness to understand ongoing interactions, and mental models to predict likely outcomes. Analytic systems that perform root-cause analysis in power grids or credit-risk models in finance primarily *explain the past*; monitoring systems such as anomaly detectors and online recommendation engines focus on *understanding the present*; time-series forecasters and large language models that predict the next token or utterance instantiate the *predicting the future* role. Modern AI architectures often blend these roles, but keeping the three questions in mind provides a useful lens for interpreting model behavior.

To connect this value-centric lens to concrete designs, we now make more precise what we mean by an intelligent system and how a design begins with a clearly stated problem and representation.

1.3 Intelligent Systems

An *intelligent system* is an artificial entity composed of both software and hardware components that:

- Acquire, store, and apply knowledge,
- Perceive and interpret environmental data to maintain situational awareness,
- Make decisions and act based on incomplete or imperfect information.

In contrast, an *intelligent machine* is usually a single embodied device (for example, a robot arm on a factory line) whose sensing, reasoning, and actuation are co-located. Intelligent systems can comprise multiple cooperating machines plus cloud services; intelligent machines are one concrete realization within that broader system-of-systems view.

This working definition is consistent with those used in cyber-physical systems literature and the IEEE Standards Association's descriptions of intelligent agents, emphasizing perception, cognition, and action as the three pillars of autonomy.[1]

[1] Compare with the IEEE Global Initiative on Ethics of Autonomous and Intelligent Systems,

Here, "knowledge" encompasses encoded data sets, learned model parameters, rule bases, and semantic ontologies that the system can query or update during operation. The hardware enables interaction with the environment (e.g., sensors, actuators), while the software performs reasoning and decision-making.

1.3.1 From value-centric questions to concrete designs

The three value-centric questions ("explain the past, understand the present, predict the future") only become actionable once a designer fixes a problem statement, a representation, and the constraints under which the system operates. Rather than treating these as separate case studies, we fold them into a compact design checklist that we reuse whenever it helps structure a design discussion:

1. **Problem definition.** State the task in operational terms. Example: "Detect stop signs quickly enough to enable safe braking." The definition should tell us which of the three value-centric roles dominates (here: understanding the present, plus explaining why braking events occur).

2. **Representation.** Decide how the world will be encoded numerically. Stop-sign detection uses camera images (matrices of intensities) plus metadata such as lane boundaries or GPS position; a financial recommender might rely on structured tabular data.

3. **Objectives and constraints.** Specify the metric to optimize (e.g., minimize false negatives) and the hard constraints (minimum stopping distance, latency budgets, regulatory rules). Practical implementations refine these with regions of interest, masking, or sensor fusion (LiDAR + camera) so the classifier only runs where a stop sign could plausibly appear.

These three ingredients determine what the intelligent system must sense, infer, and control. Once they are in place we can reason about the interacting components that implement the perception → reasoning → action loop.

1.3.2 Components of AI Systems: Thinking, Perception, and Action

AI systems can be decomposed into three interrelated components:

Perception: How the system senses and interprets environmental data, extracting features or state estimates.

Ethically Aligned Design, 1st ed., 2019.

Reasoning and Decision-Making: How the system combines models and control policies with learned value functions to plan actions or react in real time.

Action: How the system executes decisions to affect the environment.

Example: Autonomous Vehicle Perception starts with sensors (for example, cameras) whose signals are converted into numeric arrays and state estimates. Reasoning uses those estimates to classify objects (stop signs, pedestrians) and predict near-future motion. Action closes the loop by issuing steering, acceleration, and braking commands that satisfy safety constraints.

1.4 Case Study: AI-Enabled Camera as an Intelligent System

The design checklist above becomes concrete when we dissect a deployed system. Consider a networked camera that detects humans and escalates alarms in an industrial plant.

Checklist instantiated for the camera system

Problem + value role
Detect humans entering restricted zones (understand the present) and log footage for later audits (explain the past).

Representation
Images are streamed as $H \times W \times 3$ tensors; regions of interest and background models are maintained to suppress noise.

Objectives/constraints
Maintain < 200 ms end-to-end latency and $< 1\%$ false negatives; respect privacy/retention policies.

Hardware (perception/action)
CMOS (complementary metal-oxide-semiconductor) sensor, on-board DSP (digital signal processor)/accelerator, motorized pan/tilt for re-targeting.

Software (reasoning)
A YOLO-style object detector (e.g., YOLOv8 (Jocher et al., 2023)) fine-tuned on site-specific data, fused with Kalman filters for track smoothing and MPC (model predictive control) logic that commands the pan/tilt actuator or triggers alerts.

Integration
Edge inference handles immediate reactions; metadata is sent to a cloud analytics service that enriches logs and retrains models (predicting the future by anticipating recurrent intrusion times).

This decomposition highlights the same three pillars—perception, reasoning, and action—while adding the operational nuance (latency budgets, privacy constraints) that graduate-level systems must address. Many later chapters discuss building blocks that could be used in such a pipeline: convolutional networks (Chapter 11) for visual detection, recurrent models (Chapter 12) for temporal smoothing and sequence prediction, supervised learning and calibration (Chapters 3 to 4) for reliable scoring and threshold selection, fuzzy controllers (Chapters 16 to 18) for rule-based escalation policies, and evolutionary algorithms

(Chapter 19) for tuning design choices such as placement, thresholds, or hyperparameters.

An intelligent system is therefore not just the hardware or the software alone, but the *system of components* working together to perceive, reason, and act under explicit objectives.

1.5 Levels and Architectures of Intelligent Systems

Having introduced working definitions and concrete examples, we now summarize capabilities and architectural patterns that reappear across the book.

What Constitutes Intelligence in Systems? Intelligence in systems is often characterized by the perception–reasoning–action loop, augmented by learning and adaptation. A fuller capability checklist appears later in the Key Characteristics paragraph after the system vignettes.

These capabilities can be realized in various architectures, ranging from connectionist models (e.g., neural networks) to symbolic systems and hybrid approaches.

Levels of Intelligence (as an organizing lens) Intelligence is not necessarily binary (intelligent vs. non-intelligent); rather, deployed systems combine different degrees of reactivity, deliberation, and adaptation. For the purposes of this book we use a four-layer shorthand—reactive systems (level 1), deliberative planners (level 2), adaptive learners (level 3), and meta-cognitive agents that reason about their own policies (level 4)—as an informal organizing lens rather than a strict hierarchy. It is compatible with domain-specific taxonomies (e.g., SAE Levels 0–5 for automated driving)[2]. We return to this lens in the backmatter synthesis (Figure 77 and Table 10), where the full design-space map situates each model family by level, model nature, and learning signal.

Connectionist vs. agent-based/decentralized approaches Two broad paradigms in intelligent system design are:

[2]SAE Levels are defined for driving automation; we cite them here only as an example of a domain-specific capability scale, not as a claimed one-to-one mapping to the four-layer lens used in this book.

- **Connectionist Models:** Systems structured as interconnected processing units (e.g., neural networks) with defined input-output stages.
- **Agent-based or decentralized systems:** Collections of agents or modules that operate semi-independently, often with only local communication, such as swarm intelligence or evolutionary algorithms.

Both approaches have merits and limitations, and hybrid models often combine elements of each.

Example: Swarm Intelligence Swarm systems consist of multiple agents solving subproblems independently but collectively achieving a global objective. Each agent follows simple rules without a global world model, yet the emergent behavior can be intelligent. This contrasts with monolithic systems possessing explicit internal representations.

Swarm intelligence can be formalized via decentralized update laws of the form $\mathbf{x}_i(t+1) = f\big(\mathbf{x}_i(t), \{\mathbf{x}_j(t)\}_{j\in\mathcal{N}_i}\big)$, where each agent i interacts only with its neighborhood $\mathcal{N}_i$. Similar update patterns appear again in Chapter 19, but we do not focus on stability proofs in this book.

Examples of Input and Output Variables in Dynamic Systems To ground these ideas, consider input/output sketches from systems readers often encounter in labs or industry. Each pairs raw sensory cues with actuator or decision outputs.

- **Autonomous quadrotor:**
  - *Inputs:* Inertial measurement unit (IMU) rates, barometer/altimeter, camera or LiDAR features, GPS fixes.
  - *Outputs:* Motor thrust commands and attitude setpoints that regulate yaw/pitch/roll and track waypoints.
- **Smart microgrid:**
  - *Inputs:* Load forecasts, solar/wind availability, electricity prices, state-of-charge estimates for batteries.
  - *Outputs:* Dispatch setpoints (generator outputs, battery charge/discharge, demand-response signals) that balance stability, cost, and

emissions.

- **Building HVAC controller:**
  - *Inputs:* Zone temperature/CO_2/humidity sensors, occupancy estimates, outdoor weather feeds.
  - *Outputs:* Fan speeds, damper positions, valve openings, heat-pump setpoints—tunable levers to meet comfort and energy targets.
- **Robot-assisted surgery:**
  - *Inputs:* Endoscopic vision, force/torque sensing at instruments, surgeon console commands.
  - *Outputs:* Precise tool trajectories, force limits, and safety interlocks that respect tissue constraints.

These vignettes echo a common pattern: intelligent systems fuse heterogeneous sensors to produce calibrated control or decision signals under safety, comfort, or performance constraints.

Emotions as Utility Signals

From a design perspective, emotions can be viewed abstractly as changes in an agent's internal utility or value function: positive affect corresponds to utility gains, negative affect to losses, and social emotions to comparisons between agents' utilities. Artificial systems can mimic this by modulating learning rates, exploration pressure, or safety margins in response to internal "frustration" or "satisfaction" signals without presupposing rich phenomenology. This book treats this view strictly as a modeling device for embedding motivational signals into controllers; affective computing and cognitive science work with much richer state representations than we use here.

Key Characteristics of Intelligent Systems Building on the examples above, we summarize the essential capabilities that characterize an intelligent system:

1. **Sensory Perception:** The system must be able to receive and interpret

inputs from its environment, which may be in various forms such as numerical data, images, sounds, or tactile signals.

2. **Pattern Recognition and Learning:** The system should identify patterns within the input data, including hidden or subtle features, and improve its performance over time by learning from experience.

3. **Knowledge Retention:** Acquired knowledge must be stored and utilized for future decision-making.

4. **Inference from Incomplete Information:** The system should be capable of drawing conclusions and making decisions even when presented with partial or approximate data.

5. **Adaptability:** It must handle unfamiliar or novel situations by generalizing from prior knowledge and adapting its behavior accordingly.

6. **Inductive Reasoning:** The system should be able to generalize patterns from observed examples—i.e., infer general rules or hypotheses from specific data instances (e.g., learn a classifier from labeled data). This differs from applying pre-written conditional logic; induction discovers the rules, whereas conditional statements merely execute them.

Intelligent Systems as Decision Makers At the core, intelligent systems perform a mapping from inputs to outputs, where the outputs represent decisions or actions influenced by the system's internal understanding or model of the environment. Formally, if we denote the input vector by $\mathbf{x} \in \mathcal{X}$ and the output vector by $\mathbf{y} \in \mathcal{Y}$, then an intelligent system implements a function

$$\mathbf{y} = f(\mathbf{x}; \theta), \tag{1.1}$$

where θ represents internal parameters or knowledge that may evolve over time through learning. This abstraction is shared by the diverse examples seen so far: they all ingest data, transform it through a parameterized mapping, and emit decisions or control signals. In this book we primarily use the system-level language (mapping under constraints), and we use "machine" when embodiment and actuation are the point. Because the chapter alternates between these viewpoints, we briefly clarify terminology and the limits of the language we use.

1.6 Intelligent Systems and Intelligent Machines

Terminology Clarification

- **Intelligent System:** A computational system (encompassing its hardware, software, and data interfaces) that perceives its environment, processes information, and acts autonomously or semi-autonomously (with limited human oversight or shared control).
- **Intelligent Machine:** A physical instantiation of an intelligent system, often embodied as a robot or automated device.

The terms are related but not identical; intelligent machines are a subset of intelligent systems, typically emphasizing the physical embodiment.

Behavior, Not Components The word *intelligent* is inevitably anthropocentric: in practice, we judge intelligence through observed behavior and performance under constraints. Motors, sensors, and circuits are enabling components, but they are not "intelligent" on their own. Intelligence emerges from how the full system processes inputs, maintains state, and selects actions.

Intelligence is also not synonymous with optimality. Many deployed systems are approximate, noisy, or biased, yet they can still be meaningfully analyzed and improved as goal-directed agents. In this book, "intelligent" is therefore used in an engineering sense: a system that maps percepts to actions with a design intent, and that can be evaluated against explicit objectives and constraints.

Examples Robots developed by Boston Dynamics (e.g., quadrupeds) illustrate how feedback control, state estimation, and trajectory planning can produce behaviors that humans interpret as intelligent (balance recovery, robust locomotion, disturbance rejection) even though the system has no intrinsic understanding or feelings. Voice-activated assistants and robots provide a second common example: they appear intelligent because they can condition actions on language inputs, maintain limited context, and complete tasks that align with user intent.

Consciousness and Intelligence While machines can exhibit intelligent behaviors, the question of whether they possess consciousness or self-awareness remains open and is a subject of ongoing research and philosophical debate.

In this book we treat consciousness operationally:
we focus on meta-cognition (self-monitoring of one's own decision process) rather than phenomenal awareness. This keeps the discussion tied to observable, designable behaviors rather than philosophical claims.

Author's note: "subject of its own thought"

When I say *strong* machine intelligence, I mean something specific: the system can turn the lens inward. It does not just predict; it also keeps enough self-modeling machinery (confidence monitors, policy checks, explanation traces) to critique and revise *its own* reasoning. This is the machine becoming the subject of its own thought.
The rest of the book uses a simpler four-layer taxonomy (reactive → deliberative → adaptive → meta-cognitive). Keep this "subject of its own thought" lens in the background: it is why Level 4 systems demand extra care in design and governance.

1.7 Levels, Meta-cognition, and Safety

This book uses levels of intelligence as an organizing lens rather than a formal taxonomy. The four levels introduced above (reactive, deliberative, adaptive, and meta-cognitive) are meant to clarify what a system can do, what it must represent, and what kinds of failures are plausible. For a working definition of AI and intelligent systems, see Section 1.3.

Meta-cognition (Operational View) In this book, meta-cognition refers to a controller's ability to monitor, assess, and revise its own reasoning policies. In practice this can look like confidence monitors, audits of decision traces, and bounded self-correction loops rather than unconstrained self-modification.

Implications and Risks If a system can improve its own utility autonomously and rapidly, it may induce competitive dynamics in which improving one utility degrades another's. This occurs in multi-agent settings (competing organizations or robots) and in multi-objective optimization when safety objectives conflict with performance. These scenarios motivate conservative design and governance, especially as systems move from adaptive learning to self-monitoring and policy revision.

Designing Safe Intelligent Systems One practical mitigation is to require auditable decision traces, routine self-inspection/error analysis, and bounded backtracking/self-correction inside explicit, designer-defined interfaces.
Such systems can improve without uncontrolled self-modification: policy updates are gated by testable criteria, and any self-editing of code or reward functions proceeds only through approved interfaces.

Reader's guide. The remainder of this chapter is practical: who the book is for, how chapters fit together, and how to navigate recurring structure. Notation and reading conventions are collected in the front matter (see *Notation and Conventions*).

1.8 Audience, Prerequisites, and Scope

This material has been rewritten to stand on its own as a book. It surveys the design and analysis of intelligent systems along two main strands:

- data-driven models for prediction and decision making (linear models, kernels, deep networks, sequence models, Transformers);
- soft-computing and search methods (self-organizing maps, fuzzy systems, evolutionary and genetic algorithms).

The emphasis is on breadth with enough mathematical depth that you can relate ideas across chapters rather than treating each technique in isolation.

The book also maintains a deliberate dual emphasis: representation learning with neural and kernelized models on one hand, and soft-computing approaches (fuzzy systems and evolutionary optimization) on the other. This balance keeps robustness, interpretability, and optimization themes all in view rather than treating deep networks in isolation.

The assumed background is undergraduate calculus and linear algebra (vectors, matrices, eigenvalues) and basic probability and statistics. No prior dedicated AI or machine-learning course is assumed: key ideas such as losses, optimization, gradient descent, backpropagation, kernels, fuzzy operators, and evolutionary operators are introduced from first principles when they first appear. Familiarity with signals and systems, and with linear time-invariant (LTI) models in particular, is helpful for the sequence-modeling and control-oriented parts of the book; Appendix D (*Linear Systems Primer*) provides a concise refresher.

1.9 Roadmap and Reading Paths

Figure 1 summarizes the narrative arc of the book: a core supervised path (linear and logistic regression to MLPs to CNNs to RNNs and attention/Transformers), a branch through competitive learning and fuzzy inference for rule-based reasoning, and a parallel thread on optimization culminating in evolutionary computing. Early on, Chapter 2 provides a complementary symbolic-search perspective so we can contrast "intelligence via transformations" with ERM-based modeling. Chapters cross-reference one another so you can skim the path most relevant to your project and return for foundational refreshers as needed.

Readers arrive with different goals. The roadmap is intentionally a dependency graph rather than a single linear track; the following paths are common starting points:

1. **ML-focused path:** Chapters 3 to 4 → Chapters 6 to 7 → Chapters 11 to 13 → Chapter 14.
2. **Control/systems path:** Chapters 1 to 3 → Chapter 10 → Chapters 15 to 18 → Chapter 19.
3. **Soft-computing path:** Chapters 1 to 3 → Chapters 15 to 18 with optional detours to Chapter 6 and Chapter 19.

Reinforcement learning is a major pillar of modern AI, but it is not a primary focus of this book; for a dedicated treatment, see (Sutton and Barto, 2018).

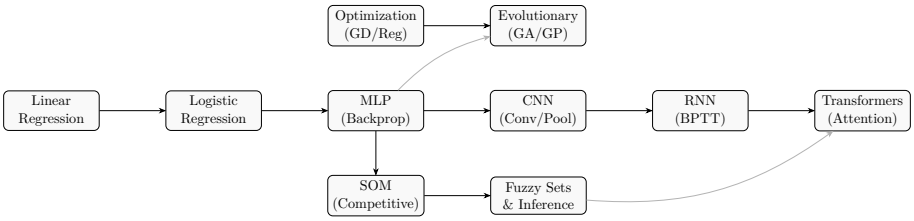


Figure 1: Roadmap of the book strands (core supervised path through attention models; SOM/fuzzy; optimization/evolutionary). It also serves as a quick prerequisite map when you want to jump ahead and later backfill foundations.

1.10 Using and Navigating This Book

- **Before each chapter:** skim the Learning Outcomes and check where it sits on the Roadmap.
- **While reading:** follow cross-referenced figures/equations and pause at the short checkpoints and worked examples.
- **After each chapter:** review the Summary and Common Pitfalls; revisit the pseudocode and attempt the Exercises.
- **Keep the front matter close:** the Notation and Conventions section defines symbols reused throughout the book; cross-references point back to it when conventions matter.

Conventions and reading aids. The front matter summarizes notation and common conventions, and it also includes a short guide to reading figures and recurring box styles.

Key takeaways

Minimum viable mastery

- Intelligent systems integrate perception, decision, and action; we judge them by behavior under constraints, not by anthropomorphic language.
- The system–machine distinction is mostly about embodiment: intelligent machines are physical instances of intelligent systems.
- "Levels" are an organizing lens; meta-cognition is treated operationally (self-monitoring and bounded self-correction), and audits are part of the design, not an afterthought.

Common pitfalls

- Treating the roadmap as a single linear syllabus rather than a dependency graph of prerequisites.
- Confusing "probability" with "decision": many later failures come from thresholds and costs, not from modeling.
- Letting notation drift: keep Appendix A close when symbols are reused in different chapters.

Ethical considerations Even a technically correct model can be harmful when it is deployed as a decision rule. When systems touch people, treat audits (slices, calibration, robustness, and constraint checks) as part of the design contract, not as optional polish.

Exercises

Exercise 1 (Easy). Pick one engineered system you know (a recommender, a robot, a control loop, a monitoring pipeline) and write its design in the book's vocabulary: *percepts*, *representation*, *actions*, *objective*, and *constraints*.
Deliverable: a one-page block diagram + 5 bullets (one per term) written concretely for your system.

Exercise 2 (Medium). Choose one plausible failure mode for that system that would *not* show up as a single headline metric drop (a silent failure). Define an audit that would expose it (slice, calibration check, constraint test, or robustness probe).
Deliverable: the failure-mode description + an explicit test procedure + the pass/fail criterion.

Exercise 3 (Medium). Use Figure 1 to justify a learning path for your system: which representation family would you start with (symbolic, supervised, neural, fuzzy, evolutionary) and why?
Deliverable: a short paragraph that names (i) the representation, (ii) the update/search operator, (iii) the feedback signal, and (iv) the first audit you would run.

If you are skipping ahead. Keep the operational vocabulary (representation, actions, objective/goal test, and audit/checks) and the roadmap (Figure 1) in mind; later chapters assume these as the organizing lens for both symbolic and data-driven tools.

Where we head next. Chapter 2 grounds this vocabulary in a compact case study: symbolic integration as transformation search. The example shows decomposition, safe rewrites, heuristic branching, backtracking, and residual checks in one place; the supervised chapters then reinterpret the same loop with losses, parameter updates, and validation audits. Keep this continuity in view: the implementation changes, but the engineering questions (state, action, objective, verification) stay the same through fuzzy reasoning and evolutionary optimization.

2 Symbolic Integration and Problem-Solving Strategies

Chapter 1 treated "intelligence" operationally: represent a problem, choose actions, and verify or correct under constraints. Here we make that concrete with a deliberately small example—symbolic integration—where the system's actions are algebraic transformations. Figure 1 places this symbolic strand alongside the data-driven path, and we use it as a rehearsal for the same audit discipline that appears later in ERM, fuzzy inference, and evolutionary search.

We study symbolic problem solving through *integration-by-transformation*: preserve meaning while rewriting an expression into a form that exposes a solution. The goal is not a catalog of tricks, but the system pattern: explicit representations, meaning-preserving actions, heuristic branching with backtracking, and a goal test that certifies correctness.

We separate transformations into two tiers: reliable moves that never change the antiderivative class (factoring constants, linear substitutions, polynomial division) and heuristic moves that may succeed only for certain structures. A practical policy is to exhaust safe steps first, then branch judiciously through the heuristic catalog while keeping enough state to backtrack.

Learning Outcomes

- Decompose symbolic integration problems into safe vs. heuristic transformations and understand when each is appropriate.
- Trace the transformation-tree search (state save/restore, heuristics, termination) and connect it to broader notions of intelligent problem solving.
- Contrast symbolic transformation search with data-driven modeling pipelines and identify what each paradigm contributes.

Design motif

Meaning-preserving moves plus a mechanical check: if $F(x)$ is proposed as an antiderivative of $f(x)$, then differentiating should recover the integrand on the declared domain, i.e., $F'(x) = f(x)$ (equivalently, $F'(x) - f(x) = 0$).

2.1 Context and Motivation

Consider the task of solving an integral of the form

$$\int f(x)\,dx,$$

where $f(x)$ may be a complicated function. Traditional approaches often rely on consulting integral tables or applying well-known formulas. For example, integrals such as

$$\int \frac{1}{x}\,dx = \ln|x| + C.$$

These are straightforward and can be solved by direct lookup or simple substitution.

However, many integrals encountered in practice do not match any entry in standard integral tables, nor do they succumb easily to elementary techniques. The integration-by-transformation view treats this as a search problem: propose meaning-preserving rewrites, backtrack when a branch gets harder, and verify candidates mechanically by differentiation.

Aside: the Risch algorithm

The Risch algorithm (Risch, 1969) decides whether an elementary antiderivative exists for a large class of integrands by reducing the problem to algebra over differential fields. Unfortunately its implementation is intricate, requires case explosions, and still leaves many useful nonelementary functions unresolved. Practical computer algebra systems therefore augment Risch-style decision procedures with heuristic transformations—the focus of this chapter—to keep runtimes bounded and to return human-readable answers when possible.

2.2 Problem Decomposition and Transformation

A key insight in tackling complex integrals is to *reduce* the problem into manageable subproblems. This involves applying *transformations* to rewrite the integral into a form that is either directly solvable or closer to known forms.

Safe Transformations We define *safe transformations* as invertible substitutions that allow back-substitution: if $u = \phi(x)$ and F_u is an antiderivative of

$T[g](u)$, then $F_u \circ \phi$ differentiates back to $g(x)$. We write F for antiderivatives of f-type terms and G for antiderivatives of g-type terms, so $F'(x) = f(x)$ and $G'(x) = g(x)$. Safe transformations are algebraic substitutions or factorings that survive reversal. Examples include:

- **Constant factor extraction:** If G is an antiderivative of g, then ag has antiderivative aG; differentiating confirms $\frac{d}{dx}[aG(x)] = ag(x)$.
- **Linear substitution:** Let $u = ax + b$ with $a \neq 0$. Differentiating gives $du = a\,dx$ and hence $dx = \frac{du}{a}$; substituting shows that

$$\int f(ax+b)\,dx = \frac{1}{a}\int f(u)\,du. \tag{2.1}$$

  This is the standard change-of-variables formula.
- **Polynomial division:** If $p(x)$ and $q(x)$ are polynomials with $\deg p \geq \deg q$, then perform polynomial division:

$$\frac{p(x)}{q(x)} = s(x) + \frac{r(x)}{q(x)},$$

  where $\deg r < \deg q$. Linearity of integration lets us integrate $s(x)$ term-by-term, while the proper fraction $\frac{r(x)}{q(x)}$ can be addressed via partial fractions or further substitutions, yielding an equivalent antiderivative.

These transformations are *safe* because they always preserve the integral's value and simplify the problem without introducing ambiguity. When a substitution transforms an integral over $x \in [0,1]$ into $\int_0^1 u^b(1-u)^c\,du$ with $b, c > -1$, the resulting definite integral evaluates to a Beta function $B(b+1, c+1)$; the Beta identity applies to that definite integral on $[0,1]$, and it is therefore customary to fall back on it when an elementary antiderivative is unavailable.

Example: Applying Safe Transformations Suppose we have an integral of the form

$$\int a \cdot x^b(1-x)^c\,dx,$$

where a, b, c are constants. A safe transformation might be to factor out the constant a and then consider substitutions or binomial expansions that reduce

the powers to known integrals (e.g., Beta-function evaluations when b and c are integers).

Limitations of safe transformations Safe transformations preserve meaning and allow back-substitution, but they are not a complete solution strategy: after exhausting the invertible rewrites you know, you can still end up with an integrand that does not match any template you can finish. At that point the system has to make an explicit decision: stop (declare "unknown" under the current rule set), or branch into heuristic moves that may succeed but are not guaranteed.

2.3 Heuristic Transformations

When safe transformations fail to yield a solution, we turn to *heuristic transformations*, which are not guaranteed to succeed but often provide a path forward. These heuristics are based on experience, pattern recognition, and mathematical intuition.

Definition Heuristic transformations are problem-solving *tricks* or *strategies* that attempt to rewrite the integral into a solvable form by exploiting structural properties of the integrand. They may involve:

- Trigonometric identities and substitutions, e.g., using relationships among $\sin x$, $\cos x$, $\tan x$, $\cot x$, $\sec x$, and $\csc x$.
- Algebraic manipulations that simplify complicated expressions.
- Variable substitutions that transform the integral into a standard form.
- Recognizing patterns such as functions of $10x$ or other scaled arguments and applying appropriate scaling substitutions (e.g., if the integrand contains $f(cx)$, introduce $u = cx$ so that the scale factor is absorbed).

Example: Trigonometric Heuristics Consider an integral involving sine and cosine:

$$\int \frac{\sin x}{\cos x}\, dx.$$

Recognizing that $\sin x/\cos x = \tan x$, we can rewrite the integral as

$$\int \tan x \, dx = -\ln|\cos x| + C.$$

which is a standard integral with the constant of integration explicitly noted.

Similarly, if the integrand involves expressions like $\sin^2 x + \cos^2 x$, we can use the Pythagorean identity to simplify.

Heuristics as a Form of Intelligence The use of heuristic transformations reflects a form of mathematical intelligence: the ability to recognize patterns, apply non-obvious substitutions, and creatively manipulate expressions to reach a solution. Unlike safe transformations, heuristics may fail or lead to dead ends, but they expand the problem-solving repertoire beyond mechanical procedures.

Absolute values and branches

- **Square roots:** specify the sign/branch. If you drop $|\cdot|$, restrict the substitution interval so the sign is fixed (e.g., $\cos y \geq 0$ on $y \in (-\pi/2, \pi/2)$ so $\sqrt{1-\sin^2 y} = \cos y$).
- **Logarithms:** default to $\log|f(x)|$ unless you guarantee f keeps one sign on the declared domain.
- **Arctrig/hyperbolic inverses:** state principal values and any periodicity you rely on for back-substitution.

2.4 Summary of the Approach

The overall strategy for symbolic integration can be summarized as follows:

1. **Apply all safe transformations** to simplify the integral and attempt to match known solvable forms.
2. **Re-evaluate the transformed integrand** to identify structural cues (symmetry, polynomial degree, trigonometric patterns).
3. **Choose among multiple transformation paths** by comparing simple cost heuristics such as expression-tree depth, number of nonzero coefficients, or anticipated integration rules.

4. **Fallback to heuristics and backtracking** when safe transformations stall, maintaining a stack of previous states to enable systematic exploration.

Cost heuristic. Score candidates by a triple: tree depth, number of nonlinear operators, and symbol count. The nonlinearity term counts transcendental nodes (e.g., trigonometric, exponential, logarithmic), while the symbol count tallies nodes in the expression's abstract syntax tree (AST), excluding simple literals such as $-1, 0, 1$. Prefer branches that reduce or preserve this triple, and break ties toward rules with known templates (e.g., partial fractions, reduction formulas).

2.5 Heuristic Transformations: Revisiting the Integral with $1 - x^2$

Recall the integral under consideration:

$$\int \frac{4}{(1-x^2)^{5/2}}\, dx. \tag{2.2}$$

For real-valued integration we restrict attention to $x \in (-1, 1)$, ensuring the denominator $(1 - x^2)^{5/2}$ is well-defined and nonzero on the interval of interest. The antiderivative derived below is valid on this open interval; the integrand has non-integrable singularities at $x = \pm 1$.

When encountering expressions involving $1 - x^2$, a classical heuristic substitution is the trigonometric setup

$$\begin{aligned} x &= \sin y, \\ dx &= \cos y\, dy, \\ 1 - \sin^2 y &= \cos^2 y. \end{aligned} \tag{2.3}$$

which leverages the Pythagorean identity in a form that immediately exposes a secant power after substitution. This is the standard move for denominators built from $1 - x^2$: $x = \sin y$ turns $1 - x^2$ into $\cos^2 y$.

Applying this substitution transforms the integral into a trigonometric form that is often easier to handle.

Step 1: Substitution and Differential Set $x = \sin y$, so $dx = \cos y\, dy$ as in (2.3). We take $y = \arcsin x$ with $y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ so that the substitution remains bijective on $x \in (-1, 1)$. Geometrically, this is the principal branch of arcsin; on it $\cos y \geq 0$, so $\sqrt{1 - \sin^2 y} = \cos y$ has no sign ambiguity.

Substituting into (2.2) and using $dx = \cos y\, dy$ yields

$$\begin{aligned}
\int \frac{4}{(1-x^2)^{5/2}}\, dx &= \int \frac{4}{(1-\sin^2 y)^{5/2}} \cos y\, dy \\
&= \int \frac{4\cos y}{(\cos^2 y)^{5/2}}\, dy \\
&= \int \frac{4\cos y}{\cos^5 y}\, dy \qquad (\text{since } \cos y \geq 0 \text{ on } (-\tfrac{\pi}{2}, \tfrac{\pi}{2})) \\
&= 4 \int \cos^{-4} y\, dy \\
&= 4 \int \sec^4 y\, dy.
\end{aligned}$$

The exponent arithmetic is now explicit: $(\cos^2 y)^{5/2} = \cos^5 y$ on the chosen branch, and $\cos y \cdot \cos^{-5} y = \cos^{-4} y$.

Thus, the integral reduces to

$$4 \int \sec^4 y\, dy. \tag{2.4}$$

Step 2: Choosing the Next Transformation At this stage, two common safe transformations are available:

- Express $\sec^4 y$ in terms of $\tan y$, using the identity $\sec^2 y = 1 + \tan^2 y$, and then perform substitution $u = \tan y$ with $du = \sec^2 y\, dy$.
- Use reduction formulas for powers of secant directly, e.g.,

$$\int \sec^n y\, dy = \frac{\sec^{n-2} y \tan y}{n-1} + \frac{n-2}{n-1} \int \sec^{n-2} y\, dy, \quad n > 1. \tag{2.5}$$

Standard reduction formulas provide a deterministic alternative if the substitution path is judged too costly.

The choice between these paths is nontrivial, especially for an automated system. Humans often pick the substitution $u = \tan y$ intuitively because it

simplifies the integral, but a machine requires a deterministic decision rule.

Step 3: Functional Composition and Path Selection To automate the choice, the system applies the cost heuristic defined above (tree depth, nonlinearity, symbol count) to the two safe paths listed in Step 2; the substitution route typically scores lower and is tried first, with the reduction formula held as a fallback if substitution stalls.

Two safe options from here

- **(a) Substitution $u = \tan y$.** Since $\sec^4 y\,dy = \sec^2 y\,(\sec^2 y\,dy)$ and $\sec^2 y = 1 + \tan^2 y$, set $u = \tan y$, $du = \sec^2 y\,dy$:

$$\begin{aligned} 4\int \sec^4 y\,dy &= 4\int \sec^2 y\,(\sec^2 y\,dy) \\ &= 4\int (1+\tan^2 y)\sec^2 y\,dy \\ &= 4\int (1+u^2)\,du \\ &= 4\left(u + \tfrac{u^3}{3}\right) + C \\ &= 4\tan y + \tfrac{4}{3}\tan^3 y + C. \end{aligned}$$

- **(b) Reduction formula.** For even $n > 1$,

$$\int \sec^n y\,dy = \frac{\sec^{n-2} y \tan y}{n-1} + \frac{n-2}{n-1}\int \sec^{n-2} y\,dy.$$

  Applying this with $n = 4$ to the running integral (2.4) gives

$$\begin{aligned} 4\int \sec^4 y\,dy &= 4\left(\frac{\sec^2 y \tan y}{3} + \frac{2}{3}\int \sec^2 y\,dy\right) \\ &= 4\left(\frac{(1+\tan^2 y)\tan y}{3} + \frac{2}{3}\tan y\right) + C \\ &= 4\left(\tan y + \tfrac{1}{3}\tan^3 y\right) + C \\ &= 4\tan y + \tfrac{4}{3}\tan^3 y + C, \end{aligned}$$

  matching the substitution branch exactly.

Back-substitution and check With F denoting the antiderivative of f throughout this chapter, back-substitution is immediate from the chosen branch:

$$\cos y = \sqrt{1-\sin^2 y} = \sqrt{1-x^2}, \qquad \tan y = \frac{\sin y}{\cos y} = \frac{x}{\sqrt{1-x^2}}.$$

The positive square root is the correct one because $\cos y \geq 0$ on $y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Using this identity, both paths yield:

$$F(x) = 4\left[x(1-x^2)^{-1/2} + \frac{x^3}{3(1-x^2)^{3/2}}\right] + C. \tag{2.6}$$

Differentiating (2.6) term by term gives

$$\begin{aligned} F'(x) &= 4\Big[(1-x^2)^{-1/2} + x^2(1-x^2)^{-3/2} \\ &\qquad + x^2(1-x^2)^{-3/2} + x^4(1-x^2)^{-5/2}\Big] \\ &= 4(1-x^2)^{-5/2}\Big[(1-x^2)^2 + 2x^2(1-x^2) + x^4\Big] \\ &= 4(1-x^2)^{-5/2}, \end{aligned}$$

so the candidate antiderivative passes the residual check on any interior sample set from $(-1,1)$. Outside $(-1,1)$ the principal-branch integrand is complex-valued; a real continuation rewrites the integral as $\int 4(x^2-1)^{-5/2}dx$ with $x = \cosh t$.

Pattern rule For integrals of the form $\int (1-x^2)^{-k-1/2}dx$, the substitution $x = \sin y$ reduces them to $\int \sec^{2k} y\, dy$; apply the even-power reduction accordingly. This is why the $1-x^2$ pattern triggers the trigonometric branch.

Example: solving an integral via transformation trees The worked integral above already showed branch behavior (e.g., $x = \sin y$ vs. $x = \tanh u$) under domain constraints. We keep that example as a template and now make the search logic explicit: domain-incompatible heuristic branches are pruned, safe branches can converge to the same antiderivative, and residual checks provide the goal test at each accepted leaf.

2.6 Transformation Trees and Search Strategies

Definition: A **transformation tree** is a conceptual structure representing all possible sequences of transformations applied to an expression in an attempt to solve or simplify it.

- Each node corresponds to a state of the expression.
- Edges correspond to transformations (safe or heuristic).
- Leaves correspond to either solved expressions or dead ends (no solution).

Figure 2 shows the actual tree explored for $\int \frac{4}{(1-x^2)^{5/2}}\,dx$. Solid branches denote safe algebraic steps (guaranteed progress), while dashed branches illustrate heuristic substitutions that may fail and trigger backtracking. Computer algebra systems follow similar playbooks (Bronstein, 2005; Risch, 1969): a Risch-style decision core handles provably solvable cases, while a curated bank of heuristics (pattern rewrites, rational substitutions, special-function fallbacks) explores auxiliary branches with explicit depth/time budgets.

Example: For the integral problem, the root node is the original integral. From there, branches encode concrete substitutions and algebraic manipulations, and failed heuristic paths explicitly return to the previous decision point.

Safe vs. Heuristic Transformations:

- **Safe transformations** are guaranteed to preserve equivalence and progress towards a solution.
- **Heuristic transformations** may or may not lead to a solution; they are attempts that carry risk but can be beneficial.

Backtracking: If a branch leads to no solution, the system must backtrack to a previous node and try alternative transformations. This requires the ability to:

- *Freeze* the current state before branching.
- *Restore* previous states upon failure (e.g., by pushing serialized expression trees and associated metadata onto a stack for later reinstatement).

In practice this corresponds to pushing serialized expression trees (i.e., deep copies of the tree structure together with any transformation metadata) onto a stack so they can be reinstated after unsuccessful exploratory steps.

2.7 Algorithmic Outline for Symbolic Problem Solving

The general algorithm for solving symbolic problems such as integrals can be summarized as follows:

1. **Define the goal:** For example, express the integral in terms of known functions from a table.
2. **Enumerate transformations:** List all possible safe and heuristic transformations applicable to the current expression.
3. **Apply safe transformations:** Attempt all safe transformations and check if the problem is solved.
4. **If not solved, apply heuristic transformations:** Attempt heuristic transformations to explore alternative paths; common template hits include $\int f'(x)/f(x)\,dx \to \log|f(x)| + C$ and $\int r'(x)e^{r(x)}\,dx \to e^{r(x)} + C$.
5. **Branch and backtrack:** For each transformation, branch the search tree. If a branch fails, backtrack and try other branches.
6. **Use heuristics to guide search:** For example, use functional composition depth or cost metrics to prioritize branches.
7. **Terminate cleanly:** Stop when a closed-form antiderivative is found, or when depth/time budgets are exceeded without success; optional numeric residual tests can accept approximate solutions.

Note: This approach resembles a *greedy search* with backtracking, but it does not guarantee an optimal or even successful solution in all cases.

Transformation-tree search (pseudocode)

```
function SolveIntegral(f0,
                       domain=(-1,1),
                       depth_limit=8,
                       time_limit=2s,
                       eps_abs=1e-6,
```

```
                        eps_rel=1e-6,
                        samples=24):
    stack <- [(f0, domain, empty_history, depth=0)]
    start <- clock()
    best  <- {status="fail",
              F=None,
              residual=None,
              history=[],
              domain=domain}
    while stack not empty:
        current, dom, history, depth <- pop(stack)
        if clock() - start > time_limit
           or depth > depth_limit:
            continue
        if passes_residual_test(current, f0, dom,
                                eps_abs, eps_rel, samples):
            res = residual(current, f0, dom, samples)
            return {status="closed_form",
                    F=current,
                    residual=res,
                    history=history,
                    domain=dom}
        safe, heuristic <- enumerate_transforms(current, dom)
        for T in safe:
            g, new_dom <- apply(T, current, dom)
            push(stack,
                 (g, new_dom, history + [T], depth+1))
        for H in heuristic (ordered by cost)
             when depth+1 <= depth_limit:
            g, new_dom <- apply(H, current, dom)
            push(stack,
                 (g, new_dom, history + [H], depth+1))
    return best
```

The pseudocode mirrors the narrative: record the original integrand f_0, track the current domain/branch, apply safe transforms eagerly, explore heuristic branches within time/depth budgets, and only accept a candidate antiderivative once a sampled residual check (absolute or relative) passes on points inside the declared domain; return a clear status/history/domain either way.

Residual test implementation. At each candidate F, sample K points inside the current domain but away from singularities and branch points, and form $R = \max_i |F'(x_i) - f_0(x_i)|$. Accept if $R \le \varepsilon_{\text{abs}}$ or $R/(1 + \max_i |f_0(x_i)|) \le \varepsilon_{\text{rel}}$. Automatic differentiation or a high-order finite difference (step $h = O(\sqrt{\varepsilon_{\text{machine}}})$) keeps the check numerically stable. As substitutions shrink the domain, shrink the sample set and log the updated interval alongside the history.

Termination policies and numeric fallbacks

- **Budgeting:** Cap depth, number of heuristic branches, and runtime (e.g., depth limit $D = 8$, two-second wall clock). When limits are reached, report "no elementary antiderivative within budget D" rather than looping forever.
- **Residual checks:** Differentiate candidate antiderivatives symbolically and numerically. Sample points inside the declared domain (away from poles) and accept only if $\max_i |F'(x_i) - f(x_i)| \le \varepsilon_{\text{abs}}$ or the relative tolerance passes; otherwise prune or refine before returning.
- **Numeric escape hatch:** Switch to adaptive quadrature (e.g., Gauss–Kronrod) once symbolic attempts fail; return both the numeric estimate and the failed transformation history so users can adjust heuristics, noting that the numeric value is not a closed form.
- **Domain reminders:** When substitutions shrink domains (e.g., $x = \sin y$ enforces $|x| \le 1$), log the restriction and branch choice so the numeric fallback samples within the valid range and the report is reproducible.

Transformation toolkit (safe vs. heuristic)

Preconditions. Keep domains and branch choices explicit. For example, a square-root substitution may require $x \in (-1, 1)$. Assume principal branches unless noted.

- **Constant factor. Safe:** $\int a\, g(x)\, dx = a \int g(x)\, dx$ with no domain change. **Heuristic:** complete the square before trig substitutions, and track any branch cuts you introduce.
- **Linear substitution. Safe:** $u = ax + b,\ dx = du/a$ with $a \neq 0$, invertible on the stated interval. **Heuristic:** trig substitutions such as $x = \sin u$, $x = \tan u$, or $t = \tan(x/2)$, with stated branch intervals.
- **Polynomial division / partial fractions. Safe:** split an improper rational function into polynomial plus proper fraction while keeping denominators nonzero on the domain. **Heuristic:** rationalizing substitutions such as $x = 1/u$ or $x = u^2$ to expose symmetry, while avoiding new zeros or poles.
- **Log-derivative pattern. Safe:** $\int f'(x)/f(x)\, dx = \log|f(x)| + C$ when $f(x) \neq 0$ on the declared domain. **Heuristic:** template lookups such as Beta/Gamma forms or exponential-times-polynomial motifs, with parameter checks.

Worked example: Beta template vs. numeric fallback Consider the Beta integral

$$I(a, b) = \int_0^1 x^{a-1}(1-x)^{b-1}\, \mathrm{d}x, \qquad a, b > 0.$$

Safe transformations (factor constants, recognize the Beta template) immediately identify the elementary value $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$. By contrast, the perturbed integral

$$\int_0^1 x^{a-1}(1-x)^{b-1} \log(1+x)\, \mathrm{d}x$$

fails the template check after all safe moves. The solver therefore (i) records the unmet template, (ii) pushes a heuristic branch such as differentiation under the integral sign, and (iii) if the branch exceeds time/depth budgets, falls back to adaptive quadrature with the reported residual $|I_{\text{numeric}} - I_{\text{candidate}}|$. This concrete pattern—try Beta/Gamma reduction, else return a certified numeric answer—embodies the policy described in the termination box.

Failure path with certified numeric residual. Setting $a = \frac{3}{2}, b = 2$ in the perturbed integral above illustrates the full fallback. Safe moves reduce the plain Beta integral to $B(3/2, 2) = 4/15$, but the extra $\log(1 + x)$ term triggers every heuristic branch (integration by parts, differentiation under the integral sign, series expansion) without yielding a closed form before the default depth limit $D = 8$. The solver then hands the integrand to an adaptive Gauss–Kronrod routine, which returns $I_{\text{numeric}} \approx 0.0915453885$ with an internal error certificate $< 3 \times 10^{-7}$; this is a certified quadrature value rather than a closed form. The residual check

$$|I_{\text{numeric}} - I_{\text{previous refine}}| \leq 3 \times 10^{-7}$$

is attached to the report along with the failed transformation history, making it explicit that no elementary antiderivative was located within the allotted budget even though a numerically reliable answer exists.

2.8 Discussion: What this example illustrates

Under the operational framing in Chapter 1, the integrator exhibits several ingredients associated with intelligent problem solving: it maintains an explicit state (the current expression), chooses actions (transformations), manages contingencies (branching and backtracking), and verifies results with a crisp goal test (differentiate and check the residual). At the same time, it is limited: it does not learn new transformations from data, and its effectiveness depends on a human-designed library of moves and heuristics.

Not every heuristic is helpful. For instance, applying $x = \tan y$ to $\int (1 + x^2)^{3/2}\, dx$ looks attractive because $1 + \tan^2 y = \sec^2 y$, yet it transforms the problem into $\int \sec^5 y\, dy$, which is more complicated than the original integral. In a transformation-tree implementation this branch simply backtracks and explores alternatives (e.g., $x = \tanh u$ for $|x| < 1$ or $x = \cosh u$ for $|x| > 1$), underscoring

why explicit search discipline and residual checks are essential.

This contrast helps position the data-driven chapters that follow, where the system's "actions" are parameter updates guided by loss functions and validation checks.

Connection to statistical learning. Symbolic integration is a clean playground for thinking about representations, action sequences, and verification. In data-driven modeling, the objects change (datasets, models, and losses), but the system-level pattern is similar: choose a hypothesis class, optimize an objective under resource constraints, and validate that the result generalizes.

Connection: transformation search vs. empirical risk minimization

- **Goal test:** residual check $\max_{x \in S} |F' - f| \leq \varepsilon$ vs. performance on held-out data.
- **Inductive bias:** safe/heuristic precedence vs. model class and regularization that shape what is learnable.
- **Budget:** depth/time limits vs. compute/epoch budgets and early stopping.

Key takeaways

Minimum viable mastery

- Symbolic integration is a compact example of a goal-driven system: represent state, apply meaning-preserving actions, and verify outcomes.
- Safe moves encode guaranteed transformations; heuristic moves trade certainty for coverage and require backtracking discipline.
- Residual checks act as a crisp goal test: differentiate a candidate and measure whether it agrees with the original integrand on the declared domain.

Common pitfalls

- Losing the system-level point in calculus detail: always name the state, action, heuristic, and goal test.
- Ignoring domains/branches: substitutions can shrink domains, introduce branch cuts, and invalidate a "correct" algebraic rewrite.
- Treating heuristics as proofs: heuristic branches must be verified (or backtracked), not trusted.

Exercises

Exercise 1 (Easy). Implement a small rule set of *safe transformations* (constant extraction, linear substitution, polynomial division).
Deliverable: for each proposed antiderivative F, report a residual check $F'(x) - f(x)$ on a declared domain.

Exercise 2 (Medium). Add one heuristic transformation (e.g., integration by parts, a trig identity, or a pattern match) and implement backtracking when a branch stalls.
Deliverable: a trace showing (state, chosen action, backtrack points) on 3 integrals.

Exercise 3 (Medium). Design one adversarial example where a heuristic rewrite *looks* plausible but fails a domain/branch condition.
Deliverable: the counterexample + the exact check that catches it.

If you are skipping ahead. Keep the pattern vocabulary: safe vs. heuristic moves, explicit budgets, and residual/verification checks. The data-driven chapters reuse the same discipline (objective, constraints, and validation) even though the "actions" become parameter updates; the fuzzy and evolutionary chapters reuse it again when tuning operators and rule systems under constraints.

Where we head next. For the data-driven thread (datasets, objectives, diagnostics, classification), continue to Chapters 3 and 4. For nonlinear function classes and nonconvex training dynamics, continue through Chapters 5 to 6. Later, when you reach fuzzy inference and evolutionary search, reuse this chapter's core habit: define admissible moves, track budgets, and verify outcomes against explicit criteria.

3 Supervised Learning Foundations

Chapter 2 illustrated a non-statistical lens: solve problems by transformation search, with explicit goal tests. Here we shift to the data-driven lens. Figure 1 marks this as the core supervised strand, but the continuity is intentional: we still specify state, define actions, set a budget, and verify outcomes.

When a mapping is known, we use the formula: Celsius and Fahrenheit are linked by a simple rule, and many physical laws give direct input–output relationships. The problems that motivate machine learning do not. The mapping is unknown, messy, or only partially understood, so we settle for an approximation. That approximation might be statistical (learned from data), rule-based (encoded from experience), biologically inspired (neural computation), behavioral (fuzzy rules), or evolutionary (search over candidate solutions). This chapter focuses on the statistical, data-driven strand: supervised learning.

In this lens, supervised learning is prediction and inference: given evidence $\mathbf{x}$, estimate an output y that you can act on or audit. Other modeling goals exist (summarizing structure, compressing representations, discovering clusters), but supervised learning is the cleanest place to learn how to fit models, compare alternatives, and check whether apparent success is real or just memorization.

Supervised learning begins with three commitments: choose a functional form that can plausibly approximate the mapping, collect paired examples of inputs and outputs, and define a quantitative measure of "how wrong" a prediction is. Once those are in place, training adjusts model parameters so predictions align with observed outputs.

The word *fitting* is meant literally. In classical curve fitting—and in practical settings like sensor calibration—we choose parameters so a predicted curve (or surface) passes near measured points. Keep the camera thread from Chapter 1 in mind: a camera system is useful because it can predict something actionable from what it senses. A simple example is exposure calibration: we collect scenes with known reference targets, measure raw sensor readouts $\mathbf{x}$, and learn parameters that map those readouts to a correction y so the system produces consistent brightness across conditions. The same pattern repeats at higher levels (object detection scores, tracking signals, alert decisions): the details change, but the core act is the same—use paired input/output examples to fit parameters that make predictions reliable.

This chapter builds the supervised-learning toolkit around that central act. We first make the pieces explicit (data, models, and losses), then show what training looks like when it succeeds and when it fails (underfitting vs. overfitting). Next we formalize empirical risk minimization (ERM) and the main "anti-memorization" tools (regularization and validation). Finally, we work through linear regression as the first fully transparent case study where the entire pipeline

is visible end to end.

Learning Outcomes

- Formalize datasets, hypotheses, and empirical risk minimization (ERM) with consistent notation used in Chapters 3 to 4.
- Compare common regression/classification losses and regularizers, understanding when to prefer each.
- Diagnose under/overfitting with data splits, learning curves, and bias–variance reasoning; use these diagnostics to guide model selection and regularization.

Design motif

Data → model → objective → audit. This workflow shows up repeatedly in later chapters, even when the models become deeper and the optimization less forgiving.

Before we turn the "fitting" story into equations, let us fix the handful of symbols we will reuse for several pages. The goal is not to introduce new notation, but to keep the derivations readable while the ideas are still new.

- Data $X \in \mathbb{R}^{N\times d}$ with rows $\mathbf{x}_i^\top$; targets $y_i \in \mathbb{R}$ for regression and $y_i \in \{0,1\}$ for binary classification. The affine map $y_{\pm 1} = 2y-1$ switches to $\{-1,+1\}$ when margin-based expressions are convenient.
- Parameters θ (model-specific), weights $\mathbf{w} \in \mathbb{R}^d$; predictions carry hats: $\hat{y}_i = h_\theta(\mathbf{x}_i)$, $\hat{\mathbf{y}} = X\mathbf{w}$.
- The loss $\ell(\hat{y}, y)$ is the teacher's grading rubric; the objective aggregates losses over data and adds regularization. Parameters are learned from data; hyperparameters (e.g., λ in regularization) are chosen by validation.
- Noise uses ε; residuals use $e = y - \hat{y}$. Vectors are bold lowercase, matrices bold uppercase; scalars are italic.
- Bias absorption (when used): augmented feature $\tilde{\mathbf{x}} = [\mathbf{x}; 1]$ with corresponding augmented weights.

3.1 Problem Setup and Notation

We observe a dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ drawn independently and identically distributed (i.i.d.) from an unknown distribution $\mathcal{P}$ on the input–output space $\mathcal{X} \times \mathcal{Y}$. A hypothesis (model) $h_\theta : \mathcal{X} \to \mathcal{Y}$ with parameters θ produces predictions $\hat{y}_i = h_\theta(\mathbf{x}_i)$. A pointwise loss function $\ell(\hat{y}, y)$ measures the penalty incurred by predicting $\hat{y}$ when the true label is y.

The *population risk* and *empirical risk* associated with h_θ are

$$\begin{aligned} R(h_\theta) &= \mathbb{E}_{(\mathbf{x},y)\sim\mathcal{P}}\big[\ell\big(h_\theta(\mathbf{x}), y\big)\big], \\ \hat{R}_N(h_\theta) &= \frac{1}{N}\sum_{i=1}^{N} \ell\big(h_\theta(\mathbf{x}_i), y_i\big). \end{aligned} \tag{3.1}$$

Because $\mathcal{P}$ is unknown, learning algorithms minimize empirical proxies of $R(h_\theta)$. We posit a model family h_θ and use data to choose parameter values that make its predictions behave like the measured input–output pairs. In practice we do this on a *training set* (the data used to fit parameters), and we reserve held-out data to check whether the fitted model is trustworthy; Section 3.6 makes these evaluation protocols precise. Audit is not an external add-on: in engineering terms, it is part of the model contract.

3.2 Fitting, Overfitting, and Underfitting

Fitting is the act of choosing parameters θ so a model's predictions match observed data. Concretely, we pick a loss ℓ, evaluate it on examples $(\mathbf{x}_i, y_i)$, and use an optimization method to search for parameters that make the aggregate loss small. This is what practitioners usually mean by *training*.

It helps to picture training as repeated adjustment under feedback. You make a prediction, measure the mistake with a loss, update parameters to reduce that mistake, and repeat. In sensor calibration, this feels familiar: if your measured output is consistently off, you change a gain or offset; if it is noisy, you adjust how aggressively you trust any one reading. Supervised learning packages that intuition into a general recipe that can be reused across problems.

The goal is not to "fit the training set" as an end in itself. A good fit is one that holds up on new data: the fitted model should behave sensibly on inputs it has not seen. When fitting fails, it tends to fail in one of two recognizable ways.

Underfitting. The model family is too rigid for the task, the features do not contain enough information, or the optimization did not do its job. This is the student who cannot solve the practice problems before the exam: the mismatch is obvious even on the training set. The remedy is to change the representation or the hypothesis class, improve the data, or fix the optimization.

Overfitting. The model is flexible enough to match the training set by memorizing its quirks. This is the student who memorizes the worked examples so well that a small twist on the exam causes failure. Overfitting can look like success until you test on held-out data.

What we aim for. We want a well-fitted model: low training error and comparable validation/test error. The tools below are designed to keep that distinction visible: objectives (losses and regularizers), validation protocols (splits and cross-validation), and diagnostics (learning curves and bias–variance reasoning).

Figure 3 is the baseline bias-variance diagnostic used throughout the chapter.

3.3 Empirical Risk Minimization and Regularization

To make the informal idea of "fitting" mathematically precise, we choose an objective and minimize it. The supervised-learning baseline is *empirical risk minimization*: minimize the average loss on the training data (Vapnik, 1998).

This is the first place where the chapter's opening promises become concrete. If we do not know the true input–output law, we still need a disciplined way to compare candidate models and to say whether one parameter choice is better than another. The loss plays the role of the teacher's rubric, and ERM is the simplest way to aggregate that rubric over many examples: instead of arguing about one example at a time, we ask for parameters that perform well on average across the dataset.

The *empirical risk minimizer* (ERM) selects

$$\hat{\theta}_{\mathrm{ERM}} = \arg\min_{\theta} \; \hat{R}_N(h_\theta). \tag{3.2}$$

To mitigate overfitting, we often add a regularizer $\Omega(\theta)$ with strength $\lambda \geq 0$:

$$\hat{\theta}_\lambda = \arg\min_{\theta} \; \hat{R}_N(h_\theta) + \lambda\,\Omega(\theta), \qquad \Omega(\theta) \in \{\|\theta\|_2^2,\ \|\theta\|_1, \ldots\}. \tag{3.3}$$

For the common L2 penalty, this is the same mathematical move as Tikhonov regularization in inverse problems: stabilize fitting by penalizing large parameter norms (Tikhonov, 1963).

Regularization is not an arbitrary penalty. It is the mathematical version of a teaching move: if a student can memorize every worked example, you change the exercises so memorization is less effective and understanding is rewarded. Regularization plays the same role. It makes some parameter settings expensive, which pushes learning toward explanations that generalize better.

In supervised learning, this matters because a model can drive training loss down in ways that do not survive contact with new data. Regularization is one of the main tools we use to "push back" against memorization: we still fit the data, but we also express a preference for solutions that are stable, simple, or structured in ways that match the problem.

Ridge and lasso. Two penalties show up so often that they have become part of the basic vocabulary:

- **Ridge (L2)** adds $\|\theta\|_2^2$, which shrinks weights smoothly and stabilizes solutions when features are correlated.
- **Lasso (L1)** adds $\|\theta\|_1$, which tends to set some weights to exactly zero, yielding sparse models and a form of feature selection.

The difference is easiest to remember geometrically: L2 has round level sets, while L1 has corners, and corners create exact zeros.

Regularization: L1/L2 and scaling

- **Why regularize?** Flexible models can fit training data by effectively memorizing idiosyncrasies (noise, quirks of the sample) rather than capturing stable structure. Regularization makes such memorization expensive and rewards explanations that survive on held-out data.
- **L2 (ridge)** shrinks weights smoothly, is rotationally invariant, and works well when features are dense and correlated.
- **L1 (lasso)** promotes sparsity, effectively performing feature selection when many coefficients should be zero.
- **Why the names?** "Ridge" refers to the ridge-like valleys that appear in least-squares objectives under multicollinearity; the L2 penalty lifts the valley floor and stabilizes the solution. "LASSO" is an acronym for *Least Absolute Shrinkage and Selection Operator*.
- **Why L1 vs. L2 feels different:** the L2 penalty discourages large coefficients but rarely drives them exactly to zero, while the L1 penalty creates corners in the geometry that tend to set some coefficients to exactly zero.
- **Standardization** (zero mean, unit variance) is essential before applying L1/L2/elastic-net so the penalty treats all dimensions comparably.
- With an intercept term, centering y makes the algebra cleaner; ridge and lasso still apply directly once features are scaled.

Figure 4 summarizes the geometric reason L1 and L2 penalties behave differently.

Figure 5 provides the coefficient-path view used when tuning sparsity strength.

3.4 Elastic-net paths and cross-validation

Pure L1 or L2 penalties rarely dominate modern workflows; the *elastic net* mixes them to balance sparsity and stability:

$$\hat{\theta}_{\alpha,\lambda} = \arg\min_{\theta} \hat{R}_N(h_\theta) + \lambda\Big(\alpha\|\theta\|_1 + \tfrac{1-\alpha}{2}\|\theta\|_2^2\Big), \qquad \alpha \in [0,1]. \tag{3.4}$$

Setting $\alpha = 1$ recovers the lasso, $\alpha = 0$ yields ridge, and intermediate values trace a solution path that tends to group correlated features while still pruning irrelevant ones. In practice we standardize the features once, draw a logarithmic grid of λ values, and run K-fold cross-validation for each pair (α, λ). The "one-standard-error" rule selects the largest λ whose validation error is within one standard error of the minimum. It gives a stable operating point and avoids over-interpreting tiny validation differences.

3.5 Common Loss Functions

Loss functions make the teacher signal quantitative: they decide what counts as a small mistake, what counts as a large one, and which kinds of errors matter most. For binary classification with labels $y \in \{-1, +1\}$ and margin $z = y\,f(\mathbf{x})$, two standard losses are

$$\ell_{\text{hinge}}(y, z) = \max\big(0, 1 - z\big), \qquad \ell_{\text{logistic}}(y, z) = \log\big(1 + e^{-z}\big).$$

Here $y \in \{-1, +1\}$; when labels are instead coded as $y \in \{0, 1\}$ (common in probability-of-class formulations), the margin expression uses $y_{\pm 1} = 2y - 1$ to map between codings. Figure 6 visualizes these curves together with the squared hinge so you can match the algebra to the margin geometry. For regression with residual $e = y - \hat{y}$, we frequently use

$$\ell_{\text{sq}}(e) = \tfrac{1}{2}e^2, \qquad \ell_{\text{abs}}(e) = |e|.$$

The Huber loss interpolates between these: it is quadratic when $|e| \le \delta$ and linear beyond that threshold (here the plot uses $\delta = 1$), reducing sensitivity to outliers while remaining smooth around the origin.

Figure 7 compares common regression losses used in this section.

Table 1: Common losses and typical use (reference for Chapters 3 to 5). Match the loss to your modeling assumptions (noise, margins) and to the downstream decision metric you care about.

Loss	**Convex?**	**Typical use**
Squared error $\frac{1}{2}e^2$	Yes	Regression when Gaussian noise is plausible; differentiable everywhere.
Absolute error $\lvert e \rvert$	Yes	Robust regression with Laplacian noise assumptions; non-differentiable at 0.
Huber (quadratic $\rightarrow$ linear)	Yes	Regression when moderate outliers are present; smooth near zero.
Logistic (binary cross-entropy)	Yes	Probabilistic classification; pairs naturally with sigmoid.
Hinge / squared hinge	Yes	Margin-based classifiers (SVMs, large-margin perceptrons).

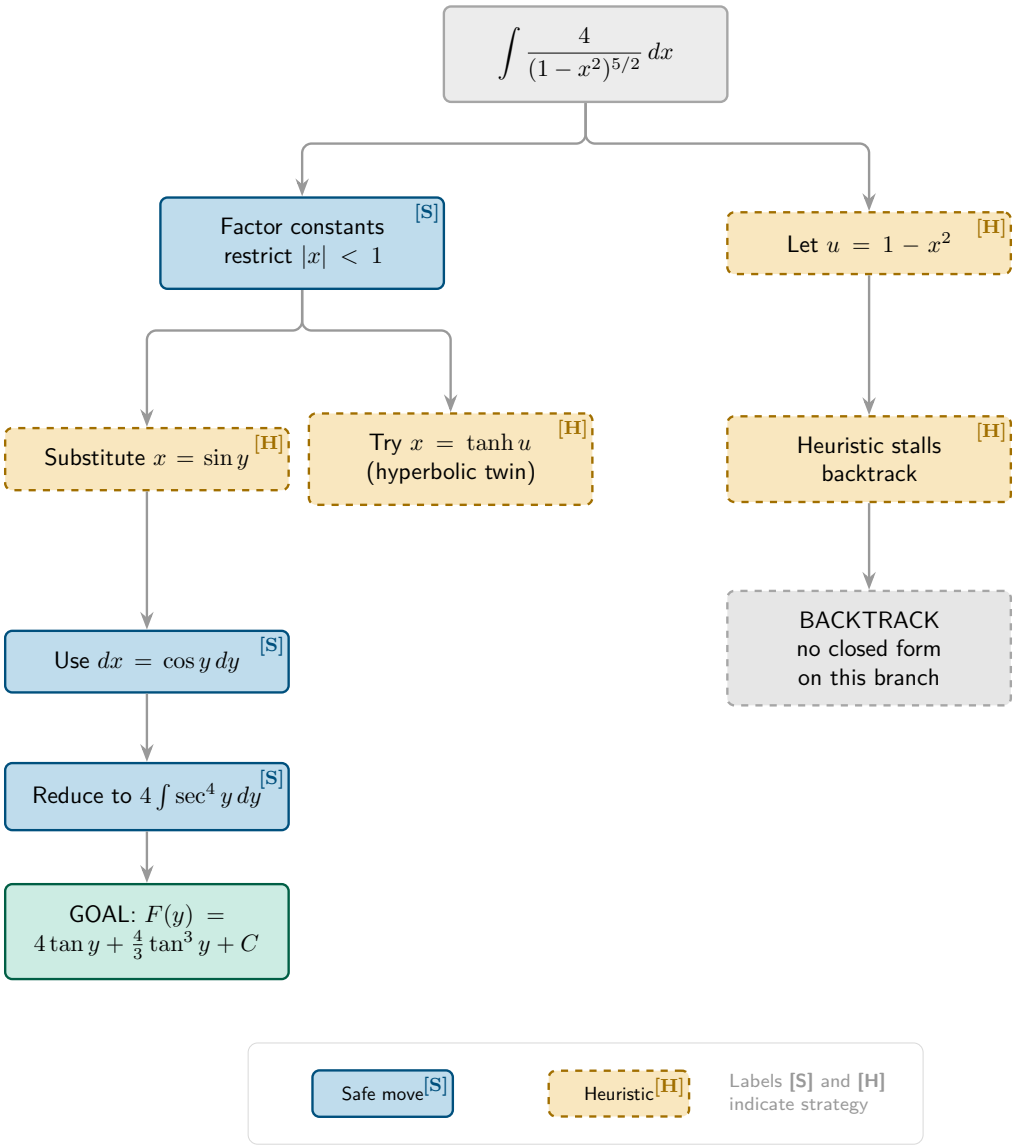


Figure 2: Transformation tree for the running example integral. Badges **[S]**/**[H]** mark safe vs. heuristic moves; the left branch shows the sequential safe path, while the right branch terminates in explicit backtracking. The tree view makes solved and dead-end leaves visible.

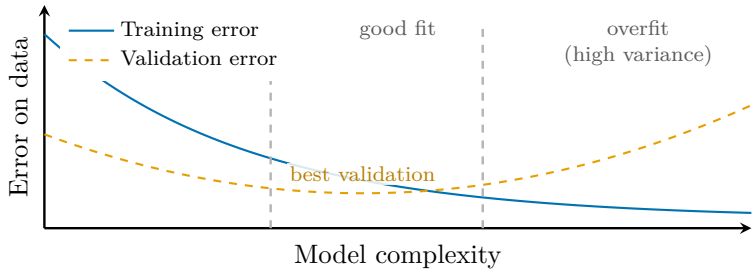


Figure 3: Schematic underfitting and overfitting as a function of model complexity. Training error typically decreases with complexity, while validation error often has a U-shape; regularization and model selection aim to operate near the minimum of the validation curve.

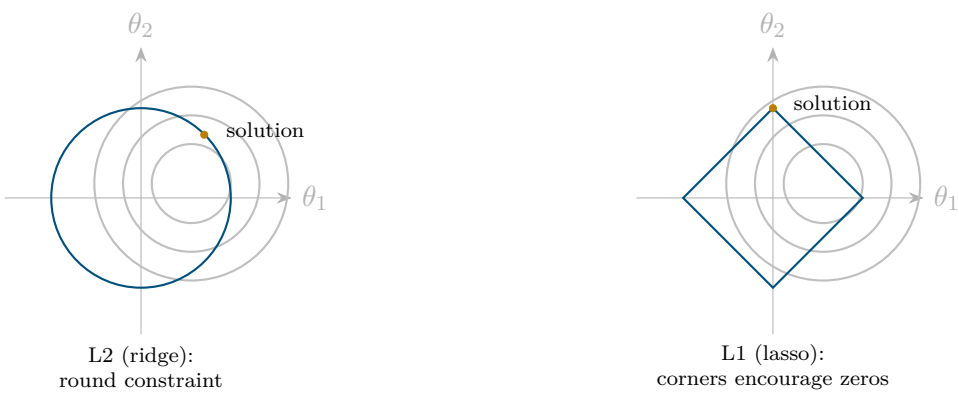


Figure 4: Why L1 promotes sparsity (geometry). Minimizing loss subject to an L2 constraint tends to hit a smooth boundary; an L1 constraint has corners aligned with coordinate axes, so tangency often occurs where some coordinates are exactly zero.

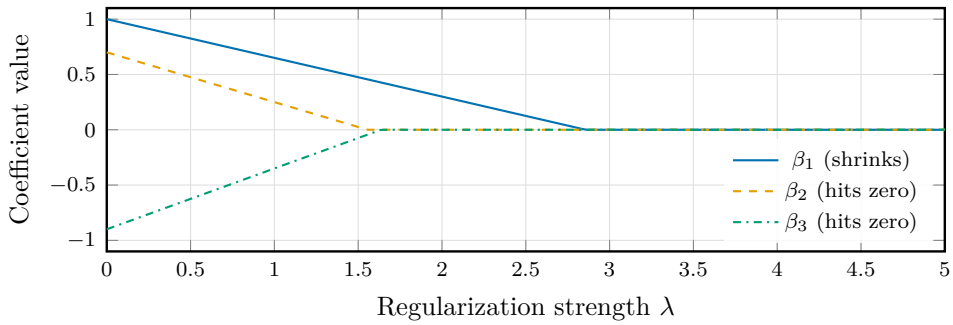


Figure 5: Illustrative lasso path as the regularization strength increases. Coefficients shrink, and some become exactly zero, yielding sparse models; the shrinkage order depends on feature correlation and scaling.

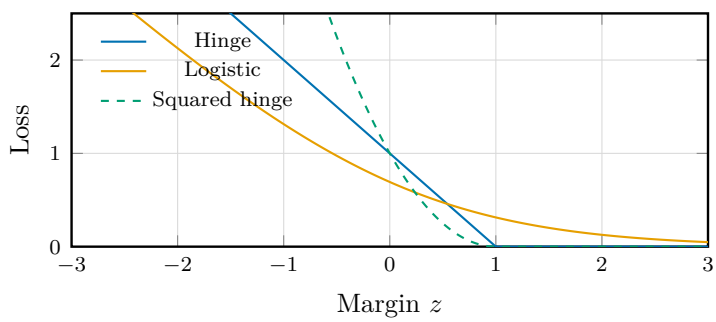


Figure 6: Classification losses as functions of the signed margin z. The curves highlight how different losses treat confident mistakes and near-boundary points.

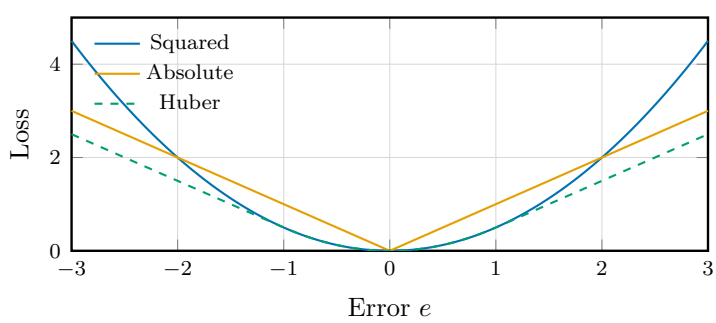


Figure 7: Regression losses versus prediction error. The Huber loss transitions from quadratic to linear, reducing sensitivity to outliers while staying differentiable near zero.

3.6 Model Selection, Splits, and Learning Curves

Up to this point, we have defined what it means to fit: choose a model family, pick an objective (loss plus any regularizer), and tune parameters to reduce that objective on observed examples. The next question is how to choose among competing model families, hyperparameters, and training procedures without fooling ourselves. Model selection is the discipline of making those choices using validation data, while keeping one final dataset split untouched so that the reported performance remains honest.

In other words, this is where the chapter's "audit" step becomes operational: we decide what to trust by checking performance on data the model has not been allowed to fit.

Practical workflows allocate data into training, validation, and test portions. Training data are used to fit parameters; validation data guide choices such as hyperparameters and model families; and the test set provides an unbiased audit once those choices are fixed. The key habit is the separation of roles: training is where you allow the model to "learn" (and potentially overfit), validation is where you decide what kind of learning you trust, and the test set is the final audit.

Risk & audit

- **Leakage:** avoid split mistakes (duplicates, near-duplicates, time leakage) that inflate validation accuracy.
- **Metric mismatch:** align the loss you optimize with the metric you report (and the decision you must make).
- **Overfitting signals:** track training vs. validation curves and use learning curves to diagnose data hunger vs. excess capacity.
- **Distribution shift:** audit performance by slice (population, device, lighting, region) rather than relying on one aggregate score.
- **Calibration:** check reliability when probabilities drive actions (thresholds, alerts, resource allocation).
- **Reporting discipline:** log data split policy, seeds, and selection criteria; Appendix B defines the book-wide template.

Proper scoring rules and calibration

- **Log loss (cross-entropy)** and the **Brier score** are *proper* scoring rules: in expectation, they are minimized by predicting the true class probability (Brier, 1950; Gneiting and Raftery, 2007).
- **Brier** is squared error in probability space; it penalizes confident mistakes less harshly than log loss and is often paired with reliability diagrams.
- **Log loss** heavily punishes overconfident errors (loss $\to \infty$ as predicted probability $\to 0$ on the true class), so it is a natural objective when probabilities will be thresholded downstream.
- **Practical tip:** train with log loss, but monitor both log loss and Brier score on validation data to catch calibration issues early.

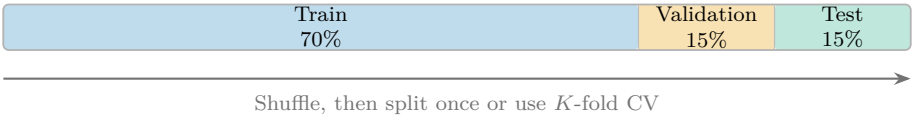


Figure 8: Dataset partitioning into training, validation, and test segments. Any resampling scheme should preserve disjoint evaluation data; when classes are imbalanced, shuffle within strata so each split reflects the overall class mix.

Figure 8 lays out the split protocol for the train/validation/test workflow.

A concrete toy task

As a reference point, keep in mind one small binary classification problem: a two-moons toy with a standard train/validation/test split. It is deliberately simple, but it is rich enough to reveal the recurring failure modes (memorization, metric mismatch, split leakage) and the recurring remedies (regularization, validation, and diagnostics).

To make the workflow concrete, Figure 9 summarizes the standard ERM pipeline from dataset to model selection.

Figure 11 anchors the calibration and capacity diagnostics in this aside.

Learning curves plot training and validation error against the number of training examples, revealing underfitting or overfitting regimes; Figure 10 shows the basic shapes to watch for.

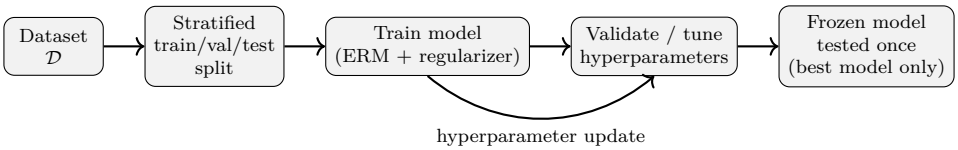


Figure 9: Mini ERM pipeline (split once, iterate train/validate, then test only the best model on the held-out set). This keeps tuning decisions separate from final reporting.

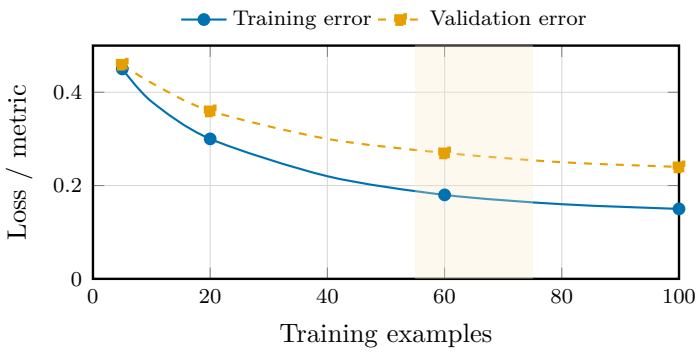


Figure 10: Illustrative learning curves reveal under/overfitting: the validation curve flattens while additional data continue to decrease training error only marginally. A shaded patience window marks when early stopping would halt if no validation improvement occurs.

Data-leakage checklist

- Split data before any preprocessing or feature selection.
- Fit scalers/imputers/dimensionality-reduction transforms on the training fold only; reuse fitted parameters on validation/test (or within each CV fold via pipelines).
- Respect temporal order for time-series; avoid target/future-derived features.
- Wrap preprocessing + model in a pipeline for cross-validation so transformers refit inside each fold.

Bias–variance at a glance

- **High bias (underfit):** train and validation errors both plateau high and together; add capacity/features or reduce regularization.
- **High variance (overfit):** train error low, validation error high/diverging; add data, strengthen regularization, or use early stopping.
- **Well fit:** train/validation track closely and decrease or level off at low error; further gains require better data or priors.

Learning curves explain *why* the train/validation split is useful: they show whether more data, more capacity, or more regularization is the lever that actually moves the validation error. Once you can read these curves, a natural next question is what happens as we scale up data and model size. The aside below summarizes two modern empirical patterns that are best treated as guidance, not as a recipe.

Aside: scaling laws and double descent

The simplest story is the classical bias–variance picture: as model capacity grows, training error falls, and validation error often has a U-shape. In modern overparameterized models, that picture can be incomplete. You may see *double descent*: after the classical U-shape, error can decrease again once model size exceeds the interpolation threshold (Belkin et al., 2019). You may also hear *scaling laws*: in some regimes, validation loss decreases roughly as a power law of compute, data, and model size (Kaplan et al., 2020; Hoffmann et al., 2022).
Treat both as diagnostics rather than guarantees. Use them to decide whether to collect more data, shrink or expand a model, or regularize more aggressively, but still make final choices by comparing validation curves. Do not chase the interpolation peak as a goal.

Regularization trades model complexity for generalization; Figure 12 depicts the effect of ridge penalties on the weight norm.

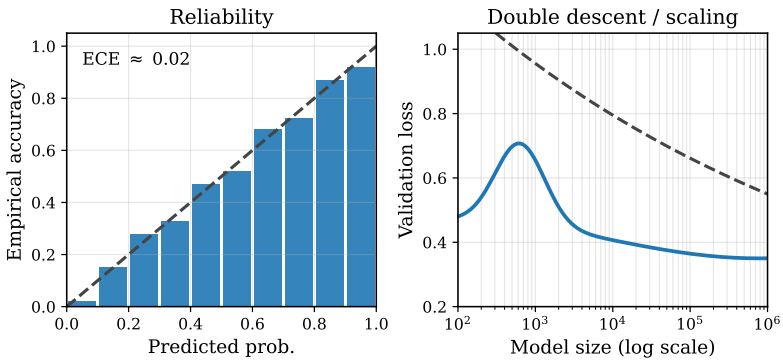


Figure 11: Calibration and capacity diagnostics. Left: reliability diagram with binned predicted probabilities vs. empirical accuracy; Expected Calibration Error (ECE) measures deviation from the diagonal. Right: illustrative double-descent risk vs. model size (log-scale on the x-axis); the dashed line sketches a scaling-law trend for thinking about capacity and regularization. *Source: author-generated.*

3.7 Linear regression: a first full case study

Up to this point, the supervised-learning pipeline has been described in abstract terms: a dataset, a hypothesis class, an objective, and an audit. Linear regression keeps the same pipeline but makes each component explicit enough to inspect end to end.

A useful habit, before you commit to any model family, is to ask whether there is any signal to model in the first place. If the relationship were deterministic (like Celsius $\leftrightarrow$ Fahrenheit), there would be nothing to learn. In supervised learning we assume the relationship is statistical: the same input can map to different outputs because of noise, missing variables, or genuine uncertainty. For simple problems, a scatter plot and a correlation coefficient can reveal whether a linear trend is even plausible. For high-dimensional data, analogous sanity checks (feature scaling, collinearity) help you decide whether linear regression is a sensible starting point or merely a baseline.

In this section, the coefficients $\boldsymbol{\beta}$ are the knobs we turn. "Learning" means estimating $\boldsymbol{\beta}$ from $(\mathbf{X}, \mathbf{y})$ so predictions align with observed targets under the chosen objective. Because least squares is convex with a closed form, repeating the fit on the same data returns the same solution (up to numerical tolerance). That determinism is the teaching benefit: it lets us isolate the roles of loss

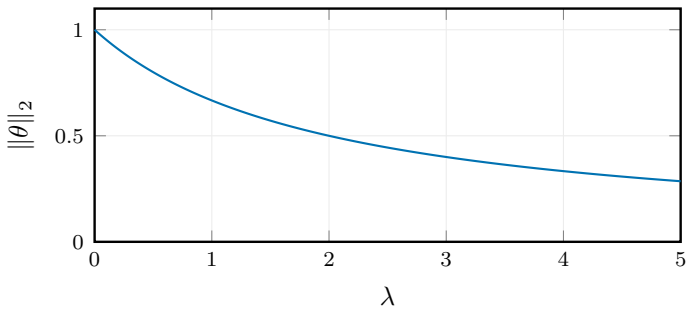


Figure 12: Ridge regularization shrinks parameter norms as the penalty strength increases. This controls variance without forcing sparsity.

design, optimization, regularization, and validation before moving to models where those roles are harder to disentangle.

Given inputs $\mathbf{x}_i \in \mathbb{R}^d$ and continuous targets $y_i \in \mathbb{R}$, the linear model predicts

$$\hat{\mathbf{y}} = \mathbf{X}\boldsymbol{\beta}, \qquad \mathbf{X} \in \mathbb{R}^{N \times d}. \tag{3.5}$$

Equivalently, $\hat{y}_i = \mathbf{x}_i^\top \boldsymbol{\beta}$ for each data point. The vector $\boldsymbol{\beta}$ is the set of adjustable parameters: *fitting* the model means choosing $\boldsymbol{\beta}$ so that predictions align with observed outputs. The residual $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$ captures what the model fails to explain on the data at hand.

A common way to formalize "measurement scatter" is to write

$$y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma^2), \tag{3.6}$$

where ε_i is observation noise (sensor noise, unmodeled effects, annotation noise, etc.). Under this assumption,

$$p(y_i \mid \mathbf{x}_i, \boldsymbol{\beta}) = \mathcal{N}(\mathbf{x}_i^\top \boldsymbol{\beta}, \sigma^2), \tag{3.7}$$

and (assuming i.i.d. observations) the likelihood factorizes:

$$p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}) = \prod_{i=1}^{N} p(y_i \mid \mathbf{x}_i, \boldsymbol{\beta}). \tag{3.8}$$

Maximizing the (log) likelihood is equivalent to minimizing the negative log-likelihood, and for Gaussian noise that becomes (up to constants and a scale factor $1/(2\sigma^2)$) the familiar sum of squared errors:

$$-\log p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}) = \frac{1}{2\sigma^2} \sum_{i=1}^{N} (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2 + \text{const.} \tag{3.9}$$

This is the simplest example of a recurring theme: if you propose an "educated guess" model for how data are generated, training often becomes "minimize a loss".

To fit $\boldsymbol{\beta}$, we need a grading rubric. Squared error is the standard starting point because it is smooth and strongly penalizes large mistakes:

$$\mathcal{L}(\boldsymbol{\beta}) = \tfrac{1}{2}\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2. \tag{3.10}$$

The gradient is simple,

$$\nabla_{\boldsymbol{\beta}} \mathcal{L}(\boldsymbol{\beta}) = \mathbf{X}^\top (\mathbf{X}\boldsymbol{\beta} - \mathbf{y}), \tag{3.11}$$

so gradient descent is explicit: $\boldsymbol{\beta} \leftarrow \boldsymbol{\beta} - \eta \mathbf{X}^\top (\mathbf{X}\boldsymbol{\beta} - \mathbf{y})$. This same loop reappears later when the model is no longer linear and the loss is no longer quadratic.

Least squares is convex and satisfies the normal equations:

$$\mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}. \tag{3.12}$$

When $\mathbf{X}^\top \mathbf{X}$ is invertible, the solution can be written explicitly as

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}. \tag{3.13}$$

Geometrically, the prediction $\hat{\mathbf{y}}$ is the orthogonal projection of $\mathbf{y}$ onto the column space of $\mathbf{X}$. In code, solve the linear system (QR/SVD) rather than forming $(\mathbf{X}^\top \mathbf{X})^{-1}$ explicitly; collinearity can make $\mathbf{X}^\top \mathbf{X}$ poorly conditioned even when the mathematics is correct.

Where overfitting enters. With raw features, a linear model may underfit; with aggressive feature expansions (polynomials, splines, kernels, learned

features), the same least-squares machinery can overfit. This is where the earlier tools matter. Regularization (ridge, lasso, elastic net) makes memorization harder; validation selects hyperparameters; learning curves diagnose whether error is limited by bias, variance, or data.

Ridge and lasso in one line. Ridge adds $\lambda\|\boldsymbol{\beta}\|_2^2$ to the objective, shrinking coefficients and stabilizing solutions when features are correlated; lasso uses $\|\boldsymbol{\beta}\|_1$ and tends to drive some coefficients to exactly zero. The ridge shrinkage behavior is visualized in Figure 12.

The discipline of supervised learning is reusable across models: define a dataset and a hypothesis class, choose an objective (loss plus regularizer), optimize it, and then audit generalization with clean train/validation/test separation. In Chapter 4, we apply this toolkit to classification, where the loss becomes a Bernoulli negative log-likelihood (cross-entropy) and the evaluation tools expand (confusion matrices, ROC/PR curves, and calibration).

Key takeaways

Minimum viable mastery

- Supervised learning chooses a hypothesis class and fits parameters by minimizing empirical risk, then audits generalization on held-out data.
- Overfitting is a training success but a deployment failure; regularization and validation protocols are the practical defenses.
- Learning curves and bias–variance reasoning are diagnostics: they help decide whether to add data, change capacity, or adjust regularization.

Common pitfalls

- Tuning on the test set (or peeking repeatedly) turns the test set into training data.
- Leakage through preprocessing: fit scalers/encoders/imputers on the training split only, then apply to val/test.
- Over-interpreting a single metric: use learning curves and slice audits, not only one headline number.

Exercises

Exercise 1 (Medium). Implement ridge regression two ways: (i) a closed-form solve (QR/SVD) and (ii) gradient descent.
Deliverable: validation curves as λ varies, plus one paragraph comparing the two implementations.

Exercise 2 (Medium). Create a controlled overfitting experiment (polynomial degree or noisy features).
Deliverable: learning curves + a bias/variance diagnosis and a concrete regularization choice justified by the curves.

Exercise 3 (Easy). Demonstrate leakage by fitting preprocessing on the full dataset (incorrect) vs. training split only (correct).
Deliverable: the test error gap + a one-paragraph explanation of why the error is not "real performance."

If you are skipping ahead. Keep the ERM loop and split hygiene close: model class, loss/regularizer, optimizer, and a clean train/val/test protocol. Those ideas are assumed immediately in Chapter 4 and reused throughout the neural chapters.

Where we head next. Chapter 4 extends this ERM setup from continuous targets to discrete labels: outputs become class probabilities, the loss becomes cross-entropy, and evaluation adds confusion matrices, ROC/PR curves, and threshold selection. Keep this chapter's audit loop close; the same split hygiene and validation logic are reused in later neural chapters and remain relevant when tuning fuzzy and evolutionary systems.

4 Classification and Logistic Regression

Chapter 2 illustrated intelligence as transformation search with explicit goal tests. Chapter 3 introduced the data-driven view: represent a task with data, choose a hypothesis class, minimize empirical risk under regularization, and audit performance on held-out data. This chapter extends that toolkit from regression to classification, and Figure 1 places it on the core supervised path.

Learning Outcomes

After this chapter, you should be able to:

- Derive the logistic log-likelihood and its gradient.
- Explain the NLL (cross-entropy) connection and convexity.
- Extend to softmax regression for multiclass problems.

Design motif

Same loop, different likelihood. Logistic regression keeps the linear score, but models a Bernoulli outcome; the loss becomes a negative log-likelihood (binary cross-entropy).

4.1 From regression to classification

Linear regression models a continuous target y and, under a Gaussian noise model, yields a closed-form solution via the normal equations (Chapter 3, Section 3.7). Classification changes the output space: y is a discrete label, and the model predicts class probabilities rather than raw responses. The ERM pipeline is unchanged, but the likelihood/loss pair changes to Bernoulli plus negative log-likelihood (binary cross-entropy), and optimization is typically iterative.

Throughout this chapter we use $y \in \{0, 1\}$ by default, switching to $y_{\pm 1} = 2y - 1$ only when margin-based expressions are convenient. Predictions carry hats (e.g., $\hat{y}$), ε denotes noise, and $e = y - \hat{y}$ denotes residuals. For a refresher on data splits, learning curves, and the bias–variance vocabulary, see Chapter 3; here we focus on the logistic-specific modeling and diagnostics.

Worked example: a toy decision boundary

On a two-moons binary classification toy, the ideas in this chapter have a visible effect:

- Logistic regression gives a calibrated probabilistic baseline (linear score + sigmoid) that is easy to diagnose.
- The nonlinearity we will need later is *not* in the optimizer but in the representation: Chapter 6 introduces multilayer features and Chapter 7 supplies the training engine that lets those features improve the decision boundary.

4.2 Classification problem statement

With that setup in place, we now formalize the classification target. Unlike regression, where y is continuous, classification predicts a *discrete label*. We start with the binary case $y \in \{0, 1\}$, where the goal is to estimate a probability

$$\pi(\mathbf{x}) = P(y = 1 \mid \mathbf{x}),$$

and then produce a decision by thresholding $\pi(\mathbf{x})$ (or by comparing class probabilities when there are more than two classes). For multiclass problems the label belongs to one of K classes,

$$y \in \{c_1, c_2, \ldots, c_K\},$$

and the goal is to estimate $P(y = c_k \mid \mathbf{x})$ for each k; we return to the softmax extension later in the chapter.

4.3 Bayes Optimal Classifier

A fundamental result in statistical pattern recognition is that the *Bayes classifier* is the optimal classifier in terms of minimizing the expected classification error. The Bayes classifier assigns $\mathbf{x}$ to the class:

$$\hat{y} = \arg \max_{c_k \in \{c_1, \ldots, c_K\}} P(y = c_k \mid \mathbf{x}).$$

Using Bayes' theorem, the posterior probability can be expressed as

$$P(y = c_k \mid \mathbf{x}) = \frac{P(\mathbf{x} \mid y = c_k) P(y = c_k)}{P(\mathbf{x})}. \tag{4.1}$$

Here

- $P(\mathbf{x} \mid y = c_k)$ is the *class-conditional likelihood*,
- $P(y = c_k)$ is the *prior probability* of class c_k,
- $P(\mathbf{x}) = \sum_{j=1}^{K} P(\mathbf{x} \mid y = c_j) P(y = c_j)$ is the marginal likelihood of the input.

Equation (4.1) provides a principled way to compute the posterior probabilities, and thus the optimal classification rule.

Challenges in Practice Despite its theoretical appeal, the Bayes classifier is rarely used directly: the class-conditional densities and priors are usually unknown, and estimating them reliably (especially in high dimensions) is itself a modeling problem. Practical classifiers therefore approximate Bayes' rule by making simplifying assumptions (a generative model) or by learning the posterior $P(y \mid \mathbf{x})$ directly (a discriminative model).

Running example: a two-cluster dataset To keep the discussion concrete, we reuse a small toy dataset in Figure 13 consisting of two Gaussian clusters. Under a simple generative assumption (equal covariances and similar priors), Figure 14 visualizes the Bayes-optimal decision boundary: it is linear in this linear discriminant analysis (LDA) setting, while unequal covariances yield a quadratic boundary. This running example will anchor the geometric intuition (what a decision boundary looks like) before we turn to discriminative models that learn $\pi(\mathbf{x})$ directly.

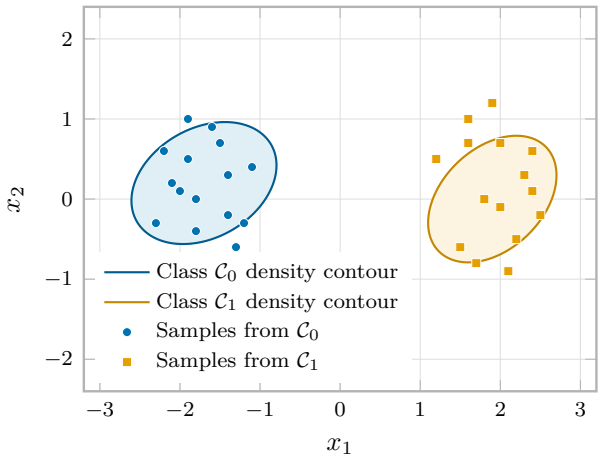


Figure 13: Synthetic binary dataset built from two anisotropic Gaussian clusters; shaded ellipses hint at the underlying density while the scattered samples are reused throughout the running examples. It helps build intuition for how decision boundaries relate to class geometry before fitting discriminative models.

As shown in Figure 16, linear scores (logits), the binary cross-entropy (BCE) loss shape, and regularization strength appear in one compact diagnostic view.

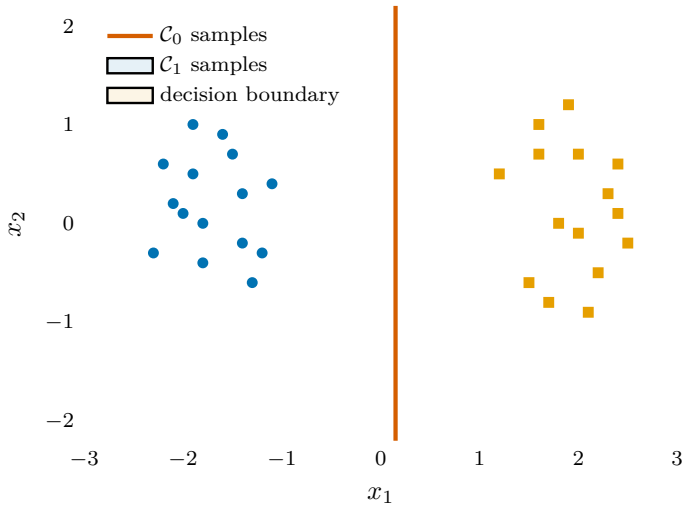


Figure 14: Bayes-optimal boundary for two Gaussian classes. Equal covariances and similar priors (LDA setting) yield a linear separator; unequal covariances yield a quadratic boundary. The boundary is near the equal-posterior line (vertical, pink); left/right regions map to predicted classes R0 and R1.

Figure 15 provides the low-data MAP-versus-MLE comparison used in this discussion.

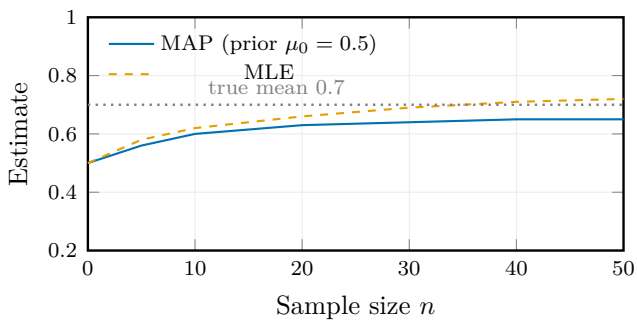


Figure 15: MAP estimates interpolate between the prior mean and the data-driven MLE. As the sample size grows, the MAP curve approaches the true mean; with little data, the prior dominates.

Naive Bayes Approximation One classical workaround is the Naive Bayes classifier, which assumes that the components of $\mathbf{x}$ are conditionally independent given the class label. Under this assumption,

$$P(\mathbf{x} \mid y = c_k) = \prod_{j=1}^{p} P(x_j \mid y = c_k),$$

making the computation and estimation of the likelihood tractable. It is important to remember that this factorization is justified only under the conditional independence assumption; when the features are strongly correlated, Naive Bayes can suffer because the assumption is violated.

4.4 Logistic Regression: A Probabilistic Discriminative Model

One widely used approach to classification, especially for binary problems, is *logistic regression.* Logistic regression models the posterior probability $P(y = 1 \mid \mathbf{x})$ directly as a function of $\mathbf{x}$, without explicitly modeling the class-conditional densities.

Logistic regression at a glance

Objective: Minimize binary cross-entropy (negative log-likelihood) between true labels $y \in \{0, 1\}$ and predicted probabilities $\pi(\mathbf{x}) = \sigma(\boldsymbol{\beta}^\top \tilde{\mathbf{x}})$.
Key hyperparameters: Regularization type/strength (L2 or L1, penalty λ; many libraries use $C = 1/\lambda$), feature scaling, optimization settings (step size, iterations).
Defaults: Standardize features; use L2 regularization with moderate strength; start with a 0.5 decision threshold and adjust only if class costs are asymmetric.
Common pitfalls: Strong collinearity between features, severe class imbalance, and uncalibrated probability outputs if the model is over-regularized or trained on a biased sample.

Binary Classification Setup Binary labels and $\pi(\mathbf{x})$ were defined in Section 4.2; here we focus on modeling the log-odds with a linear predictor.

Linear Model for the Log-Odds Logistic regression assumes that the *log-odds* (also called the *logit*) of the positive class is a linear function of the input features. Introducing the augmented feature vector $\tilde{\mathbf{x}} = [1, x_1, \ldots, x_p]^\top$ and parameter vector $\boldsymbol{\beta} = [\beta_0, \beta_1, \ldots, \beta_p]^\top$, we write

$$\begin{aligned} \log \frac{\pi(\mathbf{x})}{1-\pi(\mathbf{x})} &= \boldsymbol{\beta}^\top \tilde{\mathbf{x}}, \\ \pi(\mathbf{x}) &= \frac{1}{1+\exp\left(-\boldsymbol{\beta}^\top \tilde{\mathbf{x}}\right)}. \end{aligned} \tag{4.2}$$

This implies that the posterior probability $\pi(\mathbf{x})$ is the *logistic sigmoid* applied to the linear predictor.

Author's note: why "logistic" and why "regression"?

Logistic is the link function: a sigmoid takes a real score and turns it into a probability in $[0, 1]$. And *regression* is what remains linear: the *log-odds* (the logit) is modeled as a linear function of the features.
The model outputs a probability. The class label comes later, as a decision rule: threshold the probability in the binary case (or compare probabilities in the multiclass extension).

4.4.1 Likelihood, loss, and gradient

For data $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$ with $y_i \in \{0, 1\}$, define $p_i = \pi(\mathbf{x}_i) = \sigma(\boldsymbol{\beta}^\top \tilde{\mathbf{x}}_i)$. Under a Bernoulli model, the likelihood factorizes as

$$p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}) = \prod_{i=1}^N p_i^{y_i} (1-p_i)^{1-y_i}, \tag{4.3}$$

so the log-likelihood is

$$\log p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}) = \sum_{i=1}^N \Big(y_i \log p_i + (1-y_i) \log(1-p_i) \Big). \tag{4.4}$$

Maximizing the log-likelihood in (4.4) is equivalent to minimizing the negative log-likelihood (binary cross-entropy). With the design matrix $\mathbf{X} = [\tilde{\mathbf{x}}_1^\top; \ldots; \tilde{\mathbf{x}}_N^\top]$

and vector $\mathbf{p} = (p_1, \ldots, p_N)^\top$, the gradient of the negative log-likelihood is

$$\nabla_{\boldsymbol{\beta}}\Big(-\log p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta})\Big) = \mathbf{X}^\top(\mathbf{p} - \mathbf{y}). \tag{4.5}$$

The Hessian has the form $\mathbf{X}^\top\mathbf{W}\mathbf{X}$ where $\mathbf{W} = \operatorname{diag}(p_i(1 - p_i)) \succeq 0$, which makes the objective convex and explains why second-order methods work well for moderate feature dimensions.

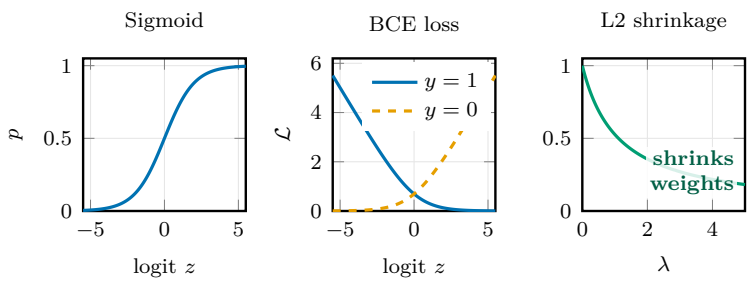


Figure 16: The sigmoid maps logits to probabilities (left). The binary cross-entropy (negative log-likelihood) penalizes confident wrong predictions sharply (middle). Regularization typically shrinks parameter norms as the penalty strength increases (right).

Optimization geometry (why iterative solvers) Unlike linear regression, logistic regression does not have a closed-form solution for $\boldsymbol{\beta}$, even though the objective is convex. In practice we therefore rely on iterative solvers (gradient methods, quasi-Newton methods, or Newton/iteratively reweighted least squares (IRLS) in moderate dimensions). Figure 17 is a convex quadratic toy that reminds us what an optimization trajectory looks like when we minimize a smooth objective: step size and conditioning shape how quickly iterates contract toward the minimizer.

Geometry of the logistic surface. The decision rule is linear in feature space even though the posterior itself is smoothly varying. Figure 18 depicts this duality: the white hyperplane slices the space into two half-spaces while the probability "ramp" shows how margins translate into calibrated confidences.

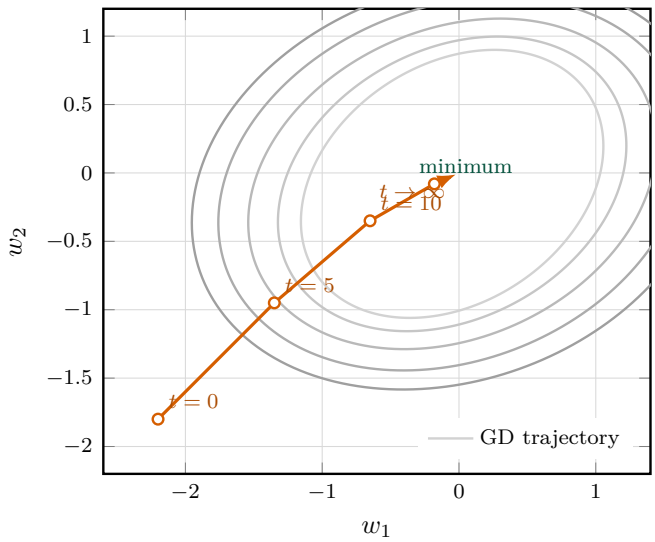


Figure 17: Gradient-descent iterates contracting toward the minimizer of a convex quadratic cost (schematic). Ellipses are level sets; arrows show the "steepest descent along contours" direction. Poor conditioning stretches the ellipses and slows convergence.

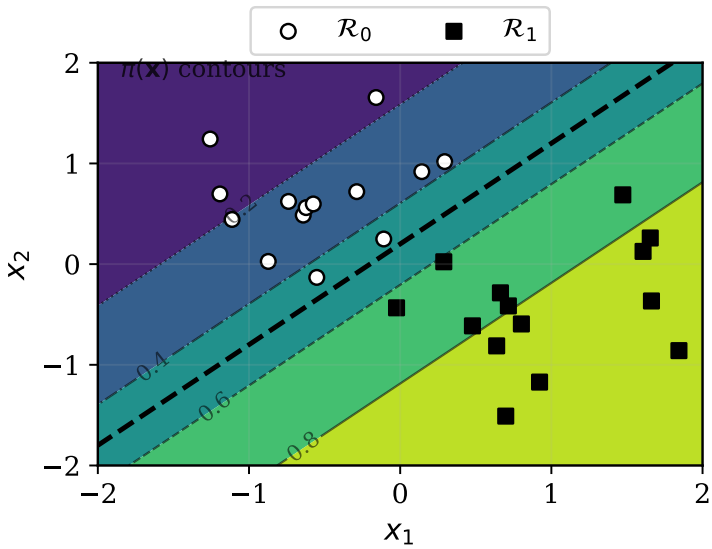


Figure 18: Illustrative logistic-regression boundary. The dashed line marks the linear decision boundary at probability 0.5; labeled contours show how the posterior varies smoothly with margin, enabling calibrated decisions and adjustable thresholds. *Source: author-generated.*

4.5 Probabilistic Interpretation: MLE and MAP

The ERM view in Chapter 3 treats learning as minimizing an average loss plus (optionally) a regularizer. The probabilistic view arrives at the same objective from a different direction:

- **MLE** maximizes the data likelihood under a chosen observation model (for logistic regression: Bernoulli with $p_i = \sigma(\boldsymbol{\beta}^\top \tilde{\mathbf{x}}_i)$) (Fisher, 1922).
- **MAP** maximizes the posterior, which multiplies the likelihood by a prior $p(\boldsymbol{\beta})$. In optimization form, MAP adds a penalty $-\log p(\boldsymbol{\beta})$, which is exactly regularization (Bishop, 1995).

Two common priors explain the two penalties that appear most often in practice: a zero-mean Gaussian prior yields an L2 (ridge) penalty, while a Laplace prior yields an L1 (lasso) penalty. The L2 penalty also has an older life as a stabilizer for ill-posed inverse problems (Tikhonov regularization) (Tikhonov, 1963). The schematic below illustrates the MLE→MAP idea on a simple mean-estimation problem: with little data, the prior matters; with enough data, MAP approaches MLE.

4.6 Confusion Matrices and Derived Metrics

Abbreviation note. ROC = receiver operating characteristic; PR = precision–recall; AUC = area under the curve; ECE = expected calibration error. We use AUROC/AUPRC for the corresponding AUCs.

Risk & audit

- **Threshold choice:** a high AUC can still yield a poor operating point; pick thresholds using a validation objective that matches the cost.
- **Class imbalance:** report PR curves (or per-class metrics) when positives are rare; audit confusion matrices, not just accuracy.
- **Probability quality:** logits can be miscalibrated; use reliability diagrams/ECE when probabilities are consumed downstream.
- **Feature shortcuts:** strong apparent accuracy can come from spurious correlates; sanity-check with slices and perturbations.
- **Regularization:** tune λ with held-out data; do not report the best test result after repeated tuning.

Once we have a probabilistic classifier, we need diagnostics that quantify performance on held-out data. For multi-class prediction, the confusion matrix C_{ij} records the number of examples with true class i predicted as j. From C we compute accuracy, per-class precision/recall, and aggregate metrics. *Macro-averaged* precision/recall first evaluate the metric per class and then average them uniformly, whereas *micro-averaged* precision/recall pool all true/false positives across classes before computing the ratio (equivalent to weighting each example equally). Visual inspection (Figure 20) helps diagnose systematic errors across classes.

On highly imbalanced problems accuracy and AUROC can be misleading; prefer class-balanced metrics (macro-F1) and AUPRC. Figure 19 collects ROC and PR curves on one page so you can choose operating points explicitly.

Imbalance and thresholds

Use class or sample weights (e.g., inverse prevalence) inside the loss, and pick thresholds via ROC/PR curves or explicit cost ratios rather than defaulting to 0.5. With symmetric priors but asymmetric costs, predict class 1 when the logit exceeds $\log(c_{10}/c_{01})$; for rare positives, report PR-AUC alongside AUROC.

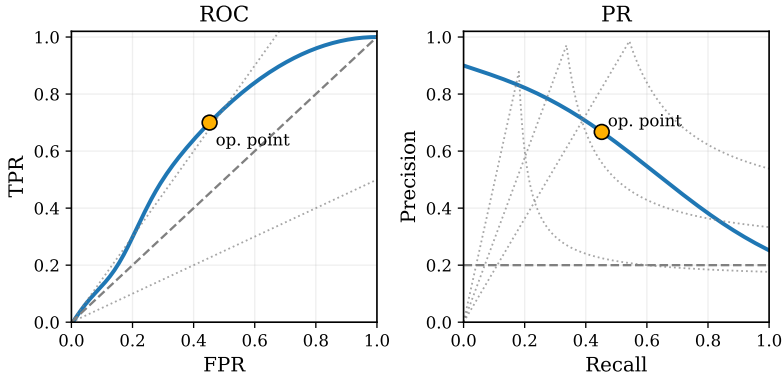


Figure 19: ROC and PR curves with an explicit operating point. Left: ROC curve with iso-cost lines; right: PR curve with a class-prevalence baseline and iso-F1 contours. Together they visualize threshold trade-offs and calibration quality. *Source: author-generated.*

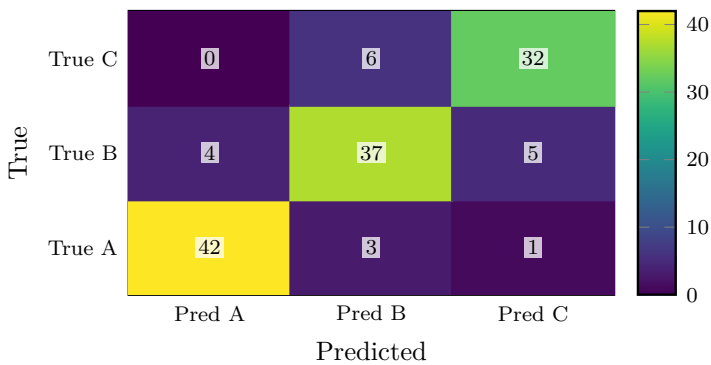


Figure 20: Confusion matrix for a three-class classifier; diagonals dominate, indicating strong accuracy with modest confusion between classes B and C.

Key takeaways

Minimum viable mastery

- Logistic regression models class probability with a sigmoid link and maximizes a concave log-likelihood (equivalently minimizes a convex negative log-likelihood); there is no closed-form solution.
- ROC and PR curves provide threshold-independent evaluation; AUC summarizes performance.
- Proper feature scaling and regularization improve convergence and generalization.

Common pitfalls

- Confusing scores with decisions: choose thresholds using costs and PR/ROC trade-offs, not a default 0.5.
- Reporting only AUC: check calibration when probabilities are used downstream (reliability/ECE).
- Letting imbalance hide failure: stratify splits and audit per-class performance, not only overall accuracy.

Probability calibration

Discrimination metrics (ROC/PR, AUC) say how well a classifier ranks examples but not how reliable its probabilities are. Calibration methods such as Platt scaling and temperature scaling adjust the logits so that predicted probabilities match empirical frequencies (e.g., 0.8 scores correspond to $\approx 80\%$ positives), often measured via Expected Calibration Error (ECE) and inspected with reliability diagrams (Platt, 1999; Guo et al., 2017).

Table 2: Handling class imbalance for logistic models (Chapter 4 reference table). A compact menu of resampling, weighting, and thresholding strategies for rare positives.

Tactic	**When/why**
Stratified splits (and K-fold)	Preserve class ratios in train/validation/test to avoid optimistic validation scores.
Class weighting / cost-sensitive loss	Multiply the cross-entropy (or hinge loss) by per-class weights so minority errors matter more. Useful when collecting more data is difficult.
Resampling (over/undersampling, SMOTE)	Balance the dataset prior to training. Helps tree ensembles and linear models; pair with cross-validation to avoid overfitting. Use simple baselines (logistic/SVM) as a tie-break to detect overfitting.
Threshold tuning	Choose a decision threshold based on PR curves or cost ratios rather than default 0.5; report PR-AUC when positives are rare.

Exercises

Exercise 1 (Easy). Implement logistic regression from scratch (no library solver): forward logits, sigmoid, cross-entropy, and gradients. Verify the gradient on a tiny dataset with a finite-difference check.
Deliverable: the worst relative gradient error + the exact loss convention you used (sum vs. mean over batch).

Exercise 2 (Medium). Build an imbalanced binary dataset and treat thresholding as part of the model. Plot ROC and PR curves, pick a decision threshold that satisfies a stated constraint (e.g., max false-negative rate), and report calibration (reliability diagram or ECE).
Deliverable: ROC/PR plots + the chosen threshold and the achieved constraint + one calibration diagnostic.

Exercise 3 (Medium). Run a slice audit: define 2–3 meaningful slices (by a feature range, a subgroup, or a covariate bin), and report how performance and calibration differ across slices. If one slice is worse, propose one concrete remedy (re-weighting, data collection, or a different feature).
Deliverable: a small slice table + one paragraph describing the most actionable failure mode you found.

If you are skipping ahead. Keep three ideas: (i) Bayes-optimal boundaries are a conceptual anchor, (ii) logistic regression is the first practical probabilistic baseline, and (iii) evaluation depends on costs, thresholds, and calibration. Those recur throughout the neural and deployment chapters and remain important whenever later chapters output scores that must be turned into decisions.

Where we head next. Logistic regression is still linear at the boundary. Chapter 5 introduces trainable neuron models, and stacking nonlinear units then removes that linear ceiling and sets up multilayer networks plus backpropagation. Carry this chapter's decision discipline forward: probability quality, threshold choice, and class-imbalance handling remain core checks even when the model family changes.

Part I takeaways

- Intelligence as engineered self-correction: represent state, choose actions, verify outcomes.
- Two recurring toolkits: safe vs. heuristic moves (search) and ERM (model–loss–optimize–audit).
- Classification as a probabilistic decision problem: Bayes optimality and calibrated scores precede thresholds.
- Reading paths are a dependency graph: later chapters reuse diagnostics, notations, and audit habits introduced here.
- **Unifying lens:** this Part fixes the representation and feedback signal, and the rest is choosing an update rule plus audits that prevent silent failure.

Part II: Neural networks, sequence modeling, and NLP

How to read Part II (rolling window)

- **One loop, many architectures:** define a loss, run a forward pass, cache intermediates, backprop gradients, validate, then iterate.
- **Two bottlenecks reappear:** expressivity (can the model represent the boundary?) and trainability (can gradients reach the parameters?).
- **Continuity checks:** watch shapes and caching in feedforward models, watch masking in sequence models, and keep calibration/slice audits alongside accuracy.

Part II bridge: from ERM to trainable architectures

Part I treated learning as disciplined self-correction: represent the problem, define what "wrong" means (loss), improve systematically (optimization), and verify with honest splits and diagnostics. In Part II we keep that loop, but we make a second design choice explicit: we choose architectures that make learning possible at scale.

The guiding question becomes: "given this computation graph, can signals flow through it during both the forward pass (information) and the backward pass (gradients)?" That is why we begin with the perceptron, build to multi-layer networks and backpropagation, then add structure that matches common data: convolution for images, recurrence and associative memory for state and recall, and attention for content-based retrieval in language and other sequences.

Read the repeated ideas as purposeful: the ERM loop stays fixed, while the model class changes the kinds of generalization you can get and the kinds of bugs you can accidentally introduce (shape and caching mistakes, masking mistakes, and evaluation leakage).

Payoffs to watch for: (i) backprop as reusable local error signals, (ii) inductive bias as the reason CNNs, RNNs, and attention behave differently under the same ERM procedure, and (iii) auditing beyond accuracy as the difference between a model with a low loss and a system that makes good decisions.

5 Introduction to Neural Networks

Chapter 4 established linear and logistic models as strong baselines, but their decision boundaries stay linear in the original feature space. We now shift from a statistical lens to a biological abstraction: simple neurons whose composition yields nonlinear decision surfaces. This chapter introduces neuron models, perceptrons, activation functions, and the first learning rules (perceptron and Adaline); Figure 1 marks this as the start of the neural strand.

Learning Outcomes

- Write the forward computation of a perceptron ($\mathbf{w}^\top \mathbf{x} + b$) and interpret its decision boundary geometrically as a hyperplane.
- Implement (or trace) the perceptron and Adaline update rules and explain what triggers an update.
- Explain, with XOR as the motivating counterexample, why a single linear separator is not enough and why later chapters introduce smooth activations and gradient-based training.

Design motif

Biology as engineering abstraction: start with simple units, make the update rule explicit, and use geometry to build intuition before the algebra gets deep.

5.1 Biological Inspiration

Neural networks borrow a simple but powerful idea from biology: complex behavior can emerge from many simple units interacting through many simple connections. The goal is not to model the brain in detail, but to steal an engineering abstraction we can write down and implement: signals flow through a network of units, and learning adjusts connection strengths so the overall system changes its behavior.

Neurons and Neural Activity A biological neuron can be viewed as a processing unit that receives multiple inputs, integrates them, and produces an output if certain conditions are met. In the simplified picture we use here, the key parts are:

- **Dendrites:** Receive incoming signals from other neurons.
- **Cell body (soma):** Integrates incoming signals.
- **Axon:** Transmits the output signal to other neurons.
- **Synapses:** Junctions where signals are transmitted between neurons.

Incoming signals can excite or inhibit the neuron. When the combined influence crosses a threshold, the neuron "fires" and sends an impulse down the axon to connected neurons. Real neurons are richer than this sketch (timing, frequency, and graded effects matter), but the abstraction is enough to motivate an artificial unit that combines inputs, applies a nonlinearity, and emits an output.

Complexities and unknowns (and what we steal anyway). Real neural tissue is a biophysical system: spikes, timing, chemistry, adaptation, and many interacting feedback loops. We do not pretend to model that faithfully here. Instead, we borrow an engineering abstraction that has two virtues: (i) it is composable, and (ii) it gives us a learning rule we can write down, debug, and scale.

So what do we *keep* from biology? The idea that many small units, each doing a simple computation, can be wired into a larger system whose behavior is richer than any single unit. And what do we *let go* of? The detailed mechanisms (spike timing, ion channels, and biological realism) that matter for neuroscience but are not necessary to build useful learning machines.

A good analogy is circuit design: you can build powerful systems out of simple, idealized components without simulating electron physics at every wire. In the same spirit, we keep the compositional story (many units, many connections) and insist that the update rule is explicit and learnable. This is a recurring theme in the book: we look to nature for candidate mechanisms behind intelligent behavior, describe them in scientific terms, and then simplify them until they become concrete engineering problems we can test and improve.

5.2 From Biological to Artificial Neural Networks

Artificial neural networks (ANNs) are computational models inspired by the structure and function of biological neural systems. The goal is to create systems that can process information, learn from data, and perform tasks that require intelligence.

Key Features of Artificial Neural Networks To design an ANN that captures essential aspects of biological neural processing, several features must be considered:

1. **Architecture:** The arrangement and connectivity of neurons within the network. This includes the number of layers, the pattern of connections (e.g., feedforward, recurrent), and the flow of information.
2. **Signal Propagation:** How input signals are transmitted through the network, transformed by neurons, and produce outputs.
3. **Learning Mechanism:** The method by which the network adjusts its parameters (e.g., weights) based on data to improve performance. This involves capturing and retaining knowledge from experience.
4. **Activation Dynamics:** The rules governing neuron activation, including how neurons decide to fire based on inputs, the degree of activation, and inhibition mechanisms.

Historical Context The concept of artificial neural networks dates back to the early 1940s, with pioneering work that attempted to mathematically model neuron behavior. Over the past eight decades, ANNs have evolved significantly, leading to a variety of architectures and learning algorithms. This evolution reflects ongoing efforts to better approximate biological intelligence and to address practical challenges in computation and learning.

Where this fits in Part II. This chapter starts the neural strand by introducing the perceptron neuron (defined below) as the basic trainable unit: a weighted sum followed by an activation/threshold, together with the first learning rules (mistake-driven and gradient-driven). The point is not to model biology literally; the point is to steal an engineering abstraction we can write down, debug, and scale.

Part II local roadmap: what stays the same, what changes

What stays the same. We still follow the empirical risk minimization (ERM) loop: choose a representation, define a loss, optimize, then verify with honest evaluation and diagnostics. As we proceed, you will find that this theme generalizes across many of the algorithms we discuss.

What changes (the design space). Instead of choosing one linear model, we build systems by composing many simple units. Used on its own, a perceptron-like unit still induces a linear decision boundary in the original feature space (as in logistic regression). Its importance is that it is *composable*: by changing how units are connected, what nonlinearities they apply, and what signals drive learning, we can move from a single linear boundary to richer nonlinear decision surfaces when the task demands it. Many of these design choices are loosely inspired by how we think biological systems process information (as sketched in the section that follows), but the target remains an engineering model, not a faithful brain simulation.

How the rest of Part II builds.

- **Units → multilayer systems:** Chapter 6 chains units into multilayer networks; Chapter 7 shows how to compute all gradients efficiently (the training engine).
- **Architectures as inductive bias:** later chapters introduce structured connectivity (e.g., convolution, recurrence, attention) that changes what patterns are easy to represent and what information can flow.
- **Audit hooks:** keep learning curves, calibration, and slice checks alongside accuracy. A good loss is not automatically a good decision.

5.3 Neural Network Architectures

Neural networks can be broadly categorized based on the flow of information through their structure. Understanding these architectures is crucial for designing and analyzing neural models that mimic biological neural systems.

Feedforward Neural Networks Feedforward neural networks (FNNs) are characterized by a unidirectional flow of information from input to output layers without any cycles or loops. The information propagates forward through successive layers of neurons, each layer transforming the input received from the previous layer.

Conceptually, this can be thought of as a cascade of neuron activations where each neuron receives input signals, processes them, and passes the output to the next layer. This architecture aligns with the idea that sensory information in biological systems is processed in a hierarchical manner.

Shapes and convention. We use the row-major (deep-learning) convention from the start. A single example is a row vector $\mathbf{a}^{(l)} \in \mathbb{R}^{1\times n_l}$, a mini-batch stacks examples by rows $\mathbf{A}^{(l)} \in \mathbb{R}^{B\times n_l}$, and weights map features by right multiplication $\mathbf{W}^{(l)} \in \mathbb{R}^{n_{l-1}\times n_l}$. Biases $\mathbf{b}^{(l)} \in \mathbb{R}^{n_l}$ broadcast across the batch: $\mathbf{Z}^{(l)} = \mathbf{A}^{(l-1)}\mathbf{W}^{(l)} + \mathbf{1}_B(\mathbf{b}^{(l)})^\top$, where $\mathbf{1}_B \in \mathbb{R}^{B\times 1}$ is an all-ones column vector. For a concrete check, if $B = 4$, $n_{l-1} = 3$, and $n_l = 5$, then $\mathbf{A}^{(l-1)} \in \mathbb{R}^{4\times 3}$, $\mathbf{W}^{(l)} \in \mathbb{R}^{3\times 5}$, and $\mathbf{Z}^{(l)} \in \mathbb{R}^{4\times 5}$. We reserve $\phi(\cdot)$ for kernel feature maps (Appendix C).

With that convention fixed, if we denote the input vector as $\mathbf{x}$, the output of layer l as $\mathbf{a}^{(l)}$, and the weight matrix connecting layer $l-1$ to layer l as $\mathbf{W}^{(l)}$, the feedforward operation is:

$$\begin{aligned} \mathbf{z}^{(l)} &= \mathbf{a}^{(l-1)}\mathbf{W}^{(l)} + \mathbf{b}^{(l)} \\ \mathbf{a}^{(l)} &= f(\mathbf{z}^{(l)}). \end{aligned} \tag{5.1}$$

where $\mathbf{b}^{(l)}$ is the bias vector and $f(\cdot)$ is the activation function applied element-wise.

Recurrent Neural Networks In contrast, recurrent neural networks (RNNs) allow information to flow in cycles, enabling feedback connections. This means that the network's state at a given time depends not only on the current input but also on previous states, effectively creating a form of memory.

The recurrent architecture is more flexible and biologically plausible since neurons can influence each other bidirectionally and inputs/outputs can be introduced or sampled at various points in the network. This allows modeling of

temporal sequences and dynamic behaviors. A simple recurrent update is

$$\mathbf{h}_t = f(\mathbf{x}_t \mathbf{W}_{xh} + \mathbf{h}_{t-1} \mathbf{W}_{hh} + \mathbf{b}_h), \qquad \mathbf{y}_t = \mathbf{h}_t \mathbf{W}_{hy} + \mathbf{b}_y,$$

with the full treatment deferred to Chapter 12.

5.4 Activation Functions

Activation functions determine how the input to a neuron is transformed into an output signal, effectively controlling the neuron's excitation level. They play a critical role in enabling neural networks to model complex, nonlinear relationships.

Biological Motivation In biological neurons, excitation occurs when the combined chemical signals exceed a certain threshold, triggering an action potential (a "fire"). Similarly, artificial neurons use activation functions to decide whether to activate (fire) based on their input.

Common Activation Functions Activation functions map the aggregated input z to a neuron's output $y = f(z)$; they inject nonlinearity and control gradient flow during learning. Different choices trade off biological plausibility, numerical stability, and ease of optimization.

- **Step Function (Heaviside)**:

$$f(x) = \begin{cases} 1 & x > 0 \\ 0 & x \leq 0 \end{cases}$$

  Models a binary firing behavior but is not differentiable, limiting its use in gradient-based learning.

- **Sign Function**:

$$f(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$$

  Allows for inhibitory (negative) outputs, mimicking excitatory and inhibitory neuron behavior. We adopt the convention $f(0) = 0$; some au-

thors either leave $\operatorname{sgn}(0)$ undefined or set it to $+1$, so it is helpful to state the choice explicitly.

- **Linear Function**:
$$f(x) = x$$
Useful in some contexts but cannot model nonlinearities alone.

- **Sigmoid Function**:
$$f(x) = \frac{1}{1 + e^{-x}}$$
Smoothly maps inputs to $(0, 1)$, differentiable, and historically popular. Because sigmoid outputs saturate near 0 and 1, gradients can become small in deep stacks; later chapters discuss practical workarounds and alternatives.

- **Hyperbolic Tangent (tanh)**:
$$f(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$
Maps inputs to $(-1, 1)$, zero-centered, often preferred over sigmoid.

- **ReLU (Rectified Linear Unit)**:
$$f(x) = \max(0, x)$$
Computationally efficient and helps mitigate vanishing gradient problems.

Notation note: activations and thresholds

In this chapter we use $f(\cdot)$ as a generic placeholder for an activation function; when we need the logistic sigmoid specifically we write $\sigma(\cdot)$. Elsewhere in the book, $\phi(\cdot)$ denotes a kernel feature map. For thresholded functions we adopt $H(0) = 1$ (Heaviside) and $\operatorname{sgn}(0) = 0$ by convention. These choices do not affect continuous models but keep examples consistent.

5.5 Learning Paradigms in Neural Networks

When building a neural network, whether feedforward or recurrent, the fundamental process involves producing an output, comparing it with a target, and then adjusting the network parameters based on the error. This iterative process is the essence of *learning*. We distinguish several learning paradigms depending on the availability and nature of the target information:

Supervised Learning In supervised learning, the network is provided with input-output pairs. The network produces an output for a given input, compares it to the known target output, computes an error, and updates its parameters to reduce this error. This requires labeled data and is the most common learning paradigm in practice.

Unsupervised Learning In unsupervised learning, there is no explicit target output. The network must discover patterns or structure in the input data by itself. This often involves competition among different patterns, where some patterns become dominant and reinforce themselves, while others are suppressed. The network evolves until it reaches an equilibrium state where the learned representations stabilize. Beyond competitive learning, unsupervised methods encompass clustering, density estimation, dimensionality reduction, autoencoders, and modern self-supervised objectives—any setting where structure is inferred directly from the inputs.

Reinforcement Learning Reinforcement learning (RL) models learning from interaction with feedback. An agent with policy $\pi(a \mid s)$ selects actions, collects rewards, and updates π to improve expected return. Full RL treatments appear later; here the point is that not all learning is supervised, and neural-network controllers are common in modern RL.

5.6 Fundamentals of Artificial Neural Networks

The foundational model of artificial neural networks dates back to McCulloch and Pitts (1943), who proposed a simple neuron model capturing essential features of biological neurons.

McCulloch-Pitts Neuron Model Consider a single neuron with multiple binary inputs $x_i \in \{0, 1\}$, $i = 1, \ldots, n$. Each input is associated with a weight w_i, which can be positive (excitatory) or negative (inhibitory). The neuron computes a weighted sum of its inputs:

$$S = \sum_{i=1}^{n} w_i x_i. \tag{5.2}$$

The output y of the neuron is determined by comparing S to a threshold θ:

$$y = \begin{cases} 1, & \text{if } S \geq \theta, \\ 0, & \text{otherwise.} \end{cases} \tag{5.3}$$

Key characteristics of this model include:

- **Binary inputs:** Inputs are either active (1) or inactive (0).
- **Excitatory and inhibitory weights:** Weights $w_i > 0$ excite the neuron, while $w_i < 0$ inhibit it.
- **Thresholding:** The neuron fires (outputs 1) only if the weighted sum exceeds the threshold.

Interpretation This simple neuron can be viewed as a linear classifier that partitions the input space into two regions separated by the hyperplane defined by the equation

$$\sum_{i=1}^{n} w_i x_i = \theta.$$

The learning task reduces to finding appropriate weights w_i and threshold θ that correctly classify inputs.

Excitation and Inhibition The neuron can be excited or inhibited depending on the sign and magnitude of the weights. For example:

- If all $w_i > 0$, the neuron is purely excitatory.
- If some $w_i < 0$, those inputs inhibit the neuron.

- The balance of excitation and inhibition determines the neuron's response.

In biological circuits inhibition is carried by specialized interneurons, whereas here a negative weight is an abstract shortcut; the sign simply indicates whether an input pushes the weighted sum above or below the threshold. Artificial neurons are function approximators; similarity to biology is inspirational, not mechanistic.

Learning Objective In this model, learning can be interpreted as adjusting the weights w_i and threshold θ to achieve desired input-output mappings. The challenge is to find these parameters such that the neuron outputs 1 for inputs belonging to a certain class and 0 otherwise.

5.7 Mathematical Formulation of the Neuron Output

To summarize, the neuron output is given by

$$y = f\left(\sum_{i=1}^{n} w_i x_i - \theta\right), \tag{5.4}$$

where $f(\cdot)$ is the activation function, which in the McCulloch-Pitts model is a Heaviside step function:

$$f(z) = \begin{cases} 1, & z \geq 0, \\ 0, & z < 0. \end{cases}$$

We explicitly set $f(0) = 1$; other texts sometimes use $f(0) = \frac{1}{2}$, so documenting the convention avoids confusion when comparing derivations. It is also common to absorb the threshold into an augmented weight vector by defining $x_0 = 1$ and $w_0 = -\theta$, yielding a pure inner product $\mathbf{w}^\top \mathbf{x}$ that we will reuse in the perceptron section.

This model laid the groundwork for later developments in neural networks, including the introduction of differentiable nonlinearities that enable gradient-based learning.

5.8 The McCulloch–Pitts Neuron: Logical Computation and Limits

Recall the MP neuron definition in (5.2)–(5.3) (equivalently (5.4)); here we focus on logic-gate examples and the limitations that motivate learnable variants.

Example: AND and OR gates

- **AND:** For inputs x_1, x_2, set $w_1 = w_2 = 1$ and $\theta = 2$. The unit outputs 1 only when both inputs are 1, matching the AND truth table.
- **OR:** Keep $w_1 = w_2 = 1$ but set $\theta = 1$. The unit outputs 1 when at least one input is 1, matching OR.

This demonstrates how the MP neuron can implement simple logical functions by appropriate choice of weights and threshold.

Limitations of the MP model Despite its conceptual simplicity, the MP neuron has significant limitations:

- **No learning mechanism:** The weights and threshold must be manually assigned or guessed. There is no algorithmic way to adjust parameters based on data.
- **Limited computational power:** The MP neuron can only represent linearly separable functions. Complex patterns requiring nonlinear decision boundaries cannot be modeled.
- **Binary inputs and outputs:** The model is restricted to binary signals, limiting its applicability to real-valued data.

These limitations motivated the development of more sophisticated neuron models and learning algorithms.

5.9 From MP Neuron to Perceptron and Beyond

The MP neuron laid the groundwork for subsequent models that introduced learning capabilities and continuous-valued inputs and outputs.

Perceptron model The perceptron, introduced by Rosenblatt in 1958, extends the MP neuron by incorporating a learning algorithm to adjust weights

based on training data. The perceptron output is

$$y = \begin{cases} 1 & \text{if } \mathbf{w}^\top \mathbf{x} + b \geq 0, \\ 0 & \text{otherwise,} \end{cases} \tag{5.5}$$

where $\mathbf{x}$ is the input vector, $\mathbf{w}$ the weight vector, and b the bias (threshold). Here $\mathbf{w}^\top \mathbf{x}$ is simply the scalar inner product; under the book's row-major convention you can read it as $\mathbf{x}\mathbf{w}$ as well.

Targets and encodings

We switch between labels in `{0,1}` (probability view) and labels in `{-1,+1}` (margin view). Convert with `y_pm = 2*y01 - 1` and `y01 = (y_pm + 1)/2`. Perceptron updates below use the `{-1,+1}` encoding.

The perceptron learning rule iteratively updates weights to reduce classification errors, enabling the model to learn from data rather than relying on manual parameter selection. The signed-margin derivation below yields the update used in practice; the induced separating hyperplane and signed distance are illustrated in Figure 21.

Perceptron update from the signed margin. Let $d_i = y_i(\mathbf{w}^\top \mathbf{x}_i + b)$ be the signed margin. If $d_i \geq 0$ the example is correctly classified; if $d_i < 0$ the example is misclassified. A common perceptron criterion is

$$J(\mathbf{w}, b) = -\sum_{i \in \mathcal{M}} d_i = -\sum_{i \in \mathcal{M}} y_i(\mathbf{w}^\top \mathbf{x}_i + b),$$

where $\mathcal{M}$ is the set of misclassified examples. Taking a gradient step on J yields

$$\mathbf{w} \leftarrow \mathbf{w} + \eta y_i \mathbf{x}_i, \qquad b \leftarrow b + \eta y_i,$$

which is exactly the perceptron update. In augmented form, set $x_0 = 1$ and $w_0 = b$, and the update becomes $\mathbf{w} \leftarrow \mathbf{w} + \eta y_i \mathbf{x}_i$. Geometrically, each mistake nudges the hyperplane so the signed distance d_i increases.

Worked example: learning an OR gate by mistake-driven updates

We use labels $y \in \{-1, +1\}$ (so OR has targets $(-1, +1, +1, +1)$ for $(0,0), (0,1), (1,0), (1,1)$), learning rate $\eta = 1$, and initialize $\mathbf{w} = (0,0)$, $b = 0$. Cycle through the four inputs in that fixed order; whenever $y(\mathbf{w}^\top \mathbf{x} + b) \leq 0$, apply $\mathbf{w} \leftarrow \mathbf{w} + y\mathbf{x}$, $b \leftarrow b + y$.

The sequence of updates (after each mistake) is:

step	(x_1, x_2), y	(w_1, w_2)	b
1	$(0,0), -1$	$(0,0)$	-1
2	$(0,1), +1$	$(0,1)$	0
3	$(1,0), +1$	$(1,1)$	1
4	$(0,0), -1$	$(1,1)$	0
5	$(0,0), -1$	$(1,1)$	-1
6	$(0,1), +1$	$(1,2)$	0
7	$(0,0), -1$	$(1,2)$	-1
8	$(1,0), +1$	$(2,2)$	0
9	$(0,0), -1$	$(2,2)$	-1

After these 9 updates, predicting $+1$ when $\mathbf{w}^\top \mathbf{x} + b \geq 0$ (and -1 otherwise) with $\mathbf{w} = (2,2)$, $b = -1$ labels all four OR inputs correctly.

Perceptron convergence theorem. If a training set is linearly separable with margin $\gamma > 0$, the perceptron learning algorithm is guaranteed to find a separating hyperplane after at most $(R/\gamma)^2$ updates, where R bounds the input norms. Rescaling features changes R and γ, so standardizing inputs tightens the bound. When the data are not separable the algorithm can cycle forever; Section 6.1 (and Chapter 6) therefore emphasize feature scaling, bias terms, and the move to differentiable multilayer models to handle nonlinear problems.

Perceptron convergence theorem (proof sketch)

Assume there exists a unit vector $\mathbf{w}^\star$ such that $y_i\,\mathbf{w}^\star\cdot\mathbf{x}_i \geq \gamma$ for all i and that $\|\mathbf{x}_i\| \leq R$. Let $\mathbf{w}^{(t)}$ denote the perceptron weights after t mistakes, with $\mathbf{w}^{(0)} = \mathbf{0}$. Each mistake updates:

$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} + y_i\mathbf{x}_i.$$

1. **Progress along the separator.** The inner product with w* grows by at least gamma each mistake:

$$\begin{aligned}\mathbf{w}^{(t+1)} \cdot \mathbf{w}^\star &= \mathbf{w}^{(t)} \cdot \mathbf{w}^\star + y_i\,\mathbf{x}_i \cdot \mathbf{w}^\star \\ &\geq \mathbf{w}^{(t)} \cdot \mathbf{w}^\star + \gamma.\end{aligned}$$

   Thus after T mistakes, the dot product with w* is at least T*gamma.

2. **Bounding the norm.** The squared norm grows slowly:

$$\begin{aligned}\|\mathbf{w}^{(t+1)}\|^2 &= \|\mathbf{w}^{(t)}\|^2 + \|\mathbf{x}_i\|^2 + 2y_i\mathbf{x}_i \cdot \mathbf{w}^{(t)} \\ &\leq \|\mathbf{w}^{(t)}\|^2 + R^2,\end{aligned}$$

   because the mistake condition implies $y_i\mathbf{x}_i \cdot \mathbf{w}^{(t)} \leq 0$. Inductively, $\|\mathbf{w}^{(T)}\|^2 \leq TR^2$.

3. **Combine via Cauchy–Schwarz.**

$$\begin{aligned}T\gamma &\leq \mathbf{w}^{(T)} \cdot \mathbf{w}^\star \\ &\leq \|\mathbf{w}^{(T)}\|\|\mathbf{w}^\star\| \leq \sqrt{T}\,R,\end{aligned}$$

   which implies $T \leq (R/\gamma)^2$.

Therefore the perceptron halts after finitely many mistakes on separable data. If the data are not separable, some $\gamma > 0$ cannot be found, and the above argument no longer applies; hence the need for multilayer networks.

Universal Approximation Theorem (UAT)

Statement (Cybenko, 1989; Hornik, 1991): A feedforward network with a single hidden layer containing a *sufficient* (finite) number of units with a non-constant, bounded, and monotone-increasing continuous activation function can approximate any continuous function $f : \mathbb{R}^n \to \mathbb{R}$ on a compact set $K \subset \mathbb{R}^n$ to any desired accuracy $\epsilon > 0$.
Formally, for any $\epsilon > 0$ there exist weights and biases such that

$$\sup_{\mathbf{x} \in K} \left| \hat{f}(\mathbf{x}) - f(\mathbf{x}) \right| < \epsilon.$$

What it does *not* say. The theorem is purely existential: it guarantees that an approximating network *exists* but says nothing about how wide the hidden layer must be, how to find the weights, or how well gradient-based training will locate the solution. In practice, depth (multiple layers) often requires far fewer total parameters than the width needed for a single-layer approximation. The perceptron limitations you just saw are a reminder: expressivity alone does not imply trainability.

Common perceptron pitfalls

- **Feature scaling:** Large-magnitude features dominate updates; standardize inputs first.
- **Random seed sensitivity:** Different initial weights can lead to drastically different separating hyperplanes.
- **Non-separable data:** Without slack variables or kernels the perceptron will not converge; diagnose this before training indefinitely. XOR is the canonical counterexample.

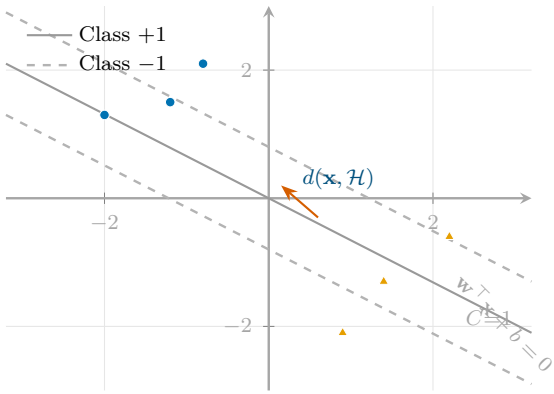


Figure 21: Perceptron geometry. Points on opposite sides of the separating hyperplane receive different labels, and signed distance to the boundary controls both prediction and update magnitude. Compare with Figure 18 in Chapter 4: both are linear separators, but logistic smooths the boundary into calibrated probabilities.

Geometry intuition: beyond XOR (two triangles)

XOR is the canonical non-separable toy, but the limitation is geometric, not specific to a truth table. A second picture to keep in mind is *two interleaved triangles*: one class occupies the vertices of a large triangle, while the other occupies the vertices of a smaller triangle rotated inside it. No single line separates the two sets; any separating boundary must "bend" or be assembled from multiple linear pieces.

Why this matters. A perceptron draws exactly one hyperplane, so it cannot represent such boundaries. Chapter 6 restores expressivity by composing units so the overall decision boundary can be piecewise-linear (or curved) while still being trainable by gradients.

Adaline model The Adaptive Linear Neuron (Adaline), developed in the 1960s, further improves on the perceptron by using a linear activation function and minimizing a continuous error function (mean squared error). This allows the use of gradient descent for training, leading to more stable convergence.

Adaline weight update (derivation) Adaline uses a linear output $y = \mathbf{w}^\top \mathbf{x} + b$ and the squared error

$$E = \tfrac{1}{2}(t - y)^2.$$

The gradient is $\partial E/\partial \mathbf{w} = -(t - y)\mathbf{x}$ and $\partial E/\partial b = -(t - y)$, so the update is

$$\mathbf{w} \leftarrow \mathbf{w} + \eta(t - y)\mathbf{x}, \qquad b \leftarrow b + \eta(t - y).$$

Unlike the perceptron, Adaline updates on every example and scales the step by the residual $t - y$; this is the first explicit appearance of gradient-based weight optimization in the neural narrative.

The perceptron and Adaline models are limited to linearly separable problems. To overcome this, multilayer perceptrons (MLPs) with hidden layers were introduced; Chapter 6 and Chapter 7 develop the mechanics in full.

Perceptron vs. logistic regression

Linear score $s(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b$. The perceptron predicts $\mathbb{1}[s \geq 0]$ and updates $\mathbf{w} \leftarrow \mathbf{w} + \eta y_i \mathbf{x}_i$ (and $b \leftarrow b + \eta y_i$) only on mistakes, with $y_i \in \{-1, +1\}$. Logistic regression predicts $\sigma(s)$, minimizes cross-entropy $-\sum_i y_i \log \sigma(s_i) + (1 - y_i)\log(1 - \sigma(s_i))$, and steps by $\sum_i (\sigma(s_i) - y_i)\mathbf{x}_i$. Prefer logistic when calibrated probabilities and smooth optimization are needed (Chapter 4).

Author's note: what a single perceptron cannot express

A single perceptron makes one global, all-or-none decision: one hyperplane, one threshold, one set of weights shared across every example. That simplicity is the point of the model, and it is also the source of the limitation.

Many real problems need *communities* of units that can specialize. Different hidden units respond to different regions, features, or patterns, and the model combines those responses into a richer decision surface. Multi-layer networks do not only add parameters; they add internal structure that lets different parts of the model "care about" different parts of the data.

Perceptron learning loop (pseudocode)

```
function TrainPerceptron(X, y_pm1, eta, epochs):
    # X: N x d features, y in {-1,+1}
    w <- 0_d
    b <- 0
    for epoch in 1..epochs:
        for i in 1..N:
            s <- dot(w, X[i]) + b
            yhat <- +1 if s >= 0 else -1
            if yhat != y_pm1[i]:
                w <- w + eta * y_pm1[i] * X[i]
                b <- b + eta * y_pm1[i]
    return w, b
```

The logic is intentionally simple: compute a signed score, threshold it, and update only on mistakes. Logistic regression keeps the same linear score but replaces the hard threshold with a probabilistic link and a smooth objective (Chapter 4); multilayer networks replace the single separator with learned intermediate features (Chapters 6 to 7).

Key takeaways

Minimum viable mastery

- The perceptron and Adaline turn threshold units into trainable classifiers by updating weights from data.
- Geometry (hyperplanes and signed distance) explains predictions and update magnitude.
- Logistic regression keeps the same linear score but learns calibrated probabilities via a smooth loss (Chapter 4).
- Nonlinear tasks (e.g., XOR) require multilayer networks and backpropagation (Chapters 6 to 7).

Common pitfalls

- Expecting a single hyperplane to solve nonconvex structure: without hidden units you cannot express XOR-like logic.
- Mixing label codings ($\{-1, +1\}$ vs. $\{0, 1\}$) without adjusting the loss/update formulas.
- Treating linear scores as probabilities: calibrated probabilities require a probabilistic model/loss (e.g., logistic).

Exercises

Exercise 1 (Easy). Train a perceptron on a 2D linearly separable dataset.
Deliverable: a plot of the data + the learned separating line, and a short note stating the label encoding you used.

Exercise 2 (Easy). Repeat Exercise 1 with Adaline (squared error + gradient descent).
Deliverable: one learning curve and a comparison of convergence behavior versus the perceptron.

Exercise 3 (Medium). Build the XOR dataset and show, empirically, the failure mode of a single hyperplane.
Deliverable: a plot + a one-paragraph explanation of why this forces the move to multi-layer models in Chapter 6.

If you are skipping ahead. Remember the two bottlenecks that force multilayer networks: expressivity (nonlinear boundaries) and trainability (smooth gradients). Chapter 6 and Chapter 7 assume these motivations.

Where we head next. Perceptrons are intentionally simple: hard thresholds and uniform updates. Their strengths (linear separation) and limits (for example XOR) motivate multilayer models. Chapter 6 continues this thread by chaining units, defining a loss, and asking the key training question: *how should weights change to improve performance?* That question leads directly to the chain rule and smooth activations. Chapter 7 then scales the same cache-and-chain-rule accounting to arbitrary depth and efficient implementation.

6 Multi-Layer Perceptrons: Challenges and Foundations

Chapter 5 introduced the perceptron and Adaline: single units that learn by updating weights from data, with Adaline giving our first explicit glimpse of gradient descent on a smooth performance function.

We now take the smallest step beyond a single unit. We connect two neurons in series, treat the whole diagram as one trainable model, and choose a simple performance function that tells us when the output is moving in the right direc-

tion. This is the step where the simple unit becomes itself a building block. The central question is practical: *if performance improves when the output moves one way, how should each weight move?*

As soon as we try to compute those derivatives, a key obstacle appears: hard thresholds do not carry useful gradient information. Replacing them with smooth activations fixes the learning signal and gives us a clean update story. We keep the network small enough that every intermediate quantity is visible and derive the update rules once in full; scaling the same logic to deeper models is then a matter of organization.

Learning Outcomes

By the end of this chapter, you should be able to:

- Write a complete forward pass for a two-neuron chain, including intermediate quantities (p_1, y_1, p_2, y_2).
- Choose a scalar performance function P, explain what "improving performance" means, and sanity-check one tiny update numerically.
- Use the chain rule to derive the gradients for $w_2, b_2, \mathbf{w}_1, b_1$ and interpret δ terms as reusable local error signals.
- Explain why hard thresholds break derivative-based learning and why smooth activations restore a usable learning signal.

Design motif

Start small. Keep every intermediate quantity visible. Let the chain rule explain how the learning signal moves, and use quick numerical checks to catch sign and accounting mistakes early.

6.1 From a single unit to the smallest network

A short map: building a trainable network. The chapter follows one tight loop:

- **Build:** write the forward computation (a two-neuron chain).
- **Judge:** define a scalar performance function P (we use squared error).
- **Move:** update parameters using the direction given by derivatives (gradi-

ent descent).

- **Fix:** Take out the threshold and make the unit differentiable so the learning signal can pass through it.

Once you can execute this loop for two neurons, scaling to many neurons is mostly accounting; we return to the organization when we scale up.

How this chapter fits the workflow. The objective-and-audit loop still applies; what changes is the *representation*. Once a model has internal computations, you have to keep track of intermediate variables and make sure the learning signal can actually pass through them. Before you scale up, do a few small sanity checks: confirm the forward pass, confirm the gradient signs, and check one tiny numerical update.

- From Chapter 3: diagnostics (learning curves, bias–variance) tell you *what* is going wrong.
- From Chapter 4: a linear probabilistic baseline tells you *how far* you can go without nonlinear features.
- Here: we build the smallest multi-layer network and derive its update rules, so the mechanics feel concrete.

Function estimation as the unifying view. Learning is about building a usable input→output mapping from examples. The difference from linear models is not the goal, but the *representation*: a neural network builds the mapping by composing simple units, so intermediate signals become part of the model. Each unit implements its own local mapping (from its input to its pre-activation and activation), and these local maps accumulate into a successful nonlinear mapping $f : X \to Y$. In this chapter we keep the accounting minimal—one tiny network and a simple squared-error objective—so we can focus on how the update rules emerge.

From one unit to a chain of units. A single unit (the perceptron) computes a weighted sum and then applies an activation:

$$y = f(p), \qquad p = \mathbf{w}^\top \mathbf{x} + b, \tag{6.1}$$

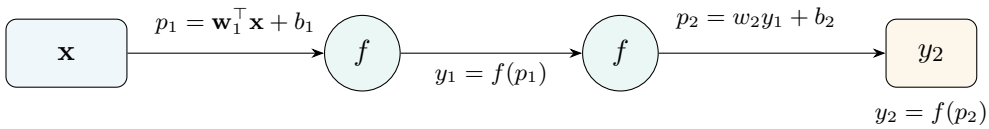


Figure 22: The minimal neural network used in this chapter is a two-neuron chain. The first unit produces an intermediate signal, and the second unit maps that signal to the final output.

where $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{w} \in \mathbb{R}^n$, and b is a bias. With one unit, the decision boundary $\mathbf{w}^\top \mathbf{x} + b = 0$ is a hyperplane, so a single unit can only produce linear separations; XOR is the canonical reminder of that limitation.

The smallest step beyond a single unit is a *two-neuron chain*: one unit feeds another. Write

$$\begin{aligned} p_1 &= \mathbf{w}_1^\top \mathbf{x} + b_1, \quad y_1 = f(p_1), \\ p_2 &= w_2 y_1 + b_2, \quad y_2 = f(p_2). \end{aligned} \tag{6.2}$$

This is the point where the unit becomes a building block: an intermediate signal y_1 is computed, reused, and then transformed again. Complexity now comes not only from what each block does in isolation, but from how the blocks connect, and how a change in one propagates to the other. We will reuse Figure 22 as our tracking diagram when we apply the chain rule.

Author's note: why a network changes the story. With one perceptron, every example pushes on the same single separator. But if you add a second unit, you are no longer forcing the entire problem through one shared knife-edge: you create intermediate signals that can specialize. Some units become sensitive to one pattern, others to another, and a later unit learns how to combine them.

I like to read those intermediate signals as learned transformations whose job is simple: make the final separation easier than it looked in the raw input space.

A checklist of what we must settle (and why). A diagram becomes a learning machine only after we make four choices explicit:

- **Parameters:** which numbers we are allowed to change (weights and biases).
- **Performance:** a single scalar P that measures how wrong the current

output is.

- **Update:** a rule that changes parameters to reduce P.
- **Differentiability:** if the update is derivative-based, the path from parameters to P must be differentiable so the chain rule can assign credit (and blame).

We keep the network tiny so you can see all four ingredients at the same time.

Bias as a learned threshold. A threshold can be written as a bias. If $p = \mathbf{w}^\top \mathbf{x} - \theta$, define $b = -\theta$ and write $p = \mathbf{w}^\top \mathbf{x} + b$. In practice we append $x_0 = 1$ and treat b as an extra weight w_0; the algebra is identical. Intuitively, the bias controls *where* the unit switches, while the weights control *which directions* in input space push the unit toward switching. This simple trick lets us treat the bias exactly like another weight in both notation and updates.

6.2 Performance: what are we trying to improve?

Once we have a forward computation, we need a performance function that tells us whether the output is good. For one training example with target t, a simple choice is the squared error

$$P = \frac{1}{2}(y_2 - t)^2. \tag{6.3}$$

The factor $\frac{1}{2}$ makes derivatives cleaner. If you prefer to *maximize* a score rather than minimize an error, you could take $-\frac{1}{2}(y_2 - t)^2$ instead. The math below is identical up to a sign. We will minimize P.

Why a square? The signed error $e = y_2 - t$ can be positive or negative. Squaring removes the sign, penalizes large mistakes more than small ones, and keeps P smooth. That smoothness matters: if P changes continuously with the weights, derivatives can give a reliable direction for improvement. It is also easy to interpret and explain.

Author's note: one objective is enough for the first derivation

I start with squared error for one reason: it keeps the algebra short, so the chain rule is the only moving part you have to watch. The lesson here is not "squared error is always right." The lesson is learning how to take a forward computation and turn it into correct weight updates.
Once that story is clear, you can swap the objective and redo only the small set of derivatives that touch the output layer.

A geometric intuition. For a fixed input, the performance becomes a surface over the weights. If the surface is bowl-shaped, the best parameters sit near the bottom. Learning is then "navigation": move the weights in a direction that reduces performance. This is why it is preferable to use an algorithm that follows the local slope rather than guessing directions coordinate by coordinate. Figure 23 visualizes why moving in one vector direction is typically more efficient than changing one coordinate at a time.

6.3 Gradient descent: how do weights move?

We now ask: how should $\mathbf{w}_1, w_2, b_1, b_2$ change to reduce P? The standard answer is gradient descent:

$$\theta \leftarrow \theta - \eta \nabla_\theta P, \tag{6.4}$$

where θ stands for any parameter and $\eta > 0$ is the step size. Geometrically, you can picture the performance surface as a landscape: the gradient points uphill, so we step in the opposite direction to descend toward a minimum. The step size controls how far we move; too large can overshoot, too small can crawl.

For a weight vector, the update is a vector step:

$$\Delta \mathbf{w} = -\eta \nabla_{\mathbf{w}} P. \tag{6.5}$$

This is the "move the weights in the right direction" story made precise: we do not guess the direction; we compute it from the derivative of performance. Importantly, we update *all* weights at once (a vector step), not one coordinate at a time.

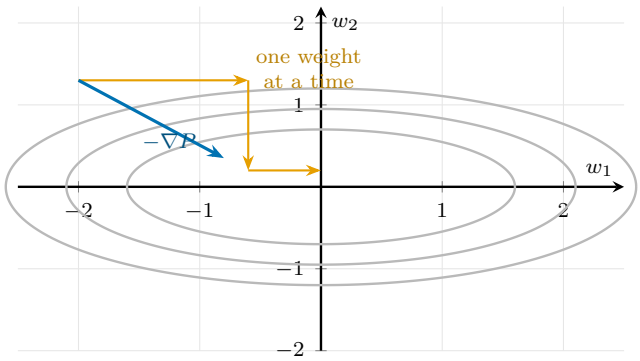


Figure 23: Think of performance as a surface over the weights. Gradient descent moves in one vector step (blue), whereas coordinate-wise updates can zig-zag (orange).

Step size is a design choice. The gradient gives a direction; the step size η sets how far you move. Too large and you bounce around (or diverge); too small and learning crawls. Many practical tricks are really ways of managing this tradeoff while keeping updates stable. In more uneven landscapes with multiple valleys, step size also affects *which* valley you settle into: aggressive steps can jump across good regions, while timid steps can get you stuck early.

6.4 Why hard thresholds block learning

At this point the story is simple: define P, compute ∇P, and update weights. The catch is that computing ∇P requires derivatives through the activation. When we apply the chain rule to the forward-pass equations in (6.2), factors like $f'(p_1)$ and $f'(p_2)$ appear immediately. Figure 24 makes this derivative contrast concrete.

If f is a hard threshold (a step function), it is discontinuous and non-differentiable at the threshold. That breaks the gradient story: $f'(p)$ either does not exist or is zero almost everywhere, so derivatives cannot guide learning. This is the core reason we replace thresholds with *smooth, differentiable activations.*

Absorbing the threshold. Folding the threshold into a bias simplifies notation, but it does not remove the real issue: a hard step is still discontinuous. When you try to compute how a weight changes performance, the chain rule

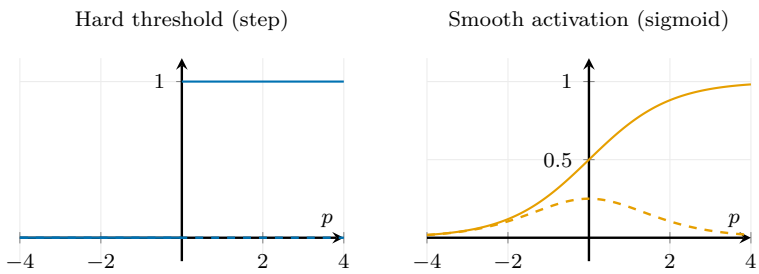


Figure 24: Hard thresholds block gradient-based learning because the derivative is zero almost everywhere. A smooth activation like the sigmoid provides informative derivatives across a wide range of inputs.

immediately introduces a derivative of the activation—terms of the form $f'(p)$ appear. With a step function, that derivative is either undefined (at the threshold) or zero almost everywhere, so the learning signal cannot pass through the unit in a useful way. This is why we replace the step with a smooth activation.

6.5 Differentiable activations and the sigmoid trick

A classic choice is the logistic (sigmoid) function

$$\sigma(p) = \frac{1}{1 + e^{-p}}. \tag{6.6}$$

It maps real inputs to $(0, 1)$ and is differentiable everywhere. The key identity is

$$\sigma'(p) = \sigma(p)\,[1 - \sigma(p)]. \tag{6.7}$$

This is a useful trick: the derivative is a function of the *output* itself. If $y = \sigma(p)$ is already computed in the forward pass, then $\sigma'(p) = y(1 - y)$ is immediately available in the backward pass. No extra exponentials are needed.

Author's note: the derivative is already in the forward pass

In practice you do not want to recompute expensive expressions during learning. With a sigmoid, once you have the forward output $y = \sigma(p)$, you also have its slope for free: $\sigma'(p) = y(1 - y)$.
That tiny identity hints at the bigger pattern: the forward pass is not only for predictions. It is also where you produce (and store) the intermediate values the backward pass will reuse. This idea, as intuitive as it might feel, is one of the reasons multi-layer perceptrons became trainable at scale and why modern deep learning can exist as an engineering discipline.

Derivation sketch. Let $\beta = \sigma(\alpha) = (1 + e^{-\alpha})^{-1}$. Differentiate and rewrite the result in terms of β:

$$\frac{d\beta}{d\alpha} = \frac{e^{-\alpha}}{(1 + e^{-\alpha})^2} = \left(\frac{1}{1 + e^{-\alpha}}\right)\left(1 - \frac{1}{1 + e^{-\alpha}}\right) = \beta(1 - \beta).$$

The key point is that once you have computed β on the forward pass, you can compute the slope from β alone, without recomputing exponentials.

6.6 Deriving weight updates for the two-neuron network

The diagram in Figure 22 is also a derivative map: to update a weight, follow how a small change in that weight would flow forward to the output and then back to the performance. The chain rule turns that story into algebra.

We now compute the gradients from the forward-pass equations in (6.2) using the chain rule. First note the easy derivatives:

- $\dfrac{\partial P}{\partial y_2} = y_2 - t.$
- $\dfrac{\partial y_i}{\partial p_i} = f'(p_i).$
- $\dfrac{\partial p_2}{\partial w_2} = y_1$ and $\dfrac{\partial p_2}{\partial y_1} = w_2.$
- $\dfrac{\partial p_1}{\partial \mathbf{w}_1} = \mathbf{x}.$

For the second-layer parameters, the chain rule gives

$$\begin{aligned}\frac{\partial P}{\partial w_2} &= \frac{\partial P}{\partial y_2}\frac{\partial y_2}{\partial p_2}\frac{\partial p_2}{\partial w_2} \\ &= (y_2 - t)\, f'(p_2)\, y_1,\end{aligned} \tag{6.8}$$

and similarly

$$\frac{\partial P}{\partial b_2} = (y_2 - t) f'(p_2). \tag{6.9}$$

For the first-layer parameters, the effect of the second layer appears explicitly through the chain rule:

$$\begin{aligned}\frac{\partial P}{\partial \mathbf{w}_1} &= \frac{\partial P}{\partial y_2}\frac{\partial y_2}{\partial p_2}\frac{\partial p_2}{\partial y_1}\frac{\partial y_1}{\partial p_1}\frac{\partial p_1}{\partial \mathbf{w}_1} \\ &= (y_2 - t)\, f'(p_2)\, w_2\, f'(p_1)\, \mathbf{x},\end{aligned} \tag{6.10}$$

with bias derivative

$$\frac{\partial P}{\partial b_1} = (y_2 - t) f'(p_2) w_2 f'(p_1). \tag{6.11}$$

To make the reuse explicit, define local error terms:

$$\delta_2 := \frac{\partial P}{\partial p_2} = (y_2 - t) f'(p_2). \tag{6.12}$$

Then $\partial P/\partial w_2 = \delta_2 y_1$ and $\partial P/\partial b_2 = \delta_2$. The first layer receives an error signal propagated backward:

$$\delta_1 := \frac{\partial P}{\partial p_1} = \delta_2 w_2 f'(p_1), \tag{6.13}$$

so $\partial P/\partial \mathbf{w}_1 = \delta_1 \mathbf{x}$ and $\partial P/\partial b_1 = \delta_1$.

This is the central lesson: once we compute a local error term, it can be reused across many gradients. That reuse is what makes multi-layer training efficient and is why deeper networks remain tractable.

Worked example: one numerical gradient step (sanity check)

Take a single input $\mathbf{x} = [1, -1]^\top$, target $t = 1$, sigmoid activation $f = \sigma$, and parameters $\mathbf{w}_1 = [0.8,\, 0.2]^\top$, $b_1 = 0$, $w_2 = 1$, $b_2 = 0$.

Forward: $p_1 = 0.6$, $y_1 = \sigma(p_1) \approx 0.646$; $p_2 = y_1$, $y_2 = \sigma(p_2) \approx 0.656$; $P = \frac{1}{2}(y_2 - t)^2 \approx 0.059$.

Backward: $\sigma'(p) = y(1 - y)$, so $\delta_2 = (y_2 - t)\sigma'(p_2) \approx -0.078$ and $\delta_1 = \delta_2 w_2 \sigma'(p_1) \approx -0.018$. Thus $\nabla_{\mathbf{w}_1} P = \delta_1 \mathbf{x} \approx [-0.018,\, +0.018]^\top$ and $\nabla_{w_2} P = \delta_2 y_1 \approx -0.050$.

Update: with $\eta = 0.5$, gradient descent increases w_2 slightly (since the gradient is negative) and nudges $\mathbf{w}_1$ in a direction that increases y_2 toward the target.

6.7 From two neurons to multi-layer networks

Nothing essential changes for deeper networks; we simply apply the same chain rule repeatedly. For a layer l with pre-activations $\mathbf{p}^{(l)}$ and weights $\mathbf{W}^{(l+1)}$, the error signal satisfies

$$\boldsymbol{\delta}^{(l)} = \left(\boldsymbol{\delta}^{(l+1)} (\mathbf{W}^{(l+1)})^\top\right) \circ f'(\mathbf{p}^{(l)}), \tag{6.14}$$

where $\circ$ denotes element-wise multiplication. This recursion is the basic backward update pattern in multi-layer networks; the only real challenge is organizing the computation so you can evaluate it efficiently.

Author's note: what changes when you scale up

Going from two neurons to many layers does not change the idea. It is still the chain rule and the same local derivatives. What changes is the *organization*.

You run a forward pass that stores what you will need, then a backward pass that reuses those stored values to compute every gradient efficiently over a whole batch. Later we turn that into an implementable algorithm and show the accounting that keeps it reliable.

6.8 Summary

- A two-neuron chain is the smallest network that goes beyond a single perceptron.
- Learning starts by defining a performance function and asking how weights change it.
- Gradient descent uses derivatives to choose the correct update direction.
- Hard thresholds obstruct gradients; smooth activations fix the problem.
- The sigmoid derivative $\sigma'(p) = \sigma(p)(1 - \sigma(p))$ is a convenient identity because it reuses the output.
- The two-neuron derivation already contains the backward-recursion pattern used in deep networks.

Derivation closure: implement, cache, fail-fast

- **Implement:** treat the chapter equations as a forward function plus a scalar loss; write them once in vector form before batching.
- **Cache:** keep $(\mathbf{x}, p_1, y_1, p_2, y_2)$ from the forward pass so each gradient term is a local reuse, not a re-derivation.
- **Fail-fast checks:** run finite-difference gradient checks on a tiny example, then track train/validation divergence to catch update-sign or step-size mistakes early.
- **Handoff:** the same cache-then-backward discipline scales directly; only the index/shape accounting grows.

Key takeaways

If you remember only a few things. The summary in Section 6.8 is the core loop: a tiny computation graph, a scalar loss, and gradients that tell you how to move weights. The pitfalls below are the places people usually lose a sign, a factor, or a cached value. **Common pitfalls**

- Treating a step function like a differentiable activation and then trying to "repair" gradients by hand.
- Dropping a cached intermediate (or reusing the wrong one) and silently breaking the chain rule.
- Mixing up pre-activations and activations; when you batch, also keep shapes explicit so you do not rely on accidental broadcasting.

Exercises

Exercise 1 (Easy). Implement finite differences for the two-neuron chain and compare to your analytic gradients.
Deliverable: the worst relative error across parameters + the exact ϵ you used and why it was not too large/small.

Exercise 2 (Medium). Replace the smooth activation with a hard threshold and attempt to train with the same gradient-descent recipe. Explain, in one paragraph, *exactly* what breaks (mathematically and numerically) and which workaround you would adopt in practice (smooth surrogate, straight-through estimator, or a different optimization view).
Deliverable: one training trace (loss vs. steps) + the paragraph explanation.

Exercise 3 (Medium). Train a tiny MLP on XOR and treat sensitivity as a debugging signal: run 10 random initializations under the same hyperparameters, then diagnose the variance.
Deliverable: a plot of decision regions for 2 successful and 2 failed runs + one paragraph linking the outcomes to initialization scale and activation saturation. **If you are reading for implementation.** Be able to read a forward graph and a backward recursion: local derivatives, cached activations, and a scalar loss driving parameter updates. That is exactly what gets scaled up in deeper networks.

Where we head next. Chapter 7 answers the practical scaling question: how to go from the smallest network in this chapter to networks with many layers and many units per layer. It uses the same ideas, but organizes them into a general recipe—a forward pass that caches intermediate quantities, and a backward pass that reuses those caches to compute gradients efficiently for deep, batched models.

7 Backpropagation Learning in Multi-Layer Perceptrons

Building on the two-neuron derivation in Chapter 6, we scale the same idea to an L-layer multi-layer perceptron (MLP). Conceptually it is still the chain rule; what changes is the accounting. We compute layerwise error signals (the δ's), reuse them to obtain every weight and bias gradient, and organize the work into a backward sweep you can implement and debug. The goal is practical: you should be able to trace one forward pass, write down one backward pass, and know exactly which stored intermediate values each gradient depends on. By the end of this chapter, the backpropagation algorithm should feel like a concrete procedure rather than a black box.

Learning Outcomes

- Derive the layerwise $\boldsymbol{\delta}$ recursion for an L-layer MLP and use it to write gradients for $\mathbf{W}^{(l)}$ and $\mathbf{b}^{(l)}$.
- Keep a consistent "shape ledger" for activations, pre-activations, and error signals so vectorized code matches the algebra.
- Run one tiny numeric trace and one finite-difference gradient check to confirm your implementation before scaling up.
- Recognize the two common output-layer patterns: squared error (general) and softmax + cross-entropy (simplified δ).

Design motif

Treat the network as a computation you can trace. Cache what the forward pass computes, then sweep backward once and reuse the same local error signals to update every parameter.

7.1 Context and Motivation

Multi-layer networks raise a specific question: *How do we update the weights across multiple layers when the only explicit error signal is at the output?* In a single-layer perceptron, the output error touches the weights directly; in a deep network, a change in one layer propagates through subsequent layers and alters the output in a nonlinear, intertwined way.

Shallow networks (one hidden layer) already move beyond linear separability, but more complex tasks demand deeper hierarchies of features. The multi-layer perceptron stacks these layers to learn richer decision boundaries, and backpropagation is the mechanism that makes that depth trainable.

Intuition: backprop as a controlled ripple effect

Changing one early weight can only affect the loss by moving the intermediate quantities it feeds into. Backpropagation does not introduce a new idea; it is the chain rule written in an efficient order: run the forward pass once, cache the intermediates you will need, then flow sensitivities backward so each layer gets its own local error signal. Identities like $\sigma'(p) = y(1-y)$ (from Chapter 6) are valuable because they let you compute a derivative from cached forward values.

Implementation lens. A practical MLP rarely updates one coordinate at a time; gradients are treated as full vectors so every weight moves coherently. Backpropagation is what turns a multilayer diagram into a trainable model: it reuses local error signals so you can compute *all* gradients efficiently, making it practical to learn hidden representations (not just tune a final linear layer) while keeping the same validation→audit discipline (learning curves, early stopping, and slice checks—performance broken down by meaningful subgroups/conditions).

7.2 Problem Setup

Consider a multi-layer perceptron with layers indexed by $l = 0, 1, \ldots, L$, where $l = 0$ is the input layer and L is the output layer. Each layer l contains neurons indexed by i, and the output of neuron i in layer l is denoted by $a_i^{(l)}$. The input to this neuron before activation is denoted by $z_i^{(l)}$. The weights connecting neuron i in layer $l-1$ to neuron j in layer l are denoted by $w_{ij}^{(l)}$.

The forward pass through the network is given by:

$$\begin{aligned} z_j^{(l)} &= \sum_i a_i^{(l-1)} w_{ij}^{(l)} + b_j^{(l)}, \\ a_j^{(l)} &= f\big(z_j^{(l)}\big). \end{aligned} \tag{7.1}$$

where $b_j^{(l)}$ is the bias term for neuron j in layer l, and $f(\cdot)$ is the activation

function, typically nonlinear (e.g., sigmoid, ReLU). Equation (7.1) makes it explicit that we sum over every incoming neuron i in layer $l-1$ to form the affine pre-activation $z_j^{(l)}$.

7.3 Loss and Objective

To keep the algebra uncluttered (and aligned with Chapter 6), we use a simple squared-error objective. Let the network output be $\mathbf{a}^{(L)}$ and let $\mathbf{t}$ be the target. For classification, $\mathbf{t}$ is often one-hot (a vector with a 1 at the correct class index and 0 elsewhere). If you want the output to behave like a probability vector, you normally pair that target with a sigmoid/softmax output and cross-entropy; we keep squared error here so the chain rule is the only moving part. For a single example, a standard choice is the half-squared error:

$$\mathcal{L} = \frac{1}{2} \sum_k \left(t_k - a_k^{(L)} \right)^2 . \tag{7.2}$$

It is often helpful to name the *error* explicitly. Define the componentwise output error as $e_k := a_k^{(L)} - t_k$ and the error vector $\mathbf{e} := \mathbf{a}^{(L)} - \mathbf{t}$. Then $\mathcal{L} = \frac{1}{2} \sum_k e_k^2 = \frac{1}{2} \|\mathbf{e}\|_2^2$, and the derivative you need to start the backward pass is $\partial \mathcal{L} / \partial a_k^{(L)} = e_k$ (squared error removes the sign inside $\mathcal{L}$, but the sign reappears in the derivative). When you train with a mini-batch, you typically use the *mean* loss over the batch; this introduces a factor $1/B$ in the gradients, which you can treat as part of the effective step size. The goal of learning is to adjust the weights $\{w_{ij}^{(l)}\}$ to minimize $\mathcal{L}$. Later in this chapter, we briefly note how common alternatives (notably cross-entropy with sigmoid/softmax outputs) simplify the output-layer error term; the backprop recursion itself does not change.

Challenges in weight updates With more than one layer, the error you measure at the output does not tell each earlier weight what to do directly. Each weight affects the loss only through a chain of intermediate quantities: it shifts a pre-activation, changes an activation, and that change fans out through downstream units.

One way to phrase the bottleneck is *credit assignment*. If a small change in $w_{ij}^{(l)}$ would make the loss smaller, we want that weight to move in that direction; if it would make the loss larger, we want it to move the other way. The chain

rule gives both the sign and the magnitude, but only if we compute the right intermediate sensitivities.

Backpropagation is the organization that makes those sensitivities reusable: it defines a local error signal (the δ's), computes them once from the output back toward the input, and then turns them into gradients for every weight and bias.

7.4 Notation for Layers and Neurons

To formalize this, we introduce the following notation:

- l: layer index, with $l = 0$ representing the input layer, and $l = L$ the output layer.
- i: neuron index in layer $l - 1$.
- j: neuron index in layer l.
- k: neuron index in layer L (output layer).
- $a_i^{(l)}$: activation of neuron i in layer l.
- $z_j^{(l)}$: weighted input to neuron j in layer l.
- $w_{ij}^{(l)}$: weight from neuron i in layer $l - 1$ to neuron j in layer l.
- $b_j^{(l)}$: bias of neuron j in layer l.
- $f(\cdot)$: activation function.

These definitions carry directly into the forward-pass recap below, where we chain the affine map and nonlinearity across layers.

Notation handoff. In Chapter 6 we wrote a unit as $y = f(p)$, where p is the pre-activation and y is the post-activation output. In this chapter we use the standard deep-learning pair $a = f(z)$: $p \leftrightarrow z$ and $y \leftrightarrow a$. The calculus is unchanged; the notation just scales cleanly to many layers and batch matrix form. If you jump into chapters out of order, keep Appendix A nearby for symbol overloads.

7.5 Forward Pass Recap

The forward pass computes activations layer by layer:

$$z_j^{(l)} = \sum_i a_i^{(l-1)} w_{ij}^{(l)} + b_j^{(l)},$$
$$a_j^{(l)} = f\big(z_j^{(l)}\big).$$

The output layer activations $a_k^{(L)}$ are compared to the targets t_k to form a loss (e.g., Equation (7.2)), and backpropagation propagates that output error backward through the layers to compute every weight gradient efficiently.

Mini example (aligned with this chapter): mean squared error (MSE) backprop on a two-layer MLP

```
# Shapes: X in R^{B x d}, W1 in R^{d x h}, W2 in R^{h x 1}
# Activations: H = f(Z1), y = f(Z2)  (use a smooth f)
def step_mse(X, t, params, eta):
    W1, b1, W2, b2 = params
    B = X.shape[0]
    # Forward pass
    Z1 = X @ W1 + b1
    H  = f(Z1)
    Z2 = H @ W2 + b2
    y  = f(Z2)
    # Loss (mean squared error over batch)
    P = 0.5 * ((y - t)**2).mean()
    # Backward pass (delta = dP/dZ)
    delta2 = (y - t) * fprime(Z2) / B
    grad_W2 = H.T @ delta2
    grad_b2 = delta2.sum(axis=0)
    delta1 = (delta2 @ W2.T) * fprime(Z1)
    grad_W1 = X.T @ delta1
    grad_b1 = delta1.sum(axis=0)
    # GD step
    return (W1 - eta * grad_W1, b1 - eta * grad_b1,
            W2 - eta * grad_W2, b2 - eta * grad_b2)
```

The elementwise product `*` mirrors the Hadamard (element-by-element) product notation from Equation (7.1). The key technical point is the `fprime` factors: this is exactly where hard thresholds break the learning signal.

Shape ledger for an L-layer MLP (batch size B)

If your gradients look "plausible" but training fails, check shapes first. For each layer:

- $\mathbf{A}^{(l-1)} \in \mathbb{R}^{B\times n_{l-1}},\ \mathbf{Z}^{(l)}, \boldsymbol{\delta}^{(l)} \in \mathbb{R}^{B\times n_l}$
- $\mathbf{W}^{(l)} \in \mathbb{R}^{n_{l-1}\times n_l},\ \mathbf{b}^{(l)} \in \mathbb{R}^{n_l}$
- $\partial L/\partial \mathbf{W}^{(l)} = (\mathbf{A}^{(l-1)})^\top \boldsymbol{\delta}^{(l)}/B \in \mathbb{R}^{n_{l-1}\times n_l}$
- $\partial L/\partial \mathbf{b}^{(l)} = \text{batch_mean}(\boldsymbol{\delta}^{(l)}) \in \mathbb{R}^{n_l}$

Layers share this structure; convolutional/sequence models reuse the same calculus with different indexing and shape conventions.

7.6 Backpropagation: Recursive Computation of Error Terms

Our goal is the gradient of the loss with respect to each weight $w_{ij}^{(l)}$ connecting layer l to layer $l+1$, i.e., the weight from neuron i in layer l to neuron j in layer $l+1$.

We will continue with the squared-error loss from Chapter 6:

$$\mathcal{L} = \frac{1}{2}\sum_k (t_k - a_k^{(L)})^2. \tag{7.3}$$

where t_k is the target output and $a_k^{(L)}$ is the activation of output neuron k. Other losses change only a few local derivatives (most notably at the output layer), but the backprop recursion and overall structure are the same.

To update the weights using gradient descent, we need to compute

$$\frac{\partial \mathcal{L}}{\partial w_{ij}^{(l)}}.$$

By the chain rule, we have

$$\frac{\partial \mathcal{L}}{\partial w_{ij}^{(l)}} = \frac{\partial \mathcal{L}}{\partial z_j^{(l+1)}} \cdot \frac{\partial z_j^{(l+1)}}{\partial w_{ij}^{(l)}}. \tag{7.4}$$

where $z_j^{(l+1)}$ is the weighted input to neuron j in layer $l+1$:

$$z_j^{(l+1)} = \sum_i a_i^{(l)} w_{ij}^{(l)} + b_j^{(l+1)}.$$

Here $a_i^{(l)}$ is the activation of neuron i in layer l, and $b_j^{(l+1)}$ the bias term.

Since $z_j^{(l+1)}$ is linear in $w_{ij}^{(l)}$, we have

$$\frac{\partial z_j^{(l+1)}}{\partial w_{ij}^{(l)}} = a_i^{(l)}.$$

Thus,

$$\frac{\partial \mathcal{L}}{\partial w_{ij}^{(l)}} = \delta_j^{(l+1)} a_i^{(l)}, \tag{7.5}$$

where we define the *error term*

$$\delta_j^{(l+1)} := \frac{\partial \mathcal{L}}{\partial z_j^{(l+1)}}.$$

Collecting the $\delta_j^{(l+1)}$ for all neurons in layer $l+1$ forms a vector $\boldsymbol{\delta}^{(l+1)}$ with the same dimension as $z^{(l+1)}$, ensuring the gradient $\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(l)}}$ has the same shape as the weight matrix.

Interpretation. The term $\delta_j^{(l+1)}$ measures how sensitive the loss is to changes in the pre-activation $z_j^{(l+1)}$. Our task reduces to computing these δ terms for all neurons in the network.

7.6.1 Output layer error terms

For the output layer L, the activation of neuron k is

$$a_k^{(L)} = f\big(z_k^{(L)}\big),$$

where $f(\cdot)$ is the activation function.

The error term for output neuron k is

$$\begin{aligned}\delta_k^{(L)} &= \frac{\partial \mathcal{L}}{\partial z_k^{(L)}} \\ &= \frac{\partial \mathcal{L}}{\partial a_k^{(L)}} \frac{\partial a_k^{(L)}}{\partial z_k^{(L)}} \\ &= \left(a_k^{(L)} - t_k\right) f'\left(z_k^{(L)}\right),\end{aligned} \tag{7.6}$$

where f' denotes the derivative of the activation function evaluated element-wise. With squared error, this is the entire story: $\partial \mathcal{L}/\partial a_k^{(L)} = a_k^{(L)} - t_k$, and the chain rule contributes the extra $f'(z_k^{(L)})$ factor.

For cross-entropy paired with a sigmoid or softmax output, the derivative of the log-likelihood removes that extra activation-derivative factor at the output, leaving $\delta_k^{(L)} = a_k^{(L)} - t_k$ (with δ still defined as $\partial \mathcal{L}/\partial z$). This simplification is one reason those loss/output pairs are so common: you avoid carrying an additional $f'(z)$ term in the output-layer learning signal.

7.6.2 Hidden layer error terms

For a hidden neuron j in layer l, the error term $\delta_j^{(l)}$ depends on the error terms of the neurons in the next layer $l+1$ to which it connects. Using the chain rule,

$$\begin{aligned}\delta_j^{(l)} &= \frac{\partial \mathcal{L}}{\partial z_j^{(l)}} \\ &= \sum_k \frac{\partial \mathcal{L}}{\partial z_k^{(l+1)}} \frac{\partial z_k^{(l+1)}}{\partial z_j^{(l)}} \\ &= \sum_k \delta_k^{(l+1)} \frac{\partial z_k^{(l+1)}}{\partial z_j^{(l)}}.\end{aligned} \tag{7.7}$$

Since

$$z_k^{(l+1)} = \sum_m a_m^{(l)} w_{mk}^{(l)} + b_k^{(l+1)},$$

and $a_j^{(l)} = f\big(z_j^{(l)}\big)$, we have

$$\frac{\partial z_k^{(l+1)}}{\partial z_j^{(l)}} = w_{jk}^{(l)} f'\big(z_j^{(l)}\big).$$

Substituting into (7.7) yields

$$\delta_j^{(l)} = f'\big(z_j^{(l)}\big) \sum_k w_{jk}^{(l)} \delta_k^{(l+1)}. \tag{7.8}$$

For sigmoid activations f, the derivative simplifies to $f'(z_j^{(l)}) = a_j^{(l)}(1 - a_j^{(l)})$; other activations require substituting their respective derivatives in (7.8).

Summary: Backpropagation recursion Backpropagation is reverse-mode automatic differentiation (AD) on the network graph. A forward pass stores intermediate values (pre-activations and activations); a backward pass reuses those stored values to compute all gradients efficiently. Frameworks (PyTorch, JAX, TensorFlow) automate the mechanics, but the manual story is what lets you sanity-check shapes, signs, and scaling when training does something surprising.

The step-by-step output-layer derivation and the hidden-layer recursion are already established in Sections 7.6.1 and 7.6.2. The next step is to see how those same gradients are accumulated in batch and stochastic settings.

7.7 Batch and Stochastic Gradient Descent

Given a training set of N examples $\{(\mathbf{x}^{(n)}, \mathbf{t}^{(n)})\}_{n=1}^N$, the weight updates can be computed in different ways:

- **Batch gradient descent:** Compute the gradient over the entire dataset and update weights once per epoch (shown here for a single output unit; the multi-layer case uses the same idea with layerwise δ's and cached activations):

$$\Delta w = -\frac{\eta}{N} \sum_{n=1}^{N} \delta^{(n)} \mathbf{x}^{(n)}.$$

- **Stochastic gradient descent (SGD):** Update weights after each training example using the instantaneous gradient $-\eta\, \delta^{(n)} \mathbf{x}^{(n)}$. The updates

are noisy, but that noise can be useful: it often keeps learning moving in flat regions and can help you avoid getting stuck too early.

Optimizer and stability notes

Start with plain SGD first; it makes it obvious whether the gradients you derived are doing the right thing. If training oscillates, add momentum; if step-size sensitivity dominates, try an adaptive method like AdamW (see Chapter 11). For regularization, tie this back to the L2 discussion in Chapter 3: in SGD, "weight decay" is weight shrinkage toward zero; in AdamW, that shrinkage is applied as a separate decay step rather than being mixed into the adaptive gradient. If gradients occasionally spike (common in deep or ill-conditioned nets), clipping is a practical safety valve. For classification, pair cross-entropy with a numerically stable softmax (log-sum-exp / subtract the max logit), as discussed in Chapter 11.

Implementation pattern (modern practice): softmax + cross-entropy

```
# Shapes: X in R^{B x d}, Y in R^{B x c} (one-hot)
#         W1 in R^{d x h}, W2 in R^{h x c}
# Note: stable softmax:
#   subtract row-wise max logit before exp.
def step_ce_softmax(X, Y, params, eta, wd=1e-4, p_drop=0.1):
    W1, b1, W2, b2 = params
    B = X.shape[0]
    # Forward pass
    Z1 = X @ W1 + b1
    H1 = relu(Z1)
    mask1 = (np.random.rand(*H1.shape) > p_drop).astype(H1.dtype)
    # Inverted dropout
    H1 = H1 * mask1 / (1 - p_drop)
    Z2 = H1 @ W2 + b2
    Yhat = softmax(Z2)
    # Backward pass: for softmax + cross-entropy, delta2 = Yhat - Y
    delta2 = (Yhat - Y) / B
    grad_W2 = H1.T @ delta2 + wd * W2
    grad_b2 = delta2.sum(axis=0)
    delta1 = (delta2 @ W2.T) * relu_deriv(Z1)
    # Dropout backprop: reuse mask + scale
    delta1 = delta1 * mask1 / (1 - p_drop)
    grad_W1 = X.T @ delta1 + wd * W1
    grad_b1 = delta1.sum(axis=0)
    # SGD step
    return (W1 - eta * grad_W1, b1 - eta * grad_b1,
            W2 - eta * grad_W2, b2 - eta * grad_b2)
```

This snippet uses the same δ recursion as the squared-error derivation; the main change is the output layer, where softmax + cross-entropy yields $\delta^{(L)} = \hat{\mathbf{Y}} - \mathbf{Y}$ (up to batch scaling). Dropout and weight decay are optional here; they are shown only to indicate where they enter the forward and backward passes.

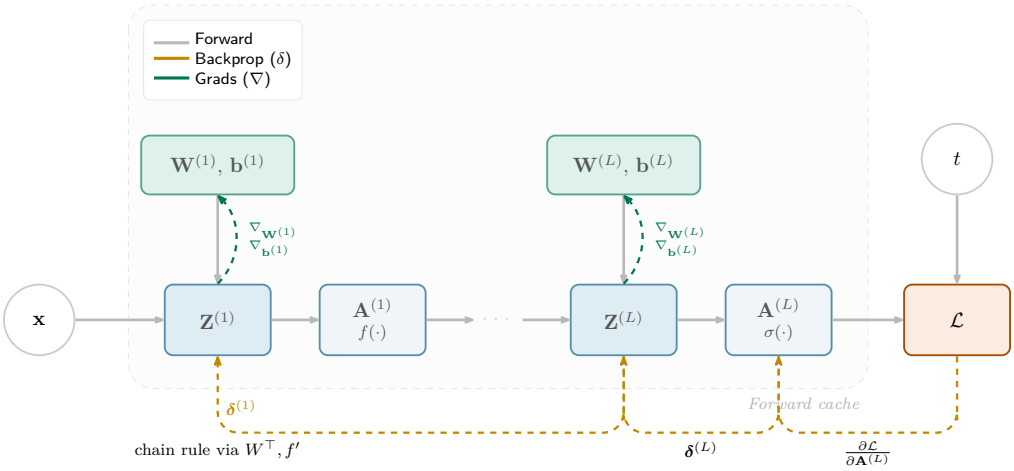


Figure 25: Computational graph for a feedforward network. Backpropagation is reverse-mode automatic differentiation (AD): the forward sweep caches intermediate values, and the reverse sweep propagates deltas while accumulating weight/bias gradients from those cached values.

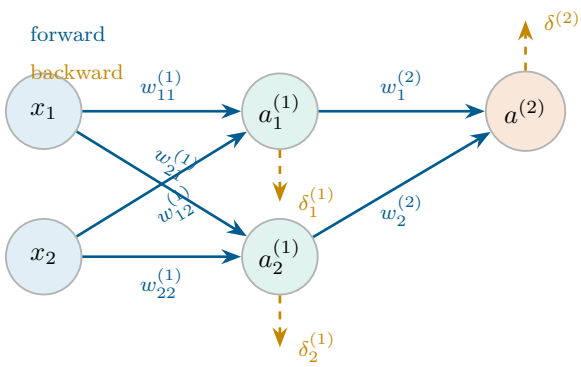


Figure 26: Forward (blue) and backward (orange) flows for a two-layer MLP. Cached activations and layerwise deltas travel along these arrows; backward signals use next-layer weights and activation derivatives.

Debugging and gradient-check checklist

- **Overfit a tiny batch:** Drive the loss near zero on a handful of samples; if you cannot, the issue is almost always code or setup, not generalization.
- **Finite differences on a tiny net:** Fix seeds, perturb one parameter, and compare analytic vs. numerical gradients. Do this before running long experiments.
- **Track per-layer gradient norms:** $\|\nabla \mathbf{W}^{(l)}\|$ that vanish (all zeros) or explode are early warnings.
- **Assert shapes and broadcasts:** Verify $\mathbf{Z}^{(l)}, \mathbf{A}^{(l)}, \boldsymbol{\delta}^{(l)}$ shapes and bias broadcasting explicitly.
- **Sanity baselines:** For a one-layer linear model, backprop should match the closed-form gradient you already know.

7.8 Backpropagation Algorithm: Brief Numerical Check

For a quick sanity check, take a tiny 2–2–1 network with sigmoid output and cross-entropy loss. Using

$$\mathbf{W}^{(1)} = \begin{bmatrix} 0.5 & -0.3 \\ 0.8 & 0.2 \end{bmatrix}, \quad \mathbf{b}^{(1)} = [0.1, -0.2],$$
$$\mathbf{W}^{(2)} = \begin{bmatrix} 0.7 \\ -0.4 \end{bmatrix}, \quad \mathbf{b}^{(2)} = 0.05,$$
$$\mathbf{x} = [0.6, -1.2], \quad t = 1,$$

the forward pass yields

$$z^{(1)} = [-0.56, -0.62], \quad a^{(1)} = [0.3635, 0.3498], \quad z^{(2)} = 0.1646, \quad a^{(2)} = 0.5411,$$

with loss $\mathcal{L} \approx 0.6142$. The cross-entropy output error is $\delta^{(2)} = a^{(2)} - t = -0.4590$. Backpropagating gives

$$\delta^{(1)} = [-0.0743, 0.0418], \quad \nabla_{\mathbf{W}^{(2)}} = [-0.1669, -0.1605]^\top, \quad \nabla_{\mathbf{b}^{(2)}} = -0.4590,$$

and

$$\nabla_{\mathbf{W}^{(1)}} = \begin{bmatrix} -0.0446 & 0.0251 \\ 0.0892 & -0.0501 \end{bmatrix}, \quad \nabla_{\mathbf{b}^{(1)}} = [-0.0743, 0.0418].$$

Finite-difference checks on the same network match to numerical precision, validating the implementation.

Aside: squared-error loss (alternative) The remainder of this subsection sketches the classic squared-error backprop derivation as a separate reminder; it is *not* a continuation of the cross-entropy numerical check above.

For one output unit with activation $a = \sigma(z)$ and target t, define the scalar error $e = a - t$ and squared error $\mathcal{L}_{\text{SE}} = \frac{1}{2}e^2$. The output-layer error signal (still defined as $\delta = \partial\mathcal{L}/\partial z$) is

$$\delta^{(L)} = (a - t)\,\sigma'(z), \qquad \sigma'(z) = \sigma(z)\big(1 - \sigma(z)\big).$$

For a hidden unit j in layer l, the same chain-rule logic gives

$$\delta_j^{(l)} = f'(z_j^{(l)}) \sum_k w_{jk}^{(l)}\, \delta_k^{(l+1)},$$

i.e., "local slope times the weighted sum of downstream error signals," with $w_{jk}^{(l)}$ the weight from unit j (layer l) to unit k (layer $l+1$).

Weight update rule (with momentum). Once you have δ's, turning them into weight updates is straightforward. For a weight $w_{ij}^{(l)}$ that connects unit i in layer $l-1$ to unit j in layer l, the gradient is proportional to "input times local error": $a_i^{(l-1)}\delta_j^{(l)}$. With momentum, one common update is

$$\Delta w_{ij}^{(l)}(n) = -\eta\, a_i^{(l-1)}(n)\, \delta_j^{(l)}(n) + \gamma \Delta w_{ij}^{(l)}(n-1), \tag{7.9}$$

followed by $w_{ij}^{(l)}(n) = w_{ij}^{(l)}(n-1) + \Delta w_{ij}^{(l)}(n)$. Here η is the step size and $\gamma \in [0, 1)$ is the momentum coefficient. (For the first hidden layer, $a_i^{(0)}$ is just the input feature x_i.) The index n denotes the update step (for example, the current training example in SGD, or the current mini-batch update).

A practical reading of η and γ. The learning rate sets the basic step scale; momentum averages recent gradient directions so updates do not zig-zag as much across narrow valleys. If training oscillates, reduce η; if it is painfully slow along a consistent direction, momentum can help.

One update step (what happens in one mini-batch)

1. **Initialize:** choose an initialization scale (Xavier is a common default for sigmoid/tanh; He is a common default for ReLU-family activations) and set biases (often zero).
2. **Forward pass:** compute and cache $z^{(l)}$ and $a^{(l)}$ for each layer.
3. **Loss + output signal:** compute $\mathcal{L}$ and $\delta^{(L)} = \partial\mathcal{L}/\partial z^{(L)}$.
4. **Backward pass:** propagate $\delta^{(l)}$ from $l = L - 1$ down to the first hidden layer.
5. **Gradients:** form $\nabla_{\mathbf{W}^{(l)}}\mathcal{L}$ and $\nabla_{\mathbf{b}^{(l)}}\mathcal{L}$ from cached activations and δ's.
6. **Update:** apply your optimizer step (SGD, momentum, Adam) and repeat for the next mini-batch.

Mini-batch backprop with explicit regularization

Inputs: mini-batch $\{(\mathbf{x}_b, \mathbf{t}_b)\}_{b=1}^{B}$, learning rate η, L2 coefficient λ, dropout keep probability $q = 1 - p$.

1. **Forward pass:** propagate activations layer by layer. If you use dropout in a hidden layer, draw a mask $\mathbf{m} \sim \text{Bernoulli}(q)$, apply $\tilde{\mathbf{a}} = \mathbf{m} \odot \mathbf{a}/q$, and cache $\mathbf{m}$ for the backward step.
2. **Backward pass:** compute $\nabla_{\mathbf{W}^{(\ell)}}\mathcal{L}$ using cached activations (and cached dropout masks) so dropped units contribute zero gradient.
3. **Update block (per layer):**

$$\mathbf{g}_\ell = \frac{1}{B}\nabla_{\mathbf{W}^{(\ell)}}\mathcal{L} + \lambda\mathbf{W}^{(\ell)}, \qquad \mathbf{W}^{(\ell)} \leftarrow \mathbf{W}^{(\ell)} - \eta\,\mathbf{g}_\ell.$$

Biases skip the weight-decay term. With Adam or SGD+momentum, $\mathbf{g}_\ell$ is what you feed into the optimizer update so regularization stays explicit.

Remarks Track both training and validation curves: a flat training loss usually points to an optimization issue (step size, activation saturation, initialization), while a widening train/validation gap points to capacity or regularization. Shuffle each epoch, checkpoint the best validation model, and let an early-stopping rule end runs before you overfit. In essence, the objective is to build confidence that your design decisions generalize out-of-sample; these curves give you a simple way to describe that generalization quantitatively.

7.9 Training Procedure and Epochs

An epoch is one complete pass over the training set. Most implementations shuffle the data and iterate over mini-batches: each update is cheaper than a full-batch gradient, and the resulting noise is often helpful for making progress.

1. Shuffle the training examples (or shuffle within class strata for imbalanced problems).
2. For each mini-batch: run a forward pass, compute the loss, run a backward pass, and update parameters.
3. At the end of the epoch: evaluate on a validation split, checkpoint the best model, and apply an early-stopping rule if needed.

After each epoch, look at both training and validation curves. A training loss that keeps falling while validation stalls is your cue to regularize, stop, or change capacity.

Common habits that prevent wasted runs.

- Start by overfitting a tiny batch and running a gradient check (the checklist box earlier in the chapter).
- Log the random seed and the full optimizer recipe; without it, "it did not train" is not a reproducible diagnosis.
- If you change one thing (activation, initialization, step size), keep everything else fixed so you can attribute cause and effect.

7.10 Role and Design of Hidden Layers

Hidden layers are where an MLP earns its flexibility. They do not just add parameters; they add intermediate representations, so different units can specialize and a later layer can combine those specializations. A *hidden layer* is any layer whose activations are not directly compared to the target: it sits between the input and output, and its outputs are internal signals the network learns to shape into useful features.

Design questions you actually have to answer.

- **Depth vs. width:** do you want a few wide layers or many narrow layers?
- **Capacity vs. data:** how much flexibility can your dataset support before you overfit?
- **Activation choice:** will gradients flow (ReLU family), or will they saturate (sigmoid/tanh) for the scale of your pre-activations? This connects directly to the smooth-activation discussion in Section 6.5.

A simple, defensible starting point.

- Pick one or two hidden layers and a moderate width, then let validation performance tell you whether you need more capacity.
- Choose an activation that keeps gradients alive (ReLU is a standard choice; if you see many dead units, a leaky ReLU is a simple alternative).
- Use early stopping and weight decay as your first line of defense against overfitting.

Trade-offs to keep in mind.

- **Too much capacity:** training loss drops quickly, validation does not; the model memorizes details you did not intend it to learn.
- **Too little capacity:** both training and validation plateau early; the model cannot represent the function you are asking for.

Example Execution Before you scale this to larger models, run one tiny network end-to-end and check each ingredient: compute the forward values, compute the output δ, push it one layer back, and confirm one gradient numerically. The explicit numbers in Section 7.8 can serve as a reference trace when you debug your own implementation.

Remarks on Convergence and Practical Considerations Backpropagation gives the *right* gradients; whether those gradients turn into learning depends on a handful of design and optimization choices. When training stalls or becomes unstable, the usual levers are:

- **Initialization scale:** keep pre-activations in a regime where derivatives are not all near zero (or wildly large).
- **Activation choice:** saturation (sigmoid/tanh) and dead regions (ReLU) show up directly as weak gradients.
- **Step size and schedule:** too large diverges, too small crawls; schedules and momentum smooth the path.
- **Regularization:** weight decay and dropout trade training fit for generalization.
- **Optimizer details:** momentum/Adam change the effective step direction and can rescue poorly conditioned problems.

Later deep-learning chapters revisit these choices in larger architectures; for now, it helps to see the canonical activation functions side by side.

Comparing canonical nonlinearities With the backprop derivation in place, it helps to compare the standard nonlinearities side by side. Figure 27 overlays the step, sigmoid, tanh, and ReLU curves so saturation regions and derivative behavior are visually apparent in one view.

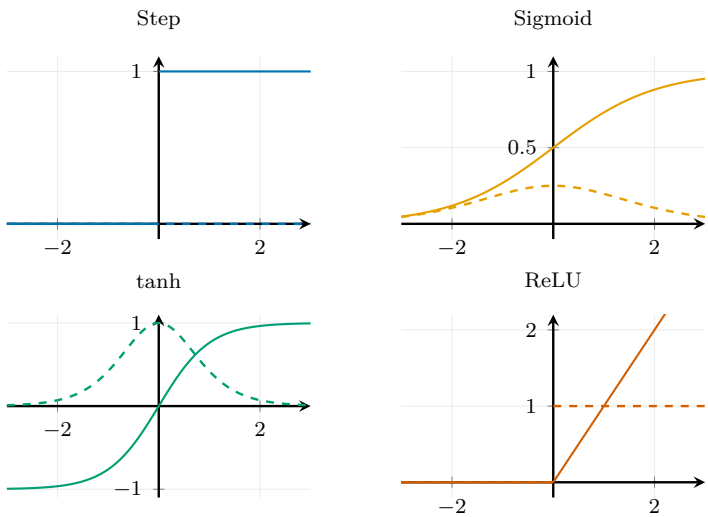


Figure 27: Canonical activation functions on a common axis. Solid curves show the activation; dashed curves show its derivative.

For quick reference:

Activation	$f(z)$	$f'(z)$	Reuse from forward pass?
Sigmoid	$\sigma(z)$	$\sigma(z)\big(1-\sigma(z)\big)$	Yes ($a(1-a)$)
tanh	$\tanh(z)$	$1-\tanh^2(z)$	Yes ($1-a^2$)
ReLU	$\max(0,z)$	$\mathbf{1}_{z>0}$	Yes (sign of pre-activation)

For ReLU, take derivative 0 at the origin.

Trade-offs Some activations match the intuition of a smooth "firing curve"; others are chosen mainly because they train well. Sigmoid and tanh saturate at large magnitude inputs, which slows gradients in deep networks. ReLU avoids saturation on the positive side but can produce "dying ReLUs" when biases push units negative and the gradients become zero; if many units stall, use He initialization, reduce the learning rate, or swap to a leaky ReLU with a small negative slope (e.g., 0.01).

With these stability levers in mind, we now turn to a simple function-estimation case study.

7.11 Case Study: Learning the Function $y = x \sin x$

Consider the problem of training an MLP to approximate the function

$$y = x \sin x.$$

Setup.

- Generate a dataset of input-output pairs $\{(x_i, y_i)\}$ where $y_i = x_i \sin x_i$.
- Use this dataset to train an MLP regressor.
- Evaluate the network's ability to generalize by testing on inputs not seen during training.

What to look for.

- **Capacity vs. fit:** if the model is too small, the best-fit curve will miss structure; if it is too large for your data density, it will fit noise or interpolation artifacts.
- **Activation effects:** saturated activations can make learning look "stuck" even when the model has enough parameters.
- **Data density:** uniform sampling over $[-3\pi, 3\pi]$ reveals whether your model interpolates smoothly between points or produces oscillatory artifacts.

Practical notes.

- This is a regression problem, not a classification problem.
- The target is nonlinear and oscillatory; underfitting and overfitting are both easy to see in a plot.
- If you report performance, report it as a reproducible experiment (seed, split, optimizer recipe). Avoid quoting a single "typical" number unless you can regenerate it.

7.12 Applications of Backpropagation-Trained Networks

MLPs show up in two roles. Sometimes they are the whole model, as in small function-approximation and classification tasks. More often they are trainable

blocks inside larger systems: the same affine→nonlinearity pattern appears in convolutional networks, recurrent networks, and attention models. Later chapters return to this connection at scale. In all cases, the training signal still arrives as a loss at the output, and backpropagation is what assigns that loss to each parameter through the stored intermediate values from the forward pass.

7.13 Limitations and Known Failure Modes of MLPs

MLPs are flexible, but they require care to train reliably. The main limitations show up as training instability, sensitivity to design choices, and difficulty diagnosing silent implementation mistakes. Training can be sensitive: small changes to the random initialization, data order, or step-size settings can lead to noticeably different outcomes, especially when the problem is poorly conditioned.

- **Run-to-run variability:** different initializations and data orders can lead to different solutions. Treat seeds and splits as part of the experiment, not as incidental details.
- **Optimization fragility:** step size, activation choice, and initialization scale directly affect gradient flow. When learning stalls, use the gradient norms and the activation-derivative picture in Figure 27 as your first diagnostic.
- **Silent implementation bugs:** shape/broadcasting mistakes can produce gradients that look plausible but are wrong. This is why the shape ledger and finite-difference checks in this chapter are part of the basic implementation discipline.

7.14 Summary

At this point you have the full backprop workflow: store computations from the forward pass, compute $\delta^{(L)}$, propagate δ's backward, and assemble gradients with the shapes kept honest. To close the derivation, it helps to write the same story once more in a compact matrix form.

Backpropagation algorithm recap (matrix form). One compact way to write the forward pass is:

$$\mathbf{Z}^{(l)} = \mathbf{A}^{(l-1)}\mathbf{W}^{(l)} + \mathbf{1}(\mathbf{b}^{(l)})^\top, \quad \mathbf{A}^{(l)} = f^{(l)}(\mathbf{Z}^{(l)}),$$

where $\mathbf{W}^{(l)}$ and $\mathbf{b}^{(l)}$ are the weights and biases of layer l, $\mathbf{A}^{(l-1)}$ is the previous layer activation (rows are samples), $\mathbf{1} \in \mathbb{R}^B$ is an all-ones vector that broadcasts the bias across the batch, and $f^{(l)}$ is the (possibly layer-specific) activation function.

Define the error signal at layer l as:

$$\boldsymbol{\delta}^{(l)} = \frac{\partial \mathcal{L}}{\partial \mathbf{Z}^{(l)}}.$$

Then the chain rule gives a reusable backward recursion:

$$\begin{aligned} \boldsymbol{\delta}^{(L)} &= \nabla_{\mathbf{A}^{(L)}} \mathcal{L} \odot f^{(L)\prime}(\mathbf{Z}^{(L)}), \\ \boldsymbol{\delta}^{(l-1)} &= \left(\boldsymbol{\delta}^{(l)} (\mathbf{W}^{(l)})^\top\right) \odot f^{(l-1)\prime}(\mathbf{Z}^{(l-1)}), \quad l = L, \ldots, 2, \end{aligned} \tag{7.10}$$

where $\odot$ denotes element-wise multiplication and $f^{(l)\prime}$ is the derivative of the activation function at layer l.

Finally, turn those δ's into parameter gradients:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(l)}} &= (\mathbf{A}^{(l-1)})^\top \boldsymbol{\delta}^{(l)}, \\ \frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(l)}} &= \mathbf{1}^\top \boldsymbol{\delta}^{(l)}. \end{aligned} \tag{7.11}$$

If your loss is defined as a mean over the batch, divide these batch sums by B; the shape ledger earlier in the chapter uses that mean convention.

These equations are the whole story: cache forward values, propagate δ's backward, then assemble gradients. Figure 26 complements the algebra by showing how cached activations (forward arrows) line up with backward error signals in a simple two-layer network.

Key takeaways

If you remember only a few things

- Backpropagation is the chain rule organized: cache forward values, propagate δ's backward using Equation (7.10), then assemble gradients with Equation (7.11).
- The δ's are not mysterious: they are sensitivities ($\partial\mathcal{L}/\partial\mathbf{Z}^{(l)}$) that you reuse to update every parameter.
- For softmax + cross-entropy, the output-layer error simplifies to $(\hat{\mathbf{Y}} - \mathbf{Y})/B$; a small finite-difference check can confirm this identity, and it is used in the modern-practice code pattern earlier in the chapter.
- When training misbehaves, debug like an engineer: overfit a tiny batch, run a finite-difference check, then inspect per-layer gradient norms and shapes.

Common pitfalls

- Dropping a transpose: $(\mathbf{W}^{(l)})^\top$ is easy to miss and produces gradients that look "reasonable" but train the wrong model.
- Mixing up pre-activations and activations: $f'(\mathbf{Z})$ depends on what you cache.
- Losing the batch scaling: decide whether you average over the batch ($/B$) and do it consistently.
- Silent broadcasting mistakes: bias terms and per-example deltas can accidentally align in NumPy/PyTorch and hide a bug.

Practical early stopping and checkpointing

- Keep a validation split separate from the training mini-batches, and record $L_{\text{val}}^{(e)}$ after each epoch.
- Stop when L_{val} has not improved for k consecutive epochs (a patience of $k \in [5, 10]$ is a reasonable first try). If the curve is noisy, require a small minimum improvement before you reset patience.
- Always checkpoint the parameters that achieved the best validation score and restore them before testing. If training is noisy, averaging the last few checkpoints can stabilize performance.

Derivation closure: implement, cache, fail-fast

- **Implement:** write one clean layer routine $(\mathbf{A}^{(l-1)}, \mathbf{W}^{(l)}, \mathbf{b}^{(l)}) \mapsto (\mathbf{Z}^{(l)}, \mathbf{A}^{(l)})$, then reuse it for every layer.
- **Cache:** store what backprop needs later (at minimum $\mathbf{A}^{(l-1)}$ and $\mathbf{Z}^{(l)}$). Do not recompute forward quantities during the backward pass.
- **Fail fast:** gradient-check one tiny network and overfit one tiny batch before you trust results on a real dataset.
- **Make runs comparable:** record seeds, optimizer, schedule, and stopping rule. The reporting checklist in Appendix B helps keep comparisons honest.

Backpropagation (dense MLP) pseudocode

```
function MLPTrainStep(X, Y, params, eta):
    # X: B x d_in, Y: B x d_out; params: {W[l], b[l]} for l=1..L
    # Forward (cache everything the backward pass will reuse)
    A[0] <- X
    for l in 1..L:
        Z[l] <- A[l-1] @ W[l]
        Z[l] <- Z[l] + b[l]   # b broadcast across the batch
        A[l] <- f_l(Z[l])
    # Backward (delta[l] = dL/dZ[l])
    dA <- dLoss_dA(A[L], Y)
    delta[L] <- dA * fprime_L(Z[L])
    for l in L..1:
```

```
        gradW[l] <- transpose(A[l-1]) @ delta[l] / B
        gradb[l] <- sum_rows(delta[l]) / B
        if l > 1:
            dAprev <- delta[l] @ transpose(W[l])
            delta[l-1] <- dAprev * fprime_{l-1}(Z[l-1])
    # Update
    for l in 1..L:
        W[l] <- W[l] - eta * gradW[l]
        b[l] <- b[l] - eta * gradb[l]
    return params
```

This matches the matrix-form recap: the forward pass produces activations and caches; the backward pass propagates layerwise error signals and assembles gradients; the update step is an optimizer choice (plain SGD shown here). The most common implementation bugs are shape errors and missing transposes, so keep the shape ledger and one tiny gradient check close.

Exercises

Exercise 1 (Easy). For the pseudocode box above, write the *cache contract*: list exactly what must be stored during the forward pass so the backward pass can be computed without re-running forward computations.
Deliverable: a one-page "shape + cache ledger" (names, shapes, and where each cached tensor is used).

Exercise 2 (Medium). Implement the 2-layer MLP train step (forward + backward) and run a finite-difference gradient check on a tiny batch. Use the same loss convention as the chapter (mean over batch) and report the worst relative error across parameters.
Deliverable: the max relative error + the first bug you found (transpose, batch scaling, or a cached value) and how you diagnosed it.

Exercise 3 (Medium). Create a controlled "gradient flow" stress test: keep the dataset fixed and vary only (i) activation (sigmoid/tanh/ReLU) and (ii) initialization scale. Track per-layer gradient norms over the first 50 updates and relate the curves to the saturation/derivative picture in Figure 27.
Deliverable: one plot of gradient norms by layer + one paragraph explaining which failure mode you observed (vanishing, exploding, or silent plateaus) and why.

If you are jumping chapters. Keep one habit: run a tiny gradient check before scaling up.

Where we head next. Chapter 8 introduces radial basis function networks, an alternative nonlinear route where many hidden features are fixed and the output layer can be solved more directly. This provides a clean contrast to end-to-end backpropagation and a different training decomposition to compare against.

8 Radial Basis Function Networks

Building on the multilayer perceptron (MLP) architecture (Chapter 6) and its training machinery (Chapter 7), this chapter introduces radial basis function

networks (RBFNs): three-layer models with fixed nonlinear bases and a linear readout. Figure 1 places this as the kernel/prototype branch alongside the MLP path.

Learning Outcomes

- Explain the architecture and training stages of RBF networks (center selection, width tuning, linear solve).
- Relate RBF solutions to linear estimators (normal equations, pseudoinverse, Wiener filtering) and know when ridge regularization is needed.
- Compare RBFNs to kernelized methods and other nonlinear classifiers to choose appropriate models in practice.

Design motif

Make the nonlinearity explicit: use a fixed (or lightly tuned) basis expansion in the hidden layer, then learn the output weights with linear-algebra tools.

8.1 Overview and Motivation

We start with the three-layer picture and the basis-function intuition, then write the same story in matrix form so training becomes a regularized least-squares solve. Near the end, we connect the finite-basis picture to the kernel viewpoint. (The short Wiener-filter box is optional context if you want the signal-processing parallel.)

Unlike MLPs, which learn weights in every layer, an RBFN separates the job into two parts. The hidden layer provides a fixed nonlinear feature map (set by centers and widths). The output layer then learns a linear combination of those features. That split is the main story here: choose the basis, then solve for the readout.

The key idea is that "nonlinear" can come from the representation rather than the readout. Radial basis functions act like localized, kernel-style features; once you lift inputs into that feature space, the output layer is still linear, and the weights are responsible for finding the separating border. This is closely related to the kernel trick used in SVMs; Appendix C collects the classical

kernel viewpoint.

Chapter 3 frames this as a bias–variance tuning problem (capacity, regularization, and diagnostics via learning curves). Kernel methods such as kernel ridge regression and support vector machines (SVMs) interpret the same trade-off through an RBF kernel matrix; here we keep the bases explicit, then connect to the dual/kernel view later in the chapter.

Running example: XOR as a representation problem. XOR is the smallest reminder that a single line in the input space is not always enough. The point of RBF networks is not that the output layer becomes complicated; it stays linear. The complication is pushed into the feature map: the hidden units apply localized, kernel-like transforms, and the output weights are responsible for finding the separating border in that transformed space. We will return to XOR twice: first to see the feature map, then to see how the linear solve chooses the readout weights.

Centers from clustering (a practical default). The hidden layer of an RBF network is easiest to understand when its centers are viewed as k-means prototypes: pick a coverage of the input space that reflects the data distribution, assign widths accordingly, and let the output layer learn the linear weights on top of those prototypes. Unsupervised clustering up front makes the later supervised solve far more stable.

8.2 Architecture of RBFNs

The RBFN consists of three layers:

Notation and shapes

We denote each basis response by $\varphi_i(\mathbf{x})$; stacking them yields $\mathbf{G}(\mathbf{x}) \in \mathbb{R}^M$ with entries $G_i(\mathbf{x}) = \varphi_i(\mathbf{x})$. For a dataset of N samples, the corresponding design matrix $\mathbf{\Phi} \in \mathbb{R}^{N \times M}$ stacks one transformed sample per row, with entries $\mathbf{\Phi}_{ji} = \varphi_i(\mathbf{x}_j) = G_i(\mathbf{x}_j)$. This matches the design-matrix convention used in Chapter 3.

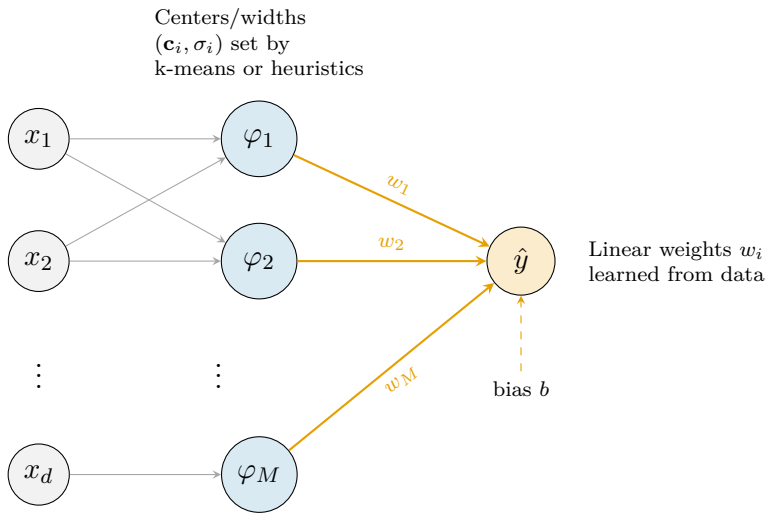


Figure 28: RBFN architecture. Inputs feed radial units parameterized by centers and widths; a linear readout with weights and bias is trained by a regression or classification loss. In the classical pipeline only the output weights are learned, while centers and widths come from clustering or spacing heuristics; variants also fine-tune centers and widths end-to-end.

Figure 28 highlights the split between fixed radial features and a trained linear readout.

A picture to keep in mind Once you have the architecture in mind, it helps to visualize what the hidden layer *does*. In one dimension, you can literally draw the bases as overlapping Gaussian bumps; the model output is a weighted sum of those bumps. Figure 29 is the mental model we will reuse as we introduce centers, widths, and the final linear solve.

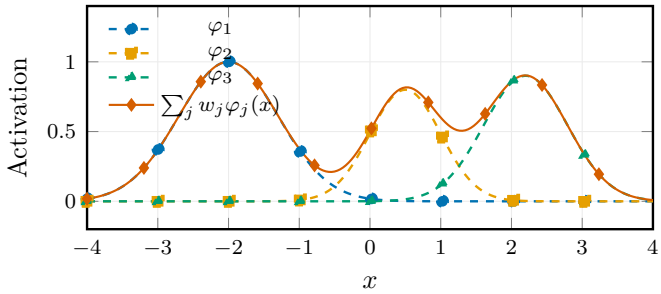


Figure 29: Localized Gaussian basis functions (dashed) and their learned output-weighted combination $\sum_j w_j \varphi_j(x)$ (solid). Overlapping bumps allow RBF networks to interpolate complex signals smoothly.

Figure 30 compares center-placement strategies used during initialization.

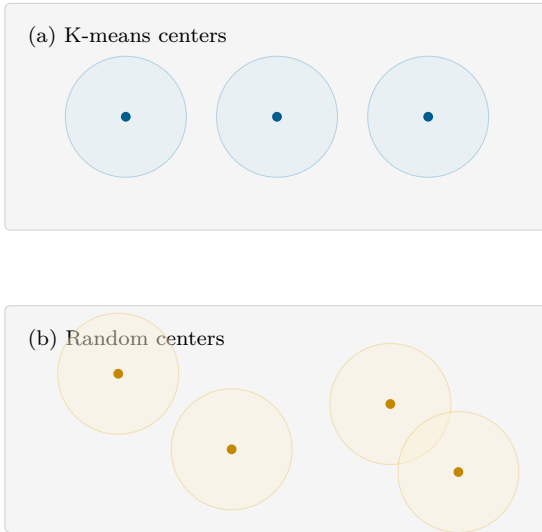


Figure 30: Center placement and overlap (schematic). K-means prototypes tend to tile the data manifold more evenly than random picks, giving more uniform receptive-field overlap. Random centers can leave gaps or create excessive overlap, which affects the width (sigma) choice and the conditioning of the later linear solve.

Later in the chapter we contrast this finite-basis ("primal") view with the kernel ("dual") view and show how Nyström-style approximations fit into the same story.

Figure 28 plus the overview above summarize the three-layer flow. Next we write the same story algebraically, so the training step becomes a linear solve with shapes you can check. The key distinction is that the input-to-hidden layer connections do not have trainable weights; instead, the hidden layer units themselves perform nonlinear transformations of the input.

8.2.1 Mathematical Formulation

Let the input vector be $\mathbf{x} \in \mathbb{R}^n$. The hidden layer computes the vector

$$\mathbf{G}(\mathbf{x}) = \begin{bmatrix} G_1(\mathbf{x}) \\ G_2(\mathbf{x}) \\ \vdots \\ G_M(\mathbf{x}) \end{bmatrix} \in \mathbb{R}^M.$$

where each $G_i(\mathbf{x})$ is a radial basis function centered at some point $\mathbf{c}_i \in \mathbb{R}^n$; stacking all M responses into $\mathbf{G}(\mathbf{x})$ makes it clear that M controls the dimensionality of the transformed feature space.

The output layer then computes

$$\mathbf{y}(\mathbf{x}) = \mathbf{W}^\top \mathbf{G}(\mathbf{x}) + \mathbf{b}, \tag{8.1}$$

where $\mathbf{W} \in \mathbb{R}^{M \times K}$ is the weight matrix connecting the hidden layer to the output layer, and $\mathbf{b} \in \mathbb{R}^K$ is a bias vector.

Interpretation: The hidden layer maps the input $\mathbf{x}$ into a new feature space via nonlinear functions G_i, and the output layer performs a linear combination of these features to produce the final output.

8.3 Radial Basis Functions

The functions $G_i(\mathbf{x})$ are typically chosen to be radially symmetric functions centered at $\mathbf{c}_i$, such as Gaussian functions:

$$G_i(\mathbf{x}) = \varphi\left(\|\mathbf{x} - \mathbf{c}_i\|\right) = \exp\left(-\frac{\|\mathbf{x} - \mathbf{c}_i\|^2}{2\sigma_i^2}\right), \tag{8.2}$$

where σ_i is the width (spread) parameter controlling the receptive field of the i-th basis function.

Other choices of radial basis functions are possible, but the Gaussian is the most common due to its smoothness and locality properties.

Normalized RBFs. Some texts normalize the hidden responses as $\tilde{G}_i(\mathbf{x}) = G_i(\mathbf{x}) / \sum_j G_j(\mathbf{x})$ to smooth predictions when center density is uneven; the linear readout then uses $\tilde{\mathbf{G}}(\mathbf{x})$ in place of $\mathbf{G}(\mathbf{x})$.

8.4 Key Properties and Advantages

- **Nonlinear transformation without weights:** The input-to-hidden layer mapping is fixed by the choice of centers $\{\mathbf{c}_i\}$ and widths $\{\sigma_i\}$, not by trainable weights.
- **Linear output layer:** Training reduces to finding the optimal weights $\mathbf{W}$ in a linear model, which can be done efficiently using linear regression

techniques.

- **Universal approximation:** With sufficiently many radial basis functions placed densely over a compact domain (and with nondegenerate widths), RBFNs can approximate any continuous function to arbitrary accuracy (Park and Sandberg, 1991; Micchelli, 1986).
- **Interpretability:** Each hidden unit corresponds to a localized region in input space, making it easier to understand which prototypes influence a given prediction.

Curse of dimensionality. In high dimensions Euclidean distances concentrate, so widths and center counts must scale with dimension; kernel ridge regression or learned features (e.g., CNNs) often dominate for images/audio.

8.5 Transforming Nonlinearly Separable Data into Linearly Separable Space

Some datasets are not linearly separable in the original input space. A nonlinear feature map can move the same points into a space where a single linear boundary is enough.

Consider a nonlinear transformation function $g(\cdot)$ applied to the input vector $\mathbf{x} \in \mathbb{R}^n$, producing a transformed vector $\mathbf{g}(\mathbf{x}) \in \mathbb{R}^m$. The goal is to find a weight vector $\mathbf{w} \in \mathbb{R}^m$ such that the linear combination $\mathbf{w}^\top \mathbf{g}(\mathbf{x})$ separates the classes.

Example setup (XOR).

- Inputs: $\mathbf{x} \in \{0,1\}^2$ with the four corners $(0,0), (0,1), (1,0), (1,1)$.
- Two radial units centered at $\mathbf{c}_1 = (0,0)^\top$ and $\mathbf{c}_2 = (1,1)^\top$.
- Feature map: $\mathbf{g}(\mathbf{x}) = [g_1(\mathbf{x}), g_2(\mathbf{x})]^\top$ with Gaussian g_i.
- Linear readout: $y = \mathbf{w}^\top \mathbf{g}(\mathbf{x})$.

Assumptions:

- For simplicity, set $\sigma^2 = 1$ (so $2\sigma^2 = 2$) in the Gaussian activation.
- Use two centers: $\mathbf{c}_1 = (0,0)^\top$ and $\mathbf{c}_2 = (1,1)^\top$.

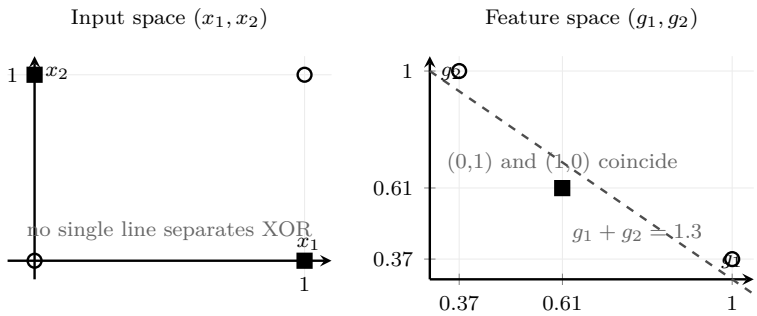


Figure 31: XOR before and after an RBF feature map. Left: in (x_1, x_2), no single line separates the labels. Right: in (g_1, g_2), the transformed points are linearly separable; one valid separating border is $g_1 + g_2 = 1.3$ (equivalently $w = [1, 1]$, $b = -1.3$).

- Use a Gaussian radial basis function (RBF):

$$g_i(\mathbf{x}) = \exp\left(-\frac{\|\mathbf{x} - \mathbf{c}_i\|^2}{2\sigma^2}\right).$$

Transformation Results: Applying the transformation to the inputs yields new points in the g_1-g_2 space. For example, the input $\mathbf{x} = (0, 0)$ maps to $(g_1, g_2) = (1, e^{-1})$, and $\mathbf{x} = (1, 1)$ maps to $(e^{-1}, 1)$. The two off-diagonal corners $(0, 1)$ and $(1, 0)$ map to the same feature point $(e^{-1/2}, e^{-1/2})$. In this feature plane the XOR labels become linearly separable; for instance, the separator $g_1 + g_2 = 1.3$ places the diagonal corners on one side and the off-diagonal corners on the other (Figure 31).

8.6 Finding the Optimal Weight Vector w

Given the transformed data $\mathbf{g}(\mathbf{x})$ and desired outputs $\mathbf{d}$, we want to find $\mathbf{w}$ that minimizes the squared error between the predicted output and the target. Using the design matrix $\mathbf{\Phi}$ defined in the notation box, the model predicts $\hat{\mathbf{d}} = \mathbf{\Phi}\mathbf{w}$, and the least-squares objective is

$$J(\mathbf{w}) = \|\mathbf{d} - \mathbf{\Phi}\mathbf{w}\|^2. \tag{8.3}$$

Differentiating (8.3) with respect to $\mathbf{w}$ and setting the gradient to zero yields

$$\mathbf{\Phi}^\top \mathbf{\Phi}\, \mathbf{w} = \mathbf{\Phi}^\top \mathbf{d}. \tag{8.4}$$

When $\mathbf{\Phi}^\top \mathbf{\Phi}$ is well conditioned, the closed-form solution is $\mathbf{w}^\star = (\mathbf{\Phi}^\top \mathbf{\Phi})^{-1} \mathbf{\Phi}^\top \mathbf{d}$; in practice we almost always add ridge regularization as described in the training section below.

Conditioning and capacity. When M is large and Gaussians overlap heavily, $\mathbf{\Phi}^\top \mathbf{\Phi}$ can become ill-conditioned. Ridge regularization (adding λI) stabilizes the solve and controls variance, mirroring the bias–variance trade-off from Chapter 3. Choosing M, σ, and λ together is essential for good generalization; Chapter 3's learning-curve diagnostics apply directly, and kernel methods (e.g., kernel ridge regression or SVMs) interpret the same trade-off via RBF kernels.

8.7 The Role of the Transformation Function $g(\cdot)$

The nonlinear map $g(\cdot)$ and its role in constructing $\mathbf{\Phi}$ were defined in the transformation example above; here we focus on its parameters and how to choose it.

Two parameters characterize $g(\cdot)$:

- $\mathbf{c}_i$: the center of the i-th basis function.
- σ_i: the width or spread parameter controlling the receptive field of the basis function.

Choosing $g(\cdot)$: The choice of $g(\cdot)$ is crucial. It defines how the input space is mapped into the feature space where linear separation is possible. A common rule-of-thumb for Gaussian widths is to set σ so that neighboring centers at average spacing $\bar{r}$ overlap with height $\exp\bigl(-\bar{r}^2/(2\sigma^2)\bigr) \approx 0.5$–$0.7$; too small σ fragments the boundary, too large washes out locality.

8.8 Examples of Kernel Functions

1. Inverse Distance Function:

$$g(r) = \frac{1}{r+\epsilon}, \quad \epsilon > 0,$$

where $r = \|\mathbf{x} - \mathbf{c}\|$. This function decreases as the distance increases but can become unbounded near zero, potentially causing numerical instability.

2. Gaussian Radial Basis Function:

$$g(r) = \exp\left(-\frac{r^2}{2\sigma^2}\right).$$

This function is smooth, bounded, and has a clear interpretation as a localized receptive field (the region of input space that activates the unit strongly) centered at $\mathbf{c}$ with width σ. It is the most commonly used kernel in RBF networks.

Why "radial"? Why a Gaussian?

An RBF unit is called *radial* because its response depends primarily on distance from a center: points at the same radius (in the chosen metric) produce the same activation. The Gaussian basis is popular because it is smooth, has a clear center, and its width parameter σ directly controls locality: large σ makes each unit "see" broadly (risking underfit), while small σ makes units highly local (risking overfit and poor conditioning). The practical art is to pick centers that cover the data and then tune σ (and ridge λ) by validation, as in Figure 32.

8.9 Interpretation of the Width Parameter σ

The parameter σ controls the spread of the basis function. Conceptually, increasing σ broadens the Gaussian bell, while decreasing σ produces a narrow spike around the centroid.

- $\sigma = 1$: The function is broad, covering a large region of the input space.
- $\sigma = 0.3$: The function is narrow and sharply peaked around the centroid.

Choosing σ appropriately is critical for the network's performance:

- If σ is too large, the basis functions overlap excessively, leading to smooth but potentially underfitting models.
- If σ is too small, the basis functions become too localized, which may cause overfitting and poor generalization.

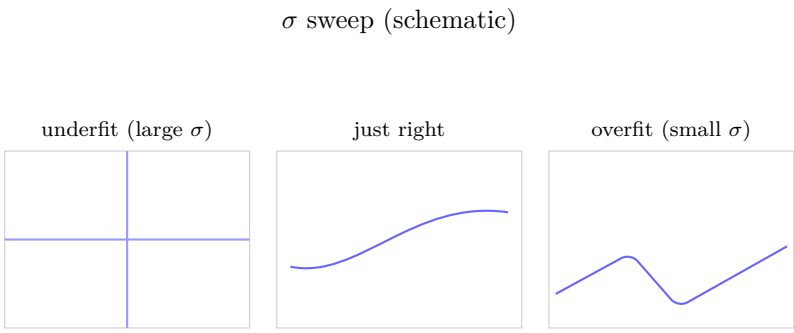


Figure 32: Schematic illustration of how the width parameter σ influences decision boundaries: too-large σ underfits (bases overlap heavily and wash out locality), intermediate σ captures the boundary, and too-small σ can produce fragmented regions. For a computed XOR example, see Figure 34.

8.10 Effect of σ on Classification Boundaries

Consider a one-dimensional dataset with two classes (e.g., red and blue points). Projecting a sample x through the Gaussian basis functions produces feature activations

$$\varphi_i(x) = \exp\left(-\frac{(x - c_i)^2}{2\sigma^2}\right),$$

which serve as localized similarity measures to each center c_i. When σ is large, many points activate the same basis functions with comparable strength, leading to smooth decision boundaries after the linear output layer. When σ is small, only points very close to a center elicit large activations, yielding sharply varying boundaries that can overfit noise. Visualizing $\varphi_i(x)$ for several centers illustrates how tuning σ controls the flexibility of the classifier.

We return to this tuning picture again later with an XOR-style toy example, where the decision boundary is easy to visualize.

Notation note. In this chapter we write radial basis functions as $\varphi_i(\cdot)$ and use $\mathbf{\Phi}$ for the associated design matrix. When we need a generic kernel feature map, we use $\phi(\cdot)$ (consistent with Appendix C); when probability density functions are needed, we write them as $p(\cdot)$. This avoids overloading a single symbol.

8.11 Radial Basis Function Networks: Parameter Estimation and Training

Recall that in Radial Basis Function (RBF) networks, the hidden layer neurons compute outputs based on radial basis functions centered at certain points $\mathbf{c}_i$ with spread parameters σ_i. The output is a linear combination of these nonlinear transformations. The key challenge is to determine the parameters:

$$\{\mathbf{c}_i, \sigma_i, w_i\}_{i=1}^{M},$$

where M is the number of hidden neurons.

Finding the Centers $\mathbf{c}_i$: A natural approach to find the centers is to use clustering algorithms on the input data. For example, if we decide to have M hidden neurons, we run a clustering algorithm (e.g., K-means) to find M centroids:

$$\mathbf{c}_1, \mathbf{c}_2, \ldots, \mathbf{c}_M.$$

These centroids represent typical data points around which the radial basis functions are centered. This approach ensures that the radial basis functions cover the input space effectively.

Determining the Spread Parameters σ_i: The spread parameters control the width of each radial basis function. One can initialize all σ_i to a common value or assign different values based on the data distribution. A practical rule-of-thumb is

$$\sigma \approx \frac{d_{\max}}{\sqrt{2M}},$$

where $d_{\max}$ is the maximum pairwise distance between centers and M the number of RBF units; this ensures neighboring receptive fields overlap without collapsing to a constant function. After setting this global width, refine to per-center widths by setting each σ_i proportional to the average distance between the center $\mathbf{c}_i$ and its nearest neighboring centers. Anisotropic variants scale each dimension separately but follow the same principle of matching the local density of prototypes.

Training the Output Weights w_i: Given fixed centers and spreads, the output weights w_i can be found by minimizing the squared error between the network output and the target values. The network output for an input $\mathbf{x}$ is:

$$\hat{y}(\mathbf{x}) = \sum_{i=1}^{M} w_i \varphi_i(\mathbf{x}).$$

where

$$\varphi_i(\mathbf{x}) = \exp\left(-\frac{\|\mathbf{x} - \mathbf{c}_i\|^2}{2\sigma_i^2}\right).$$

The training problem reduces to solving the linear system:

$$\min_{\mathbf{w}} \|\mathbf{y} - \mathbf{\Phi}\mathbf{w}\|^2, \tag{8.5}$$

where $\mathbf{y}$ is the vector of target outputs and $\mathbf{\Phi}$ is the design matrix with entries $\mathbf{\Phi}_{ji} = \varphi_i(\mathbf{x}_j)$. When $\mathbf{\Phi}^\top\mathbf{\Phi}$ is well-conditioned, the ordinary least-squares solution is

$$\mathbf{w}^\star = (\mathbf{\Phi}^\top\mathbf{\Phi})^{-1}\mathbf{\Phi}^\top\mathbf{y}.$$

Dual viewpoint: RBFN vs. kernel ridge regression

Fixing the RBF centers and widths makes the hidden layer a finite basis expansion. Training restricts itself to the M coefficients $\mathbf{w}$ and resembles kernel ridge regression with a truncated basis. In the dual view, kernel ridge regression solves

$$\min_{\boldsymbol{\alpha}} \|\mathbf{y} - \mathbf{K}\boldsymbol{\alpha}\|^2 + \lambda \boldsymbol{\alpha}^\top \mathbf{K}\boldsymbol{\alpha},$$

where $\mathbf{K}_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$ uses the same Gaussian kernel. Setting $M = N$ and letting the RBF centers coincide with the training points recovers this dual form exactly. Finite M acts like a Nyström approximation: $\boldsymbol{\Phi}\mathbf{w}$ projects onto a subset of kernel features using $M < N$ landmark bases, rather than expanding on all N training points.
Numerically, $\boldsymbol{\Phi}^\top \boldsymbol{\Phi}$ can be ill-conditioned if the bases overlap excessively or if centers cluster tightly; kernel ridge has the same issue via $\mathbf{K}$.
Regularization is therefore essential: add λI before inversion,

$$\mathbf{w}^\star = (\boldsymbol{\Phi}^\top \boldsymbol{\Phi} + \lambda I)^{-1} \boldsymbol{\Phi}^\top \mathbf{y},$$

mirroring the $\lambda \boldsymbol{\alpha}^\top \mathbf{K}\boldsymbol{\alpha}$ term in the dual problem. Larger λ damps coefficients when σ is large (heavy overlap) or when data are noisy, while smaller λ preserves sharper fits at the cost of conditioning. Choosing λ via cross-validation keeps both primal (RBFN) and dual (kernel ridge) systems stable.

Figure 33 summarizes the primal and dual viewpoints side by side and highlights where a finite basis acts like a Nyström approximation.

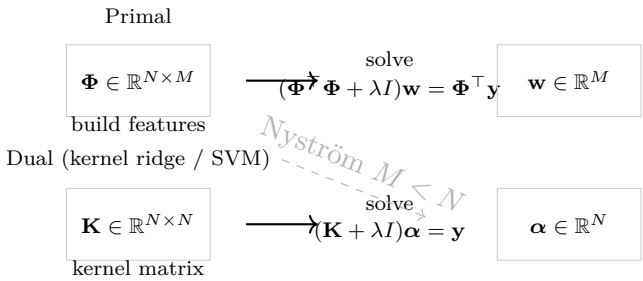


Figure 33: Primal (finite basis) vs. dual (kernel ridge) viewpoints. Using as many centers as data points recovers the dual form; using fewer centers corresponds to a Nyström approximation. The same trade-off appears in kernel methods through the choice of kernel and effective rank.

To improve numerical stability or control model complexity, a Tikhonov (ridge) regulariser can be added,

$$\mathbf{w}_\lambda^\star = (\mathbf{\Phi}^\top\mathbf{\Phi} + \lambda I)^{-1}\mathbf{\Phi}^\top\mathbf{y}, \qquad \lambda > 0,$$

or more generally one can use the Moore–Penrose pseudoinverse $\mathbf{\Phi}^+$ when $\mathbf{\Phi}^\top\mathbf{\Phi}$ is singular, yielding $\mathbf{w}^\star = \mathbf{\Phi}^+\mathbf{y}$. A quick dimensional sanity check is that $\mathbf{\Phi} \in \mathbb{R}^{N\times M}$, $\mathbf{w} \in \mathbb{R}^M$, and $\mathbf{y} \in \mathbb{R}^N$; all matrix products above respect these shapes.

Iterative Optimization of σ_i and w_i: Since both σ_i and w_i affect the network output, an alternating optimization procedure can be employed:

1. Initialize σ_i (e.g., all equal or based on data heuristics).
2. Fix σ_i and find w_i by solving the linear least squares problem (8.5).
3. Fix w_i and update σ_i to minimize the error, possibly using gradient-based methods or heuristics.
4. Repeat steps 2 and 3 until convergence or error criteria are met.

Note that the spreads σ_i can be scalar or vector-valued (anisotropic), allowing different widths in each input dimension:

$$\sigma_i = [\sigma_{i1}, \sigma_{i2}, \ldots, \sigma_{id}],$$

where d is the input dimension.

Summary of the Training Algorithm:

1. Use clustering (e.g., K-means) to find centers $\mathbf{c}_i$ (or sample centers uniformly at random).
2. Set widths σ_i via a rule-of-thumb (global σ from average center spacing or per-cluster covariance).
3. Build $\mathbf{\Phi}$ with entries $\mathbf{\Phi}_{ji} = \varphi_i(\mathbf{x}_j)$; choose a small grid of λ values and solve $(\mathbf{\Phi}^\top\mathbf{\Phi} + \lambda I)\mathbf{w} = \mathbf{\Phi}^\top\mathbf{y}$.
4. Evaluate on a validation set and pick (σ, λ, M) that minimizes validation loss; for classification, cross-entropy or hinge losses are also feasible with the same design matrix $\mathbf{\Phi}$.

Practical RBFN training (pseudocode)

```
Input: X, y, M, center_method=kmeans, sigma_rule, lambda_grid
Centers = center_method(X, M)
sigma = sigma_rule(Centers)
Phi = build_design_matrix(X, Centers, sigma)   # NxM
for lambda in lambda_grid:
    w_lambda = solve((Phi^T Phi + lambda I) w = Phi^T y)
    val_err[lambda] = validation_loss(Phi_val, y_val, w_lambda)
lambda_star = argmin val_err
Predict: yhat(x) = phi(x, Centers, sigma)^T w_lambda_star
```

Worked toy (classification, XOR-like). Consider four points and XOR labels

$$\mathbf{x}_1 = (0,0),\ \mathbf{x}_2 = (0,1),\ \mathbf{x}_3 = (1,0),\ \mathbf{x}_4 = (1,1), \qquad \mathbf{t} = [0,1,1,0].$$

Choose $M = 4$ centers at the data and set a global σ from the mean inter-center distance (here $\sigma \approx 0.8$). Build $\mathbf{\Phi}$ with entries $\mathbf{\Phi}_{ji} = \exp\big(-\|\mathbf{x}_j - \mathbf{c}_i\|^2/(2\sigma^2)\big)$ and solve $(\mathbf{\Phi}^\top\mathbf{\Phi} + \lambda I)\mathbf{w} = \mathbf{\Phi}^\top\mathbf{t}$ over a small grid $\lambda \in \{10^{-4}, 10^{-3}, 10^{-2}\}$. The best λ yields a linear separator in the lifted $\mathbf{\Phi}$-space that classifies XOR. Widening σ makes bases overlap heavily and can wash out locality (very smooth boundaries and low margins), while shrinking σ makes the model extremely local and can

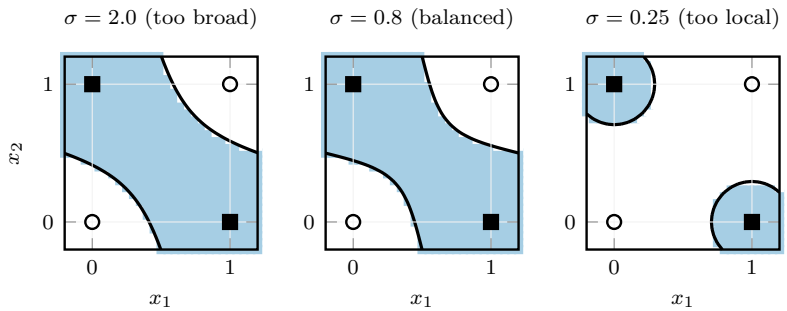


Figure 34: Effect of σ on an RBFN XOR boundary (4 centers at the data corners, ridge $\lambda = 10^{-3}$, threshold 0.5). Too-large σ makes bases overlap heavily, producing a very smooth, low-contrast boundary; intermediate σ yields a cleaner separation; too-small σ makes the model extremely local, producing small "islands" around prototypes.

create small islands that fit the training points but generalize poorly. Ridge helps whenever $\mathbf{\Phi}^\top \mathbf{\Phi}$ is poorly conditioned (typically when bases overlap heavily or centers cluster).

Figure 34 shows one such boundary for a fixed choice of centers, width, and ridge regularization.

8.12 Remarks on Radial Basis Function Networks

Advantages:

- **Training speed:** Once centers and spreads are fixed, training reduces to a linear least squares problem with a closed-form solution, which is computationally efficient.
- **Universal approximation:** RBF networks can approximate any continuous function on a compact domain to arbitrary accuracy given sufficient neurons, provided the centers cover the domain and the widths are chosen to avoid degeneracy (Micchelli, 1986; Park and Sandberg, 1991).
- **Interpretability:** Centers correspond to representative data points, making the network structure more interpretable.
- **Applications:** RBF networks have been successfully applied in control systems, communication systems, chaotic time series prediction (e.g., weather and power load forecasting), and decision-making tasks.

- **Flexible losses:** Squared loss is standard for regression; logistic or hinge losses pair naturally with the fixed design matrix for classification.

Disadvantages:

- **Parameter selection:** Choosing the number of neurons M, centers $\mathbf{c}_i$, and spreads σ_i is nontrivial and often requires heuristics or cross-validation.
- **Scalability:** The number of radial units required can grow quickly with input dimensionality, increasing computation and storage costs.
- **Center determination:** Identifying good centers (via clustering or other heuristics) can be computationally expensive and sensitive to noisy data.

Sidebar (optional): Wiener filtering in one paragraph

The Wiener filter is the same least-squares projection story in signal-processing notation. In the Wiener case, you write the linear estimator as $y(t) = \mathbf{w}^\top \mathbf{x}(t)$ and the normal equations involve second-order statistics $\mathbf{R} = \mathbb{E}[\mathbf{x}\mathbf{x}^\top]$ and $\mathbf{p} = \mathbb{E}[d\,\mathbf{x}]$, giving $\mathbf{R}\mathbf{w}^\star = \mathbf{p}$. In an RBF network, you replace the raw input $\mathbf{x}$ by fixed nonlinear features $\mathbf{\Phi}$ (built from radial units), and the same idea becomes $(\mathbf{\Phi}^\top \mathbf{\Phi} + \lambda I)\mathbf{w}^\star = \mathbf{\Phi}^\top \mathbf{y}$. The point of the sidebar is just this mapping: both are "fit a linear readout to fixed features," and conditioning/regularization control how stable that fit is.

8.13 Preview: Unsupervised and Localized Learning

In Chapter 9, we move from supervised RBF models to unsupervised, self-organizing methods. Self-organizing maps (SOMs) and Hopfield-style associative memory discover structure in data (clusters, manifolds) without labeled targets, complementing the supervised architectures covered so far.

Key takeaways

Minimum viable mastery

- Localized Gaussian bases + linear readout give an interpretable nonlinear model; center/width/regularization choices control bias–variance.
- Primal RBFNs and kernel ridge regression are two views of the same estimator (full vs. truncated basis); regularization cures conditioning.
- RBFNs bridge learned-feature models (MLPs) and kernel methods (SVMs/GPs); they form a strong baseline for localized decision boundaries.

Common pitfalls

- Width selection: too small memorizes, too large collapses to a linear model; validate σ and λ.
- Poor conditioning: without regularization, solves can be numerically unstable even when the math is correct.
- Confusing kernels with bases: a full kernel method scales differently than a truncated (primal) basis expansion.

Exercises

Exercise 1 (Medium). Train an RBFN on a 2D nonlinear classification task (e.g., two moons). Sweep M, σ, and a small λ grid; plot decision boundaries and validation curves.
Deliverable: one figure with 3 boundaries (underfit / best / overfit) + the chosen (M, σ, λ) and a short bias–variance diagnosis.

Exercise 2 (Easy). Treat regularization as numerical hygiene: for one fixed dataset and basis, plot the condition number of the linear system versus λ and show how predictions become stable as λ increases from 0.
Deliverable: condition-number vs. λ plot + one paragraph explaining why "mathematically correct" can still be numerically wrong at $\lambda = 0$.

Exercise 3 (Medium). Demonstrate the primal/dual equivalence: use all data points as centers with $\lambda > 0$ and verify that the primal RBFN predictions match kernel ridge regression with an RBF kernel.
Deliverable: a short derivation relating $\mathbf{w}$ and $\boldsymbol{\alpha}$ + a numerical check that the two prediction vectors agree to tight tolerance.

If you are skipping ahead. Keep the idea that "nonlinear" can still be linear-in-parameters once a basis is fixed. That perspective is reused in kernel methods and shows up again when we discuss architectural bias (CNNs) and representation choices.

Where we head next. Chapter 9 moves from supervised objectives to unlabeled competitive learning and prototype organization. Chapter 10 then revisits recurrence through an energy-based lens, setting up later sequence-model chapters.

9 Introduction to Self-Organizing Networks and Unsupervised Learning

Chapter 8 kept nonlinearity close to linear algebra: once you fix a set of basis functions, you lift the inputs into a feature space and (often) solve a regularized least-squares problem for the output weights. Now we switch the question. Instead of "how do I fit targets?", we ask: when labels are missing, expensive, or

not even the right interface, what structure is already present in the inputs?

This chapter opens the unsupervised neural thread with *Self-Organizing Maps* (SOMs), also known as Kohonen maps. You learn a set of prototypes and place them on a grid so that, as training progresses, units that are neighbors on the grid tend to respond to similar inputs. Chapter 10 then studies Hopfield networks as an energy-based associative memory model. Both operate without explicit targets, but they use that freedom differently: SOMs emphasize organization and visualization, while Hopfield emphasizes retrieval dynamics.

Learning Outcomes

- Describe how competitive learning, cooperation, and annealing interact in SOM training.
- Monitor SOM quality via quantization error (QE), topographic error (TE), and interpret U-matrices (unified distance matrix plots).
- Connect SOMs to broader unsupervised techniques (clustering, dimensionality reduction) and know when to use each.

Design motif

Competition plus cooperation: pick a winner, then let its neighbors learn too, so the map becomes both a clustering device and a visualization.

9.1 Overview of Self-Organizing Networks

Self-organizing networks aim to discover structure in input data without labels. The most prominent example is the *Self-Organizing Map* (SOM), introduced by Teuvo Kohonen (Kohonen, 1982). SOMs are used for clustering and for building a visualization-friendly map of a high-dimensional dataset.

The idea is fairly intuitive. If the data have a meaningful notion of similarity in the original space, then repeated *competition* (pick the best-matching unit) and *cooperation* (update its neighbors as well) can organize a fixed 2D *grid* of prototype vectors so that nearby grid locations tend to represent nearby regions of the data. The map is not a perfect geometric embedding; it is an engineered bias that often produces a useful, inspectable picture.

SOMs are trained without labeled outputs; the only signal is input similarity under your chosen distance. Learning is *competitive* (each input picks a best-

matching unit) but also *cooperative* (neighbors of the winner move too). That neighborhood coupling is the deliberate bias: it encourages nearby grid locations to represent similar inputs, so the map can be read both as a set of prototypes and as a visualization.

Historical intuition: two sheets and topographic neighborhoods

A useful way to picture SOMs is as two coupled "sheets": an input space and a fixed 2D grid of units. Each input is connected (in principle) to the whole grid, but learning makes some regions respond strongly (excitation) while others respond weakly (inhibition). The payoff is a *topographic* map: inputs that are far apart in the original space can end up near one another on the grid if they are statistically similar under the features the SOM has learned.

Author's note: tie SOMs back to clustering and dimensionality reduction

SOMs live in the same ecosystem as clustering and dimensionality reduction: they learn prototypes without labels and simultaneously organize those prototypes on a low-dimensional grid. Treat the update rules as a carefully annealed clustering algorithm whose output just happens to be arranged on a grid for interpretability.

The neighborhood influence is usually controlled by a kernel (often Gaussian) whose amplitude decays with grid distance and shrinks as training progresses, so early updates promote global organization while later updates refine only the closest units. Figure 35 juxtaposes these two time scales: the left panel shows why coarse early steps help traverse the energy landscape quickly, while the right panel compares two decaying learning-rate schedules commonly used when training SOMs.

Two practical SOM diagnostics show up repeatedly: the U-Matrix (a neighbor-distance heat map) and component planes (feature-wise maps). We will use both later as quick audits for whether the map is organizing structure (rather than just shuffling prototypes). A compact definition is: for each unit i with prototype $\mathbf{w}_i$, the U-Matrix value is the average distance to its grid

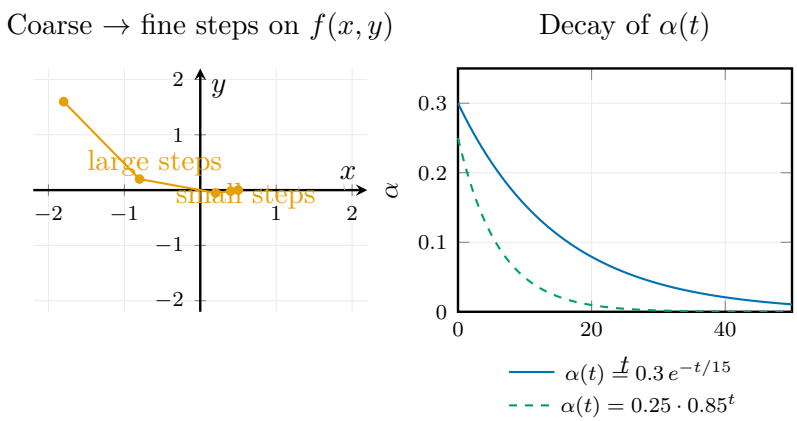


Figure 35: Learning-rate scheduling intuition (schematic). On a smooth objective (left), large initial steps quickly cover ground and roughly align prototypes, while a decaying step-size refines the solution near convergence. Right: common exponential and multiplicative decays used in SOM training.

neighbors $\mathcal{N}(i)$,

$$U(i) = \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \|\mathbf{w}_i - \mathbf{w}_j\|,$$

where $\mathcal{N}(i)$ is typically the 4-neighbor or 8-neighbor set on the 2D grid, excluding unit i itself.

Before delving into the mathematical formulation and algorithmic details of SOMs, it is important to review two foundational concepts that underpin their operation: *clustering* and *dimensionality reduction*.

9.2 Clustering: Identifying Similarities and Dissimilarities

Clustering is the process of grouping a set of objects such that objects within the same group (cluster) are more similar to each other than to those in other groups. Formally, given a dataset $\mathcal{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N\}$ where each $\mathbf{x}_i \in \mathbb{R}^d$ is represented by a feature vector, the goal is to partition the data into K clusters $\{C_1, C_2, \ldots, C_K\}$ such that:

- **Intra-cluster similarity** is maximized: points within the same cluster are close to each other.
- **Inter-cluster dissimilarity** is maximized: points in different clusters are far apart.

In the classical formulation used here (e.g., for K-means), the clusters form a partition of $\mathcal{X}$: they are disjoint and their union equals the entire dataset.

Example: Think of clustering as "discovering operating modes" from measurements. Your $\mathbf{x}_i$ could be a feature vector extracted from vibration spectra, network traffic, or sensor logs. Without labels, you still want the algorithm to separate one regime from another because the points inside a regime are genuinely similar in the feature space.

K-means Clustering: A classical and widely used clustering algorithm is *K-means*, which operates as follows:

1. Initialize K cluster centroids $\{\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_K\}$ randomly.
2. For each data point $\mathbf{x}_i$, assign it to the cluster with the nearest centroid:

$$c_i = \arg\min_k \|\mathbf{x}_i - \mathbf{m}_k\|_2.$$

   Here $\|\cdot\|_2$ denotes the Euclidean norm.

3. Update each centroid as the mean of all points assigned to it:

$$\mathbf{m}_k = \frac{1}{|C_k|} \sum_{\mathbf{x}_i \in C_k} \mathbf{x}_i.$$

   Here $|C_k|$ is the number of points in cluster C_k.

4. Repeat steps 2 and 3 until convergence (i.e., cluster assignments no longer change significantly).

K-means is an unsupervised learning method because it does not require labeled data; it discovers clusters purely based on feature similarity.

Keep the K-means update in mind: SOMs reuse the same prototype-moving idea, but add a neighborhood on a fixed 2D grid so the prototypes are not only learned, but also *organized* for inspection.

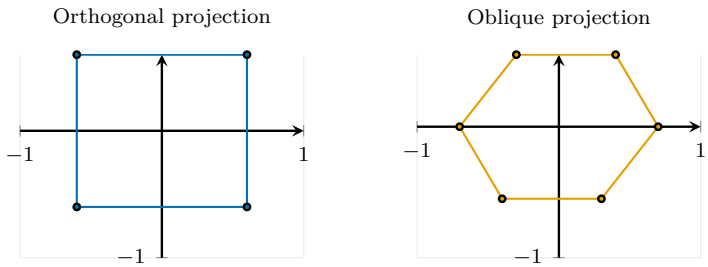


Figure 36: Classical MDS intuition (schematic). Projecting a cube onto a plane via an orthogonal map yields a square (left), whereas an oblique projection along a body diagonal produces a hexagon (right). The local adjacency of vertices is preserved even though metric structure is distorted.

9.3 Dimensionality Reduction: Simplifying High-Dimensional Data

Dimensionality reduction is what you do when the ambient dimension is too large to see and reason about directly. You accept some information loss, but you try to preserve the relationships you care about (variance, distances, neighborhoods) so the reduced view remains useful. This matters for:

- **Visualization:** Humans can easily interpret data in two or three dimensions.
- **Computational efficiency:** Reducing dimensions can simplify subsequent processing.
- **Noise reduction:** Eliminating irrelevant or redundant features.

Example: Consider a three-dimensional cube. Depending on its orientation, a linear projection (matrix multiplication by $P : \mathbb{R}^3 \to \mathbb{R}^2$ with matrix representation in $\mathbb{R}^{2\times 3}$) onto a two-dimensional plane can look like different shapes: a square arises from an orthogonal projection onto a face, whereas a hexagon appears under an oblique projection along a body-diagonal. This highlights that while the combinatorial adjacency (which vertices are connected) is preserved under such a projection, Euclidean lengths and angles are inevitably distorted. Figure 36 illustrates these two views.

Common techniques include Principal Component Analysis (PCA), which preserves directions of maximum variance, and classical Multidimensional Scal-

ing (MDS), which reconstructs a configuration whose pairwise distances match the original ones as closely as possible (via double-centering the squared-distance matrix and an eigen-decomposition). Nonlinear methods such as t-distributed stochastic neighbor embedding (t-SNE) or Uniform Manifold Approximation and Projection (UMAP) emphasize local neighborhoods but typically sacrifice global distance fidelity. SOMs will give us a different kind of reduction: a *discrete* map built from learned prototypes on a 2D grid, not a continuous coordinate chart.

9.4 Feature Mapping: From Prototype Vectors to Grid Coordinates

The dimensionality-reduction goals were covered in Section 9.3; here we make explicit what kind of "mapping" a SOM actually learns. A SOM does not return a continuous coordinate system like PCA or classical MDS. Instead, it learns a *finite set* of prototype vectors $\{\mathbf{w}_i\}$ arranged on a low-dimensional grid with fixed coordinates $\{\mathbf{r}_i\}$. Each input $\mathbf{x} \in \mathbb{R}^n$ is mapped to the grid by choosing its best matching unit (BMU),

$$c(\mathbf{x}) = \operatorname*{argmin}_i \|\mathbf{x} - \mathbf{w}_i\|_2^2,$$

and then using $\mathbf{r}_{c(\mathbf{x})}$ (or a neighborhood-smoothed variant) as the reduced representation. In that sense, the map $f : \mathbb{R}^n \to \mathbb{R}^m$ is *implicit*: it is realized by nearest-prototype assignment plus fixed grid coordinates.

This "discrete embedding" view is the reason SOMs are useful for visualization and clustering at the same time: nearby grid locations tend to represent nearby regions of the input space, but grid distance is only an *approximate* proxy for geometry. Read the map primarily as neighborhood structure and qualitative ordering, not as a metric-preserving chart.

How to read a SOM map as a feature representation

- **Out-of-sample mapping:** a new point maps to its BMU index $c(\mathbf{x})$ and grid coordinate $\mathbf{r}_{c(\mathbf{x})}$.
- **What transfers from PCA/MDS:** you can still ask whether neighborhoods are preserved and whether the map is stable across runs.
- **What does not transfer:** do not treat grid distance as true Euclidean distance in data space; the representation is discrete and only approximately geometry-preserving.

9.5 Self-Organizing Maps (SOMs): Introduction

Self-Organizing Maps (SOMs), also known as Kohonen maps, sit at a practical intersection: they behave like a prototype-based clustering method, but the prototypes are arranged on a 2D grid so you can inspect how the dataset organizes itself. Unlike supervised neural networks, SOMs learn without explicit target outputs or labels. Instead, they organize a bank of prototype vectors so that nearby units on the grid tend to represent similar inputs.

SOM at a glance

What it learns: Prototype vectors $\mathbf{w}_i$ in input space, arranged on a fixed 2D grid so that neighboring units tend to represent neighboring regions of the data (topographic mapping).
Knobs you actually tune: Map size/topology, the learning-rate schedule $\alpha(t)$, the neighborhood width $\sigma(t)$ and its decay, and the distance metric (typically squared Euclidean).
A practical starting point: A 2D rectangular grid, squared Euclidean distance, exponential decays for $\alpha(t)$ and $\sigma(t)$, and a number of units comparable to (or slightly larger than) the number of clusters you expect to resolve.
Common ways to fool yourself: Using a map that is too small, shrinking $\alpha(t)$ or $\sigma(t)$ too quickly (the map freezes before it organizes), and reading grid distance as if it were a true metric in data space.

Historical Context The concept of SOMs traces back to early models of self-organizing topographic maps, such as the two-sheet formulation of Willshaw and von der Malsburg (1976). Teuvo Kohonen later formalized and popularized the algorithmic framework in Kohonen (1982) (see also Kohonen, 2001).

Basic Architecture Architecturally, a SOM pairs an input vector $\mathbf{x} \in \mathbb{R}^n$ with a usually two-dimensional grid of units. Each unit i has (i) a fixed grid coordinate $\mathbf{r}_i = [u_i, v_i]^\top$ with $u_i, v_i \in \mathbb{Z}$, and (ii) a prototype (codebook) vector $\mathbf{w}_i \in \mathbb{R}^n$. The coordinates $\mathbf{r}_i$ define proximity *on the grid* and enter the neighborhood function (Section 9.16); the prototypes $\mathbf{w}_i$ are what you compare to data points when you pick the winner.

Each output neuron therefore possesses a weight vector of the same dimensionality as the input, so evaluating the match between an input and the map amounts to comparing the input against every stored prototype. The neurons then compete; the closest (best matching) unit "wins" and its neighbors are allowed to adapt by nudging their weight vectors toward the input, while distant units remain unchanged during that update. The resulting organization produces a discrete map that preserves qualitative ordering; it approximates the topology of the input space without providing a continuous Euclidean embedding.

Key Concept: Topographic Mapping The fundamental idea is simple: inputs that are similar in the original space should activate units that are close to each other on the map. In practice, that means if $\mathbf{x}_1$ and $\mathbf{x}_2$ are close in $\mathbb{R}^n$, then their best matching units should end up near each other on the 2D grid. The neighborhood update is the mechanism that encourages this: when one unit wins, its neighbors are pulled in the same direction, so "similar inputs" repeatedly shape the same local region of the map.

If you want a compact formal statement, let $\mathcal{N}_\epsilon(\mathbf{x}) = \{\mathbf{z} \mid \|\mathbf{z} - \mathbf{x}\|_2 < \epsilon\}$ be a small neighborhood in input space. SOM training aims (approximately) to map $\mathcal{N}_\epsilon(\mathbf{x})$ into a small neighborhood around the BMU on the grid (see Kohonen, 2001 for details and caveats). The guarantee is not exact; the engineering goal is a stable qualitative ordering that makes the map readable.

9.6 Conceptual Description of SOM Operation

1. **Initialization:** The weight vectors $\mathbf{w}_i$ are initialized, often randomly or by sampling from the input space.

2. **Competition:** For a given input $\mathbf{x}$, find the best matching unit (BMU) or winning neuron:
$$c = \operatorname*{argmin}_i \|\mathbf{x} - \mathbf{w}_i\|_2^2, \tag{9.1}$$
that is, the BMU index c minimizes the squared Euclidean distance between $\mathbf{x}$ and the candidate prototype $\mathbf{w}_i$. Minimizing the squared distance yields the same winner as minimizing the unsquared norm but streamlines gradient derivations, so we retain the squared form for consistency with later update rules.

3. **Cooperation:** Define a neighborhood function $h_{ci}(t)$ that determines the degree of influence the BMU has on its neighbors in the output grid. This function decreases with the distance between neurons c and i on the map and with time t.

4. **Adaptation:** Update the weight vectors of the BMU and its neighbors to move closer to the input vector:
$$\mathbf{w}_i(t+1) = \mathbf{w}_i(t) + \alpha(t) h_{ci}(t)\big(\mathbf{x} - \mathbf{w}_i(t)\big), \tag{9.2}$$
where $\alpha(t)$ is the learning rate, which decreases over time, and the effective width of $h_{ci}(t)$ likewise shrinks so that large-scale ordering occurs early and fine-tuning occurs later (see Section 9.16).

Author's note: distance is part of the model

The BMU rule is only as sensible as your distance. Euclidean distance is the default because it makes the update rule simple, but it assumes features are commensurate. In practice you almost always normalize inputs (and sometimes whiten them) so that one coordinate does not dominate the match by sheer scale. If your notion of "similar" is directional rather than magnitude-based, cosine distance is often a better choice; if covariance is strongly anisotropic, a Mahalanobis distance can be justified.

Tiny numeric step (online update)

Input $\mathbf{x} = [0.2, 0.8]$, two map units with weights $\mathbf{w}_1 = [0.1, 0.9]$, $\mathbf{w}_2 = [0.7, 0.3]$, coordinates $\mathbf{r}_1 = [0, 0]$, $\mathbf{r}_2 = [1, 0]$, $\alpha = 0.5$, $\sigma = 1$. BMU $c = 1$ (closest to $\mathbf{x}$). Neighborhoods: $h_{11} = 1$, $h_{21} = \exp(-1/2) \approx 0.607$. Updates:

$$\mathbf{w}_1 \leftarrow [0.15,\ 0.85], \quad \mathbf{w}_2 \leftarrow [0.548,\ 0.452].$$

Even the neighbor moves toward $\mathbf{x}$, illustrating cooperation.

This iterative process causes the map to self-organize, with neurons specializing to represent clusters or features of the input space.

9.7 Mathematical Formulation of SOM

Let the input space be $\mathcal{X} \subseteq \mathbb{R}^n$, and the output map be a grid of neurons indexed by i, each with weight vector $\mathbf{w}_i \in \mathbb{R}^n$.

Best Matching Unit (BMU) We reuse the BMU definition in (9.1); the same squared-distance criterion carries into the formal derivation here.

Neighborhood Function A common choice for the neighborhood kernel is the Gaussian function

$$h_{ci}(t) = \exp\left(-\frac{\|\mathbf{r}_c - \mathbf{r}_i\|^2}{2\sigma^2(t)}\right), \tag{9.3}$$

where $\mathbf{r}_i$ denotes the grid coordinates of neuron i and $\sigma(t)$ is the neighborhood radius that decreases monotonically with t. A common schedule is an exponential decay:

$$\sigma(k) = \sigma_0 e^{-k/\tau}, \tag{9.4}$$

where k counts updates and τ sets how quickly the neighborhood shrinks. Early in training $\sigma(t)$ is large, encouraging broad cooperation; as $\sigma(t)$ shrinks, only neurons near the BMU continue to adapt (Figure 37).

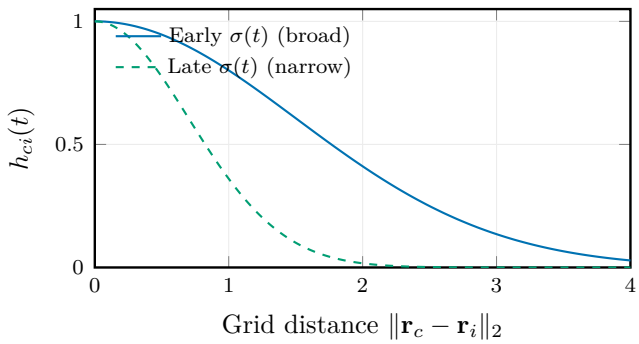


Figure 37: Gaussian neighborhood weights in SOM training (schematic). Early iterations use a broad kernel so many neighbors adapt; later iterations shrink the neighborhood width $\sigma(t)$ so only units near the BMU update.

9.8 Kohonen Self-Organizing Maps (SOMs): Network Architecture and Operation

So far we have described SOMs as an algorithmic loop: find the BMU, update the BMU and its neighbors, and anneal the step size and neighborhood width over time. If you prefer to think in terms of a network diagram, this subsection restates the same story as a concrete architecture you can trace and implement.

Network Structure The map is a fixed 2D grid of units. Each unit i stores a prototype vector $\mathbf{w}_i$ in input space and has a fixed grid coordinate $\mathbf{r}_i$. Every input is compared against every stored prototype.

Mapping and Competition For a given input $\mathbf{x}$, the units compete by similarity (or distance) between $\mathbf{x}$ and $\mathbf{w}_i$. The closest unit wins; this is the BMU rule in (9.1).

Weight Update Rule Only the winning unit and its neighbors on the grid update; the update rule is (9.2). The important qualitative effect is that one input does not just move one prototype; it moves a small *patch* of prototypes in the same direction, which is what makes the grid organize into a readable map rather than a bag of unrelated cluster centers.

9.9 Example: SOM with a 3×3 Output Map and 4-Dimensional Input

To make the symbols concrete, consider a SOM whose inputs live in $\mathbb{R}^4$ and whose output map is a 3×3 grid. Each unit $i \in \{1, \ldots, 9\}$ stores a prototype vector $\mathbf{w}_i \in \mathbb{R}^4$. For a single input $\mathbf{x} = [x_1, x_2, x_3, x_4]^\top$, the job of the map is to (i) decide which prototype matches best, then (ii) move that prototype *and a small neighborhood around it* toward $\mathbf{x}$.

Feedforward Computation For a given input $\mathbf{x}$, each neuron computes a similarity score. Two common choices are:

$$\begin{aligned} y_i &= \mathbf{w}_i^\top \mathbf{x} && \text{(dot-product similarity)}, \\ d_i &= \|\mathbf{x} - \mathbf{w}_i\|_2^2 && \text{(squared Euclidean distance)}. \end{aligned} \tag{9.5}$$

In both expressions $\mathbf{w}_i$ and $\mathbf{x}$ are column vectors, so $\mathbf{w}_i^\top \mathbf{x}$ is a scalar similarity score while d_i computes the squared Euclidean distance.

When using dot products we select the neuron with the maximum y_i; when using distances we equivalently select the neuron with the minimum d_i (or the maximum of $-d_i$):

$$c = \begin{cases} \arg\max_i y_i, & \text{dot-product similarity}, \\ \arg\min_i d_i, & \text{squared Euclidean distance}. \end{cases}$$

In practice the distance form is the most common in SOM code. If you do use dot-product similarity, normalize inputs (and often prototypes) so that "largest dot product" corresponds to your notion of "closest."

Weight Initialization and Update Weights $\mathbf{w}_i$ are typically initialized randomly (often by sampling from the data) or by a PCA-style seeding if you want a stable orientation. One training step then has a predictable shape: compute the match scores (usually d_i), pick the BMU c, compute neighborhood weights from the grid distances $\|\mathbf{r}_i - \mathbf{r}_c\|$, and apply the update in (9.2) to the BMU and a small neighborhood patch. For a fully worked numeric update with actual numbers, see the earlier *Tiny numeric step (online update)* box; the point of the present 3×3 example is to keep the dimensions explicit so you can track what

gets computed where.

This process repeats over many inputs, gradually organizing the map such that neighboring neurons respond to similar inputs, effectively performing a topology-preserving dimensionality reduction.

The grid coordinates $\mathbf{r}_i \in \mathbb{Z}^2$ introduced for the neighborhood kernel serve as the geometry of the output map; distances such as $\|\mathbf{r}_i - \mathbf{r}_c\|_2$ determine how strongly each neuron responds when c wins. Broad kernels (large $\sigma(t)$) encourage global ordering early in training, whereas shrinking $\sigma(t)$ confines adaptation to local neighborhoods so that fine-grained structure emerges. Alternative kernel shapes (e.g., Epanechnikov, bubble) can be used, though Gaussians provide smooth decay and convenient derivatives.

SOM training is typically stochastic: each input triggers an update, so the map continuously refines prototypes as data arrive. Batch variants exist, but online updates capture streaming data and mirror Kohonen's original algorithm.

9.10 Key Properties of Kohonen SOMs

Three properties are worth keeping in mind.

First, the SOM output grid is fixed by design. You choose the grid size up front, and that choice acts like the *resolution* of your map: a larger grid gives you more prototypes and finer visual detail, but it also needs enough data (and enough training) to populate the map meaningfully. A smaller grid forces a coarser organization and can merge behaviors that you might prefer to separate.

Second, learning is driven by competition. Each input selects a best-matching unit (BMU), and the BMU's prototype vector is the "closest representative" under your similarity rule. If you collapse the neighborhood to a winner-only update (i.e., $\sigma(t) \to 0$), the algorithm becomes a form of online vector quantization: prototypes still move toward the data, but the *layout* of prototypes on the grid has no reason to be ordered.

Third, SOMs add cooperation on top of winner-takes-all by updating the BMU and its neighbors together. That cooperation is the mechanism that creates smooth organization on the grid rather than a shuffled bag of prototypes: broad neighborhoods early encourage global ordering, and shrinking neighborhoods later refine local structure without tearing the map.

9.11 Winner-Takes-All Learning and Weight Update Rules

Recall that in competitive learning networks, the neuron with the highest discriminant value for a given input $\mathbf{x}$ is declared the *winner*. This subsection analyzes the classical *winner-takes-all* (WTA) principle in which only the winning neuron updates its weights, while all others remain unchanged. In the SOM setting discussed earlier, a softened variant is used in which the winner and its grid neighbors update together.

Discriminant Function and Similarity Measures The discriminant measures in Equation (9.5) are the same ones used in the earlier example; we reuse them here, favoring the distance form when deriving updates.

Weight Update Rule Once the winning neuron c is identified, WTA is the SOM update (9.2) with a collapsed neighborhood: set $h_{ci}(t) = 1$ for $i = c$ and $h_{ci}(t) = 0$ otherwise. The learning rate $\alpha(t)$ still controls step size so the winner moves toward $\mathbf{x}$ gradually rather than collapsing to it in a single update.

Learning Rate Schedule The learning rate $\alpha(t)$ controls the magnitude of weight updates. It typically decreases over time to ensure convergence and stability:

$$\alpha(t+1) \leq \alpha(t), \quad \lim_{t\to\infty} \alpha(t) = 0.$$

This schedule allows large adjustments early in training (rapid learning) and fine-tuning later (stabilization). Practitioners often start with $\alpha(0)$ in the range 0.05–0.5 and decay it toward 10^{-3} or smaller so that updates remain responsive initially but become conservative as the map stabilizes.

Summary of the Competitive Learning Algorithm

1. Initialize weights $\mathbf{w}_j(0)$ randomly or heuristically.
2. For each input $\mathbf{x}$:
   (a) Compute discriminant functions $g_j(\mathbf{x})$ or distances $d_j(\mathbf{x})$.

(b) Select winning neuron:

$$c = \arg\max_j g_j(\mathbf{x}) \quad \text{or} \quad c = \arg\min_j d_j(\mathbf{x})$$

(c) Update the winning neuron's weights using (9.2) with $h_{ci}(t)$ collapsed to a winner-only update.

3. Decrease learning rate $\alpha(t)$ according to schedule.
4. Repeat until convergence or maximum iterations reached.

9.12 Numerical Example of Competitive Learning

Consider a simple example with:

- Four input vectors $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4 \in \mathbb{R}^4$.
- A competitive layer with three neurons (clusters).
- Initial learning rate $\alpha(0) = 0.3$ with multiplicative decay $\alpha(t) = 0.3 \times 0.5^t$ (ensuring $\alpha(t) > 0$).
- No neighborhood function (i.e., only the winner updates).

Initial Weights The initial weights $\mathbf{w}_j(0)$ for neurons $j = 1, 2, 3$ are:

$$\mathbf{W}(0) = \begin{bmatrix} 0.2 & 0.3 & 0.5 & 0.1 \\ 0.2 & 0.3 & 0.1 & 0.4 \\ 0.3 & 0.5 & 0.2 & 0.3 \end{bmatrix}$$

where row j contains the initial weight vector $\mathbf{w}_j(0)$ for neuron $j = 1, 2, 3$.

Input vectors Let the four inputs be

$$\mathbf{x}_1 = \begin{bmatrix} 0.1 \\ 0.3 \\ 0.4 \\ 0.2 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 0.8 \\ 0.7 \\ 0.2 \\ 0.1 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 0.2 \\ 0.2 \\ 0.1 \\ 0.9 \end{bmatrix}, \quad \mathbf{x}_4 = \begin{bmatrix} 0.4 \\ 0.6 \\ 0.5 \\ 0.3 \end{bmatrix}.$$

We walk through one update explicitly for $\mathbf{x}_1$; the remaining inputs follow the same steps.

Step 1: pick the winner for $\mathbf{x}_1$. Using squared Euclidean distance $d_j = \|\mathbf{x}_1 - \mathbf{w}_j(0)\|_2^2$,

$$d_1 = (0.2-0.1)^2 + (0.3-0.3)^2 + (0.5-0.4)^2 + (0.1-0.2)^2 = 0.03,$$
$$d_2 = \|\mathbf{x}_1 - \mathbf{w}_2(0)\|_2^2 = 0.14, \qquad d_3 = \|\mathbf{x}_1 - \mathbf{w}_3(0)\|_2^2 = 0.13.$$

So the winner is $c = \arg\min_j d_j = 1$.

Step 2: update the winner. At $t = 0$ the learning rate is $\alpha(0) = 0.3$, and only the winner updates:

$$\mathbf{w}_1(1) = \mathbf{w}_1(0) + \alpha(0)\big(\mathbf{x}_1 - \mathbf{w}_1(0)\big) = \begin{bmatrix} 0.17 \\ 0.30 \\ 0.47 \\ 0.13 \end{bmatrix}.$$

For the next input, $t = 1$ so $\alpha(1) = 0.3 \times 0.5 = 0.15$, and the same computation repeats with the new winner.

9.13 Winner-Takes-All Learning: Consolidation After the Numerical Example

The WTA selection and update were defined in Section 9.11; the numerical example above uses the same BMU criterion (9.1) and the winner-only form of the update in (9.2).

How WTA relates to SOMs. You can view WTA learning as a limiting case of SOM training where the neighborhood collapses to a single unit: $h_{ci}(t) = \mathbf{1}[i = c]$. Adding a nontrivial neighborhood $h_{ci}(t)$ is what turns pure prototype learning into a *topographic* map: neighbors are encouraged to represent similar inputs, and the grid becomes a structured visualization rather than an unstructured codebook.

Practical considerations. In both SOMs and WTA networks, input vectors are commonly normalized (e.g., zero mean and unit variance) so that distance comparisons are meaningful. Training is typically terminated when weight changes fall below a small threshold or after a prescribed number of epochs.

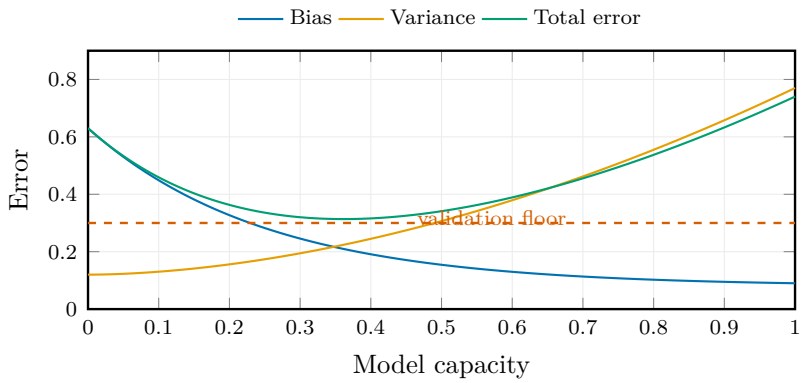


Figure 38: Illustrative bias–variance trade-off when sweeping SOM capacity (number of units or kernel width). The optimum appears near the knee where bias and variance intersect.

9.14 Regularization and Monitoring During SOM Training

Even though SOMs are inherently unsupervised, their training dynamics still benefit from the same regularization heuristics used in supervised settings. Two complementary diagnostics are especially useful in practice.

Bias–variance view. Increasing the grid resolution or keeping the kernel width large for too long can overfit local noise. Figure 38 visualizes the familiar U-shaped trade-off: the left regime underfits (high bias), whereas the right regime yields jagged maps (high variance).

Loss-landscape smoothing. Adding small cooperative penalties (e.g., weight decay between neighbors) produces smoother loss contours and accelerates convergence, as sketched in Figure 39. The penalty discourages neighboring prototypes from diverging and keeps the map topologically ordered.

Quantization vs. information preservation. Classical SOM optimizes a topology-preserving vector quantization objective; it does not include cross-entropy terms. Modern variants sometimes introduce *auxiliary* regularizers to encourage codebook utilization (e.g., entropy penalties on assignment histograms) or draw analogies to VQ-VAE. Monitoring both quantization error and an entropy-style regularizer, as in Figure 40, helps reveal when the map is

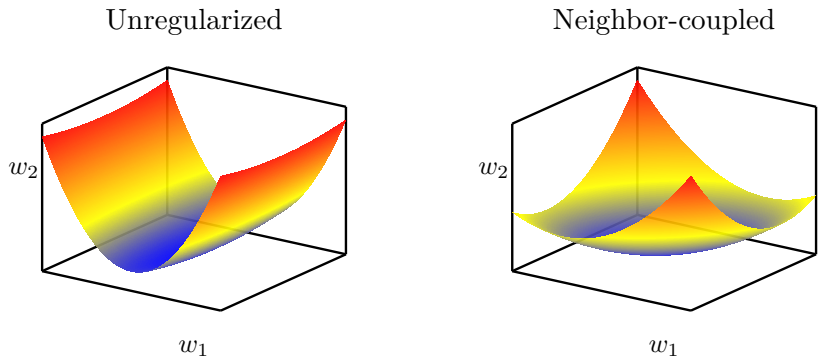


Figure 39: Regularization smooths an objective surface (schematic). Coupling neighboring prototypes (right) yields wider, flatter basins than the narrow, ill-conditioned unregularized basin (left).

collapsing to a few units or when density variations are no longer represented faithfully.

Quantization vs. topographic error. Given data points $\{\mathbf{x}_i\}$ and best-matching units $b_i = \mathrm{BMU}(\mathbf{x}_i)$, the *quantization error* is

$$\mathrm{QE} = \frac{1}{N}\sum_{i=1}^{N}\left\|\mathbf{x}_i - \mathbf{w}_{b_i}\right\|_2,$$

which measures reconstruction fidelity. The *topographic error* is the fraction of inputs whose first- and second-best BMUs are not adjacent on the grid (default: 4-neighbor connectivity), capturing topology preservation. Both metrics reappear in later figures; we monitor QE for representation quality and TE for magnification distortions.

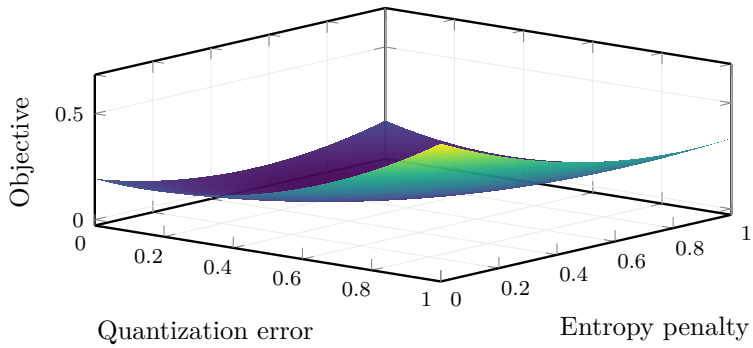


Figure 40: Illustrative objective surface combining quantization error and an entropy-style regularizer (a modern SOM variant; for example, a negative sum of $p \log p$ over unit usage). Valleys arise when prototypes cover the space evenly; ridges highlight collapse or poor topological preservation.

Tiny numeric check: QE vs. TE (they can disagree)

To see what these metrics measure, consider a 2×2 SOM grid with fixed coordinates $\mathbf{r}_1 = [0, 0]$, $\mathbf{r}_2 = [1, 0]$, $\mathbf{r}_3 = [0, 1]$, $\mathbf{r}_4 = [1, 1]$ (4-neighbor adjacency). Let the prototypes in input space be

$$\mathbf{w}_1 = [0, 0], \quad \mathbf{w}_2 = [2, 0], \quad \mathbf{w}_3 = [0, 2], \quad \mathbf{w}_4 = [0.25, 0.25].$$

Notice that $\mathbf{w}_4$ is close to $\mathbf{w}_1$ in input space, even though units 4 and 1 are diagonal neighbors on the grid.
Now evaluate four inputs, using squared Euclidean distance to pick the best and second-best matches:

$$\mathbf{x}_1 = [0.30, 0.30], \ \mathbf{x}_2 = [1.80, 0.20], \ \mathbf{x}_3 = [0.20, 1.70], \ \mathbf{x}_4 = [0.05, 0.05].$$

The BMU/second-BMU pairs are $(4, 1)$, $(2, 4)$, $(3, 4)$, $(1, 4)$. Under 4-neighbor adjacency, pairs $(4, 1)$ and $(1, 4)$ are *not* adjacent (diagonal), so $\mathrm{TE} = 2/4 = 0.5$. Meanwhile the average distance to the BMU is $\mathrm{QE} \approx 0.1962$.
The lesson is that QE is a "prototype fit" score, while TE is an "ordering" score. You can improve QE by moving prototypes toward the data while still tearing the map if nearby grid units do not represent nearby regions of the input space.

Batch SOM in practice

Online SOM updates one sample at a time: pick a best-matching unit (BMU), nudge it and its neighbors, move on. Batch SOM instead aggregates responsibilities across a dataset (or mini-batch) before shifting prototypes:

$$h_{j,i}(t) = \kappa\big(\mathrm{dist}(j, b(i)); \sigma_t\big),$$
$$\mathbf{w}_j^{(t+1)} = \frac{\sum_i h_{j,i}(t)\,\mathbf{x}_i}{\sum_i h_{j,i}(t)}.$$

Key differences:

- **Deterministic passes.** Batch updates remove stochastic noise and converge in fewer epochs on static datasets, making results reproducible (useful for dashboards/visual analytics).
- **Parallelism.** Computations collapse to matrix ops (compute BMUs, accumulate weighted sums), so GPUs/CPUs can process large mini-batches efficiently.
- **Streaming trade-off.** Online updates remain preferable when data arrive continuously or when you need the map to adapt mid-stream; batch SOM suits offline datasets.

Most modern SOM libraries expose both modes, so choose the update rule that matches your data pipeline and stability requirements.

Stopping criteria. Because stochastic updates can eventually increase topographic error, it is standard to stop training once a moving-average validation curve plateaus. Figure 41 shows the canonical trend: fast initial improvement followed by saturation.

9.15 Limitations of Winner-Takes-All and Motivation for Cooperation

While WTA is simple and effective for clustering, it has some limitations:

- Only one neuron updates per input, which can lead to slow convergence.
- The hard competition ignores relationships among neighboring neurons.

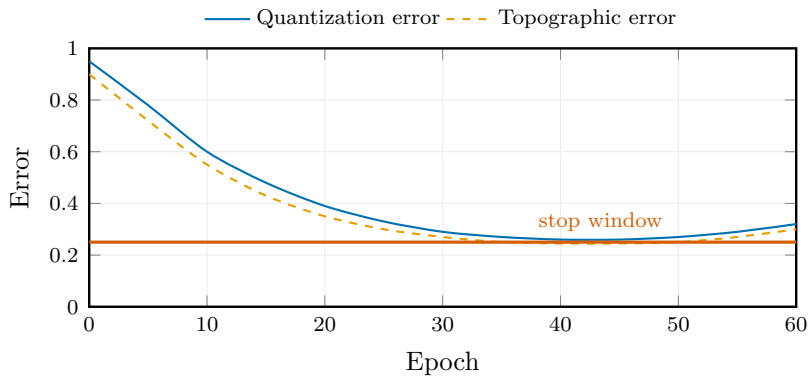


Figure 41: Illustrative validation curves used to identify an early-stopping knee. When both quantization and topographic errors flatten (shaded band), further training risks map drift.

- The resulting clusters correspond to hard assignments, so boundaries between codebook vectors are sharp with little smoothing across neighboring neurons.

There is also a very practical failure mode: you can drive the prototypes toward the data (so QE improves) while the *grid organization* never really forms. If you start with $\sigma(t) \approx 1$ from the very beginning (or you drop neighborhood updates entirely), nearby grid locations can end up representing unrelated regions of the input space. The map looks like a shuffled deck: the U-Matrix becomes speckled, and TE stays stubbornly high because the second-best match for a point often lives far away on the grid. The fix is not a new objective; it is simply the training schedule: use a broad neighborhood early so large-scale ordering can settle, then shrink $\sigma(t)$ so the map can refine without smearing away detail. More importantly for SOMs, plain WTA gives you *prototypes* but not necessarily a readable *map*. If only the winner moves, nothing forces neighboring grid units to represent neighboring regions of the data; the prototypes can end up arranged arbitrarily across the grid, and small changes in initialization can produce very different-looking maps. Cooperation is the fix: early in training you use a broad neighborhood (large $\sigma(t)$) so an input shapes a local patch rather than a single unit, which encourages global ordering; later you shrink $\sigma(t)$ so the map can refine without smearing away detail. The geometric effect of these limitations is easiest to see in Figure 42: the left panel shows the

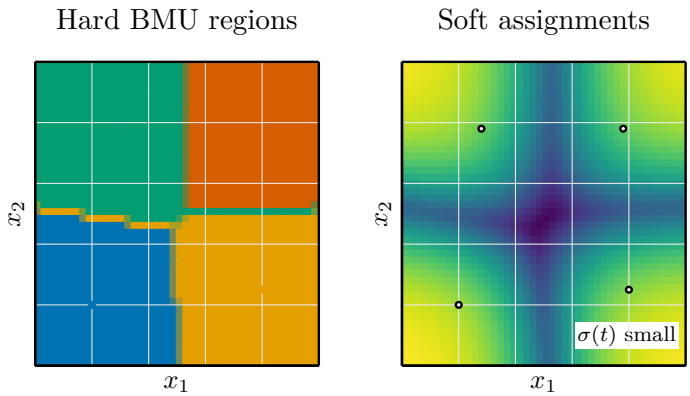


Figure 42: Voronoi-like regions induced by fixed prototypes in input space (left) and the corresponding soft assignments after sharpening the neighborhood kernel (right). Softer updates blur the decision frontiers and reduce jagged mappings between adjacent units (schematic illustration).

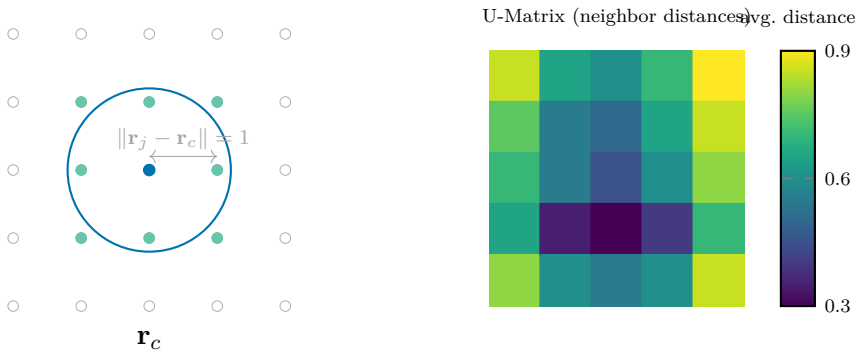
brittle Voronoi partitions created by a strict winner-takes-all rule, whereas the right panel demonstrates how shrinking the neighborhood kernel produces softer responsibilities and smoother maps.

To address these issues, the concept of *cooperation* among neurons is introduced. Instead of a single winner neuron updating its weights, a neighborhood of neurons around the winner also update their weights, albeit to a lesser extent. This idea leads to smoother mappings and better topological ordering.

9.16 Cooperation in Competitive Learning

Neighborhood Concept Consider the output layer arranged in a 2D grid of neurons. For each input $\mathbf{x}$, after determining the winning neuron c, we define a neighborhood $\mathcal{N}(c)$ consisting of neurons close to c on the grid. In practice the neighborhood weight is supplied by the kernel $h_{jc}(t)$ of (9.3), which is positive for units inside the neighborhood (and decays with the grid distance $\|\mathbf{r}_j - \mathbf{r}_c\|$) and zero for units far away.

The neighborhood size typically shrinks over time during training, starting large to encourage global ordering and gradually reducing to fine-tune local details.



SOM grid and BMU neighborhood

Figure 43: Left: a 5-by-5 SOM grid with the best matching unit (blue) and neighbors inside the Gaussian-kernel radius (green). Right: a toy U-Matrix (grayscale-safe colormap) showing average distances between neighboring codebook vectors; larger distances suggest likely cluster boundaries. Treat a U-Matrix as a qualitative boundary hint unless preprocessing and scaling are fixed.

Weight Update with Neighborhood Cooperation The grid structure and how the best matching unit (BMU) influences nearby neurons are visualized in Figure 43. The U-Matrix on the right provides a quick diagnostic for cluster boundaries during training.

Component planes are the companion view to the U-Matrix: they show one feature at a time across the grid, which helps you interpret what a boundary is *about* (Figure 44).

The weight update rule generalizes to:

$$\mathbf{w}_j(t+1) = \mathbf{w}_j(t) + \alpha(t)\, h_{jc}(t)\, (\mathbf{x} - \mathbf{w}_j(t))\,, \tag{9.6}$$

where

- $h_{jc}(t)$ is the *neighborhood function* that quantifies the degree of cooperation between neuron j and the winner c.
- $\alpha(t)$ is the learning rate at time t.

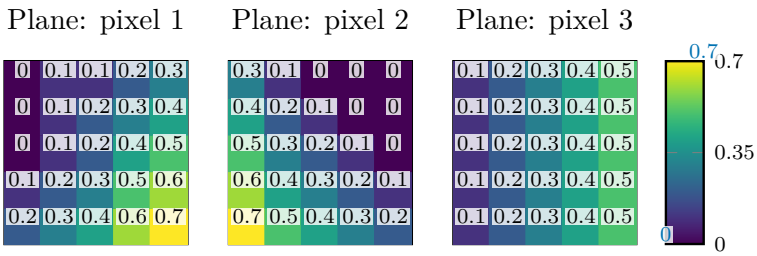


Figure 44: Component planes for three features on a trained SOM (toy data). Each plane maps one feature across the grid; aligned bright/dark regions reveal correlated features and complement the U-Matrix in Figure 43. Interpret brightness comparatively within a plane rather than as an absolute scale.

The neighborhood function satisfies:

$$h_{jc}(t) = \begin{cases} 1, & j = c \\ \in (0,1), & j \in \mathcal{N}(c), j \neq c \\ 0, & \text{otherwise} \end{cases}$$

Gaussian Neighborhood Function A common choice for $h_{jc}(t)$ is a Gaussian function based on the distance between neurons j and c on the output grid:

$$h_{jc}(t) = \exp\left(-\frac{\|\mathbf{r}_j - \mathbf{r}_c\|^2}{2\sigma^2(t)}\right), \tag{9.7}$$

where

- $\mathbf{r}_j$ and $\mathbf{r}_c$ are the coordinates of neurons j and c on the output grid.
- $\sigma(t)$ is the neighborhood radius (width) at time t, which decreases over training.

This function ensures that neurons closer to the winner receive larger updates, while distant neurons are updated less or not at all.

Interpretation The cooperative update encourages neighboring neurons to become sensitive to similar inputs, thereby preserving topological relationships in the input space. This is the key principle behind Self-Organizing Maps (SOMs).

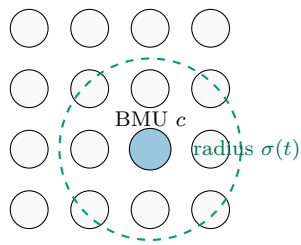


Figure 45: SOM grid with the best-matching unit (BMU) highlighted in blue and a dashed neighborhood radius indicating which prototype vectors receive cooperative updates (schematic).

9.17 Example: Neighborhood Update Illustration

Suppose the output neurons are arranged in a 2D grid as shown schematically in Figure 45, where each neuron is indexed by its grid coordinates. For an input $\mathbf{x}$, neuron c wins. The neighborhood $\mathcal{N}(c)$ might include neurons within a radius σ around c.

Each neuron j in $\mathcal{N}(c)$ updates its weight vector according to (9.6), with the magnitude of update modulated by $h_{jc}(t)$.

9.18 Summary of Cooperative Competitive Learning Algorithm

Cooperative competitive learning is the winner-takes-all loop with one extra ingredient: a neighborhood kernel on the *grid*. For each input $\mathbf{x}(t)$, you (i) compute match scores, (ii) pick the BMU c, (iii) compute neighborhood weights from grid distances $\|\mathbf{r}_j - \mathbf{r}_c\|$, and (iv) update the BMU and its neighbors with the cooperative rule. The neighborhood term is what turns "a set of prototypes" into "an organized map."

The two schedules play different roles. The learning rate $\alpha(t)$ controls how fast prototypes move; the neighborhood width $\sigma(t)$ controls *how much* the map is encouraged to be globally ordered. If $\sigma(t)$ shrinks too quickly, the map can freeze before it has a chance to organize (a common source of topology errors). If $\sigma(t)$ stays large for too long, the map can remain smooth but blurry, with weak separation between regions. This is why the monitoring tools introduced earlier (e.g., QE/TE and stability across seeds) are not decoration: they are how you tell whether the schedule is producing the behavior you intended.

9.19 Synthesis: SOM Training Rules, Annealing Schedules, and Design Choices

At this point the SOM training story is complete: you have a similarity rule (to pick a winner), a neighborhood rule (to decide who else learns), and schedules that anneal both over time. The only subtlety is that these pieces interact. A large neighborhood early is what gives you global organization; a small neighborhood late is what lets prototypes settle into fine detail without tearing the map.

Recall the weight update rule for neuron j at time step t:

$$\Delta \mathbf{w}_j(t) = \alpha(t)\, h_{j,c}(t)\, [\mathbf{x}(t) - \mathbf{w}_j(t)]\,.$$

where:

- $\mathbf{x}(t)$ is the input vector at time t.
- $\mathbf{w}_j(t)$ is the weight vector of neuron j at time t.
- c is the index of the winning neuron (best matching unit) for input $\mathbf{x}(t)$.
- $\alpha(t)$ is the learning rate, a monotonically decreasing function of time.
- $h_{j,c}(t)$ is the neighborhood function centered on the winning neuron c, also decreasing over time.

Neighborhood Function and Its Role The neighborhood function $h_{j,c}(t)$ typically takes a Gaussian form:

$$h_{j,c}(t) = \exp\left(-\frac{\|\mathbf{r}_j - \mathbf{r}_c\|^2}{2\sigma^2(t)}\right). \tag{9.8}$$

where:

- $\mathbf{r}_j$ and $\mathbf{r}_c$ are the positions of neurons j and c on the SOM grid.
- $\sigma(t)$ is the neighborhood radius, which decreases over time.

This function ensures that neurons closer to the winning neuron receive larger updates, while those farther away receive smaller or zero updates. Initially, $\sigma(t)$ is large, allowing broad neighborhood cooperation, but it shrinks as training progresses, focusing updates increasingly on the winning neuron itself.

Time-Dependent Parameters Both the learning rate $\alpha(t)$ and neighborhood radius $\sigma(t)$ decrease over time, typically following exponential decay laws:

$$\begin{aligned} \alpha(t) &= \alpha_0 \exp\left(-\frac{t}{\tau_\alpha}\right), \\ \sigma(t) &= \sigma_0 \exp\left(-\frac{t}{\tau_\sigma}\right), \end{aligned} \tag{9.9}$$

where α_0 and σ_0 are initial values, and τ_α, τ_σ are time constants controlling the decay rates.

Summary of the Six Learning Steps One run of SOM training is repetitive in a good way: you keep presenting inputs, keep finding the BMU, and keep nudging a neighborhood patch until the map stops moving. The following pseudocode is the same loop written in a way you can implement directly.

Self-Organizing Map (SOM) training pseudocode

1. **Initialize** weight vectors $\mathbf{w}_j(0)$ randomly or from samples.
2. For iteration $t = 0, \ldots, T$:
   (a) Sample an input $\mathbf{x}(t)$.
   (b) Find the best matching unit (BMU) $c = \arg\min_j \|\mathbf{x}(t) - \mathbf{w}_j(t)\|_2^2$.
   (c) Compute neighborhood coefficients $h_{j,c}(t)$.
   (d) Update every neuron:
   $$\mathbf{w}_j(t+1) = \mathbf{w}_j(t) + \alpha(t)\, h_{j,c}(t)\Big(\mathbf{x}(t) - \mathbf{w}_j(t)\Big).$$
   (e) Decay learning-rate $\alpha(t)$ and neighborhood radius $\sigma(t)$ (e.g., exponentially).

Batch SOM (deterministic pass)

1. Fix centers $\mathbf{r}_j$ on the grid; initialize $\mathbf{w}_j$ (random data or along PCA directions).
2. Repeat until convergence or max epochs:
   (a) Assign each $\mathbf{x}_n$ to its BMU c_n.
   (b) For each neuron j, update
   $$\mathbf{w}_j \leftarrow \frac{\sum_n h_{j,c_n}(t)\,\mathbf{x}_n}{\sum_n h_{j,c_n}(t)}.$$
   (c) Decay $\alpha(t), \sigma(t)$.

Batch SOM (one pass per epoch) is deterministic given the assignments; it often stabilizes faster than purely online updates.

I find it useful to tell the SOM story in *three stages*, even though the code is often written as a longer checklist. The stages are: **Initialization** (seed the prototype grid), **Competition** (pick a best-matching unit for each input), and **Cooperation** (update a neighborhood and decay $\alpha(t)$ and $\sigma(t)$). The six-step procedure in Section 9.19 is simply one way to operationalize those stages when you implement the algorithm and decide when to stop.

9.20 Applications of Kohonen Self-Organizing Maps

Kohonen SOMs show up when you want *both* a prototype model and a picture you can reason about. In practice they tend to be used in three roles:

- **Clustering / regime discovery:** Group similar points without supervision and summarize each region by a prototype vector (e.g., operating modes in telemetry features).
- **Exploratory reduction:** Map high-dimensional data onto a 2D grid index for visualization and exploratory analysis. You do not get a continuous coordinate system; you get a discrete map that is often easier to inspect.
- **Visualization diagnostics:** Use U-Matrices and component planes as "instrument panels" that reveal boundaries, smoothness, and which input

features drive the organization.

Author's note: when a SOM is the wrong tool

If your only goal is to assign points to K groups and summarize each group by a center, k-means is usually the simpler baseline. If you want a continuous low-dimensional chart with a clear linear story, start with PCA; if you need a global distance-preserving embedding, MDS is a better conceptual match. If your goal is a local-neighborhood visualization and you accept distortion of global geometry, methods like UMAP/t-SNE are often more visually dramatic.

I reach for a SOM when I want three things at once: (i) prototypes I can inspect, (ii) a fixed 2D organization that is stable enough to compare across runs, and (iii) diagnostics (U-Matrix/component planes, QE/TE) that tell me whether the picture is trustworthy. If you cannot define a sensible distance (or you cannot normalize features so distance means something), a SOM will happily organize noise.

Application example: website sessions as behavioral modes

Consider a website where the system you are trying to understand is not a motor or a circuit, but a stream of user sessions. Each session becomes a data point $\mathbf{x}$: dwell time, number of pages, scroll depth, time-to-first-interaction, device class, referrer type, returning vs. new, and a few content signals (e.g., the category of the first page visited, or an embedding of the landing-page text). You may not have labels for "intent," but you do expect recurring modes: quick bounce, comparison shopping, documentation lookup, checkout flow, support troubleshooting, and so on. A SOM is useful here because it gives you prototypes $\mathbf{w}_i$ (representative session fingerprints) *and* a 2D grid you can inspect. After training, each session maps to a best-matching unit (BMU). Dense patches of mapped sessions indicate common behavior; ridges in the U-Matrix suggest boundaries where neighboring prototypes are far apart, which often corresponds to genuinely different behaviors under your feature choices. Component planes then turn the picture into an explanation: you can see, for example, that one boundary is mostly driven by dwell time and scroll depth, while another is driven by referrer and device. In practice you read these plots as heat maps over the 2D grid: different behaviors show up as different regions you can name, compare, and inspect.

Recommendation: treat preprocessing as part of the model. Normalize continuous features, log-scale heavy-tailed counts, and be deliberate about how you represent categorical variables (one-hot, grouped buckets, or a small embedding) so Euclidean distance matches your notion of similarity.

Audit hook: rerun with a few seeds and a time-based split. If the map reorganizes completely across runs or across weeks, you are seeing instability or seasonality rather than durable structure.

Application example: mapping page archetypes and performance drivers

A second, complementary use is to map *pages* (or queries) rather than sessions. Here each data point $\mathbf{x}$ describes a URL or content item: a text embedding of the page, topic/category tags, layout signals (word count, number of images), and aggregate behavior metrics (click-through rate (CTR) from search, median dwell time, bounce rate, conversion rate, and the fraction of visits from mobile). You are effectively asking: which pages are similar in content and behavior, and where do the sharp breaks live?
A SOM turns this into a visual inventory. Pages with similar content and similar usage patterns compete for the same region of the grid, and the U-Matrix highlights boundaries between page families (e.g., "documentation-like" pages versus "marketing-like" pages). Component planes are the practical payoff: they show which variables line up with those boundaries. If one region lights up on "mobile fraction" and "bounce rate," you immediately get a hypothesis about responsiveness or page speed. If another region is coherent in "topic embedding" but not in conversion, you have found a content family that is consistent but not effective.
Recommendation: separate what you want to *organize by* (content similarity) from what you want to *diagnose* (performance metrics). A simple approach is to train the SOM on content-heavy features, then overlay performance as color/slices; otherwise the map can end up organizing primarily by whatever has the largest numeric range.
Audit hook: overlay slices (language, device, traffic source). If the map's dominant structure is "source domain" or "mobile vs. desktop," that may be a useful finding, but it is also a warning that your feature design is driving the story.

Relation to k-means and modern variants When the neighborhood is collapsed to a single winner (no neighbors), the update reduces to a k-means-style prototype move without any notion of grid organization (MacQueen, 1967); SOMs therefore sit between pure clustering (k-means) and manifold learning, adding a topographic prior that encourages neighboring units to represent similar inputs. Recent "neural-SOM" hybrids embed SOM-like updates inside deep

networks, but still rely on the same BMU search and neighborhood-weighted updates described above.

Theory notes and recipes

Convergence/magnification (theory lens): With decays $\alpha(t) \to 0$, $\sigma(t) \to 0$ and $\sum_t \alpha(t) = \infty$, $\sum_t \alpha^2(t) < \infty$, SOM updates converge under mild assumptions (Erwin et al., 1992; Cottrell and Fort, 1986). One qualitative takeaway is *magnification*: dense regions of the data tend to attract more units of the map.

A practical recipe (engineering lens): If I have no prior, I treat these as starting points, not rules. Use $5\sqrt{N}$–$10\sqrt{N}$ units when unsure; hex grids reduce anisotropy; wrap-around (toroidal) grids reduce edge effects. Initialize $\mathbf{w}_j$ from data or along the first two PCs; pick $\alpha_0 \in [0.1, 0.5]$ and decay toward $\alpha_{\min} \approx 10^{-3}$; set σ_0 to roughly the map radius and decay toward 1–1.5 cells.

Related and growing variants

Neural Gas / Growing Neural Gas (Martinetz et al., 1993; Fritzke, 1994) drop the fixed grid and instead learn neighborhood structure (and, in the growing variants, add units dynamically). **Generative topographic mapping (GTM)** (Bishop et al.) provides a probabilistic, topographic embedding with likelihoods/uncertainty. I like to remember the landscape this way: k-means gives prototypes with no organization; SOMs add a fixed organizing prior; these variants let the organization itself adapt.

Complexity and out-of-sample mapping. A full online epoch costs $O(NM)$ (N data, M units); batch passes cost similar but fewer epochs. For large M, use approximate nearest-neighbor BMU search (k-d trees or vector-index libraries such as FAISS). New points map via their BMU; optional soft responsibilities use h_{ci} for smoothing.

Theory link to other chapters. SOMs learn prototypes like the centers in Chapter 8 but add a topographic prior; quality diagnostics (QE/TE) can be tracked with the validation-curve diagnostics in Chapter 3. For task-driven em-

beddings, see Chapters 11 and 13; Hopfield (Chapter 10) contrasts with energy-based associative recall.

Quality measures and magnification. Two diagnostics are standard when reporting SOM quality:

- **Quantization error (QE):** average Euclidean distance between each input and its BMU. Lower QE indicates prototypes that better represent the data manifold.
- **Topographic error (TE):** fraction of inputs whose first- and second-best BMUs are not adjacent, quantifying topology preservation (magnification factor).

Tracking both metrics reveals whether the neighborhood decay is too slow (over-smoothing) or too aggressive (tearing the topology). Chapter 3's learning-curve plots suggest early-stopping heuristics: stop when QE/TE on a validation split flatten.

Key takeaways

- SOMs perform topology-preserving vector quantization on a discrete grid.
- A shrinking neighborhood and decaying learning rate drive coarse-to-fine organization.
- U-Matrices and quantization/topographic errors are practical diagnostics for convergence.

Minimum viable mastery.

- Given x, compute the BMU, write the weight update $w_i \leftarrow w_i + \alpha(t)h_{ci}(t)(x - w_i)$, and explain how h_{ci} enforces cooperation.
- Interpret a U-Matrix and component planes as diagnostics for neighborhood structure and convergence.
- Choose sensible $\alpha(t)$ and $\sigma(t)$ schedules and justify them using QE/TE validation curves.

Common pitfalls.

- Using a map that is too small or a neighborhood decay that is too aggressive (tearing the topology).
- Skipping input normalization, causing prototypes to track scale rather than structure.
- Treating grid distance as a true metric in data space (SOM topology is approximate, not exact geometry).
- Over-interpreting a single run without checking QE/TE plateaus and sensitivity to initialization.

Exercises

Exercise 1 (Easy). Train a small SOM on a simple 2D dataset where geometry is visible (two clusters, a ring, or a Swiss-roll projection). Visualize the grid and how input points map to BMUs; then plot the U-matrix.
Deliverable: the mapping plot + the U-matrix + one paragraph explaining what the U-matrix is revealing.

Exercise 2 (Medium). Train a 10×10 SOM on a higher-dimensional dataset (e.g., MNIST). Track QE and TE during training and show how input normalization and neighborhood annealing change both metrics.
Deliverable: QE/TE curves + one set of component-plane plots + one concrete failure mode you observed (tearing, collapse, or loss of topology).

Exercise 3 (Medium). Compare SOM prototypes to K-means centers on the same dataset. Treat the comparison as a question about *what is being optimized*: clustering distortion vs. topology preservation.
Deliverable: a side-by-side visualization (SOM vs. K-means) + a short discussion of which method is easier to audit for structure (and why).

If you are skipping ahead. Keep the notion of an "energy landscape" from this chapter in mind: Chapter 10 makes that idea explicit with a Lyapunov energy, and later attention models (Chapter 14) can be read as learned, content-based neighborhood selection.

Where we head next. Chapter 10 extends the unsupervised thread with energy-based associative memory, complementing SOM topology learning with retrieval dynamics that foreshadow later attention-style mechanisms.

10 Hopfield Networks: Introduction and Context

Chapter 9 focused on self-organizing maps and unsupervised feature maps; we now transition to another unsupervised/energy-based model: the *Hopfield network*, a recurrent system that stores patterns as attractors. Figure 1 marks this as the energy-based branch. The debug mindset carries over, but the diagnostics change: SOMs are easiest to audit through map geometry (U-matrix/component

planes), while Hopfield recall is easiest to audit through an energy trace and overlap with the intended memory.

Learning Outcomes

- Interpret Hopfield networks as energy-minimizing recurrent systems and derive their asynchronous update rule.
- Quantify capacity, recall dynamics, and pitfalls (spurious memories, bias encodings) using simple analytical bounds.
- Relate Hopfield updates to modern energy-based models and attention mechanisms to build intuition for later chapters.

Design motif

Constrain recurrence so the dynamics become a descent process: symmetric weights and an energy function turn "feedback" into "stable memory."

10.1 From Feedforward to Recurrent Neural Networks

Feedforward networks compute a single forward map: once an input has passed through the layers, the computation ends. That is the right abstraction for static input→output tasks, but it is not a natural model of memory: nothing inside the network persists unless you explicitly feed it back in.

Recurrent networks add that feedback. Cycles let the current state influence the next state, which is exactly what you want for sequences and recall—and also exactly what makes the dynamics harder to reason about. Hopfield's contribution is to keep the recurrence, but constrain it so the system settles instead of wandering.

Challenges with general recurrent networks The moment you add feedback, you inherit a dynamical system. That buys you memory, but it also means the network can (i) fail to settle and instead oscillate, (ii) end up in different states from tiny changes in the starting point, and (iii) become harder to train because learning signals must propagate through time (and that path can vanish or explode). Historically, these issues pushed many applications toward feedforward models unless recurrence was the cleanest abstraction for the task.

Then vs. now: energy-based memory in context

Classical Hopfield networks are the clean case: binary states, symmetric weights, and an explicit Lyapunov energy give you a convergence guarantee. The intuition that survives is the one we care about: memory is an energy landscape, and recall is descent toward an attractor (pattern completion). Modern systems borrow the geometry without necessarily inheriting the guarantee. Many "memory" mechanisms are differentiable, learned from data, and content-addressable at scale; the symmetry that makes Hopfield proofs work is not always present. Keep the attractor picture, but do not over-assume the math.

A useful concrete connection is modern attention: scaled dot-product attention (introduced later in Chapter 14) can be interpreted as a continuous Hopfield-style retrieval step (Ramsauer et al., 2021). Keep that in mind as you read both chapters.

Author's note: stabilizing recurrence

General recurrent networks can behave unpredictably because feedback can create cycles that oscillate or amplify small differences. Hopfield's key move was to restrict the architecture so the dynamics become a descent process: symmetric weights and no self-loops allow the network to be assigned an energy function that decreases under asynchronous updates. That single design choice turns "recurrent" from "chaotic" into "stable memory."

10.2 Hopfield's breakthrough

In 1982, John Hopfield introduced a special class of recurrent networks by putting engineering constraints on the feedback loop (Hopfield, 1982). The constraints are simple enough to implement, and each one buys you a piece of the convergence story.

First, make the weights symmetric and remove self-loops:

$$w_{ij} = w_{ji} \quad \forall i, j,$$
$$w_{ii} = 0 \quad \forall i.$$

Second, keep neuron states binary, $s_i \in \{+1, -1\}$, so the network evolves by

flipping bits rather than flowing through a continuous activation curve.

Under these choices you can define a Lyapunov energy $E(\mathbf{s})$ (a scalar function that decreases under the update) and choose an asynchronous (one-neuron-at-a-time) update rule so that each flip never increases E. That is the key move: the network becomes a descent process in a finite state space, so it must settle at a fixed point. Those fixed points are the stored patterns (and their complements), plus whatever spurious attractors appear when the network is heavily loaded.

10.3 Network Architecture and Dynamics

Consider a Hopfield network with N neurons. The state vector is $\mathbf{s} = (s_1, s_2, \ldots, s_N)^T$, where each $s_i \in \{+1, -1\}$. The symmetric weight matrix $\mathbf{W} = [w_{ij}]$ satisfies $w_{ij} = w_{ji}$ and $w_{ii} = 0$. Throughout this discussion w_{ij} denotes the weight applied to state s_j when computing the input to neuron i, so column indices correspond to presynaptic neurons.

The *local field* or *input energy* to neuron i is defined as

$$h_i(t) = \sum_{j=1}^{N} w_{ij} s_j(t).$$

The scalar $h_i(t)$ therefore represents the total input (or *local field*) accumulated at neuron i before thresholding during iteration t.

The neuron updates its state according to the sign of $h_i(t)$ relative to a threshold θ_i:

$$s_i(t+1) = \begin{cases} +1, & h_i(t) \geq \theta_i, \\ -1, & h_i(t) < \theta_i, \end{cases} \tag{10.1}$$

Typically, thresholds θ_i are set to zero or learned as part of the model.

Interpretation: The neuron "fires" (state $+1$) if the weighted sum of inputs exceeds the threshold; otherwise, it remains "off" (state -1). This binary update rule contrasts with the continuous activation functions used in feedforward networks.

10.4 Encoding conventions

Two binary encodings are common. We primarily use $s_i \in \{-1, +1\}$ because it simplifies the energy function, but many software libraries work with $x_i \in \{0, 1\}$. Define $\mathbf{s} = 2\mathbf{x} - \mathbf{1}$ and $\mathbf{x} = (\mathbf{s} + \mathbf{1})/2$; then

$$E_{\pm 1}(\mathbf{s}) = -\frac{1}{2} \sum_{i \neq j} w_{ij} s_i s_j + \sum_i \theta_i s_i$$

and

$$E_{01}(\mathbf{x}) = -\frac{1}{2} \sum_{i \neq j} w'_{ij} x_i x_j + \sum_i \theta'_i x_i + \text{const},$$

with $w'_{ij} = 4w_{ij}$ and $\theta'_i = 2\theta_i + 2\sum_{j \neq i} w_{ij}$ under the threshold convention used in $E_{\pm 1}$. The additive constant does not affect which states minimize the energy or the update dynamics. This table summarizes the correspondence:

Sign convention: threshold form vs. bias form

This chapter uses the threshold update $s_i \leftarrow \text{sign}(h_i - \theta_i)$. Under that convention the energy carries the term $+\sum_i \theta_i s_i$. If you prefer to write the same dynamics with an additive bias b_i in the local field, set $b_i = -\theta_i$; the identical energy is then written as $-\sum_i b_i s_i$. The dynamics are unchanged once the translation $b_i = -\theta_i$ is kept explicit.

	$\{-1, +1\}$ encoding	$\{0, 1\}$ encoding
State variable	$s_i \in \{-1, +1\}$	$x_i = (s_i + 1)/2$
Energy	$E_{\pm 1}(\mathbf{s})$	$E_{01}(\mathbf{x}) = E_{\pm 1}(2\mathbf{x} - \mathbf{1})$
Update rule	$s_i \leftarrow \text{sign}(h_i - \theta_i)$	$x_i \leftarrow \mathbf{1}[h'_i - \theta'_i > 0]$

Whenever an equation later in the chapter uses s_i you can translate it to x_i via this affine mapping; we call out both forms only when the distinction matters. As a concrete example, the pattern $\mathbf{x} = [1, 0, 1, 0]$ in the $\{0, 1\}$ encoding maps to $\mathbf{s} = 2\mathbf{x} - \mathbf{1} = [+1, -1, +1, -1]$; conversely $\mathbf{s} = [-1, +1, +1]$ corresponds to $\mathbf{x} = [0, 1, 1]$.

10.5 Energy Function and Stability

Hopfield defined an energy function $E : \{-1, +1\}^N \to \mathbb{R}$ associated with the network state $\mathbf{s}$:

$$E(\mathbf{s}) = -\frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N} w_{ij} s_i s_j + \sum_{i=1}^{N} \theta_i s_i. \tag{10.2}$$

Because the weights are symmetric and satisfy $w_{ii} = 0$, the double sum may equivalently be written as $\sum_{i<j} w_{ij} s_i s_j$; the $\frac{1}{2}$ factor explicitly prevents counting each unordered pair twice, so removing it would scale the energy by two. This sign is consistent with the update rule $s_i \leftarrow \text{sign}(h_i - \theta_i)$. If you switch to a bias form $h_i + b_i$, the same energy is written as $-\sum_i b_i s_i$ with $b_i = -\theta_i$.

Engineering lens: energy as an objective (not a metaphor)

It helps to read $E(\mathbf{s})$ the same way you read a loss in supervised learning: a scalar objective defined over a space of states. Here the states are discrete ($\mathbf{s} \in \{-1, +1\}^N$), so the "optimization" is not gradient descent; it is a sequence of coordinate updates (flip one bit at a time) that never increase E under Hopfield's symmetry constraints.

This is why the update rule matters. The weights define the energy landscape, and the asynchronous update is a descent procedure on that landscape. Local minima are stable memories; other minima (spurious attractors) are the price of capacity.

Hopfield network states and the energy view Encoding choices and the corresponding energy forms were laid out in Section 10.4 and Section 10.5. The key continuity idea is that Hopfield dynamics are not "mysterious recurrence": with symmetric weights, the update rule becomes a descent process on $E(\cdot)$. We make this concrete in three passes: define what "stable" means, walk one tiny numeric trace, then prove the one-step claim ($\Delta E \leq 0$) that guarantees convergence under asynchronous updates.

10.6 Energy Minimization and Stable States

The fundamental goal in Hopfield networks is to find a state $\mathbf{s}$ that minimizes the energy E. Such states correspond to stable equilibria or attractors of the network dynamics.

State update dynamics: The network updates neuron states using the sign rule in (10.1); for $\{0, 1\}$ encodings use the corresponding thresholded version. The pseudo-code below makes the asynchronous sweep explicit for later convergence arguments.

Asynchronous Hopfield update (pseudo-code)

1. Initialize $\mathbf{s}$ (e.g., noisy probe), set max sweeps T.
2. For $t = 1, \ldots, T$:
   (a) Pick a neuron index i (random order or cyclic sweep).
   (b) Compute $h_i = \sum_j w_{ij} s_j - \theta_i$.
   (c) Update $s_i \leftarrow \text{sign}(h_i)$.
3. Stop early if a full sweep causes no flips; else continue.

Each single-neuron update satisfies $\Delta E \leq 0$ by (10.3), so the loop converges to a local minimum of (10.2).

For the formal monotonicity proof and the synchronous-update caveat, see Section 10.9.

Implementation checklist (debugging a Hopfield net)

- **Weight constraints:** enforce $W = W^\top$ and $w_{ii} = 0$. If either is violated, the classical energy argument does not apply.
- **Energy definition:** implement (10.2) with the $\frac{1}{2}$ factor (or equivalently sum over $i < j$). Use one sign convention for thresholds/biases and stick to it.
- **Update scheme:** test with asynchronous single-neuron updates first. If you switch to synchronous updates, short cycles are possible (see Section 10.9).
- **Sign at zero:** choose a deterministic convention for sign(0) (e.g., keep the old state, or return +1) so runs are reproducible.
- **Diagnostics:** on a tiny network where you know the stored patterns, plot $E(\mathbf{s}(t))$ and overlap $m^{(\mu)}(\mathbf{s}(t))$. Under asynchronous updates and symmetric weights, E should never increase.

10.7 Example: Energy Calculation and State Updates

Consider a Hopfield network with three neurons, bipolar states $s_i \in \{-1, +1\}$, zero thresholds, and the symmetric weight matrix

$$W = \begin{bmatrix} 0 & 3 & -4 \\ 3 & 0 & 2 \\ -4 & 2 & 0 \end{bmatrix}.$$

Let the initial state be $\mathbf{s} = (1,\ 1,\ -1)$. Using the energy definition with the $\frac{1}{2}$ factor to avoid double counting, we obtain

$$\begin{aligned} E(\mathbf{s}) &= -\frac{1}{2}\sum_{i=1}^{3}\sum_{j=1}^{3} w_{ij} s_i s_j \\ &= -\frac{1}{2}\Big[2\cdot 3\cdot (1)(1) + 2\cdot(-4)\cdot(1)(-1) + 2\cdot 2\cdot(1)(-1)\Big] = -5. \end{aligned}$$

State update attempts: One way to check whether $\mathbf{s} = (1, 1, -1)$ is stable is to try each single-neuron flip and recompute E. Flipping s_1 gives $E(-1, 1, -1) =$

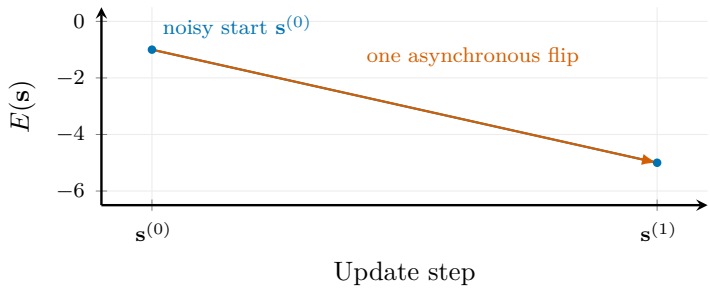


Figure 46: Hopfield energy decreases monotonically under asynchronous updates. Starting from a noisy probe $\mathbf{s}^{(0)}$, single-neuron flips move downhill until a stable memory is recovered.

9; flipping s_2 gives $E(1, -1, -1) = -3$; and flipping s_3 gives $E(1, 1, 1) = -1$. Each move raises the energy relative to -5, so no asynchronous update would accept it: the state is a local minimum.

For clarity, Table 3 reports the energy change for each single flip relative to the current state.

Flip	New state	ΔE	Accept?
$s_1 \leftarrow -1$	$(-1,\ 1,\ -1)$	$+14$	No
$s_2 \leftarrow -1$	$(1,\ -1,\ -1)$	$+2$	No
$s_3 \leftarrow +1$	$(1,\ 1,\ 1)$	$+4$	No

Table 3: Single-neuron flips from $(1, 1, -1)$; all raise the energy, so the state is a local minimum.

Because every single-neuron flip raises the energy, the state $(1, 1, -1)$ is a stable local minimum for this network. If the network is perturbed slightly (for instance, by flipping s_3 to $+1$ to create the noisy pattern $(1, 1, 1)$), the next asynchronous update flips it back. In this state the local field at neuron 3 is $h_3 = -4(1) + 2(1) = -2$, so the update rule sets $s_3 \leftarrow -1$, returning to $(1, 1, -1)$. The energy drops from -1 to -5, which is the basic content-addressable recall behavior.

As Figure 46 shows, the energy trace never increases when single neurons flip asynchronously, so the descent depicted here is exactly what the formal convergence proof below captures.

10.8 Energy Function and Convergence of Hopfield Networks

The figure and pseudo-code above already suggest the main claim: under the symmetry constraints that make $E(\cdot)$ well defined, asynchronous updates are guaranteed to move "downhill" (or stay flat) in energy. This is the formal reason Hopfield networks converge to a fixed point: the state space is finite, and you cannot decrease a bounded quantity forever.

In this subsection we prove the monotonicity step that the rest of the convergence story rests on. The only assumptions you should keep in view are the ones you would implement anyway: symmetric weights with zero diagonal, and an asynchronous update rule (one neuron at a time). When these assumptions are violated (e.g., synchronous updates or asymmetric weights), the energy argument no longer guarantees convergence.

We use the energy in (10.2); the goal is to show that each single-neuron update never increases it.

Goal: Show that asynchronous updates of neuron states always decrease (or leave unchanged) the energy E, guaranteeing convergence to a local minimum.

10.8.1 Energy Change Upon Updating a Single Neuron

Consider updating neuron i from old state s_i^{old} to new state s_i^{new}. All other neuron states s_j for $j \neq i$ remain fixed. The change in energy is $\Delta E = E_{\text{new}} - E_{\text{old}}$.

Using (10.2), write out the energies explicitly:

$$E_{\text{old}} = -\frac{1}{2}\sum_{k=1}^{N}\sum_{l=1}^{N} w_{kl} s_k^{\text{old}} s_l^{\text{old}} + \sum_{k=1}^{N} \theta_k s_k^{\text{old}},$$
$$E_{\text{new}} = -\frac{1}{2}\sum_{k=1}^{N}\sum_{l=1}^{N} w_{kl} s_k^{\text{new}} s_l^{\text{new}} + \sum_{k=1}^{N} \theta_k s_k^{\text{new}}.$$

Since only s_i changes, and weights are symmetric with zero diagonal $w_{ii} = 0$,

the difference simplifies to

$$
\begin{aligned}
\Delta E &= E_{\text{new}} - E_{\text{old}} \\
&= -\frac{1}{2}\sum_{j=1}^{N}\left(w_{ij}s_i^{\text{new}}s_j + w_{ji}s_js_i^{\text{new}}\right) + \theta_i s_i^{\text{new}} \\
&\quad + \frac{1}{2}\sum_{j=1}^{N}\left(w_{ij}s_i^{\text{old}}s_j + w_{ji}s_js_i^{\text{old}}\right) - \theta_i s_i^{\text{old}} \\
&= -\sum_{j=1}^{N} w_{ij}s_j\left(s_i^{\text{new}} - s_i^{\text{old}}\right) + \theta_i\left(s_i^{\text{new}} - s_i^{\text{old}}\right) \\
&= -\left(s_i^{\text{new}} - s_i^{\text{old}}\right)\left(\sum_{j=1}^{N} w_{ij}s_j - \theta_i\right).
\end{aligned}
\tag{10.3}
$$

Define the *local field* h_i at neuron i as

$$
h_i = \sum_{j=1}^{N} w_{ij}s_j - \theta_i. \tag{10.4}
$$

Then,

$$
\Delta E = -(s_i^{\text{new}} - s_i^{\text{old}})h_i.
$$

Under the asynchronous sign update, a neuron flips only when the new state agrees with the sign of h_i; therefore $s_i^{\text{new}} - s_i^{\text{old}}$ has the same sign as h_i, and the product above is non-negative. Hence every accepted flip satisfies $\Delta E \leq 0$.

Numeric check (single flip). With two neurons, weights $w_{12} = w_{21} = 1$, thresholds $\theta_i = 0$, and current state $(s_1, s_2) = (1, -1)$, the local field at neuron 1 is $h_1 = 1 \cdot (-1) = -1$. The update rule sets $s_1^{\text{new}} = \text{sign}(h_1) = -1$, so $s_1^{\text{new}} - s_1^{\text{old}} = -2$ and $\Delta E = -(-2)(-1) = -2 < 0$, confirming the energy drop predicted by (10.3).

Modern Hopfield views and attention

Recent work (Krotov and Hopfield, 2016, 2020; Ramsauer et al., 2021) revisits Hopfield networks as dense associative memories with continuous states and softmax interactions that are closely related to Transformer attention. In this view the stored patterns play the role of keys and values, the current state or query probes the landscape, and the update rule resembles a softmax-weighted average over memories, minimizing an energy of the form $\log \sum_\mu \exp\big(\beta \mathbf{s}^\top \mathbf{m}^\mu\big)$ with inverse temperature β. The useful connection for us is the geometry: content-addressable lookup as motion in an energy landscape. Do not confuse that bridge with the classical guarantee above: the Lyapunov proof in this chapter relies on binary states, symmetric weights, and asynchronous flips.

Continuous Hopfield networks (bridge). Modern extensions replace the binary sign activation with a smooth, often softmax-like update that keeps neuron states continuous. Storing P real-valued patterns $\{\mathbf{m}^\mu\}$, a common update takes the form

$$\mathbf{s}^{(t+1)} = \mathrm{softmax}\Big(\beta M^\top \mathbf{s}^{(t)}\Big)\, M,$$

where M stacks the memory vectors and β controls sharpness. Read this as "soft" associative recall: the update produces a weighted combination of stored patterns, and that picture lines up closely with attention in Chapter 14. Unlike the classical binary case, you should not assume strict monotone descent unless the model is built to have a Lyapunov function.

Combining the sign update in (10.1) with (10.3) yields $\Delta E \leq 0$ for each asynchronous flip, so we omit the redundant case split here.

10.9 Asynchronous vs. Synchronous Updates in Hopfield Networks

Recall from the previous discussion that the Hopfield network energy function decreases monotonically with each asynchronous update of a single neuron state. This guarantees convergence to a local minimum of the energy landscape. In contrast, fully synchronous updates (flipping all neurons at once) can lead to oscillations or short cycles rather than convergence, which is why the classical

convergence proofs assume asynchronous updates.

Why asynchronous updates? With symmetric weights you can still write the same energy $E(\mathbf{s})$, but the classic monotonicity proof is for *single-neuron* (asynchronous) updates. If you update all neurons simultaneously, two-cycles can appear.

For a concrete example, take two neurons with $\theta_1 = \theta_2 = 0$ and $w_{12} = w_{21} = 1$. Start from $(s_1, s_2) = (1, -1)$. A synchronous update computes both new states from the old pair:

$$s_1 \leftarrow \text{sign}(w_{12}s_2) = \text{sign}(-1) = -1, \qquad s_2 \leftarrow \text{sign}(w_{21}s_1) = \text{sign}(1) = 1,$$

so the state becomes $(-1, 1)$. Repeating the synchronous update maps $(-1, 1)$ back to $(1, -1)$. The network therefore oscillates in a 2-cycle, even though the energy

$$E(s_1, s_2) = -\frac{1}{2}\sum_{i,j} w_{ij}s_is_j = -w_{12}s_1s_2$$

stays constant across the two states.

To ensure the classical convergence guarantee, use *asynchronous* updates (one neuron at a time) under symmetric weights with zero diagonal. Synchronous updates can still work in practice, but the energy argument no longer rules out short cycles.

10.10 Storage Capacity of Hopfield Networks

A key question is: *How many memories can a Hopfield network reliably store and recall?*

$$p \approx 0.138\, n,$$

is the classical rule-of-thumb for the number of random patterns that can be stored with low error in a network of n neurons (McEliece et al., 1987). The important point is not the constant itself; it is that the capacity is only a small fraction of n. Both ξ^μ and its complement $-\xi^\mu$ are fixed points, and odd mixtures of stored patterns become spurious states as p/n grows. This low capacity is why Hopfield networks are not used as practical storage devices despite their elegant associative-memory behavior.

Stochastic updates (bridge to Boltzmann machines). A stochastic variant replaces the hard sign in (10.1) with probabilistic flips (e.g., Gibbs sampling). With symmetric weights this defines a Boltzmann distribution whose energy matches (10.2), linking Hopfield recall to the Boltzmann/energy-based models that underlie modern probabilistic neural networks.

10.11 Improving Storage Capacity via Weight Updates

Is it possible to improve storage by choosing the weights to "bake in" the memories directly? That is exactly what the classical Hebbian construction does: use stored patterns to set W, then let the state updates perform recall.

Hebbian learning rule: Given p stored patterns $\{\xi^1, \xi^2, \ldots, \xi^p\}$, each $\xi^\mu = (\xi_1^\mu, \xi_2^\mu, \ldots, \xi_n^\mu)$ with $\xi_i^\mu \in \{+1, -1\}$, the weights are set by:

$$w_{ij} = \frac{1}{n} \sum_{\mu=1}^{p} \xi_i^\mu \xi_j^\mu, \quad w_{ii} = 0. \tag{10.5}$$

This is the classical Hebbian learning rule for Hopfield networks.

What this formula is doing: Setting $w_{ii} = 0$ removes self-feedback. The factor $1/n$ keeps weight magnitudes $O(1)$ as the network grows. The sum itself is an outer-product accumulation: it encodes pairwise correlations across the stored patterns.

10.12 Example: Weight Calculation for a Single Pattern

Consider a fundamental memory pattern:

$$\xi = (1, 1, 1, -1),$$

with no thresholds ($\theta_i = 0$).

Step 1: Compute outer product Form the matrix $\mathbf{B} = \xi\xi^\top$. Each entry $B_{ij} = \xi_i\xi_j$ captures the pairwise correlation between neurons i and j.

$$\mathbf{B} = \xi\xi^\top = \begin{bmatrix} 1 & 1 & 1 & -1 \\ 1 & 1 & 1 & -1 \\ 1 & 1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{bmatrix}.$$

Step 2: Remove diagonal terms Zero the diagonal entries to obtain the weight matrix $\mathbf{W}$ with $w_{ii} = 0$. The off-diagonal values remain the same as in $\mathbf{B}$, encoding the pairwise interactions required to store the memory pattern. Including the $1/n$ normalization from (10.5) (here $n = 4$) gives

$$\mathbf{W} = \frac{1}{4}\Big(\mathbf{B} - \operatorname{diag}(\mathbf{B})\Big) = \frac{1}{4}\begin{bmatrix} 0 & 1 & 1 & -1 \\ 1 & 0 & 1 & -1 \\ 1 & 1 & 0 & -1 \\ -1 & -1 & -1 & 0 \end{bmatrix}.$$

Sanity check: the stored pattern is a fixed point With zero thresholds, the local field is $\mathbf{h} = \mathbf{W}\xi$. For this one-pattern example,

$$\mathbf{h} = \begin{bmatrix} 3/4 \\ 3/4 \\ 3/4 \\ -3/4 \end{bmatrix},$$

so $\operatorname{sign}(\mathbf{h}) = \xi$. That is the basic content-addressable "snap-back": if you start near ξ, asynchronous updates push you toward it.

Numeric recall trace (two asynchronous flips)

Start from a probe with two wrong bits, $\mathbf{s}^{(0)} = (1, -1, -1, -1)$. With $\theta_i = 0$, the local field is $h_i = \sum_j w_{ij} s_j$, and the update rule (10.1) sets $s_i \leftarrow \text{sign}(h_i)$. One possible asynchronous update order (neuron 2, then neuron 3) gives:

Step	Update	Local field	New state
0	–	–	$(1, -1, -1, -1)$
1	$i = 2$	$h_2 = 0.25$	$(1, 1, -1, -1)$
2	$i = 3$	$h_3 = 0.75$	$(1, 1, 1, -1) = \xi$

The energy drops along the way: $E = 0.5 \to 0 \to -1.5$.

10.13 Finalizing the Hopfield Network Derivation and Discussion

We have already defined the Hebbian weights in (10.5), the update rule in (10.1), and the energy in (10.2). Here we focus on what those ingredients imply for retrieval basins and limitations in practice.

Memory Retrieval and Basins of Attraction The stored patterns $\{\xi^\mu\}$ correspond to local minima of the energy landscape. Starting from an initial state $\mathbf{s}(0)$ that is a noisy or partial version of a stored pattern, the network dynamics converge to the closest attractor, ideally retrieving the original memory or its complement $-\xi^\mu$.

For example, if the initial state is corrupted, the network will iteratively update states to reduce energy until it reaches a stable point:

$$\mathbf{s}(\infty) \in \{\xi^\mu, -\xi^\mu\}.$$

The same story can be read visually as motion toward a basin of attraction in an energy landscape (Figure 47).

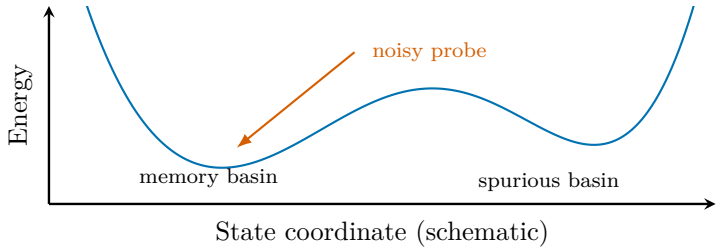


Figure 47: Energy-landscape intuition (schematic). Hopfield recall is a descent process toward a nearby basin (a stored memory), but other minima can exist and act as spurious attractors when the network is heavily loaded.

Terminology and diagnostics

Attractor. A stable fixed point of the update rule: once you reach it, further asynchronous updates do not change the state.
Basin of attraction. The set of initial states that converge to a given attractor under the chosen update scheme.
Spurious state. An attractor that is not one of the intended stored patterns (nor its complement). Spurious states are real local minima of $E(\cdot)$ created by interference among memories.
Overlap (a simple health check). For a stored pattern ξ^μ, define

$$m^{(\mu)}(\mathbf{s}) = \frac{1}{N}(\xi^\mu)^\top \mathbf{s}.$$

It ranges from $+1$ (exact match) to -1 (exact complement). Plot $E(\mathbf{s}(t))$ and $m^{(\mu)}(\mathbf{s}(t))$ versus update step to see whether recall is behaving or drifting into a spurious basin.

Limitations (what breaks first) Hopfield networks are a beautiful model to reason about, but several limitations show up quickly in practice:

- **Capacity:** with the classical Hebbian construction, reliable storage for random patterns scales like $P_{\max} \approx 0.138\,N$ for large N. Past that loading level, retrieval errors and spurious minima become common.
- **Spurious attractors:** even below capacity, the energy landscape can contain minima that are not stored patterns (or their complements). A

noisy probe can "snap" to one of these unintended fixed points.

- **Classification mismatch:** for discriminative tasks (say, digit recognition), the "nearest minimum" behavior is not the same thing as a calibrated class decision. A corrupted input can converge to the wrong stored pattern, and low energy does not imply correct label.
- **When not to use:** heavy loading (P/N large) produces a glassy landscape; scaling is $O(N^2)$ in connectivity; and the low-capacity regime makes classical Hopfield networks a poor fit for high-dimensional supervised problems compared with the ERM models in Chapter 4 or the deep models in Chapter 14.

Applications lens: memory retrieval and optimization

In the classical framing, Hopfield networks are about associative memory: you store patterns, then recover a full pattern from a partial or noisy cue (pattern completion). That is the same idea behind many information-retrieval and recognition stories: the input is a corrupted probe, and the system "snaps" to a nearby clean prototype.
There is also an optimization lens. Because $E(\mathbf{s})$ is a scalar objective over binary states, you can sometimes encode a cost function as an energy and run asynchronous updates as a heuristic descent method. This is one way Hopfield-style dynamics have been discussed for combinatorial optimization tasks (for example, variants of traveling-salesman-style objectives). The warning is the same one we give throughout this book: the algorithm will happily converge to a local minimum. Your real work is in deciding whether the energy you wrote down matches the engineering goal and whether the resulting minima are meaningful.

Example: Memory recovery (one flip) Store one pattern

$$\xi = (-1, -1, 1, -1)^T.$$

With $n = 4$, the Hebbian weights from (10.5) are

$$\mathbf{W} = \frac{1}{4}\xi\xi^\top, \qquad w_{ii} = 0,$$

which numerically becomes a single symmetric matrix

$$\mathbf{W} = \frac{1}{4}\begin{pmatrix} 0 & 1 & -1 & 1 \\ 1 & 0 & -1 & 1 \\ -1 & -1 & 0 & -1 \\ 1 & 1 & -1 & 0 \end{pmatrix}.$$

The off-diagonal entries are therefore the scaled products of pattern components (e.g., $w_{12} = w_{21} = 0.25$ and $w_{13} = w_{31} = -0.25$). More generally, every off-diagonal weight is the scaled product of the corresponding pattern entries.

Now start from a one-bit-corrupted probe $\mathbf{s}^{(0)} = [-1, -1, 1, 1]^T$, with $\theta_i = 0$. If we update neuron 4,

$$h_4 = \sum_j w_{4j} s_j = -0.75 \quad \Rightarrow \quad s_4 \leftarrow -1,$$

so in one asynchronous flip we recover $\mathbf{s}^{(1)} = \xi$. The energy drops from $E = 0$ to $E = -1.5$ under (10.2). The appearance of $-\xi$ as a fixed point is expected: the energy only depends on products $s_i s_j$, so negating all bits leaves every term unchanged.

Spurious attractors Beyond the intended memories $\{\pm\xi^\mu\}$, Hopfield networks can converge to *spurious attractors*: stable states formed by mixtures of stored patterns. These unintended minima become increasingly common as the loading factor P/N grows (here P denotes the number of stored patterns); for random patterns the practical capacity is roughly $0.138\,N$. The possibility of converging to a spurious state, or to the complemented memory rather than the original, explains why Hopfield networks are better viewed as associative memories than as discriminative classifiers.

Historical and practical significance I like Hopfield networks because they show a recurring engineering move: constrain a system until you can prove something useful, then use that proof as a debugging lens. Symmetry + no self-loops gives you an energy you can compute; asynchronous updates give you a monotone progress measure.

You will almost never deploy a classical Hopfield net as-is (capacity and

spurious minima show up fast). But the idea "memory as an objective landscape" survives, and it is the right mental model to carry into later energy-based and content-addressable mechanisms.

Connections to other chapters. Hopfield networks sit next to SOMs in spirit: both are unsupervised, but the diagnostic tools differ. SOMs give you map geometry (U-matrix/component planes); Hopfield gives you overlap and an energy trace to check whether recall is behaving. Later, attention revisits content-addressable lookup in a differentiable form.

Key takeaways

- Hopfield networks store binary patterns as attractors in an energy landscape defined by symmetric weights.
- Asynchronous updates monotonically reduce the Lyapunov energy, ensuring convergence to a fixed point.
- Capacity is limited and spurious attractors appear as the load P/N grows; treat Hopfield nets primarily as associative memories.

Minimum viable mastery.

- Write the energy $E(s)$, state the symmetry condition $W = W^\top$ with zero diagonal, and explain why asynchronous updates decrease E.
- Distinguish true memories from spurious attractors, and connect failure modes to loading P/N and correlation among stored patterns.
- Use overlap and energy traces as diagnostics when demonstrating recall under corruption.

Common pitfalls.

- Using asymmetric weights or nonzero self-connections (breaks the standard energy argument).
- Overloading the network and expecting clean recall (spurious minima dominate).
- Reporting a single cherry-picked recall trajectory instead of multiple corruption levels and multiple stored sets.

Exercises

Exercise 1 (Easy). Implement asynchronous recall for bipolar states with symmetric W and $w_{ii} = 0$. Verify empirically that energy never increases during recall.
Deliverable: one energy-vs-step plot + a one-sentence statement of the exact update rule you used.

Exercise 2 (Medium). Treat capacity as an engineering parameter: store P random patterns with Hebbian weights, corrupt one pattern by flipping k bits, and run recall. Sweep P/N and k; report when spurious attractors become common by tracking both energy and overlap $m^{(\mu)}$.
Deliverable: an overlap summary plot over the sweep + one short conclusion about what "overloading" looks like empirically.

Exercise 3 (Medium). Find a small case where synchronous updates enter a 2-cycle (or fail to converge cleanly), then rerun from the same initial state with asynchronous updates.
Deliverable: a state/energy trace showing the cycle + a short explanation of why the asynchronous regime is easier to reason about.

If you are skipping ahead. The key transferable idea is the combination of a state update rule with a scalar quantity that tracks progress (energy, loss, or validation curves). That mindset carries into deep training in Chapters 6 to 11.

Where we head next. Chapter 11 turns from associative memory to deep feedforward perception, where convolution and pooling form hierarchical features. The optimization workflow is unchanged: the ERM toolkit from Chapter 3 and the training mechanics from Chapters 6 to 7 remain central. The analogy to keep in mind is the diagnostic habit: in Hopfield networks we watched an energy trace and overlap; in deep CNNs you will watch training/validation curves and slice errors. Recurrence and attention return in Chapter 12 and Chapter 14.

11 Convolutional Neural Networks and Deep Training Tools

Chapter 10 used energy to make recurrence stable: dynamics settle, memories persist, and failure has a concrete signature. Vision problems have a different flavor. The system must form a representation that is rich enough to support decisions, yet structured enough to learn from finite data.

Convolutional networks are the simplest version of that bargain. They keep the empirical risk minimization (ERM) workflow from Chapters 3 to 4 and the gradient-training machinery from Chapter 6, but make images tractable by enforcing locality and weight sharing, so the model spends capacity on patterns that recur across the grid.

There is also a useful continuity lens from earlier chapters: a convolutional filter is a learned *local template* that is reused everywhere. This is the same high-level idea as prototypes/bases in Chapters 8 and 9—but with strict weight sharing and end-to-end gradient training, so the templates are shaped by the task loss rather than chosen or tuned in isolation.

The second thread in this chapter is pragmatic. Deep models are only useful when they train. After the CNN mechanics, we transition into the deep-training toolkit (initialization, activations, normalization, regularization, and optimizers) that will reappear in Chapters 12 to 14.

Learning Outcomes

After this chapter, you should be able to:

- Derive convolution/cross-correlation with stride and padding in 1D/2D.
- Explain receptive-field growth across layers and the role of downsampling.
- Read a CNN as a learned feature map and connect that viewpoint to linear margins (hinge/soft-margin) and earlier kernel-style ideas in Chapter 3.
- Choose losses/metrics deliberately and apply practical stabilizers (BN, dropout, weight decay, optimizers) by recognizing the failure signatures they address.

Design motif

Keep the same statistical learning loop from Chapters 3 to 4, but move a large fraction of the "bias" into the architecture: locality and weight sharing.

11.1 Motivation and map

The core backprop loop from earlier chapters scales to high-dimensional perception tasks, but naively applying dense multilayer perceptrons (MLPs) to images runs into two practical walls: *parameter count* and *inductive bias*. Images have strong local structure (edges, corners, textures), yet a fully connected layer treats every pixel as unrelated to its neighbors and spends parameters learning spatial patterns from scratch.

This is also where the training loop became practical at scale: large datasets and modern accelerators made iterative optimization feasible, and architectural priors such as locality and weight sharing made that optimization sample-efficient on image grids.

CNNs emerged as a pragmatic answer: keep end-to-end gradient training, but bake in locality and translation structure through sparse connectivity and shared weights. Early convolutional networks such as LeNet established this recipe on

document and zip-code recognition tasks (LeCun et al., 1989, 1998). The result is a model class that is both more sample-efficient and more computationally viable on large grids.

How to read this chapter.

- **Core thread (CNNs):** locality + weight sharing → convolution/pooling mechanics → channels/feature maps → end-to-end classifiers.
- **Going deeper:** depth as receptive-field growth, and why stacked small kernels can see large spatial extent.
- **Training toolkit:** the stabilizers you will reuse later (initialization, activations, normalization, regularization, optimizers).

One term we will use early is *receptive field*: the receptive field of a feature value is the region of the input grid that can influence it. The word is literal—outside that region, the unit does not "receive" signal from the input—and stacking layers grows this region.

Connection to the Ch. 1 case study. The industrial-plant camera system introduced in Section 1.4 is a direct application of this chapter: it takes a pixel grid as input, uses convolutional feature extraction to localise humans in the frame, and outputs an alarm signal when the occupancy score exceeds a calibrated threshold. Many of the tools discussed below—local receptive fields, weight sharing, pooling, and normalization—fit naturally into that pipeline.

11.2 Why fully connected layers break on images

Dense feedforward networks can be competitive on small tabular problems, but on images they become data-hungry fast for two reasons: flattening throws away the grid structure, and dense connectivity explodes parameter count. The issue is not that the model class is "wrong"; it is that the sample complexity and compute burden are badly mismatched to the structure of vision data.

An image is naturally a 2D array (or a 3D tensor with channels). For example, a single-channel (grayscale) image of size 256×256 pixels can be represented and flattened as

$$\begin{aligned} X &\in \mathbb{R}^{256\times 256}, \\ \mathbf{x} &= \mathrm{vec}(X) \in \mathbb{R}^{65{,}536}, \end{aligned} \tag{11.1}$$

where $65,536 = 256 \times 256$ is the total number of pixels.

That step is not "wrong," but it is expensive. Flattening has two major drawbacks:

- **Loss of spatial structure:** The 2D spatial relationships between pixels are ignored, which is critical for tasks like image recognition.
- **High dimensionality:** The input vector becomes very large, increasing the number of parameters and computational cost, and requiring more data to train effectively.

The punchline is not that dense networks "cannot" learn images; it is that the price is too high. Without any architectural prior, you pay in parameters, data, and compute. Convolutions and pooling supply that prior: they make the model spend capacity on local patterns and reuse those patterns across the grid.

The number of weights between the input and hidden layer is:

$$65,536 \times 100 = 6,553,600,$$

and between the hidden and output layer (assuming 4 output classes) is:

$$100 \times 4 = 400.$$

Thus, the first layer alone requires learning over 6 million parameters before we even consider deeper architectures. Coupled with the additional 400 output weights (plus biases), the optimization problem quickly becomes data-hungry and computationally expensive.

Data requirements (a sanity check). When a model has millions of free parameters and no strong architectural prior, you are asking the data to do most of the work: it must fit the mapping and also implicitly teach the model what kinds of patterns are worth representing. A crude sanity check sometimes used in practice is to compare the number of labeled examples to the number of parameters:

$$N_{\text{samples}} \geq 10 \times N_{\text{parameters}}.$$

Read this as a warning sign, not a requirement. Inductive bias (convolutions), regularization, and augmentation reduce the effective degrees of freedom, and

pretraining changes the problem entirely because you are no longer learning all features from scratch.

For the dense first layer in this example calculation, the back-of-the-envelope lands in the tens of millions of labeled images:

$$N_{\text{samples}} \gtrsim 10 \times 6{,}553{,}600 \approx 65{,}536{,}000,$$

The point is not the exact number; it is the scale mismatch between dense connectivity and the datasets most people can realistically label. Without additional structure, many very different parameter settings can fit the same training set; the architecture and the training recipe decide whether any of those settings generalize in a way you can trust.

Computational and storage constraints. Even if you could collect that many labeled examples, training cost becomes part of the model design. Each epoch is a full sweep over the dataset; each sweep costs time, memory bandwidth, and usually specialized hardware. Large models also cost money to store and to run: inference latency, memory footprint, and energy use become constraints that matter as much as accuracy. This is another reason architectural bias matters: it reduces degrees of freedom and makes it possible to spend computation on structure that transfers across images.

Overfitting risk (what to watch for). With high capacity, "training accuracy goes up" is not evidence of learning the right thing. Watch the gap between training and validation curves, check calibration (are probabilities meaningful?), and run slice audits (lighting, viewpoint, background, demographic proxies) to detect memorization or shortcut learning early.

11.3 Sparse connectivity and parameter sharing

CNNs replace dense connections with *local receptive fields*. A unit's receptive field is simply the portion of the input that can influence that unit's value through the connections that reach it. The term is borrowed from sensory systems: a neuron "receives" evidence from a region, not from everywhere at once.

Each output unit therefore connects to a small neighborhood of pixels, not the entire image. If a $k \times k$ filter scans a $H \times W$ input, the number of learned

weights is k^2 (per channel), not HW. The same filter is reused at every spatial location, so a single set of parameters detects a pattern anywhere in the image. This parameter sharing is the key to scalability: it cuts the parameter count and preserves translation equivariance.

Equivariance means: if you shift the input, the feature map shifts in the same way. By contrast, invariance means the final decision changes little (or not at all) under small shifts; pooling and downsampling are common ways to trade some equivariance for more invariance at the classifier.

Historically, dense MLPs on image benchmarks struggled to compete with classical pipelines because the parameter count and data requirements were prohibitive. CNNs reversed that trend by tying weights and focusing on local patterns while keeping the same gradient-based training loop, so capacity grows with depth and channel count rather than with the raw pixel grid.

Shape reminder

Throughout this chapter we use the row-major (deep-learning) convention for batches: inputs $X \in \mathbb{R}^{B \times d_{\text{in}}}$, weights $W \in \mathbb{R}^{d_{\text{in}} \times d_{\text{out}}}$, and biases $b \in \mathbb{R}^{d_{\text{out}}}$, with forward map $Z = XW + \mathbf{1}b^\top$. When we write single-example equations, you can read them as the same convention with $B = 1$. For convolution/cross-correlation we follow the standard deep-learning tensor convention (channels and spatial axes); the same shape logic applies once the tensors are flattened into matrix form.

Notation handoff. Here $\sigma(\cdot)$ is used only as an activation nonlinearity, while earlier statistical chapters use σ^2 for variance. Keep Appendix A nearby when moving between probability and deep-learning sections.

11.4 Convolution and pooling mechanics

For an input feature map X and filter K, the (cross-correlation) output at spatial location (i, j) is

$$(X * K)_{ij} = \sum_{u=0}^{k-1} \sum_{v=0}^{k-1} K_{uv}\, X_{i+u,\, j+v}. \tag{11.2}$$

Author's note: "convolution" vs. cross-correlation

In most deep-learning libraries, the operation called "convolution" is technically *cross-correlation*: the kernel is not flipped before sliding. The name stuck because the two operations differ only by a reversal of the kernel, and the kernel is learned anyway. What matters in practice is that the "filter" is a small learned weight matrix that is applied repeatedly across space (weight sharing). Learn the shapes, the stride/padding accounting, and the inductive bias; the terminology is imperfect but the idea is powerful.

With stride s and padding p, the output size along one axis is

$$\left\lfloor \frac{n + 2p - k}{s} \right\rfloor + 1, \tag{11.3}$$

so padding controls resolution while stride controls downsampling. Pooling then aggregates local neighborhoods (typically max or average) to build invariance and further reduce spatial size.

Convolution itself is a linear map; the nonlinearity enters when we apply an activation after each convolutional block. Stacking these linear–nonlinear stages yields hierarchical feature detectors (edges → textures → parts → objects) without abandoning the backprop training machinery.

Without padding (often called *valid* convolution), boundary pixels participate in fewer receptive fields and the spatial grid shrinks each layer. *Same* padding chooses $p = (k-1)/2$ for odd k to preserve spatial size when $s = 1$ and give edge pixels comparable influence. Larger strides reduce resolution and compute but can skip fine detail, so the padding/stride combination is a deliberate trade-off rather than a fixed rule.

11.4.1 Worked stride and padding example

Suppose an input is 6×6 and the filter is 3×3.

- **Stride $s = 1$, valid padding ($p = 0$).** Output size is $(6-3)/1 + 1 = 4$, so you get a 4×4 feature map.
- **Stride $s = 2$, valid padding.** Output size is $\lfloor (6-3)/2 \rfloor + 1 = 2$, so you get a 2×2 feature map.

- **Stride $s = 1$, same padding.** With $k = 3$, choose $p = 1$ so the output stays 6×6.

The floor in the formula is important: if $(n + 2p - k)$ is not divisible by s, the last partial window is dropped.

Worked example: 1D cross-correlation values (stride + padding)

Let the input be $\mathbf{x} = [1, 2, 0, 3, 1]$, the (unflipped) filter be $\mathbf{w} = [1, 0, -1]$, padding $p = 1$, and stride $s = 2$. The padded input is $[0, 1, 2, 0, 3, 1, 0]$ and the output length is

$$L = \left\lfloor \frac{n + 2p - k}{s} \right\rfloor + 1 = \left\lfloor \frac{5 + 2 - 3}{2} \right\rfloor + 1 = 3.$$

Sliding windows of length $k = 3$ with stride 2 yield

$$\begin{aligned} y_0 &= \langle [0, 1, 2], [1, 0, -1] \rangle = -2, \\ y_1 &= \langle [2, 0, 3], [1, 0, -1] \rangle = -1, \\ y_2 &= \langle [3, 1, 0], [1, 0, -1] \rangle = 3, \end{aligned}$$

so $\mathbf{y} = [-2, -1, 3]$.

Convolutional hyperparameters (what you choose up front).

- **Filter size** $(k \times k)$ and **number of filters** (C_{out}).
- **Stride** s and **padding** p (valid vs. same).
- **Activation** function after each convolution.
- **Pooling** type (max/average), window size, and stride (if pooling is used).

11.5 Pooling as nonparametric downsampling

Pooling reduces spatial size without learning new parameters: a max-pooling window keeps the strongest activation; average pooling keeps the mean. This is a nonparametric operation, so it can feel like "cheating" compared to learned filters, yet it often improves robustness by discarding small shifts and noise. Max pooling is the most common because it preserves the strongest feature response, but average and even median pooling appear in specialized settings. In modern

CNNs, aggressive pooling is used sparingly; strided convolutions are a common alternative when you want learned downsampling instead of a fixed aggregation.

Author's note: pooling is a design choice

Pooling is not "more correct" than strided convolutions; it is a trade-off. If you want downsampling with learned weights, use stride. If you want a fixed local summary (often more stable early in training), max pooling is a reasonable default. Avoid padding in pooling unless you need spatial alignment with a parallel branch.

11.6 Channels and feature maps

Real inputs are multi-channel. A red-green-blue (RGB) image has three channels, so a $k \times k$ filter is really $k \times k \times C_{\text{in}}$ and produces one output map. A convolutional layer applies C_{out} such filters, yielding a volume of feature maps with shape $H \times W \times C_{\text{out}}$. This is why "same" padding refers to spatial dimensions only: channel depth is set by the number of filters, not by padding.

Dimensionality accounting example

Start with an input tensor of size $50 \times 50 \times 30$. Apply 10 filters of size 3×3 with stride 1 and valid padding. The spatial size becomes $50 - 3 + 1 = 48$, so the output is $48 \times 48 \times 10$. Apply 2×2 max pooling with stride 2: the spatial size becomes $\lfloor (48 - 2)/2 \rfloor + 1 = 24$, so the pooled output is $24 \times 24 \times 10$. Flattening that volume yields $24 \cdot 24 \cdot 10 = 5760$ features for a dense classifier.

From feature maps to classifiers. Stacks of convolutional and pooling layers produce a hierarchy of feature maps. A common design is to flatten the final maps into a vector and pass them to a small dense classifier (often a softmax layer) that predicts the class label. Backpropagation updates both the dense weights and the shared convolutional filters, so the feature extractor and classifier are learned jointly.

Losses, metrics, and the "CNN as features" viewpoint

A useful way to read a CNN is as a learned feature map $\phi_\theta(\cdot)$ followed by a simple readout. The last layer is often just a linear classifier on top of the learned features, trained end-to-end by backpropagation.

Loss choice. For classification, the default is softmax cross-entropy because it turns scores into probabilities that can be audited (and sometimes calibrated) when the model is well-regularized. For regression-style outputs (counts, keypoints, continuous targets), squared error and its robust variants are common.

Margins and hinge. If you replace cross-entropy with a hinge-style objective, you are pushing for a margin in the learned feature space. This is the same geometric idea as the soft-margin story in Chapter 3; the difference is that kernels fix the feature map ahead of time, while CNNs learn ϕ_θ from data.

Metric choice. Accuracy is not enough when scores are used as probabilities. Track calibration (e.g., reliability curves and expected calibration error (ECE)) and slice performance, as emphasized in Chapter 4 and revisited in the audit boxes later in this chapter.

Multi-branch convolution blocks (Inception idea). One practical variant is to run multiple filter sizes in parallel (for example 1×1, 3×3, and 5×5) and concatenate the resulting feature maps along the channel axis. This allows the network to capture multi-scale patterns without committing to a single kernel size. When branches must line up spatially, *same* padding is used to keep all outputs the same height and width; pooling branches often pad for this alignment even though pooling by itself is usually unpadded.

At scale, this architectural prior became decisive once large labeled datasets and graphics processing units (GPUs) and training pipelines matured: the sample-efficiency and compute-efficiency gap versus classical kernel pipelines narrowed and then flipped in favor of deep convolutional stacks.

Stack depth versus receptive field. Stacking identical 3×3 filters still grows the effective receptive field: each stage wraps a thicker box around the original pixels, which explains why deep-but-narrow CNNs can capture wide

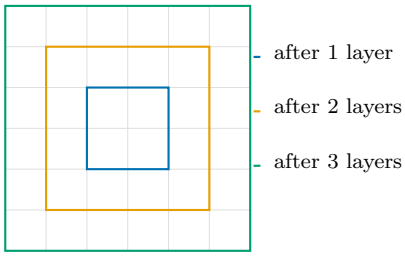


Figure 48: Receptive field growth across depth. Even with small 3×3 kernels, stacking layers expands the spatial context seen by deeper units, which is why depth can replace very large kernels.

spatial context without enormous kernels even when individual kernels remain small. Figure 48 sketches the nested growth across depth.

Quick check: receptive field growth for stacked 3×3

If each convolution uses stride 1 and a 3×3 kernel, then every new layer expands the receptive field by 2 pixels in each spatial direction. So the effective receptive field sizes go 3×3, then 5×5, then 7×7, and so on. Depth is therefore a controlled way to "see farther" without jumping to very large kernels.

With the architectural motivation in place, we now focus on how these models are trained and why deep optimization remains delicate.

11.7 Training Neural Networks: Gradient-Based Optimization

We already derived the gradient-descent training loop in Chapter 6: define a loss $\mathcal{L}$, compute gradients, then update parameters. CNNs keep the same loop. The key difference is weight sharing: a convolutional filter is applied at many spatial locations, so a single filter coefficient collects learning signal from every location where it was used.

For a weight w, the update rule is

$$w \leftarrow w - \eta \frac{\partial \mathcal{L}}{\partial w}. \tag{11.4}$$

where η is the learning rate.

Backpropagation and gradient computation The gradient $\partial\mathcal{L}/\partial w$ is computed by backpropagation: cache the forward intermediates, then apply the chain rule backward through the computation graph. In practice frameworks automate this, but the manual story is what lets you sanity-check shapes, signs, and scaling when training behaves strangely.

Preview: why deep optimization needs extra care As depth grows, gradient flow becomes fragile (vanishing/exploding gradients). The next sections make that failure mode explicit and summarize the mitigation toolkit used in modern CNN stacks.

11.8 Vanishing and Exploding Gradients in Deep Networks

Backpropagation multiplies Jacobians through many layers, so gradients can either vanish (approach zero) or explode (grow exponentially large), leading to significant training difficulties.

Mathematical intuition Consider a deep network with L layers, and focus on the error signals that get propagated backward. Let $\boldsymbol{\delta}^{(\ell)} = \partial\mathcal{L}/\partial\mathbf{z}^{(\ell)}$ denote the derivative of the loss with respect to the pre-activation at layer ℓ. A single backprop step has the form

$$\boldsymbol{\delta}^{(\ell-1)} \;=\; \left(\mathbf{W}^{(\ell)}\right)^{\top} \boldsymbol{\delta}^{(\ell)} \;\odot\; f'\left(\mathbf{z}^{(\ell-1)}\right), \tag{11.5}$$

so the early-layer signals are built from repeated multiplication by weight matrices and activation derivatives.

If the typical gain of these factors is greater than one, the signal grows as it moves backward and gradients can explode; if it is less than one, the signal shrinks and gradients can vanish. This is why initialization scale, activation choice, and normalization matter so much in deep stacks.

Consequences

- **Exploding gradients:** gradients become extremely large, causing numerical instability and making parameters diverge.
- **Vanishing gradients:** gradients become extremely small in early layers,

so those weights learn slowly and the model struggles to build useful low-level features.

Example: Activation function derivatives Consider the sigmoid activation function $\sigma(x) = \frac{1}{1+e^{-x}}$. Its derivative is

$$\sigma'(x) = \sigma(x)(1 - \sigma(x)). \tag{11.6}$$

Note that $\sigma'(x)$ approaches zero when $\sigma(x)$ is near 0 or 1, i.e., when the neuron output saturates. This saturation leads to very small gradients, exacerbating the vanishing gradient problem. The derivative is maximized at $\sigma(x) = 0.5$, where $\sigma'(x) = 0.25$, so repeatedly multiplying sigmoid derivatives can shrink gradients roughly like 0.25^L across L layers.

Training failure signatures (symptom → likely fixes)

- **Loss becomes NaN (not a number) / diverges quickly:** lower learning rate; check data normalization; add gradient clipping; verify BN statistics (or switch norm).
- **Training loss decreases but validation stalls:** audit split hygiene (Chapter 3); add regularization (weight decay, dropout); strengthen augmentation; check shortcut features via slice tests.
- **Both training and validation improve slowly:** revisit initialization and normalization; try momentum/AdamW; confirm the objective matches the metric you care about.
- **High accuracy but poor probability decisions:** audit calibration (reliability curves, ECE) and re-threshold on held-out data rather than trusting raw scores (Chapter 4).

11.9 Strategies to Mitigate Vanishing and Exploding Gradients

Weight initialization Initialization is not just a detail: it sets the scale of activations and gradients at the start of training. A practical rule is to pick the weight variance to be inversely proportional to fan-in (and sometimes fan-out), so signals neither blow up nor fade as they pass through many layers. Xavier and

He initializations are two standard choices that implement this idea for common nonlinearities.

Initializing weights: Xavier and He

Both schemes derive from a variance-preservation argument: choose the initial weight scale so that, at the start of training, the variance of activations does not explode or vanish as the signal propagates through many layers.

Xavier / Glorot initialization targets symmetric activations near zero (tanh, sigmoid in their linear regime):

$$\mathrm{Var}(w) = \frac{2}{n_{\text{in}} + n_{\text{out}}}, \quad w \sim \mathcal{U}\left[-\sqrt{\tfrac{6}{n_{\text{in}}+n_{\text{out}}}},\ \sqrt{\tfrac{6}{n_{\text{in}}+n_{\text{out}}}}\right].$$

where n_{in} and n_{out} are the number of input and output units of the layer.

He / Kaiming initialization corrects for the fact that ReLU zeros out half its inputs, so the effective fan-in is halved:

$$\mathrm{Var}(w) = \frac{2}{n_{\text{in}}}, \quad w \sim \mathcal{N}\left(0,\ \sqrt{\tfrac{2}{n_{\text{in}}}}\right).$$

Rule of thumb: use Xavier for tanh/sigmoid activations; use He for ReLU and its variants (Leaky ReLU, ELU, GELU). Without proper initialization, deep networks often train very slowly or not at all — this single choice can mean the difference between a model that converges in hundreds of epochs and one that never moves. See Chapter 7 for the gradient-flow view.

Choice of activation function Selecting activation functions whose derivatives do not vanish easily is crucial. For example:

- **ReLU (Rectified Linear Unit):** Defined as

$$\mathrm{ReLU}(x) = \max(0, x), \tag{11.7}$$

  its derivative is 1 for positive inputs and 0 otherwise. This avoids saturation in the positive regime and helps maintain gradient flow.

- **Leaky ReLU and variants:** These allow a small, non-zero gradient when the input is negative, further mitigating dead neurons and keeping

derivatives away from exact zero.

Batch normalization Batch normalization stabilizes the scale of activations during training by normalizing them (per channel) and then learning a scale and shift. The practical payoff is that fewer layers sit in saturated regimes, so gradients behave more predictably early in training. Remember that BN behaves differently at train time (mini-batch statistics) versus evaluation time (running statistics).

Gradient clipping For exploding gradients, gradient clipping limits the maximum gradient norm during backpropagation, preventing a single bad batch from producing a destructive update. It is not a substitute for good initialization and normalization, but it is a practical safety valve when training is otherwise stable.

Residual (skip) connections Even with careful initialization and normalization, plain deep networks hit a *degradation ceiling*: at some depth, accuracy on the training set itself begins to fall, indicating an optimization (not generalization) problem. He et al. (2016) solved this with a simple idea: instead of learning the desired mapping $\mathcal{H}(\mathbf{x})$ directly, let the layers learn the *residual* $\mathcal{F}(\mathbf{x}) = \mathcal{H}(\mathbf{x}) - \mathbf{x}$ and add the identity shortcut explicitly:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}) + \mathbf{x}.$$

The gradient of the loss with respect to early-layer inputs now has a direct path through the "$+\mathbf{x}$" branch, bypassing the nonlinearities:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{x}} = \frac{\partial \mathcal{L}}{\partial \mathbf{y}} \left(\frac{\partial \mathcal{F}}{\partial \mathbf{x}} + \mathbf{I} \right).$$

The identity term $\mathbf{I}$ ensures that the gradient never completely vanishes, even when the learned residual is near zero. This single architectural change allowed training of networks with 50, 100, and even 1 000+ layers and is now a universal building block of modern deep architectures.

Data augmentation: expanding effective dataset size

A well-designed augmentation pipeline can double or triple effective training-set diversity without collecting new data. Key families:

- **Geometric:** random horizontal/vertical flips, rotations (±15–30°), crops and resizes, affine shears.
- **Colour/photometric:** brightness, contrast, saturation, hue jitter; grayscale conversion; Gaussian blur.
- **Mixing augmentations: MixUp** linearly interpolates two training images and their labels ($\tilde{x} = \lambda x_i + (1-\lambda)x_j$); **CutMix** pastes a rectangular region from one image into another and mixes labels proportionally to area.
- **Test-time augmentation (TTA):** average predictions over multiple augmented views at inference to reduce prediction variance.

Key rules: (i) augmentations must be label-preserving (a flipped cat is still a cat; a horizontally flipped digit "6" may not be a "6"); (ii) training and evaluation pipelines must be kept separate; (iii) report the exact augmentation recipe alongside results — it is a model hyperparameter.

Taken together, these tools stabilize optimization; the figures below highlight dropout, normalization, and optimizer behavior in practice.

Risk & audit

- **Train/val mismatch:** augmentation, preprocessing, and normalization must match at evaluation time (except stochastic augmentation).
- **Resolution trade-offs:** downsampling can erase small objects; audit performance by scale and by class, not only overall accuracy.
- **BatchNorm regimes:** very small batch sizes can destabilize BN statistics; consider alternatives (layer/group norm) and audit sensitivity.
- **Shortcut learning:** CNNs can latch onto background cues; use perturbations, counterfactual crops, and slice audits.
- **Robustness:** report performance under shifts (lighting, camera, compression) and track calibration when scores are used as probabilities.
- **Run-to-run variance:** report seeds, spread, and stopping policy; use Appendix B as the default reporting format.

As a concrete diagnostic, Figure 49 shows how dropout can flatten the validation curve relative to the no-dropout baseline.

Batch normalization. BN accelerates convergence by keeping per-channel activations on a stable scale and then letting the network relearn the best scale/shift. Figure 50 contrasts the pre- and post-normalization activation distributions; the practical point is that gradients stay in a usable range more often. In later chapters, you will see the same stability instinct show up again in LayerNorm placement choices in very deep stacks.

Adaptive optimizers. While plain SGD remains a workhorse, Adam and related methods adapt learning rates per-parameter (Kingma and Ba, 2015). Figure 51 sketches typical loss trajectories; Adam often converges faster initially, whereas SGD+momentum can be steadier later (and sometimes attains a slightly lower asymptote after fine-tuning).

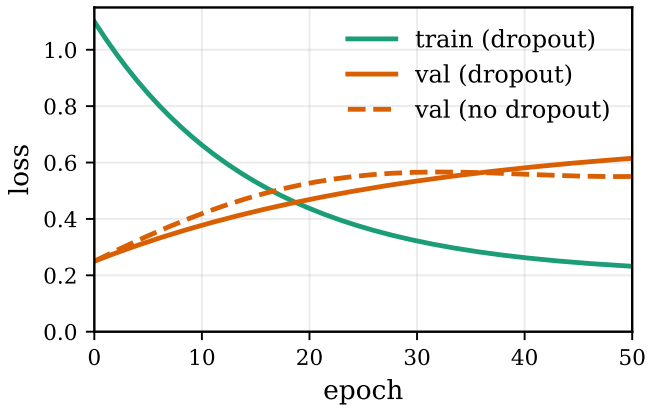


Figure 49: Illustrative dropout effect on training/validation curves. Compared to a no-dropout baseline, validation curves flatten and generalization improves; the point is the shape of the curves, not the exact values. *Source: author-generated.*

Practical optimizer notes

Mixed precision. Modern CNN stacks often run activations/gradients in FP16 or BF16 while keeping master weights in FP32. Frameworks such as PyTorch AMP/TF mixed precision insert dynamic loss scaling so gradients do not underflow; the reward is higher throughput and lower memory pressure on recent GPUs/TPUs.

AdamW vs. Adam. Decoupled weight decay (AdamW) subtracts $\eta\lambda W$ outside the adaptive-moment step, avoiding the "L2-as-gradient-scaling" behavior of classical Adam and producing more predictable regularization (Loshchilov and Hutter, 2019). In code:

```
m = beta1 * m + (1-beta1) * grad
v = beta2 * v + (1-beta2) * grad**2
W -= eta * (m_hat / (sqrt(v_hat) + eps) + lambda * W)
```

Use AdamW (or SGD+momentum) when you want clean weight-decay semantics; reserve plain Adam for rapid prototyping or when adaptive steps dominate regularization.

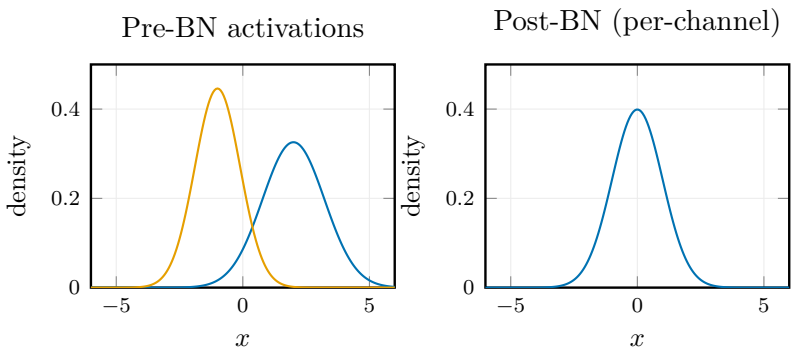


Figure 50: Batch normalization first maps each channel to a standard normal shape and then applies a learned scale and shift (γ, β); the normalized distribution shown here is the pre-affine step that stabilizes gradients.

MLP/CNN block pseudocode (schematic)

```
function ForwardBackward(params, x, y):
  # forward
  caches = []
  a = x
  for (W, b, f) in params.layers:
    z = W @ a + b
    caches.append((a, z))
    a = f(z)
  loss, delta = params.loss(a, y)

  # backward
  grads = []
  for (W, b, f), (a_prev, z_prev) in
      reversed(list(zip(params.layers, caches))):
    grads.append((delta @ a_prev.T, delta))
    delta = (W.T @ delta) * f'(z_prev)

  params.update(grads[::-1])
  return loss
```

This omits batching, convolution strides, and optimizer detail but highlights the cache-then-backprop pattern reused throughout the deep-learning chapters.

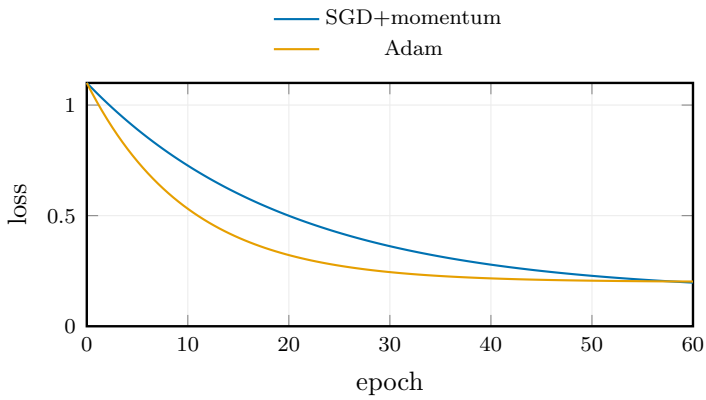


Figure 51: Schematic loss curves illustrating typical behavior: Adam often drops loss quickly early, while SGD+momentum can be steadier later.

Derivation closure: implement, cache, and fail fast

- **Implement blocks with shapes:** treat each block (conv/BN/activation/pool) as a typed transform and keep input-output shapes explicit.
- **Cache what the backward pass needs:** pre-activations, normalization statistics, and masks should be stored, not re-derived ad hoc.
- **Fail fast:** validate shape arithmetic, then run a one-batch overfit test before full training to catch broken gradients early.
- **Report like an engineer:** publish seed, augmentation recipe, and calibration metrics with every benchmark, following Appendix B.

Key takeaways

If you remember only a few things.

- Convolutions introduce sparse connectivity and parameter sharing, dramatically reducing parameters vs. fully connected layers.
- Padding and stride control spatial resolution; pooling aggregates features to build invariances.
- Batch normalization, dropout, and optimizer choice strongly influence training stability and generalization.
- Stacking small kernels expands the effective receptive field across depth.

Minimum viable mastery.

- Given (n, f, s, p), compute output shapes and parameter counts for a convolutional block.
- Explain equivariance vs. invariance and how stride/pooling/padding affect each.
- Describe what batch normalization and dropout do during training and how they change behavior at evaluation time.

Common pitfalls.

- Letting shape accounting drift (mismatched padding/stride) until the first dense layer fails.
- Forgetting train/eval mode for batchnorm/dropout, producing misleading validation curves.
- Treating pooling as mandatory or always beneficial; in some tasks, stride-2 convolutions or anti-aliasing are better choices.

Ethical considerations. CNNs are often deployed where mistakes have real cost (medical triage, access control, surveillance). Treat data provenance and slice audits as part of correctness: a high top-1 accuracy on a benchmark does not guarantee fair or safe behavior in a new population or capture setup.

Exercises

Exercise 1 (Easy). Do one full shape-and-value sanity check: hand-compute a 1D cross-correlation for a short input and kernel, then verify the output values and shapes against a short script for several (n, f, s, p) settings.
Deliverable: the hand calculation + the matching script output + a one-line rule that predicts the output length.

Exercise 2 (Medium). Test equivariance vs. invariance empirically: translate an image (or a simple synthetic pattern), record an intermediate feature map before pooling/striding, and show how it shifts; then show how pooling/stride changes the behavior.
Deliverable: two feature-map visualizations + one paragraph explaining what information pooling throws away on purpose.

Exercise 3 (Medium). Compare a plain deep CNN to the same network with identity skips (residual-style). Keep the dataset tiny so you can overfit fast, and treat the experiment as an optimization diagnosis rather than a leaderboard.
Deliverable: training curves for both models + one short explanation linking the difference to gradient flow.

If you are skipping ahead. The optimization and regularization knobs here carry almost unchanged into Chapters 12 to 14; what changes is how context is represented and propagated, not the core loss-minimization discipline.

Where we head next. Chapter 12 keeps the same training and audit discipline but shifts the modeling axis from spatial locality to temporal state and sequence dependence.

12 Recurrent Neural Networks: Dynamics, Memory, and Gating

Chapter 11 showed how architectural bias (convolution/pooling) improves sample efficiency while preserving the same optimization loop. This chapter keeps

that loop but changes the axis from space to time: sequence models need memory so present predictions can depend on prior inputs. A useful continuity is *parameter sharing*: CNNs reuse one filter across spatial locations, while recurrent neural networks (RNNs) reuse one transition rule across time steps. Figure 1 flags this as the sequential branch of the neural strand.

Learning Outcomes

- Explain why recurrent structures are needed for sequence modeling and contrast them with feedforward nets.
- Derive the forward dynamics and backpropagation-through-time (BPTT) updates for a simple (ungated) RNN cell.
- Recognize practical stabilization techniques (gradient clipping, gating, normalization) that motivate later LSTM/Transformer chapters.

Design motif

Add recurrence, then train it by unrolling time and reusing the backprop machinery from Chapter 7.

The statistical learning chapters (Chapters 3 to 4) treated learning as an engineering loop: choose a model class, define a loss, optimize it, and then audit generalization. The feedforward neural chapters then expanded the model class from linear predictors to multilayer networks (MLPs, RBF networks, and CNNs), but the basic data interface stayed the same: each example is treated as a standalone input–output pair.

Sequence problems break that interface. When order matters, the model needs a state that can carry information forward, and the training loop must account for the fact that the same parameters are reused again and again across time.

12.1 Motivation for Recurrent Neural Networks

Most of the models we have built so far make a quiet assumption about the data: examples are independent and identically distributed (i.i.d.). If you shuffle the dataset, nothing essential changes. That assumption is fine for many supervised tasks on static inputs (tabular rows, images), but it breaks for sequences.

In time series, yesterday changes what you expect at the next step. In

language, the same words in a different order can mean something else entirely, and the next word depends on the whole prefix. You can see that in predictive text: "I want to buy …" pulls you toward different completions than "Write a book about Teddy …" even before you specify what you are buying or writing about. In control, current action depends on prior state; in speech, a phoneme depends on the surrounding phonemes.

One brute-force workaround is to build a fixed history window and feed it to a standard MLP: at time t, concatenate $\mathbf{x}_t, \mathbf{x}_{t-1}, \ldots, \mathbf{x}_{t-k}$ and ask the network to predict y_t. That can work for short memory, but it scales poorly as k grows, and it does not reuse parameters in a way that reflects the repeated structure of "the same kind of step happens again and again".

Recurrent neural networks make the smallest structural change that addresses the bottleneck: add a *state* that is updated each step, and reuse the same update rule across time. The rest of this chapter is about making that idea precise, training it reliably, and recognizing the failure modes (especially gradient flow) that motivate gating and attention.

12.2 Key Idea: State and Memory in RNNs

Recurrent neural networks address this limitation by introducing a *state vector* $\mathbf{h}_t$ that summarizes information from all previous inputs up to time t. The state is updated recursively as new inputs arrive, allowing the network to maintain a form of memory.

Formally, at each time step t, the RNN receives an input vector $\mathbf{x}_t$ and updates its hidden state $\mathbf{h}_t$ according to a function f parameterized by weights θ:

$$\mathbf{h}_t = f(\mathbf{h}_{t-1}, \mathbf{x}_t; \theta) \tag{12.1}$$

The output $\mathbf{y}_t$ at time t is then computed as a function of the current state:

$$\mathbf{y}_t = g(\mathbf{h}_t; \theta') \tag{12.2}$$

Here, f and g are typically nonlinear functions implemented by neural network layers, and θ, θ' are learned parameters.

Terminology (steps and representations). Throughout this chapter, t indexes *steps* in a sequence. In a time series, $\mathbf{x}_t$ can be a measured feature vector. In other domains the raw observation at step t can be discrete (a symbol from a finite set) or structured (a set of features), but it is always converted into a numeric input vector $\mathbf{x}_t$ before it enters the recurrence. We keep the derivations in $\mathbf{x}_t$ so the same math applies across domains; Chapter 13 makes the "discrete symbols $\rightarrow$ vectors" story concrete for text.

Interpretation: The hidden state $\mathbf{h}_t$ acts as a *summary* or *encoding* of the entire input history $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_t\}$. This allows the network to make predictions that depend on order and history, not just the current input.

Parameter sharing across time. The same weights are reused at each time step. In the simplest formulation there are three learned matrices: W_{xh} (input to state), W_{hh} (state to state), and W_{hy} (state to output). When you unroll the recurrence, these matrices appear at every step, so the model learns a single transition rule rather than a separate set of parameters for each position in the sequence.

Unrolling makes the training graph T steps deep. Gradients from each time step accumulate onto the shared weights, and the repeated Jacobian products are precisely why long sequences revive the vanishing/exploding gradient issues from deep feedforward networks. Training this unrolled graph with standard backpropagation is known as *backpropagation through time* (BPTT).

Recurrent neural networks (RNNs) were among the first practical sequence models (Elman, 1990; Bengio et al., 1994). CNNs from Chapter 11 trade recurrence for parallel, spatially shared filters. Chapter 13 then supplies the representation tools and evaluation lens commonly paired with language models, and Chapter 14 revisits sequence modeling without recurrence.

12.3 Comparison with Feedforward Networks

To contrast, a feedforward network computes the output at time t as:

$$\mathbf{y}_t = \psi(\mathbf{x}_t; \theta) \tag{12.3}$$

where ψ is a nonlinear function without any dependence on past inputs. This

limits the ability of feedforward networks to model temporal dependencies unless the input vector $\mathbf{x}_t$ explicitly contains past information.

What recurrence buys you. A feedforward network reacts to the current input $\mathbf{x}_t$ unless you deliberately pack history into the input features. An RNN adds a state $\mathbf{h}_t$ that carries information forward, so predictions can depend on order and on longer histories.

Shape reminder

We keep the row-major (deep-learning) convention: $\mathbf{x}_t \in \mathbb{R}^{d_x}$, $\mathbf{h}_t \in \mathbb{R}^{d_h}$, pre-activation $\mathbf{a}_t = \mathbf{x}_t W_{xh} + \mathbf{h}_{t-1} W_{hh} + \mathbf{b}_h$, output $\mathbf{y}_t = \mathbf{h}_t W_{hy} + \mathbf{b}_y$. Column-vector formulations simply transpose the order of factors; all stability conclusions (spectral norms of Jacobian factors) carry over.

Roadmap. We start with the state-update picture, then unroll it to derive BPTT. From there we look at what breaks on long sequences (vanishing/exploding gradients) and the practical fixes (truncation, clipping, normalization, and gated cells), and we end with a short bridge to attention.

12.4 Recap: Feedforward Building Blocks

Simple RNN at a glance. A simple (ungated) RNN is trained the same way as our earlier networks: pick a loss, compute gradients, and update parameters. For sequence prediction the default loss is cross-entropy accumulated over time, with state updates of the form $\mathbf{h}_t = f(\mathbf{x}_t W_{xh} + \mathbf{h}_{t-1} W_{hh} + b_h)$ and a per-step predictive distribution $p_\theta(y_t \mid \mathbf{h}_t)$. The main design knobs are the state size and how far you backpropagate through time (full BPTT versus a truncation window). In practice, most failures are either gradient blow-up/decay on long sequences or a train/inference mismatch when teacher forcing is used without a decoding strategy that matches it.

RNNs reuse the same ingredients as multilayer perceptrons (activations, nonlinear decision boundaries, loss functions, and training heuristics) but wrap them around a temporal axis. Figure 25 from Chapter 7 highlights the canonical MLP dataflow along with common activation choices and derivatives that govern gradient flow.

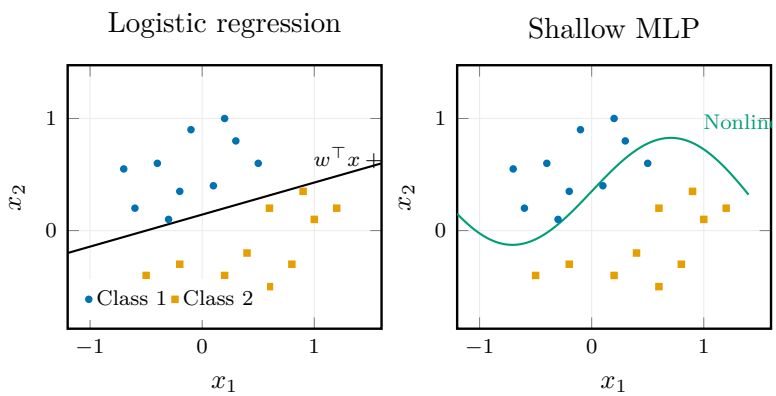


Figure 52: Decision boundaries for logistic regression (left) versus a shallow MLP (right). Linear models carve a single hyperplane, whereas hidden units can warp the boundary to follow non-convex manifolds such as the moons dataset.

Two-dimensional toy datasets remain useful for reasoning about inductive biases. Figure 52 contrasts logistic regression and a shallow MLP on the moons dataset, illustrating how additional hidden units carve nonlinear boundaries that RNN readouts later rely on when decoding the final state.

Figure 54 is the canonical unrolling view used to interpret shared-parameter BPTT updates.

Finally, Figure 53 summarizes two diagnostics: BCE geometry and the effect of learning-rate schedules/early stopping. Here BCE (binary cross-entropy) for a binary target $y \in \{0, 1\}$ and logit z is $\mathcal{L}(z, y) = \log(1 + e^{-z})$ for y=1 and $\log(1 + e^{z})$ for y=0; the *logit* z is the pre-sigmoid score so that $\sigma(z)$ yields the predicted probability. If labels are recoded to $y_\pm \in \{-1, +1\}$, the same loss is written compactly as $\log(1 + e^{-y_\pm z})$; do not mix that margin form with the $y \in \{0, 1\}$ convention used here. The middle panel contrasts a conservative schedule (smooth decay) with a more aggressive one (faster initial drop but risk of oscillation), and the right panel shows early stopping triggered when validation loss ceases to improve while training loss continues decreasing. We will reuse these when tuning sequence models, where overfitting appears as a divergence between per-step training and validation likelihood (per-token in text models).

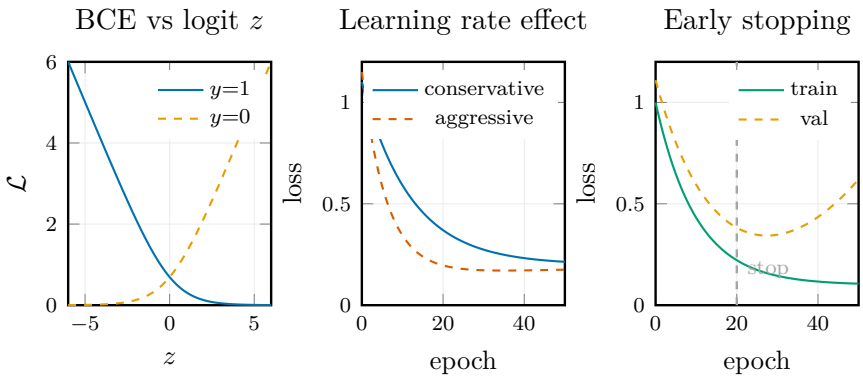


Figure 53: Binary cross-entropy geometry (left), effect of learning-rate schedules on loss (middle), and the typical training/validation divergence that motivates early stopping (right).

Author's note: treat early stopping as the default brake

Unless there is a compelling reason to run to numerical convergence, stop as soon as the validation curve flattens while the training curve keeps dropping. Checkpoint the best weights and resume only if new data or regularization changes warrant it. That simple rule prevents most runaway experiments without elaborate hyperparameter sweeps.

LayerNorm and residual RNN tips

Layer Normalization (Ba et al., 2016) stabilizes recurrent dynamics by normalizing each hidden vector $\mathbf{h}_t$ across features before applying the nonlinearity; unlike BatchNorm it works with batch size 1 and handles variable-length sequences gracefully. Residual RNN stacks (adding the input of a layer back to its output) keep gradients flowing even when depth increases, mirroring the skip-connections that make deep CNNs trainable. Together, LayerNorm + residual links curb exploding/vanishing gradients and are the default when building multi-layer RNN/LSTM stacks.

Historical note: Hopfield networks and stable recurrence

Chapter 10 covered Hopfield networks as an associative memory: a recurrent system with symmetric weights and an energy function that decreases under the update rule, so the dynamics settle into stored patterns (attractors). That is a different use of recurrence than the one in this chapter.

Here we use recurrence to model *ordered* data. The goal is not to converge to a fixed point, but to carry a useful state forward so predictions at time t can depend on earlier inputs. The shared theme is that dynamics matter; the diagnostics differ (energy trace and overlap for Hopfield, versus likelihood/perplexity and gradient health for sequence models).

12.5 Input–Output Configurations and Mathematical Formulation

RNNs appear in a few standard input–output shapes. In a many-to-one task (e.g., sentiment), the model consumes $\mathbf{x}_{1:T}$ and emits one label from the final state. In many-to-many tasks (e.g., tagging), it emits an output at every step. In one-to-many settings (conditional generation), it starts from a conditioning signal (an initial state or an encoder output) and then generates autoregressively. Bidirectional encoders run the recurrence forward and backward and concatenate states when you want both left and right history during labeling, or when you want a richer encoder representation for a decoder.

Consider an input sequence $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T\}$, where each $\mathbf{x}_t \in \mathbb{R}^d$. The RNN computes hidden states $\{\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_T\}$ and outputs $\{\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_T\}$ as follows:

$$\begin{aligned}
\mathbf{h}_0 &= \mathbf{0} && \text{(initial hidden state)}, \\
\mathbf{h}_t &= f(\mathbf{x}_t W_{xh} + \mathbf{h}_{t-1} W_{hh} + b_h), && t = 1, \ldots, T, \\
\mathbf{y}_t &= g(\mathbf{h}_t W_{hy} + b_y), && t = 1, \ldots, T.
\end{aligned} \tag{12.4}$$

Shapes and masks (batch B, time T)

Inputs $X \in \mathbb{R}^{B\times T\times d_x}$; hidden states $H \in \mathbb{R}^{B\times T\times d_h}$; logits $Y \in \mathbb{R}^{B\times T\times d_o}$. Parameters (row-major): $W_{xh} \in \mathbb{R}^{d_x\times d_h}$, $W_{hh} \in \mathbb{R}^{d_h\times d_h}$, $W_{hy} \in \mathbb{R}^{d_h\times d_o}$, biases $b_h \in \mathbb{R}^{d_h}$, $b_y \in \mathbb{R}^{d_o}$.
Padding mask $M \in \{0,1\}^{B\times T}$: loss $L = \sum_{b,t} M_{b,t}\,\mathrm{CE}(\hat{y}_{b,t}, y_{b,t}) / \sum_{b,t} M_{b,t}$, where $\mathrm{CE}(\cdot,\cdot)$ is the cross-entropy loss. Masks preview the padding/causal masks detailed in Chapter 14.

Unrolling and shared weights. The cleanest way to understand recurrence is to *unroll time*: the same cell is applied repeatedly, reusing the same parameters at each step. Training then becomes ordinary backpropagation on the unrolled computation graph, with gradients accumulated across every copy of the shared weights (backpropagation through time, BPTT).

Numeric check: shared weights mean summed gradients

It helps to see the "sum across time" in one tiny example. Consider a scalar RNN with identity activation,

$$h_t = W_{hh}h_{t-1} + W_{xh}x_t, \qquad y_t = h_t,$$

and a two-step squared-error objective $L = \frac{1}{2}\big[(y_1 - t_1)^2 + (y_2 - t_2)^2\big]$. Pick $h_0 = 1$, $W_{hh} = 0.5$, $W_{xh} = 1$, $(x_1, x_2) = (1, 2)$, and $(t_1, t_2) = (1, 2)$. The forward pass gives

$$h_1 = 1.5, \quad h_2 = 2.75, \quad e_1 = y_1 - t_1 = 0.5, \quad e_2 = y_2 - t_2 = 0.75.$$

Because W_{hh} is reused, its gradient receives contributions from *both* steps:

$$\frac{\partial L}{\partial W_{hh}} = e_1\frac{\partial h_1}{\partial W_{hh}} + e_2\frac{\partial h_2}{\partial W_{hh}} = e_1 h_0 + e_2\big(h_1 + W_{hh}h_0\big) = 2.0.$$

The second term is the one students often miss: changing W_{hh} changes h_1, which then changes h_2. A finite-difference check matches the value above.

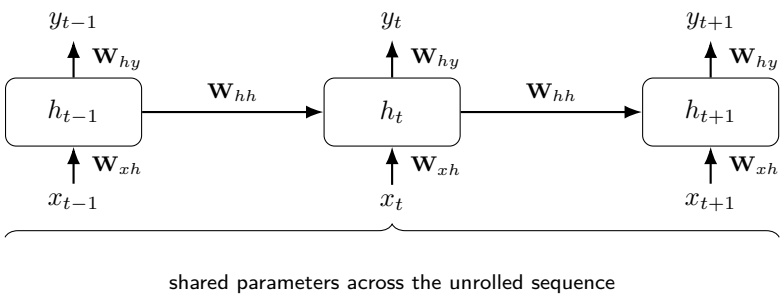


Figure 54: Unrolled RNN over three time steps. The brace marks that the same parameters $\mathbf{W}_{xh}$, $\mathbf{W}_{hh}$, and $\mathbf{W}_{hy}$ are reused at every step; backpropagation through time (BPTT) accumulates gradient contributions through all copies before updating those shared weights.

12.6 Mathematical Formulation of a Simple RNN Cell

Consider a simple RNN cell with the following update equations:

$$\mathbf{h}_t = \sigma_h \left(\mathbf{x}_t \mathbf{W}_{xh} + \mathbf{h}_{t-1} \mathbf{W}_{hh} + \mathbf{b}_h \right), \tag{12.5}$$

$$\mathbf{y}_t = \sigma_y \left(\mathbf{h}_t \mathbf{W}_{hy} + \mathbf{b}_y \right). \tag{12.6}$$

12.7 Recurrent Neural Network (RNN) Architectures and Loss Computation

Recall from previous discussions that the loss function for classification tasks often involves cross-entropy terms of the form:

$$\mathcal{L} = -\sum_i y_i \log \hat{y}_i.$$

where y_i is the true label (often one-hot encoded) and $\hat{y}_i$ is the predicted probability for class i. When $\hat{y} = y$, the loss is zero, indicating perfect prediction.

For RNNs, the total loss over a sequence is typically the sum of losses at each time step:

$$\mathcal{L}_{\text{total}} = \sum_{t=1}^{T} \mathcal{L}_t.$$

where T is the sequence length.

Forward and Backward Passes in RNNs The forward pass produces outputs $\hat{y}_t$ and the loss accumulates over time; gradients are computed via BPTT on the unrolled graph (see the unrolling discussion in Section 12.6 and the flow in Figure 55).

Truncated BPTT in practice

Unroll K steps, accumulate loss, backprop through those K steps, then detach the hidden state to stop graph growth:

```
h = h0
for t in range(T):
    h, yhat = rnn(x[t], h)
    loss += mask[t] * CE(yhat, y[t])
    if (t+1) % K == 0:
        loss.backward()
        clip_grad_norm_(model.parameters(), tau)
        opt.step(); opt.zero_grad()
        h = h.detach()  # carry state, drop graph
```

Choose K to balance memory and credit assignment (common range: 20–100 steps).

Simple (ungated) RNN cell (forward + BPTT)

```
# Forward for a sequence {x_t, y_t}
h_0 = 0
for t = 1..T:
    pre_h = h_{t-1} W_hh + x_t W_xh + b_h
    h_t  = tanh(pre_h)
    yhat_t = h_t W_hy + b_y

# Backward (BPTT with optional truncation K)
delta_pre_next = 0
for t = T..1:
    delta_y = grad_loss(yhat_t, y_t)
    grad_W_hy += h_t^T delta_y
    grad_b_y += delta_y
    delta_h = delta_y W_hy^T + delta_pre_next W_hh^T
    delta_pre = delta_h .* (1 - h_t^2)
    grad_W_hh += h_{t-1}^T delta_pre
    grad_W_xh += x_t^T delta_pre
    grad_b_h += delta_pre
    delta_pre_next = delta_pre
    if t < T-K: break  # truncated BPTT
```

Use gradient clipping (e.g., clip the global norm of parameter gradients) and prefer LayerNorm over BatchNorm inside recurrence when sequences are long to avoid exploding/vanishing gradients.

The element-wise product in the backward step corresponds to the Hadamard factors described in the derivation in Chapter 6; we write it as `.*` to align with NumPy/Matlab notation.

Code–math dictionary. In code blocks we use ASCII identifiers such as `h_t`, `W_hh`, and `b_h`; in equations the same objects appear as $\mathbf{h}_t$, $\mathbf{W}_{hh}$, and $\mathbf{b}_h$ (boldface for vectors/matrices, subscripts for time and role). Detailed algebraic derivations (forward/backward passes and gradient expressions) appear in Equations (12.5) to (12.6).

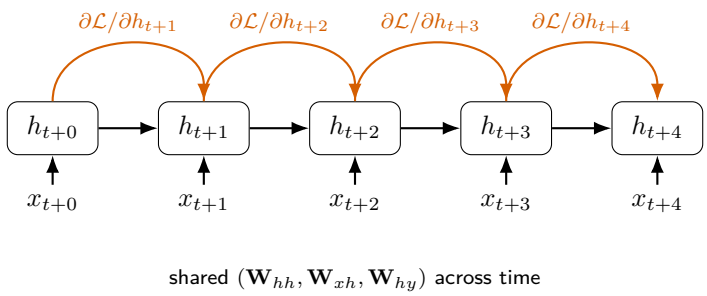


Figure 55: Backpropagation through time (BPTT): unrolled forward pass (black) and backward gradients (pink) through time.

Vanishing and Exploding Gradients Because each gradient term contains products of Jacobians such as

$$\frac{\partial \mathbf{h}_t}{\partial \mathbf{h}_{t-1}} = \operatorname{diag}\big(f'(\mathbf{a}_t)\big)\, \mathbf{W}_{hh}^{\top},$$

with pre-activation $\mathbf{a}_t = \mathbf{x}_t \mathbf{W}_{xh} + \mathbf{h}_{t-1}\mathbf{W}_{hh} + \mathbf{b}_h$ and elementwise nonlinearity f, long sequences multiply many such factors. Here $\mathbf{W}_{hh} \in \mathbb{R}^{d_h \times d_h}$ and $\operatorname{diag}\big(f'(\mathbf{a}_t)\big) \in \mathbb{R}^{d_h \times d_h}$, so the Jacobian $\partial \mathbf{h}_t / \partial \mathbf{h}_{t-1}$ is a $d_h \times d_h$ matrix. If the spectral norm of each factor is below one the product decays exponentially (vanishing); norms above one cause growth (exploding). Figure 56 illustrates both behaviors across time. Practical remedies include gradient clipping, orthogonal or unitary recurrent matrices, layer normalization, and gated architectures (LSTM/GRU) that introduce additive memory paths. Empirically, simple (ungated) RNNs often struggle to preserve dependencies beyond roughly 5–10 steps in language settings, which is why gated cells became the default for longer sequences.

Parameter Updates At each time step, the gradient of the loss with respect to parameters (e.g., weights W) depends on the chain of partial derivatives through the network states:

$$\frac{\partial \mathcal{L}}{\partial W} = \sum_{t=1}^{T} \frac{\partial \mathcal{L}_t}{\partial W}.$$

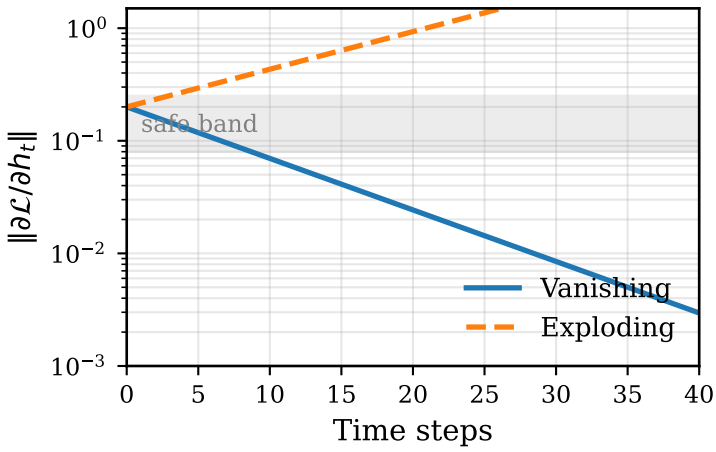


Figure 56: Vanishing (blue) versus exploding (orange) gradients on a log scale. The gray strip highlights the stability band; the inset reminds readers that repeated Jacobian products either shrink gradients (thin blue arrows) or amplify them (thick orange arrows). *Source: author-generated.*

Because of parameter sharing, the same W influences multiple time steps, and the total gradient is the sum over these contributions.

12.8 Stabilizing Recurrent Training

Gradient clipping. A practical safeguard is to clip the global norm of the gradient when it exceeds a threshold. Figure 57 shows how clipping prevents the exploding case from destabilizing optimization while leaving the vanishing regime untouched. Orthogonal or unitary initializations for the recurrent weight matrix W_{hh} are another common trick: because orthogonal matrices preserve Euclidean norms, gradients neither explode nor vanish in the very early stages of training (before nonlinearities and data-dependent effects accumulate).

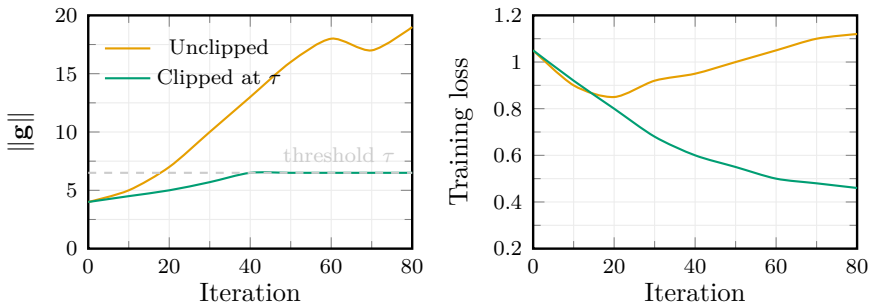


Figure 57: Gradient norms (left) explode without clipping (orange) but remain bounded when the global norm is clipped at tau (green). Training loss (right) stabilizes as a result.

Stability defaults in practice

Spectral norm $\|\mathbf{W}_{hh}\|_2 \approx 1$ keeps gradients roughly stable over tens of steps; clipping thresholds $\tau \in [0.5, 5]$ are sensible defaults when sequences are long. BatchNorm inside the recurrent loop is rarely helpful; prefer LayerNorm or gating to stabilize dynamics.

Author's note: do not fight a simple RNN. If your task needs dependencies longer than a handful of steps, do not over-tune an ungated RNN. Start with clipping and a sensible truncation window, but move to GRU/LSTM once long-range information vanishes.

Dropout in RNNs. Variational/recurrent dropout applies the same dropout mask at every time step to avoid injecting temporal noise (Gal and Ghahramani, 2016; Semeniuta et al., 2016); zoneout preserves hidden units stochastically instead of zeroing them (Krueger et al., 2017). Standard per-time-step dropout often harms sequence retention.

Teacher forcing and scheduled sampling. Sequence-to-sequence models frequently feed the ground-truth token back into the decoder during training (teacher forcing) to accelerate convergence. Figure 58 contrasts this regime with free-running inference: teacher forcing injects gold tokens at every step, whereas inference conditions the decoder on its own predictions. This mismatch is precisely what scheduled-sampling curricula aim to mitigate (Bengio et al., 2015).

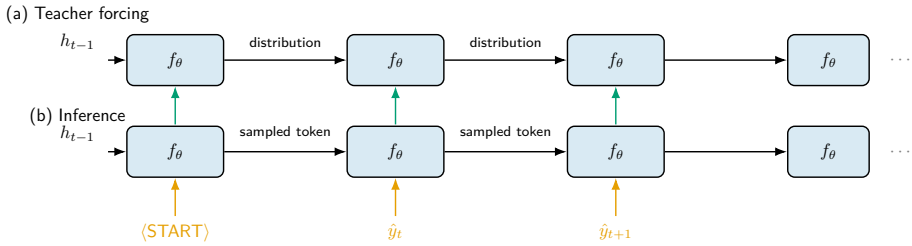


Figure 58: Teacher forcing vs. inference in a sequence-to-sequence decoder. Gold arrows show supervised targets; orange arrows highlight autoregressive feedback that motivates scheduled sampling.

Gated cells. LSTMs (Hochreiter and Schmidhuber, 1997; Gers et al., 2000) and GRUs (Cho et al., 2014) alleviate vanishing gradients by introducing additive memory paths guarded by gates. Figures 59 and 60 present the canonical cell diagrams used later in the chapter when deriving the update equations. Intuitively, LSTM forget/input/output gates control retention, writing, and exposure of the memory c_t; GRU update/reset gates interpolate between $\mathbf{h}_{t-1}$ and a candidate $\tilde{\mathbf{h}}_t$ and decide how much past information influences the candidate. In an LSTM, the state updates as $c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$ and $\mathbf{h}_t = o_t \odot \tanh(c_t)$. In a GRU, the hidden state updates as $\mathbf{h}_t = (1 - z_t) \odot \mathbf{h}_{t-1} + z_t \odot \tilde{\mathbf{h}}_t$, with the reset gate r_t shaping the candidate computation. In both figures, $\mathbf{x}_t$ denotes the input at time t, and $\mathbf{h}_{t-1}$ / c_{t-1} denote the carried state(s).

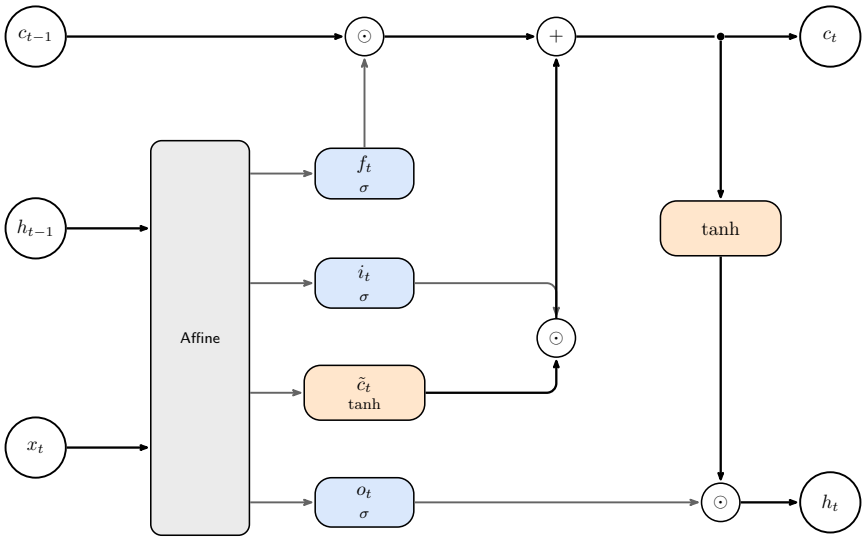


Figure 59: Long Short-Term Memory (LSTM) cell (Hochreiter and Schmidhuber, 1997; Gers et al., 2000).

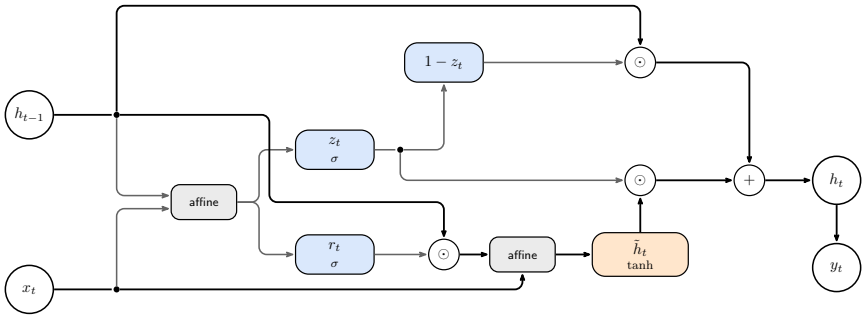


Figure 60: Gated Recurrent Unit (GRU) cell (Cho et al., 2014).

Simple RNN vs. GRU vs. LSTM

A simple (ungated) RNN has one state $\mathbf{h}_t$ and is the most parameter-efficient, but it is also the easiest to break on long sequences (vanishing/exploding gradients). A GRU adds gates that decide how much past state to keep versus overwrite, and it is often the best default when you want something robust without the full LSTM machinery. An LSTM maintains a separate cell state c_t with input/forget/output gates; it is heavier, but it tends to be the most forgiving when you truly need long-range retention.

Practical default: if you are unsure, start with a GRU (plus clipping and masking) and only simplify to a simple RNN for short-memory problems.

Implementation checklist: masks, clipping, pitfalls

Minimal training loop with masks and clipping

```
h = torch.zeros(B, d_h)
for x, y, mask in loader:         # [B, T, dx], [B, T], [B, T]
    h = h.detach()                # carry state, drop graph
    logits, h = rnn(x, h)
    # CE = per-step cross-entropy (e.g., reduction='none').
    loss = (mask * CE(logits, y)).sum() / mask.sum()
    loss.backward()
    torch.nn.utils.clip_grad_norm_(model.parameters(), tau)
    opt.step(); opt.zero_grad()
```

Variable-length sequences are padded; the mask zeros out pads in the loss. Reset h between sequences that should not share state.

Common pitfalls.

- Pads leaking into the loss (mask missing or applied after reduction).
- Hidden state carried across unrelated sequences (forgetting to reset or detach).
- No clipping on long sequences (exploding gradients is the first failure mode).
- BatchNorm inside recurrence (prefer LayerNorm for sequence models).
- Teacher forcing at training time without thinking through inference-time decoding.
- Dropout applied independently at each step when you intended variational/recurrent dropout.

12.9 From Recurrent State to Attention: Bridging Gated RNNs and Transformers

Attention mechanisms. Even with gating, long sequences can challenge fixed-size hidden states. Attention augments the decoder with a content-based

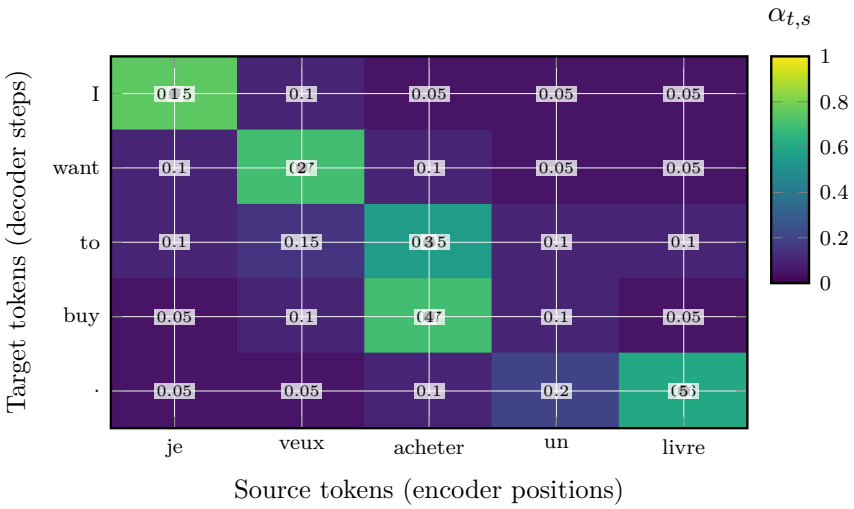


Figure 61: Attention heatmap for a translation model. Rows are target tokens (decoder steps) and columns are source tokens (encoder positions). Each cell is an attention weight; the dot in each row marks the source position receiving the most attention.

lookup into the encoder states, as visualized in Figure 61. Bright entries correspond to encoder positions that most influence each generated token.

Historical note: from gating to attention

Gating (LSTM/GRU) made long-range gradient flow workable by creating additive memory paths. Attention takes a different route: rather than forcing all history into a single state, it learns a soft retrieval rule over past representations. This shift explains why modern sequence models often prefer attention when long-range dependencies and parallel training matter most, while gated RNNs remain competitive for streaming and low-latency settings.

12.10 Synthesis: RNN Derivations, Gating Decisions, and Sequence Modelling Limits

One canonical sequence task is *next-step prediction*: use the history up to step t to predict a target at step $t + 1$. The target can be continuous (forecasting), categorical (a label at each step), or discrete from a finite set (symbols/words).

The mechanics are the same: define a per-step predictive distribution (or regression output), accumulate a loss over time, and train by BPTT on the unrolled computation graph.

12.10.1 Optional: language modeling as next-step prediction (preview)

Language modeling is the discrete-symbol case: predict the next token from the preceding prefix.

Recall that the goal is to estimate the conditional probability of a word w_t given the sequence of previous words $w_1, w_2, \ldots, w_{t-1}$:

$$P(w_t \mid w_1, w_2, \ldots, w_{t-1}).$$

This probability can be modeled using an RNN, which maintains a hidden state $\mathbf{h}_t$ that summarizes the history up to time t. A common indexing choice is: consume the current token w_t (as an embedding $\mathbf{x}_t$) and predict the *next* token w_{t+1}. This is equivalent to modeling $P(w_t \mid w_{1:t-1})$ after a one-step shift.

$$\begin{aligned} \mathbf{h}_t &= f(\mathbf{h}_{t-1}, \mathbf{x}_t; \theta), \\ P(w_{t+1} \mid w_1, \ldots, w_t) &= g(\mathbf{h}_t; \theta), \end{aligned} \tag{12.7}$$

where $\mathbf{x}_t$ is the input representation (e.g., a word embedding) of the token w_t, f is the recurrent update function parameterized by θ, and g maps the hidden state to a probability distribution over the vocabulary. Because the hidden state is computed recursively, $\mathbf{h}_t$ is a deterministic summary of the entire prefix $(w_1, \ldots, w_t)$:

$$\mathbf{h}_t = f(f(\cdots f(\mathbf{h}_0, \mathbf{x}_1), \ldots), \mathbf{x}_t).$$

What the model can remember (and what it must forget) is governed by the state dimension, the update nonlinearity, and the training objective. That is exactly why gradient flow and stabilization tricks matter in practice.

Training objective (negative log-likelihood). Given a training sequence $(w_1, w_2, \ldots, w_T)$, we typically train by minimizing the negative log-likelihood (NLL) (equivalently, maximizing the likelihood). With the next-token conven-

tion, the loss is

$$\mathcal{L}(\theta) = -\sum_{t=1}^{T-1} \log P(w_{t+1} \mid w_1, \ldots, w_t; \theta). \tag{12.8}$$

This is just cross-entropy over the vocabulary at each step: the model is rewarded for assigning high probability to the observed next token. A common reporting metric is *perplexity*, defined as exp(average NLL per token); lower perplexity means the model assigns higher probability to the observed text. If you are not working with text, you can ignore perplexity—the underlying idea is simply "negative log-likelihood per step." Chapter 13 returns to this objective in a more application-facing way (tokenization, embeddings, and audit considerations).

Self-supervised nature of language modeling A key insight is that no explicit labeling is required to train such models. The natural co-occurrence statistics of words in large corpora serve as implicit supervision. For example, the model learns that the word "juice" often follows "apple" because this pattern frequently appears in the training data. This is the essence of *self-supervised* learning in NLP, where the prediction targets are created directly from the input sequence.

Input representations. The input $\mathbf{x}_t$ is a numeric vector. In language modeling and other discrete-token problems, it is typically a token embedding produced by a learned lookup table. Chapter 13 covers where those vectors come from (training objectives, evaluation, and audit considerations).

12.10.2 Worked example: sentiment as a many-to-one decision

To make the sequence viewpoint concrete, consider a short review such as: *"This place is great."* In a sentiment classifier, we map the token sequence $(w_1, \ldots, w_T)$ to a single label $y \in \{0, 1\}$ (negative/positive). The recurrence consumes numeric inputs $\mathbf{x}_t$ (one per token) and produces a final state $\mathbf{h}_T$ that summarizes the sentence:

$$\mathbf{h}_t = f(\mathbf{h}_{t-1}, \mathbf{x}_t; \theta), \qquad \hat{y} = \sigma(\mathbf{w}^\top \mathbf{h}_T + b), \tag{12.9}$$

where $\sigma(\cdot)$ is the logistic sigmoid and $\hat{y}$ is the predicted probability of the positive class.

This setup highlights a subtle but important dependency between *representation* and *generalization*. If tokens are treated as unrelated IDs (e.g., purely one-hot indicators), then "great" and "awesome" are orthogonal inputs: the model has no built-in reason to transfer evidence across similar words unless the training objective supplies a geometry that makes them nearby. In Chapter 13, we return to this same kind of task and show how learning word vectors from neighboring words gives the model a meaningful input space before any sequence model is applied.

From sequence data to a next-step model (a minimal checklist)

A minimal sequence-model run looks like this. First decide how each step becomes a numeric input vector $\mathbf{x}_t$ (measured features, engineered features, or learned embeddings). Next decide the prediction target shape (many-to-one label, many-to-many labeling, or next-step prediction), then pick the recurrent cell f (simple RNN, GRU, or LSTM) and the output map g. Train with a per-step loss summed over time using truncated BPTT and correct padding masks. Finally, monitor validation loss and gradient norms; clipping is the first safety valve when sequences get long.

It is the same ERM loop as earlier chapters; the new wrinkle is that the loss (and therefore the gradients) accumulates across time as well as across batches. In Chapter 13, we specialize this checklist to text (discrete symbols, embeddings, next-token loss, and perplexity).

LSTM and GRU equations (compact)

LSTM (single layer):

$$\begin{aligned}
\mathbf{i}_t &= \sigma(\mathbf{x}_t\mathbf{W}_i + \mathbf{h}_{t-1}\mathbf{U}_i + \mathbf{b}_i), & \mathbf{f}_t &= \sigma(\mathbf{x}_t\mathbf{W}_f + \mathbf{h}_{t-1}\mathbf{U}_f + \mathbf{b}_f),\\
\tilde{\mathbf{c}}_t &= \tanh(\mathbf{x}_t\mathbf{W}_c + \mathbf{h}_{t-1}\mathbf{U}_c + \mathbf{b}_c), & \mathbf{o}_t &= \sigma(\mathbf{x}_t\mathbf{W}_o + \mathbf{h}_{t-1}\mathbf{U}_o + \mathbf{b}_o),\\
\mathbf{c}_t &= \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t, & \mathbf{h}_t &= \mathbf{o}_t \odot \tanh(\mathbf{c}_t).
\end{aligned}$$

GRU:

$$\begin{aligned}
\mathbf{z}_t &= \sigma(\mathbf{x}_t\mathbf{W}_z + \mathbf{h}_{t-1}\mathbf{U}_z + \mathbf{b}_z), & \mathbf{r}_t &= \sigma(\mathbf{x}_t\mathbf{W}_r + \mathbf{h}_{t-1}\mathbf{U}_r + \mathbf{b}_r),\\
\tilde{\mathbf{h}}_t &= \tanh(\mathbf{x}_t\mathbf{W}_h + (\mathbf{r}_t \odot \mathbf{h}_{t-1})\mathbf{U}_h + \mathbf{b}_h), & \mathbf{h}_t &= (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t.
\end{aligned}$$

All gates are elementwise; σ denotes the logistic sigmoid and $\odot$ the Hadamard product.

Notation note. In the sequence-model chapters, $\sigma(\cdot)$ always denotes the logistic sigmoid gate nonlinearity; when σ is used without an argument (e.g., in earlier chapters for noise scales σ^2) it refers to a standard deviation. Context distinguishes these roles; see Appendix A for the cross-chapter symbol index.

Optional refresher (padding masks and encoders). Variable-length sequences are almost always padded to a common length for batching. The padding is not data, so you must carry a mask through the loss (and through attention later) so pads contribute *zero* training signal. For encoder–decoder intuition, it helps to remember that many sequence models compress a history into a state (or a set of states) and then produce outputs from that representation; Chapter 14 revisits this idea with attention rather than recurrence.

Key takeaways

- Sequence losses add across time; shared parameters mean gradients accumulate across unrolled steps (BPTT).
- Truncated BPTT and padding masks are the engineering tools that make training feasible on long and variable-length sequences.
- Stability tools recur across architectures: clipping, gating (GRU/LSTM), and normalization; attention offers an alternative route to long-range dependencies.
- For text, the sequence is discrete and inputs are embeddings; Chapter 13 makes that concrete and introduces next-token cross-entropy and perplexity as a domain-specific diagnostic.

A good checkpoint.

- Explain BPTT, truncation, and masking for variable-length sequences.
- Describe why gates help (information and gradient flow) and when attention is a practical alternative.
- If you care about language modeling, relate cross-entropy, log-likelihood, and perplexity after reading Chapter 13.

Common pitfalls.

- Train/inference mismatch (teacher forcing without a decoding strategy that reflects it).
- Unstable gradients on long sequences (no clipping, poor initialization, or overly large learning rates).
- Incorrect padding masks that leak future tokens or contaminate the loss.

Exercises

Exercise 1 (Easy). Implement a padding-mask invariance test for a sequence model: pad the same batch to two different lengths and verify that (i) the loss is unchanged and (ii) gradients on real tokens are unchanged (to tight tolerance).
Deliverable: a short test that fails when masking is wrong + one paragraph explaining how mask bugs can masquerade as "model improvements."

Exercise 2 (Medium). Train a small time-series predictor and compare full BPTT to truncated BPTT (several window sizes K). Track training stability (gradient norms, loss oscillations) and final predictive quality under a fixed compute budget.
Deliverable: one plot showing the stability/quality trade-off across K + a short conclusion about credit assignment horizon.

Exercise 3 (Medium). Reproduce vanishing/exploding behavior by sweeping the effective scale of $\mathbf{W}_{hh}$ (or its spectral norm proxy) and testing gradient clipping as a mitigation.
Deliverable: gradient-norm traces (with/without clipping) + one paragraph linking the observation to why gating (LSTM/GRU) exists.

If you are skipping ahead. Keep the loss/perplexity diagnostics and masking discipline from this chapter. Chapter 14 changes the architecture, but the same evaluation traps and data-leakage risks remain.

Beyond gated RNNs: state-space models (S4, Mamba)

Gated RNNs (LSTM/GRU) solve vanishing gradients but remain *sequential*: computing $\mathbf{h}_t$ requires $\mathbf{h}_{t-1}$, so training cannot be parallelized across time the way Transformer attention can. Structured State-Space Models (S4) recast the recurrence as a *convolution*: the state-space equations

$$\mathbf{h}'(t) = \mathbf{A}\mathbf{h}(t) + \mathbf{B}x(t), \qquad y(t) = \mathbf{C}\mathbf{h}(t),$$

have a closed-form convolution kernel $\bar{K} = (\mathbf{CB}, \mathbf{CAB}, \mathbf{CA}^2\mathbf{B}, \ldots)$ that allows *parallel-scan* training across the full sequence in $O(L \log L)$ time instead of $O(L)$ sequential steps. At inference the recurrence form recovers constant-cost per-token generation. Mamba (Gu et al., 2023) extends S4 with input-dependent (selective) state transitions, matching Transformer perplexity on language modeling while retaining linear-time inference.
When to prefer SSMs over attention or gated RNNs: (i) very long sequences ($>$ 8k tokens) where quadratic attention cost is prohibitive; (ii) streaming or low-latency inference where constant-time recurrent steps matter; (iii) biological or physical time-series where the continuous-time state-space formulation aligns with the data-generating process.

Where we head next. Chapter 13 translates this sequence viewpoint into representation tooling: embedding objectives, neighborhood/analogy diagnostics, and deployment audits. Keep the sentiment example (Section 12.10.2) in mind: with learned token geometry, the same many-to-one task generalizes better than with raw IDs. Chapter 14 then changes architecture again, replacing recurrence with attention while preserving the same evaluation and masking discipline.

13 Neural Network Applications in Natural Language Processing

Chapter 12 framed language as prediction under context: the model must assign probability to the next token from a compressed state. That framing makes the bottleneck obvious. Tokens are discrete identifiers, while the model is a

continuous function; if the input geometry is unhelpful, the network cannot generalize systematically from one string to another.

This chapter supplies that geometry. We build word embeddings and the objectives that train them, then show how to sanity-check the resulting space (neighbors, analogies, and simple audits). Those same representation primitives become inputs to both recurrent and attention-based models; Chapter 14 reuses them when recurrence is replaced by attention.

Learning Outcomes

- Describe distributional semantics and the motivation for dense word embeddings.
- Derive and implement common embedding objectives (CBOW/skip-gram, negative sampling) and evaluate them via analogy tasks.
- Connect embedding quality to downstream architectures (RNNs, Transformers) and fairness considerations.

Design motif

Representation learning as a contract between data and objective—when you train on co-occurrence, you get both useful structure (analogies, clusters) and the biases present in the corpus.

13.1 Context and Motivation

In this chapter, we focus on neural methods for natural language processing (NLP) through the lens of representation learning. The recurring question from Chapter 12 is practical: what input space makes sequence prediction stable and data efficient? The answer is to learn vectors whose geometry captures contextual similarity.

A classic example illustrating the power of such representations is the analogy:

$$\text{king} - \text{man} + \text{woman} \approx \text{queen}.$$

This demonstrates that vector arithmetic on word embeddings can capture meaningful relationships between words. The goal is to find a vector space embedding where semantic similarity corresponds to geometric closeness.

How this chapter is organized. We start with the linguistic idea (distributional semantics) and immediately turn it into an engineering object: a lookup table of vectors that make prediction from nearby words easier. From there we build the two Word2Vec objectives (CBOW and skip-gram), show why full softmax is expensive, and introduce negative sampling as the practical workaround. We then discuss how to sanity-check an embedding space (neighbors and analogies) and close with what can go wrong in deployment (bias, leakage, and basic audits).

Notation. We use V for vocabulary size, d for embedding dimension, $\mathbf{x}_t$ for one-hot token indicators, and $\mathbf{e}_t$ for learned embedding vectors. When symbols collide with earlier chapters (e.g., d as data dimension vs. embedding size), the local meaning is authoritative; Appendix A gives the global index.

Problem statement. Given a vocabulary (corpus) of approximately 10,000 words, we want to learn a mapping from each word to a dense vector representation in a feature space of dimension d, where d is typically between 200 and 500. Formally, if the vocabulary size is V, each word w_i begins as an index (equivalently, a one-hot indicator). Our objective is to learn an embedding function

$$f : \{1, \ldots, V\} \to \mathbb{R}^d,$$

such that semantic and syntactic properties of words are preserved in the embedding space.

13.2 Warm-up: from symbols to vectors

One-hot vectors make discrete tokens compatible with linear algebra, but they do not express *similarity*: synonyms are orthogonal, and a model that treats each basis element independently has no reason to generalize from "great" to "awesome" without seeing both during training.

One-hot encoding (baseline). Fix a dictionary of size V. A token at position t can be represented as a one-hot row indicator $\mathbf{x}_t \in \{0, 1\}^{1\times V}$ with a single 1. This guarantees uniqueness, but it makes every pair of distinct words equally far apart.

Table 4: Feature-based word vectorization example. Each word is mapped to a vector of graded semantic features; fractional entries (e.g., 0.5) indicate mixed usage across contexts. Treat this as an intuition pump: real embedding coordinates are learned from data rather than hand-assigned.

Word	Gender	Royalty	Age	Person	Fruit	Title	Abstract	Sweet
man	1	0	1	1	0	0	0	0
woman	0	0	1	1	0	0	0	0
king	1	1	1	1	0	1	0	0
queen	0	1	1	1	0	1	0	0
orange	0	0	0	0	1	0	0	1
apple	0	0	0	0	1	0	0	1
monarch	0.5	1	0.5	1	0	1	0	0
royal	0	1	0.5	0	0	1	1	0

Why one-hot is not enough (two quick failures). Synonyms. A sentiment model trained to treat "great" as positive has no immediate reason to treat "awesome" as positive if the two words are orthogonal basis vectors.

Document similarity. "I enjoyed talking to the monarchs" and "I loved conversing with the royals" express the same meaning, but a bag-of-words similarity score stays low if it cannot recognize that *monarchs* and *royals* occupy similar roles.

This is the same friction point you saw in Section 12.10.2: the sequence model can only be as systematic as its input geometry.

Embedding lookup as learned features. Instead of using the identity basis, we learn an embedding table $\mathbf{W} \in \mathbb{R}^{V \times d}$ whose rows are trainable parameters. A one-hot indicator becomes a row lookup:

$$\mathbf{e}_t = \mathbf{x}_t \mathbf{W} \in \mathbb{R}^{1 \times d}.$$

This is the same linear algebra as one-hot, but now the coordinates are *learned* so that similar words end up nearby when that helps prediction.

A useful mental model is to imagine that each word is a *graded bundle of attributes*. If we could write those attributes down explicitly, each word would already be a vector:

Table 4 gives a concrete intuition for mapping symbolic tokens into graded feature vectors.

Numeric check: why the analogy arithmetic can work

Using the toy feature vectors in Table 4, the analogy is literal:

$$\mathbf{v}(\text{king}) - \mathbf{v}(\text{man}) + \mathbf{v}(\text{woman}) = \mathbf{v}(\text{queen}).$$

Real embeddings are not hand-designed, so the equality is only approximate, but the geometry is the same: a shared "relation direction" can show up as an approximately consistent offset across many word pairs. In this toy table, you can verify the identity directly from the entries.

Of course, hand-crafting features does not scale to large vocabularies, and it bakes in subjective choices. The central move of modern embeddings is to let the *training objective* discover whatever coordinates best support prediction from context.

Subword tokenization and OOV handling. Practical systems rarely operate on raw word types alone. To reduce vocabulary size and handle out-of-vocabulary (OOV) forms, they tokenize text into *subword units*. Byte Pair Encoding (BPE) and WordPiece learn a compact inventory of frequent character sequences; words are segmented into a small number of subwords that can be re-composed by the model. FastText instead augments word vectors with character n-gram embeddings, so the representation of an unseen word is the sum of its subword vectors.

13.3 Key Insight: Distributional Hypothesis

The foundational linguistic principle underlying word embeddings is the *distributional hypothesis*, often summarized by the phrase:

> *You shall know a word by the company it keeps.*

This idea, attributed to the linguist John Robert Firth, states that the meaning of a word can be inferred from the contexts in which it appears.

Example: The word *pretty* can have different meanings depending on context:

- In the collocation "pretty good," *pretty* functions as an adverb meaning "very" and modifies an adjective.

- In phrases such as "pretty image" or "pretty optics," *pretty* is an adjective meaning "attractive."

By explicitly examining the surrounding words (context windows of a few tokens to the left and right), we can infer the intended meaning: instances co-occurring with evaluative adjectives like "good" teach the "intensifier" sense, whereas contexts rich in nouns like "image" teach the "aesthetic" sense.

13.4 Contextual Meaning and Feature Extraction

The practical takeaway is that repeated context patterns reveal latent word properties. In the *pretty* example, co-occurrence with evaluative adjectives points to an intensifier sense, while co-occurrence with visual nouns points to an aesthetic sense.

This is why modern embedding models treat corpus co-occurrence as supervision.

Author's note: who chooses the features?

A natural student question is: if embeddings represent "features," who decides what the features are? In modern embedding learning, the answer is: nobody writes them down explicitly. The features emerge from the training objective. By training a model to predict nearby words (or to distinguish real context pairs from random ones), the optimization process forces the hidden representation to encode whatever properties are useful for prediction. This is best viewed as *self-supervised* learning: targets come from the text itself via context windows, rather than from human labels.

13.5 Word2Vec at a glance

The Word2Vec framework, introduced by Mikolov et al. (2013), operationalizes the distributional hypothesis through two main architectures:

1. **Continuous Bag of Words (CBOW):** Predicts the target word given its surrounding context words.

2. **skip-gram:** Predicts the surrounding context words given the target word.

Both architectures learn vectors by solving local prediction tasks rather than from hand-labeled semantic features.

13.5.1 Continuous Bag of Words (CBOW)

In CBOW, the model takes as input the context words surrounding a target word and tries to predict the target word itself. Formally, given a sequence of words $\{w_1, w_2, \ldots, w_T\}$, and a context window size n, the context for word w_t is

$$\mathcal{C}_t = \{w_{t-n}, \ldots, w_{t-1}, w_{t+1}, \ldots, w_{t+n}\}.$$

The CBOW model maximizes the probability

$$p(w_t \mid \mathcal{C}_t),$$

where the context words $\mathcal{C}_t$ are represented as one-hot vectors and combined (e.g., averaged) to form the input.

Example: Consider the sentence

"to buy an automatic car".

If we want to learn the embedding for the word *automatic*, the context might be $\{\text{to}, \text{buy}, \text{an}, \text{car}\}$. The CBOW model uses these context words to predict *automatic*.

13.5.2 skip-gram

Conversely, the skip-gram model takes the target word as input and tries to predict each of the context words. It maximizes

$$\prod_{w_c \in \mathcal{C}_t} p(w_c \mid w_t).$$

The product makes the modeling assumption explicit: every context word within the window contributes a likelihood factor. In practice we maximize the sum of log-probabilities $\sum_{w_c \in \mathcal{C}_t} \log p(w_c \mid w_t)$ so that each neighboring prediction provides an additive gradient signal.

This approach tends to perform better on infrequent words and captures more detailed semantic relationships.

This first pass is conceptual. The next sections pin down the exact parameterization and training objectives used in practice.

13.6 From lookup to objective: compact derivation path

With the conceptual picture fixed, we now run one continuous derivation path from lookup mechanics to CBOW/skip-gram training objectives. Let vocabulary size be V and embedding dimension be d. Define the embedding matrix $\mathbf{W} \in \mathbb{R}^{V\times d}$, where the i-th row $\mathbf{v}_i$ is the embedding vector for word w_i. *Convention: we treat embeddings as* ***rows****; one-hot words index rows via* $\mathbf{xW}$ *(row lookup).*

Embedding lookup as a tiny network (shapes). Fix a vocabulary of size V and an embedding dimension d (e.g., $V = 10{,}000$, $d = 300$). A token is an index $i \in \{1, \ldots, V\}$, represented as a one-hot row vector $\mathbf{x} \in \{0, 1\}^{1\times V}$. The embedding table $\mathbf{W} \in \mathbb{R}^{V\times d}$ is trainable, and the embedding for the token is the row lookup

$$\begin{aligned}
\mathbf{h} &= \mathbf{xW} \in \mathbb{R}^{1\times d}, \\
\mathbf{z} &= \mathbf{hW}_{\text{out}} \in \mathbb{R}^{1\times V}, \\
\hat{\mathbf{y}}_j &= \frac{\exp(z_j)}{\sum_{k=1}^{V} \exp(z_k)}, \quad j = 1, \ldots, V, \\
\mathcal{L} &= -\sum_{j=1}^{V} y_j \log \hat{y}_j.
\end{aligned} \tag{13.1}$$

Because $\mathbf{x}$ is one-hot, this multiplication selects exactly one row of $\mathbf{W}$; that is the whole "embedding lookup" operation.

To turn the lookup into a prediction task (CBOW or skip-gram), we add output-side vectors $\mathbf{W}_{\text{out}} \in \mathbb{R}^{d\times V}$, compute logits $\mathbf{z}$, apply a softmax to get a vocabulary distribution $\hat{\mathbf{y}}$, and then minimize cross-entropy (equivalently, negative log-likelihood) against the one-hot target $\mathbf{y}$. During learning, $\mathbf{W}$ and $\mathbf{W}_{\text{out}}$ update so that words that share contexts end up with similar vectors.

Context window and sequential input. Reusing the same toy phrase, let the context window size be 4 around the target token. To predict "automatic" in "to buy an automatic car," the context words are

to, buy, an, car.

Each context word is represented as a one-hot vector and fed into the network. Each one-hot vector shares the same embedding matrix $\mathbf{W}$; multiplying $\mathbf{xW}$ is

an efficient row lookup because $\mathbf{x}$ is one-hot.

Input Sequence Processing The same embedding lookup is applied to each context token. If $\mathbf{x}_{t+j}$ is the one-hot vector for the word at position $t+j$, then its embedding is

$$\mathbf{h}_{t+j} = \mathbf{x}_{t+j}\mathbf{W}.$$

The hidden representations $\mathbf{h}^{(i)}$ for each context word can be combined (e.g., concatenated or averaged) before passing to the output layer to predict the target word.

Dimensionality and Sparsity Note that the input vectors $\mathbf{x}_{t+j}$ are extremely sparse (one-hot), and the embedding matrix $\mathbf{W}$ is large ($10{,}000 \times 300$, for example). However, the multiplication $\mathbf{x}_{t+j}\mathbf{W}$ is efficient because it selects a single row of $\mathbf{W}$ per input word.

Interpretation of the weight matrix W. The matrix $\mathbf{W}$ can be interpreted as a lookup table: the i-th row is the embedding for word w_i, and $\mathbf{xW}$ (with one-hot $\mathbf{x}$) selects that row directly.

13.7 Word2Vec objectives in detail

With notation fixed, we now write the two Word2Vec objectives in full form: Continuous Bag of Words (CBOW) and skip-gram.

13.7.1 CBOW objective (detailed form)

In the detailed CBOW form, the model predicts center token w_t from its window context. For a sequence $w_1, w_2, \ldots, w_T$ and half-window size c, the context is $\{w_{t-c}, \ldots, w_{t-1}, w_{t+1}, \ldots, w_{t+c}\}$.

Each context token is encoded as a one-hot row in $\mathbb{R}^{1\times V}$, then mapped through $\mathbf{W} \in \mathbb{R}^{V\times d}$ to its dense embedding.

The CBOW model computes the average of the embeddings of the context words using an *input* embedding matrix $\mathbf{W}$ and predicts with a separate *output* embedding matrix $\mathbf{W}_{\text{out}}$:

$$\mathbf{h} = \frac{1}{2c} \sum_{\substack{-c\le j\le c \\ j\neq 0}} \mathbf{x}_{t+j}\mathbf{W}$$

where $\mathbf{x}_{t+j}$ is the one-hot vector for the context word at position $t+j$, and c denotes the *half-window* size (there are $2c$ context words around w_t when the document is long enough).

This hidden representation $\mathbf{h}$ is then used to predict the target word w_t via a softmax layer:

$$P(w_t \mid \text{context}) = \frac{\exp(\mathbf{h}\,\mathbf{u}_{w_t})}{\sum_{w=1}^{V} \exp(\mathbf{h}\,\mathbf{u}_w)} \tag{13.2}$$

where $\mathbf{u}_w$ is the output vector corresponding to word w. It is useful to think of the set of output vectors as the *columns* of a second matrix $\mathbf{W}_{\text{out}} \in \mathbb{R}^{d \times V}$; although $\mathbf{W}_{\text{out}}$ often starts as a copy of $\mathbf{W}^\top$, the two sets of embeddings are optimized independently during training. Many modern implementations optionally *tie* these matrices so that $\mathbf{W}_{\text{out}} = \mathbf{W}^\top$, reducing parameters and encouraging symmetry between input and output spaces.

Training maximizes corpus log-likelihood, updating both $\mathbf{W}$ and output vectors $\mathbf{u}_w$. The rows of $\mathbf{W}$ are then used as the learned embeddings.

Key Insight: Multiplying one-hot input $\mathbf{x}_{t+j}$ by $\mathbf{W}$ performs a row lookup. The embedding matrix therefore behaves as a trainable feature table.

13.7.2 skip-gram objective (detailed form)

The detailed skip-gram form reverses CBOW: use center word w_t to predict each context token w_{t+j} in window c:

$$\prod_{\substack{-c \le j \le c \\ j \ne 0}} P(w_{t+j} \mid w_t)$$

The input is the one-hot vector $\mathbf{x}_t$ representing the center word, which is projected into the embedding space via the same input embedding matrix $\mathbf{W} \in \mathbb{R}^{V \times d}$:

$$\mathbf{h} = \mathbf{x}_t \mathbf{W}$$

Each context word w_{t+j} is predicted by applying a softmax over the output vectors (again using the output-embedding matrix $\mathbf{W}_{\text{out}}$):

$$P(w_{t+j} \mid w_t) = \frac{\exp\big(\mathbf{h}\,\mathbf{u}_{w_{t+j}}\big)}{\sum_{w=1}^{V} \exp(\mathbf{h}\,\mathbf{u}_w)} \tag{13.3}$$

where $\mathbf{u}_w$ are the output vectors as before.

Training Objective: Maximize the log-likelihood of the context words given the center word over the entire corpus. Compare to the RNN language-model objective in Chapter 12: both predict nearby tokens, but here the context is a fixed sliding window rather than a learned recurrent state.

Interpretation: Skip-gram pushes words with similar neighborhoods toward similar embeddings.

13.7.3 Computational Challenges: Softmax Normalization

Both CBOW and skip-gram models require computing the softmax normalization over the entire vocabulary V, which can be very large (e.g., $V = 10,000$ or more). The denominator in equations (13.2) and (13.3) involves summing exponentials over all vocabulary words:

$$Z = \sum_{w=1}^{V} \exp(\mathbf{h}\,\mathbf{u}_w)$$

Recipe: skip-gram with negative sampling

Preprocess: tokenize (BPE/WordPiece or whitespace), lowercase if appropriate, drop rare words below a cutoff, subsample frequent words with $t \approx 10^{-5}$.
Hyperparameters: window $c = 2$–5 (often dynamic/symmetric), embedding dim $d = 100$–300, negatives $k = 5$–20, unigram noise $P_n(w) \propto f(w)^{0.75}$, learning rate on the order of 10^{-3}–10^{-2}.
Per-positive loss (one context word):
$-\log \sigma(\mathbf{h}\,\mathbf{u}_{\text{pos}}) - \sum_{i=1}^{k} \log \sigma(-\mathbf{h}\,\mathbf{u}_{\text{neg},i})$; complexity $O(k)$ vs. $O(V)$ for full softmax.

This is computationally expensive, especially when training on large corpora.

Approximate Solutions: To address this, several approximation techniques have been proposed:

- **Hierarchical softmax:** factor the softmax into a tree so each update touches only a $\log V$ path.
- **Negative sampling:** replace the full softmax with k binary logistic losses against sampled "noise" words.

13.8 Efficient Training of Word Embeddings: Hierarchical Softmax and Negative Sampling

We now expand the two approximations introduced above—hierarchical softmax and negative sampling—and use them throughout the rest of the section.

1. Hierarchical Softmax Hierarchical softmax replaces the flat softmax layer with a binary tree representation of the vocabulary. Each word corresponds to a leaf node, and the probability of a word is decomposed into the probabilities of traversing the path from the root to that leaf. This reduces the computational complexity from $O(V)$ to $O(\log V)$, where V is the vocabulary size.

The key idea is to organize words so that frequent words have shorter paths, thus further improving efficiency. During training, only the nodes along the path to the target word are updated, avoiding the need to compute scores for all words.

2. Negative Sampling Negative sampling is an alternative approximation that simplifies the objective by transforming the multi-class classification problem into multiple binary classification problems.

- For each observed word-context pair (w, c), the model aims to distinguish the true pair from randomly sampled *negative* pairs (w, c'), where c' is drawn from a noise distribution.
- Instead of computing probabilities over the entire vocabulary, the model only updates parameters for the positive pair and a small number of negative samples.

Example: Consider the sentence:

"I want to buy a big brick house in the city."

Suppose the context word is `brick`. The true target word is `house`. Negative samples might be `lion`, `bake`, or `big` (although `big` appears in the sentence, it can still be sampled as a negative example depending on the sampling strategy). Negative draws occasionally colliding with real context words is harmless. The associated losses simply push the model to separate the sampled pair unless the data provide strong evidence to the contrary.

Training Objective with Negative Sampling Define the logistic regression classifier that, given an input word vector $\mathbf{v}_w$ and an output word vector $\mathbf{v}'_c$, predicts whether the pair (w, c) is observed (label 1) or a negative sample (label 0).

The probability that the pair is observed is modeled as:

$$p(D = 1 \mid w, c) = \sigma(\mathbf{v}_w \, \mathbf{v}'_c) \tag{13.4}$$

where $\sigma(x) = \frac{1}{1+e^{-x}}$ is the sigmoid function.

The training objective for one positive pair (w, c) and k negative samples $\{c'_1, \ldots, c'_k\}$ is:

$$\log \sigma(\mathbf{v}_w \, \mathbf{v}'_c) + \sum_{i=1}^{k} \log \sigma\big(- \mathbf{v}_w \, \mathbf{v}'_{c'_i}\big) \tag{13.5}$$

where each c'_i is drawn independently from the noise distribution $P_n(c)$. A widely used practical choice is $P_n(w) \propto f(w)^{0.75}$, where $f(w)$ is the empirical unigram frequency; this slightly downweights extremely frequent words while still sampling them often enough to learn robust embeddings.

Tiny worked example (skip-gram with $k = 2$). Suppose the center word is `brick` with embedding $\mathbf{v}_{\text{brick}}$, the true context is `house` ($\mathbf{v}'_{\text{house}}$), and we sample two negatives `lion`, `bake`. We compute:

$$L = \log \sigma(\mathbf{v}_{\text{brick}} \, \mathbf{v}'_{\text{house}}) + \log \sigma(-\mathbf{v}_{\text{brick}} \, \mathbf{v}'_{\text{lion}}) + \log \sigma(-\mathbf{v}_{\text{brick}} \, \mathbf{v}'_{\text{bake}}).$$

Gradients push $\mathbf{v}_{\text{brick}}$ closer to $\mathbf{v}'_{\text{house}}$ (if the dot product is too small) and simultaneously push it away from $\mathbf{v}'_{\text{lion}}$ and $\mathbf{v}'_{\text{bake}}$. Only these three context vectors update this step, so the cost stays $O(k)$ regardless of vocabulary size.

Interpretation: The model learns to assign high similarity scores to true word-context pairs and low similarity scores to randomly sampled pairs, effectively learning meaningful embeddings without computing the full softmax. The expectation over the noise distribution is estimated by the empirical average across the k sampled negatives in (13.5). Unlike noise-contrastive estimation (NCE), negative sampling is not a consistent estimator of the normalized softmax probabilities; it is best viewed as a task-specific approximation that yields high-quality embeddings rather than calibrated class posteriors.

Backpropagation: The gradients are computed only for the positive pair and the sampled negative pairs, drastically reducing computation.

Connection to PMI (Levy & Goldberg). A useful theoretical lens relates skip-gram with negative sampling (SGNS) to pointwise mutual information (PMI). Under common choices of windowing and negative sampling distribution, SGNS implicitly factorizes a *shifted* PMI matrix such that inner products approximate:

$$\mathbf{v}_i\,\mathbf{u}_k \;\approx\; \mathrm{PMI}(i,k) - \log k, \quad \text{where} \quad \mathrm{PMI}(i,k) = \log \frac{P(i,k)}{P(i)P(k)}.$$

This connection helps explain why SGNS and GloVe often yield similar geometric regularities despite different training objectives: both methods recover statistics of co-occurrence up to monotone transformations and weighting.

13.9 Local Context vs. Global Matrix Factorization Approaches

Word embedding methods can be broadly categorized into two classes based on how they utilize context information:

1. Local Context Window Methods These methods focus on the immediate context of a word within a fixed-size window. Examples include:

- Continuous Bag-of-Words (CBOW)

- skip-gram

They learn embeddings by predicting a word given its neighbors (CBOW) or predicting neighbors given a word (skip-gram). These methods are computationally efficient and capture syntactic and semantic relationships based on local co-occurrence patterns.

2. Global Matrix Factorization Methods These methods consider the entire corpus to build a global co-occurrence matrix X, where each entry X_{ij} counts how often word i co-occurs with word j across the corpus.

- Latent Semantic Analysis (LSA) is an early example, which applies singular value decomposition (SVD) to the co-occurrence matrix.
- More recent methods include GloVe (Global Vectors), which factorizes a weighted log-count matrix (Pennington et al., 2014).

Example: Co-occurrence Matrix Suppose the vocabulary size is V. The co-occurrence matrix $X \in \mathbb{R}^{V \times V}$ is defined as:

$$X_{ij} = \text{number of times word } i \text{ appears in the context of word } j$$

This matrix is *sparse* and *large* (especially when V runs into the hundreds of thousands), so storing it explicitly or factorizing it naively can be computationally expensive.

13.10 Global Word Vector Representations via Co-occurrence Statistics

Recall that our goal is to obtain a global vector representation for words, capturing semantic relationships beyond simple one-hot encodings. Instead of encoding words individually, we leverage *co-occurrence* statistics of word pairs within a corpus to build richer embeddings.

Setup: Consider two words w_i and w_j appearing in some context window within a text corpus. We are interested in modeling the *co-occurrence* of these words, possibly mediated by a third *context* word w_k. For example, in the phrase

"big historic castle," the words "big" and "historic" are targets, and "castle" can be a context word connecting them.

Notation:

- Plain symbols w_i, w_j, w_k denote lexical items drawn from the vocabulary.
- Bold symbols denote vectors: $\mathbf{v}_i$ is the embedding of target word w_i and $\mathbf{u}_k$ the embedding of context word w_k.
- X_{ik} counts how often w_i and w_k co-occur within the chosen context window, and $X_i = \sum_k X_{ik}$ is the total number of context observations for w_i.

Goal: Define a function f that relates the co-occurrence statistics of the word pairs and context words to a scalar quantity representing their semantic association.

Visualization. Projecting the learned vectors onto two principal components typically reveals well-separated semantic clusters. Figure 62 highlights how gendered titles, fruits, and locations occupy distinct regions, reinforcing that co-occurrence-driven training captures rich lexical structure.

13.10.1 Modeling Co-occurrence Probabilities

We start by considering the conditional probability of observing a context word w_k given a target word w_i:

$$P(k|i) = \frac{X_{ik}}{X_i}. \tag{13.6}$$

This probability captures how likely the context word w_k appears near the target word w_i.

Relating to word vectors: Suppose each word w_i is represented by a vector $\mathbf{v}_i \in \mathbb{R}^d$. We want to model the relationship between $\mathbf{v}_i$, $\mathbf{u}_k$, and the co-occurrence probability $P(k|i)$.

A natural assumption is that the co-occurrence probability can be modeled as an exponential function of the inner product of the corresponding word vectors:

$$P(k|i) \propto \exp\left(\mathbf{v}_i\, \mathbf{u}_k\right). \tag{13.7}$$

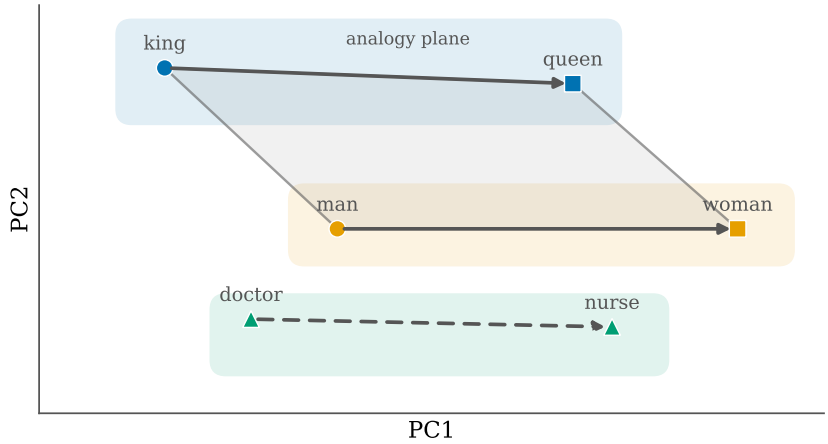


Figure 62: Analogy geometry in embedding space. The offset "v(king) - v(man) + v(woman) approx v(queen)" forms a parallelogram; a similar gender direction shifts "doctor" toward "nurse." Solid and dashed vectors highlight the shared relational direction. Points are shown after a 2D principal component analysis (PCA) projection, so directions are approximate. *Source: author-generated schematic; inspired by Mikolov et al. (2013).*

More explicitly, we can write a normalized model

$$P(k|i) \approx \frac{1}{Z_i} \exp\big(\mathbf{v}_i\, \mathbf{u}_k + b_i + b_k\big),$$

with partition function

$$Z_i = \sum_{k'} \exp\big(\mathbf{v}_i\, \mathbf{u}_{k'} + b_i + b_{k'}\big).$$

Taking logarithms on both sides and absorbing the (word-specific) normalizer into the biases gives the approximate relation

$$\log P(k|i) \approx \mathbf{v}_i\, \mathbf{u}_k + b_i + b_k, \tag{13.8}$$

where b_i and b_k are bias terms associated with words w_i and w_k, respectively. These biases account for the overall frequency or importance of each word while implicitly capturing the effect of Z_i.

Derivation: Starting from the co-occurrence counts,

$$\begin{aligned} \log X_{ik} - \log X_i = \log \frac{X_{ik}}{X_i} &= \log P(k|i) \\ &\approx \mathbf{v}_i \, \mathbf{u}_k + b_i + b_k. \end{aligned} \tag{13.9}$$

This equation suggests that the log co-occurrence counts can be approximated by a bilinear form plus biases.

13.10.2 Optimization Objective

Given the corpus co-occurrence matrix $X = [X_{ik}]$, our goal is to find word vectors $\mathbf{v}_i$, $\mathbf{u}_k$ and biases b_i, b_k that minimize the reconstruction error:

$$J = \sum_{i,k} f(X_{ik}) \left(\mathbf{v}_i \, \mathbf{u}_k + b_i + b_k - \log X_{ik} \right)^2 , \tag{13.10}$$

where f is a weighting function that controls the influence of each co-occurrence pair.

Why weighting? Many entries X_{ik} are zero or very small, which can cause numerical instability or dominate the objective. The function f is designed to:

- Downweight rare co-occurrences (small X_{ik}) to avoid overfitting noise.
- Possibly cap the influence of very frequent co-occurrences to prevent them from dominating.

A typical choice for f is:

$$f(x) = \begin{cases} \left(\frac{x}{x_{\max}} \right)^{\alpha} & \text{if } x < x_{\max}, \\ 1 & \text{otherwise}, \end{cases} \tag{13.11}$$

where $\alpha \in (0, 1)$ and $x_{\max}$ is a cutoff parameter.

13.11 Finalizing the Word Embedding Derivations

In the previous sections, we explored the formulation of word embeddings through co-occurrence statistics and matrix factorization approaches. This section closes the derivation arc and clarifies the role of bias terms and optimization strategies.

Recall the key equation relating the word vectors $\mathbf{v}_i$ and context vectors $\mathbf{u}_k$ to the co-occurrence counts X_{ik}:

$$\mathbf{v}_i \, \mathbf{u}_k + b_i + b_k = \log X_{ik}, \tag{13.12}$$

where b_i and b_k are bias terms associated with the word and context, respectively.

Symmetry and Bias Terms Initially, two separate bias terms b_i and b_k were introduced to account for asymmetries in the data. However, it is often possible to simplify the model by combining or eliminating one of the biases without loss of generality. This is because the biases can absorb constant shifts in the embeddings, and the key information lies in the relative positions of the vectors. In practice we keep both biases so that very frequent terms (e.g., stop words) can learn large offsets while rarer words keep their dot products numerically stable.

Hence, the equation can be rewritten as

$$\mathbf{v}_i \, \mathbf{u}_k = \log X_{ik} - b_i - b_k. \tag{13.13}$$

In practice, the biases b_i and b_k are learned jointly with the embeddings to best fit the observed co-occurrence statistics.

Objective Function and Optimization Let the target embeddings be rows $\mathbf{v}_i \in \mathbb{R}^{1\times d}$, the context embeddings be columns $\mathbf{u}_k \in \mathbb{R}^{d\times 1}$, and the biases scalars $b_i, b_k \in \mathbb{R}$. The scalar score $\mathbf{v}_i \, \mathbf{u}_k$ therefore measures the alignment between the target and context embeddings. The goal is to find $\{\mathbf{v}_i, \mathbf{u}_k, b_i, b_k\}$ that minimize the reconstruction error of the log co-occurrence matrix. Because raw counts span several orders of magnitude, the loss must behave like plain least squares for large X_{ik} yet dampen the influence of very small counts. Enforcing the limits $f(x) \to 0$ as $x \to 0$ and $f(x) \to 1$ for $x \geq x_{\max}$ yields the weighting scheme used by GloVe. The final weighted least-squares loss is

$$J = \sum_{i=1}^{V} \sum_{k=1}^{V} f(X_{ik}) \left(\mathbf{v}_i \, \mathbf{u}_k + b_i + b_k - \log X_{ik}\right)^2, \tag{13.14}$$

where $f(x)$ is the weighting function that downweights rare (or extremely common) co-occurrences to improve robustness. GloVe uses the piecewise definition in (13.11), so very small counts contribute little to the loss while moderately frequent pairs still influence the fit. In the original paper, practical defaults such as $\alpha \approx 0.75$ and $x_{\max} \approx 100$ were found to work well across a range of corpora (Pennington et al., 2014); keeping those guardrails explicit also explains why the same weighting recipe keeps reappearing in derived models.

Singular Value Decomposition (SVD) Connection One approach to solving this problem is to perform a low-rank approximation of the matrix $\log X$, where $X = [X_{ik}]$ is the co-occurrence matrix and the logarithm is applied elementwise (with small smoothing constants, e.g., $\epsilon = 10^{-8}$, added to avoid $\log 0$). The singular value decomposition (SVD) provides a principled method to find such a factorization:

$$\log X \approx \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r^\top, \tag{13.15}$$

where $\mathbf{U}_r \in \mathbb{R}^{V \times r}$ and $\mathbf{V}_r \in \mathbb{R}^{V \times r}$ contain the top-r singular vectors (for the desired embedding dimension $d = r$), and $\mathbf{\Sigma}_r \in \mathbb{R}^{r \times r}$ is a diagonal matrix of the corresponding singular values. The truncation rank r, often between 100 and 300 in practice, acts exactly like the embedding dimensionality knob in neural models.

By setting

$$\mathbf{v}_i = (\mathbf{U}_r)_i \mathbf{\Sigma}_r^{1/2}, \quad \mathbf{u}_k = (\mathbf{V}_r)_k \mathbf{\Sigma}_r^{1/2},$$

we obtain embeddings that approximate the log co-occurrence matrix in a least-squares sense.

Interpretation and Limitations While SVD provides a closed-form solution, it does not explicitly model the bias terms b_i, b_k or the weighting function $f(x)$. Those additional degrees of freedom allow gradient-based methods such as GloVe to better match empirical co-occurrence ratios. Biases soak up unigram frequency effects while the weighting function prevents very noisy counts from dominating the fit.

Before moving from geometry to governance, keep the boundary clear: the sections above establish how embeddings are learned and why their vector arithmetic works. The next sections address whether that learned geometry is reliable

and safe under real deployment constraints.

Risk & audit

- **Evaluation leakage:** similarity/analogy benchmarks can overlap training sources; keep a truly held-out evaluation set and treat "standard" datasets as potentially contaminated.
- **Tokenization debt:** preprocessing and vocabulary choices (case, subwords, cutoffs) change what the model can represent; version tokenizers and report them with results.
- **Frequency bias:** rare words get unstable vectors; audit neighborhoods by frequency and use subsampling/regularization so geometry is not just Zipf effects.
- **Social bias:** co-occurrence reflects social structure and stereotypes; probe for bias before using embeddings in decisions and document mitigations.
- **Privacy/memorization:** large corpora can contain sensitive strings; treat training data as a security boundary and audit downstream systems for memorization.

13.12 Bias in Natural Language Processing

An important consideration in word embedding models is the presence of bias inherited from the training corpora. Since embeddings are learned from co-occurrence patterns in text, they reflect the statistical properties of the language data, including cultural and societal biases.

Sources of Bias

- **Cultural bias:** Text corpora often contain stereotypes or skewed representations of gender, ethnicity, and other social categories (e.g., news archives that associate "nurse" more frequently with women than men).
- **Historical bias:** Older texts may reflect outdated or prejudiced views. Digitized literature from the 19th century, for instance, over-represents colonial perspectives.

- **Language-specific bias:** Different languages and dialects encode different cultural norms and connotations, such as grammatical gender or honorifics that privilege particular groups.

Impact on Embeddings For example, the well-known analogy

$$\text{king} - \text{man} + \text{woman} \approx \text{queen}$$

illustrates that many embeddings support approximately linear semantic relationships. However, these same linear structures can also reveal problematic biases, such as associating certain professions or attributes disproportionately with one gender or group.

Debiasing Techniques There is no single fix, because the "bias direction" you measure depends on the corpus, the task, and the group definitions you care about. In practice, mitigations usually fall into three buckets: post-processing (identify a direction and project it out; e.g., Hard Debiasing by Bolukbasi et al. (2016)), data-side interventions (rebalance or augment so the corpus does not teach the same stereotype as strongly), and objective-side constraints (regularize training so certain associations are discouraged). The key is to treat mitigation as iterative: apply a change, then rerun the same probes and downstream slices you used to detect the issue.

Cross-Lingual Challenges When extending embeddings to multiple languages, biases can manifest differently due to linguistic and cultural variations. For example, gender is grammatically encoded in Romance languages, so direct projection of English debiasing techniques may still leave gendered artifacts in Spanish or French embeddings. Careful consideration is required to ensure fairness and robustness across languages.

Practical bias checks

- **Dataset audit:** Inspect class balance, label sources, and sensitive attributes; check for under-represented groups and spurious correlations (e.g., profession ↔ gender cues).
- **Calibration and reliability:** Evaluate calibration (expected calibration error (ECE), reliability diagrams) overall and for key subgroups; severely miscalibrated models magnify harm when used for decision support.
- **Disaggregated evaluation:** Report accuracy, receiver operating characteristic / precision–recall (ROC/PR), and calibration metrics by subgroup rather than only aggregate scores; look for systematic performance gaps.
- **Mitigation loop:** Combine data interventions (rebalancing, augmentation) with model-side debiasing and re-evaluation; treat mitigation as an iterative, experiment-driven process.

Author's note: embeddings mix geometry and bias

Embedding spaces faithfully capture geometry (analogies, clusters) precisely because they also capture the biases present in the data. Treat every downstream use as a combination of those two facets: audit the geometry you need, but also audit the offset directions you would rather suppress. Vector arithmetic makes biases quantifiable, so put that ability to work before shipping a model.

13.13 Responsible deployment checklist

1. **Purpose & consent.** Document the use-case, decision stakes, and where humans remain in the loop; distinguish exploratory prototypes from production decision aids.
2. **Data lineage & licensing.** Track licenses for each corpus (newswire, Common Crawl, proprietary logs) and state whether downstream users may redistribute the embeddings or derived models.
3. **Privacy & security.** Scan corpora for personally identifiable information

(PII), redact when necessary, and restrict raw-data access. When embeddings leave the lab, accompany them with an acceptable-use policy and redaction guarantees.

4. **Monitoring.** Deploy subgroup-aware metrics, calibration checks, and toxicity filters in production; log drifts and institute retraining/rollback thresholds.

5. **Documentation.** Ship a short "model card" summarizing intended uses, failure modes, and evaluation data so downstream teams can reason about fit-for-purpose decisions.

Contextual embeddings and transformers. Static embeddings assign a single vector per word type, so polysemous words such as "bank" cannot adapt to their context. Transformer-based language models (e.g., BERT; Devlin et al., 2019) compute token representations conditioned on the entire sentence via multi-head self-attention, allowing each occurrence to carry a context-specific vector. The techniques developed in this chapter remain useful for lightweight models and as initialization, but modern NLP pipelines increasingly fine-tune contextual models to capture sentence-level nuance.

Wrap-up

This chapter has developed word-embedding models from co-occurrence statistics, including the role of bias terms and optimization tools such as singular value decomposition. It also emphasized a practical constraint: embeddings inherit social and cultural structure from training corpora, so bias diagnosis and mitigation are part of model quality, not an optional add-on.

Key takeaways

- Word embeddings are dense vectors learned from co-occurrence statistics (local windows or global matrices).
- Analogies and clustering arise from linear geometry in the embedding space.
- Bias in corpora propagates to embeddings; debiasing and careful datasets are important.

Minimum viable mastery.

- Explain how co-occurrence statistics induce geometry (dot products, cosine similarity, and linear offsets).
- Distinguish local-window objectives from global matrix factorization views and state when each is a good approximation.
- Identify at least one concrete bias test and one mitigation strategy, and articulate their limitations.

Common pitfalls.

- Treating nearest neighbors as meaning rather than distributional evidence (polysemy and domain shift).
- Over-interpreting analogy accuracy without controlling for frequency and evaluation set construction.
- Applying debiasing as a post-hoc patch while ignoring corpus composition and labeling practices.

Exercises

Exercise 1 (Easy). Train a small skip-gram (negative sampling) embedding model on a modest corpus and audit the geometry: nearest neighbors, cosine similarity, and one analogy-style probe (even if it fails).
Deliverable: three neighbor lists + one short note explaining one neighbor that surprised you (and what it says about distributional meaning).

Exercise 2 (Medium). Compare two embedding objectives on the same corpus: a local-window method (word2vec-style) and a global method (GloVe-style or a matrix-factorization baseline). Focus on what changes *qualitatively*: which words move, which neighborhoods stabilize, and which analogies become more/less consistent.
Deliverable: one side-by-side neighborhood comparison + one paragraph explaining the difference as an objective/weighting effect.

Exercise 3 (Medium). Run one bias audit and treat it as an engineering check, not a headline: pick a simple test (e.g., WEAT-style, neighborhood bias, or projection tests), then try one mitigation (corpus filtering, re-weighting, or a debiasing transform) and report what it changes and what it does not.
Deliverable: the bias score(s) before/after + one paragraph stating a limitation of your chosen test.

If you are skipping ahead. You can treat embeddings as feature vectors for any downstream model, but the audit mindset matters: log your corpus choices and evaluation splits so that later "fairness" or calibration conclusions have context.

The fundamental limitation of static embeddings. Every embedding learned in this chapter assigns *one* vector per word type—the same vector for "bank" regardless of whether the surrounding tokens describe a river or a financial institution. This type-level representation is a deliberate simplification: it keeps training cheap and enables the geometric structure (offset analogies, co-occurrence geometry) discussed above. But it fails for polysemous words, and it makes the representation blind to syntactic role, clause scope, and long-range discourse context. Auditing nearest neighbors (§above) reveals this directly: the

ten closest words to "bank" are a mixture of financial and geographical senses, reflecting corpus frequency rather than in-context meaning.

Contextual representations: the path forward. The solution is to replace the lookup table with a *context-sensitive encoder*: a function that maps a *full token sequence* to one vector per position, not one vector per type. ELMo (Embeddings from Language Models) introduced this idea using a bi-directional stacked LSTM: the final token representations are weighted sums of all LSTM layers, capturing different levels of syntactic and semantic abstraction depending on the downstream task. BERT (Bidirectional Encoder Representations from Transformers; Devlin et al., 2019) replaced the recurrent backbone with the self-attention mechanism, enabling fully parallel computation over the whole sequence and scaling to much larger corpora and model sizes. Both models are pre-trained on large unlabeled corpora via self-supervised objectives (masked-language modelling for BERT; next/previous-token prediction for ELMo) and then *fine-tuned* on labelled downstream data—a two-stage pipeline that re-uses the expensive pre-training across many tasks. Chapter 14 opens precisely at the point where recurrent state is replaced by attention, making contextual representation practical and scalable. Keep the distributional intuitions of this chapter in view as you read: the geometry of token embeddings persists inside transformer models, but context is now injected via query–key–value attention rather than recurrence.

Where we head next. Chapter 14 keeps the same token geometry but changes context formation: attention replaces recurrent state, so dependencies are queried directly. The same many-to-one decisions from Section 12.10.2 remain, but computational path and scaling behavior change. Keep the same audit lens as you read: tokenization choices, masking correctness, and evaluation protocol matter as much as architecture. After that, Chapter 15 pivots to soft computing (fuzzy logic and evolutionary ideas) as an alternative paradigm for reasoning under uncertainty.

14 Transformers: Attention-Based Sequence Modeling

Chapter 12 made the sequence problem explicit: stateful computation plus gradients that must flow across time. Chapter 13 then supplied the representation layer that makes those sequence objectives practical. This chapter keeps those representations but loosens the "one step at a time" constraint: instead of marching through time, we let each position look around and pull in what it needs through attention.

Learning Outcomes

After this chapter, you should be able to:

- Explain the encoder–decoder bottleneck and how attention fixes it in sequence-to-sequence problems.
- Write scaled dot-product attention and multi-head attention, and interpret them as weighted averages.
- Distinguish self-attention from cross-attention and describe where each appears, including masking (padding/causal) and positional encodings.
- Describe a Transformer block (residual paths, layer norm, FFN) and the common objectives (MLM/CLM) behind the major families (BERT/GPT/encoder–decoder).

Design motif

Make information flow explicit. Attention is a controlled mixing operation; masks enforce which interactions are allowed; residual paths and normalization keep optimization stable as models deepen and context windows grow.

Author's note: models, world models, and language models (an opinionated lens)

A *model* predicts outcomes from captured information. In the chapters so far, that "outcome" ranged from a class label, to a time-step forecast, to a retrieved memory pattern.

A *world model*, as I use the phrase here, is aspirational: it is the idea of a general-purpose internal model that can represent situations and dynamics well enough to support planning, counterplanning, and "what-if" reasoning. Whether we can build such systems reliably is part of the larger research program.

Language models are a simplified and very constrained training interface to this aspiration: at training time the next action is a token predicted from previous tokens. My opinion is that this setup can *suggest* a kind of internal world structure (because the output must remain internally and externally consistent to be useful), but it does not *guarantee* that a faithful world model has formed. Fluency is not proof of understanding; it is a behavior you still have to audit.

14.1 From encoder–decoder bottlenecks to attention

Sequence-to-sequence (seq2seq) problems have a natural story: read an input sequence and produce an output sequence. Translation is the canonical example. In the classical encoder–decoder picture, the encoder reads the source sentence and produces a representation; the decoder then uses that representation to generate the target sentence token by token.

We will keep the term *token* from Chapter 13: a token is a discrete symbol index that gets mapped to a vector. In text it might be a word or subword; in other sequence problems it can be any event ID you embed into a feature vector.

The practical bottleneck is the fixed-vector squeeze. If the decoder is only given one summary vector, it is forced to carry *everything* about the source through time, even when the next output token only depends on a small part of the input. Attention is the engineering fix: at each decoding step, compute a weighted average of the encoder states and let that mixture act as the context for the next prediction. In other words, the decoder is allowed to look back at the encoder's "memory" and ask which source positions matter *right now*.

Transformers (Vaswani et al., 2017) take that idea seriously and push it further. They remove recurrence and make "look around and mix" the core operation inside a layer. This allows parallel computation across positions and makes long-range interactions a first-class design choice rather than a side effect of how well information survives through many recurrent steps.

Seq2seq with attention (cross-attention). In an encoder–decoder RNN, the encoder produces a sequence of hidden states $\{\mathbf{h}_j\}_{j=1}^{S}$ from the input $\{\mathbf{x}_j\}$, and the decoder produces a sequence of states $\{\mathbf{s}_t\}_{t=1}^{T}$ while generating the output $\{y_t\}$. A convenient probabilistic view of translation is

$$p(\mathbf{y} \mid \mathbf{x}) = \prod_{t=1}^{T} p(y_t \mid y_{<t}, \mathbf{x}) \,. \tag{14.1}$$

The "one-vector bottleneck" appears when the decoder is forced to rely on a single summary of the entire input. Attention replaces that with a *per-step context*:

$$\begin{aligned} e_{tj} &= a(\mathbf{s}_{t-1}, \mathbf{h}_j), \\ \alpha_{tj} &= \mathrm{softmax}_j(e_{tj}), \\ \mathbf{c}_t &= \sum_{j=1}^{S} \alpha_{tj}\, \mathbf{h}_j. \end{aligned} \tag{14.2}$$

Here $\mathrm{softmax}_j$ denotes a softmax over the source positions j; explicitly,

$$\alpha_{tj} = \frac{\exp(e_{tj})}{\sum_{j'=1}^{S} \exp\big(e_{tj'}\big)}.$$

This normalization makes the weights nonnegative and ensures they sum to one across j. The scoring function $a(\cdot,\cdot)$ can be a dot product, a bilinear form, or a small MLP, and $\mathbf{c}_t$ is the context the decoder uses when predicting y_t. This style of encoder–decoder attention traces back to early neural machine translation work (Bahdanau et al., 2015). That is the core move: different output positions can consult different parts of the source. In modern terminology this is *cross-attention*: queries come from the decoder state, while keys and values come from the encoder states. This is the bridge from encoder–decoder RNNs to

Transformers: the attention math stays; what changes is that Transformers build the states $\mathbf{h}_j$ and $\mathbf{s}_t$ without recurrence. This is distinct from decoder self-attention, where queries, keys, and values all come from the decoder sequence itself.

14.2 Scaled Dot-Product Attention

Author's note: attention is a weighted average

RNNs and gated cells give you a way to carry *memory* forward, but memory is not the same thing as *relevance*. When you read a page, some words are glue and some words carry the meaning you need right now; your mind does not treat the whole history as equally important at every step. I like to read attention as the engineering version of relevance: a weighted retrieval operator, not a mystical reasoning module. A query asks what is needed, keys advertise what each candidate is about, and values carry the content that gets mixed. Similarity scores between queries and keys become nonnegative weights that sum to one; the output is the weighted sum of the value vectors.

A tiny analogy: keys, values, and weighted retrieval

Think of a tiny "database" with keys $\{70, 80\}$ and values $\{1000, 1500\}$. If your query is exactly 70, you retrieve 1000. If your query is 75, there is no exact match, so you can do a soft retrieval: score each key by a similarity such as

$$s_i = \frac{1}{|k_i - q| + \epsilon},$$

then normalize those scores so they sum to one and use them as weights on the values. Here $|70 - 75| = |80 - 75| = 5$, so $s_1 \approx s_2 \approx 0.2$, the normalized weights are $(0.5, 0.5)$, and the weighted average gives $0.5 \cdot 1000 + 0.5 \cdot 1500 = 1250$.

This is only an intuition pump: Transformers do not use scalar keys like "70." They learn vector keys and queries and score them in a learned feature space. But the weighted-average mechanism is exactly the same.

At the per-position level, the computation is

$$\mathbf{z}_i = \sum_j \alpha_{ij}\, \mathbf{v}_j, \qquad \alpha_{ij} \geq 0, \quad \sum_j \alpha_{ij} = 1,$$

Given query, key, value matrices $\mathbf{Q} \in \mathbb{R}^{n_q \times d_k}$, $\mathbf{K} \in \mathbb{R}^{n_k \times d_k}$, and $\mathbf{V} \in \mathbb{R}^{n_k \times d_v}$, the basic attention operation is

$$\mathrm{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}. \tag{14.3}$$

Here n_q is the sequence length of the queries and n_k the sequence length of keys/values. We keep the same sequence-first convention used in Chapters 12 and 13: rows index time positions (a "token dimension") and columns index features, while batch elements are processed independently. The $1/\sqrt{d_k}$ factor stabilizes gradients by keeping logits in a reasonable range.

It is often helpful to name the attention-weight matrix

$$\mathbf{A} = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right),$$

where the softmax is applied *row-wise* (each query position gets a distribution over keys). Then the output is simply $\mathbf{Z} = \mathbf{A}\mathbf{V}$. The whole pipeline is differentiable, so the projection matrices that create Q, K, V can be learned by backpropagation.

Where do Q, K, V come from? Start from token vectors stacked into a matrix $\mathbf{X} \in \mathbb{R}^{n \times d_{\text{model}}}$ (one row per position). For a *single head*, learned projections produce

$$\mathbf{Q} = \mathbf{X}\mathbf{W}^Q, \qquad \mathbf{K} = \mathbf{X}\mathbf{W}^K, \qquad \mathbf{V} = \mathbf{X}\mathbf{W}^V, \tag{14.4}$$

with $\mathbf{W}^Q, \mathbf{W}^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$ and $\mathbf{W}^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$. In self-attention, $\mathbf{X}$ is the same sequence for all three. In cross-attention (seq2seq), $\mathbf{Q}$ typically comes from decoder states while $\mathbf{K}, \mathbf{V}$ come from encoder states; the math is unchanged. Multi-head attention simply runs this projection-and-attend pattern several times in parallel with separate $\mathbf{W}_i^Q, \mathbf{W}_i^K, \mathbf{W}_i^V$.

Worked example: 2-token causal self-attention (one head)

For this toy calculation we bypass the learned projection matrices and set $Q = K = V = \mathbf{I}_2$. In a trained Transformer, separate learned linear maps produce Q, K, and V from the token embeddings. Let $d_k = d_v = 2$. Without masking, the scaled scores are

$$S = \frac{QK^\top}{\sqrt{d_k}} = \frac{1}{\sqrt{2}}\mathbf{I}_2.$$

A *causal mask* forbids looking into the future: for query index i, mask out all keys with index $j > i$ by setting those logits to $-\infty$ (so they vanish after the softmax).
Row 1 can only attend to itself, so $\alpha_{1\cdot} = [1, 0]$. Row 2 sees logits $[0,\ 1/\sqrt{2}]$, so

$$\begin{aligned} u &= e^{1/\sqrt{2}}, \\ \alpha_{2\cdot} &= \mathrm{softmax}\Big([0,\ 1/\sqrt{2}]\Big) = \left[\frac{1}{1+u},\ \frac{u}{1+u}\right] \\ &\approx [0.330238,\ 0.669762]. \end{aligned}$$

Because $V = \mathbf{I}_2$, the attention output equals the weight matrix: $\mathrm{Attn}(Q, K, V) = \alpha$.

Shape ledger

We treat mini-batches as $\mathbf{X} \in \mathbb{R}^{B\times n\times d_{\text{model}}}$ (batch, sequence, features), and h always denotes the number of attention heads. After the linear projections each head carries $\mathbf{Q}, \mathbf{K} \in \mathbb{R}^{B\times h\times n\times d_k}$, $\mathbf{V} \in \mathbb{R}^{B\times h\times n\times d_v}$, and the attention weights live in $\mathbb{R}^{B\times h\times n\times n}$. Reading dimensions in this order avoids confusion when mixing frameworks; $h \cdot d_k = d_{\text{model}}$ (often $d_v = d_k$). FFN inner widths typically 2–4$\times d_{\text{model}}$.

Complexity and memory

Naive attention is $O(n^2 d_{\text{model}})$ compute and $O(n^2)$ memory per head/layer for the attention map; this dominates long sequences. FlashAttention reduces activation I/O but keeps the quadratic arithmetic; sparse/linear variants reduce the n^2 factor by trading exactness for structure (see Longformer/BigBird/Reformer/Performer/Linformer). Causal/padding masks do not change complexity, only which entries participate.

14.3 Self-attention vs. cross-attention

The same equations serve two distinct roles. *Self-attention* means Q, K, V come from the same sequence. This is the retrieval story turned inward: each position forms a query and pulls a weighted average over values from the whole sequence, producing a *contextualized* representation of that token. In decoder-only generation, self-attention is typically *causal*: the mask enforces that position t can only use positions $\leq t$.

This is why the same surface form can behave differently depending on what surrounds it. In the phrase "red flag," the useful information is often the idiom (a warning sign), not the literal color red or a physical flag. In "red coat," red is literal. Self-attention gives the model a mechanism to build token representations that reflect these different roles by mixing in different neighbors with different weights. Self-attention by itself does not know what "before" and "after" mean: without positional information the operation is permutation-equivariant. That is why we explicitly add positional encodings in Section 14.5.

Cross-attention is the seq2seq bridge. Here the queries come from the decoder states, but the keys and values come from the encoder outputs. In translation terms: each output position asks a question (query) and then pulls a weighted average over the source-side memory (keys/values) to decide what to emit next.

14.4 Multi-Head Attention (MHA)

One attention head gives you one similarity space. Multi-head attention gives you several in parallel: each head learns its own projections and can focus on different relations at the same time (local vs. global cues, syntactic vs. semantic signals, or different parts of an image-like grid). You should not read heads as guaranteed "modules" with fixed roles; the point is capacity and parallel views

under the same weighted-average mechanism.

One intuition I find useful in language is that several relations matter at once. Some languages encode gender; some emphasize agreement; and word order can differ substantially across languages (subject–verb–object versus other patterns). During translation you may need to track several of these constraints simultaneously while still resolving local collocations. Multi-head attention gives the model multiple learned "views" at the same time: each head has its own projection matrices and can compute a different attention pattern over the same sequence.

Multiple heads attend in parallel after learned linear projections:

$$\begin{aligned} \text{head}_i &= \text{Attn}(\mathbf{X}\mathbf{W}_i^Q,\, \mathbf{X}\mathbf{W}_i^K,\, \mathbf{X}\mathbf{W}_i^V), \\ \text{MHA}(\mathbf{X}) &= [\text{head}_1; \ldots; \text{head}_h]\, \mathbf{W}^O, \end{aligned} \tag{14.5}$$

with $\mathbf{W}_i^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $\mathbf{W}_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $\mathbf{W}_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$, and output projection $\mathbf{W}^O \in \mathbb{R}^{(hd_v) \times d_{\text{model}}}$. For cross-attention, the formula is the same but keys/values come from encoder states: $\text{Attn}(\mathbf{X}_{\text{dec}}\mathbf{W}_i^Q,\, \mathbf{X}_{\text{enc}}\mathbf{W}_i^K,\, \mathbf{X}_{\text{enc}}\mathbf{W}_i^V)$. Figure 63 bundles scaled dot-product attention, multi-head concatenation, and the residual pre-LN block so the entire signal path is visible at a glance.

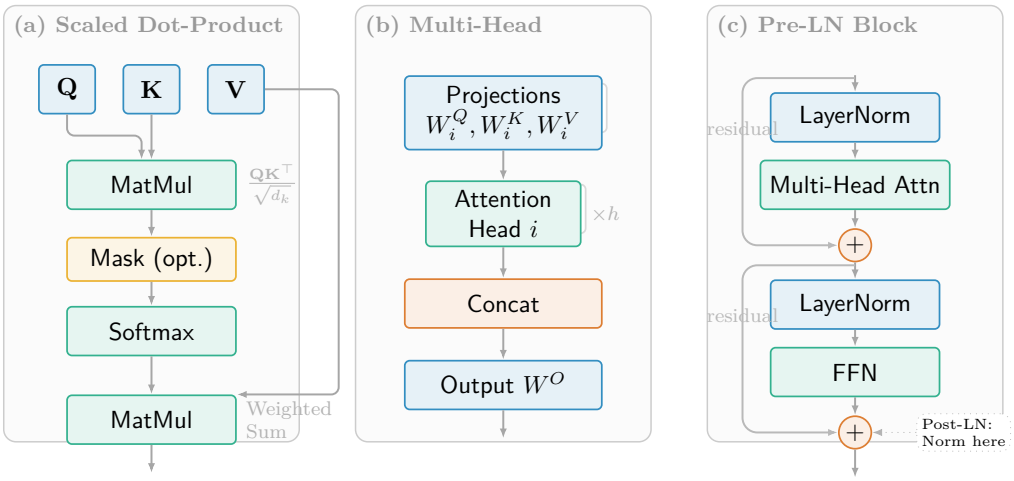


Figure 63: Reference schematic for the Transformer. Left: scaled dot-product attention. Center: multi-head concatenation with an output projection. Right: pre-LN encoder block combining attention, FFN, and residual connections; a post-LN variant simply moves each LayerNorm after its residual add (dotted alternative, not shown).

Figure 64 provides a quick visual summary of positional encoding, key/value (KV) cache reuse, and Low-Rank Adaptation (LoRA): a parameter-efficient tuning trick where the base weights stay fixed and you learn a small low-rank update.

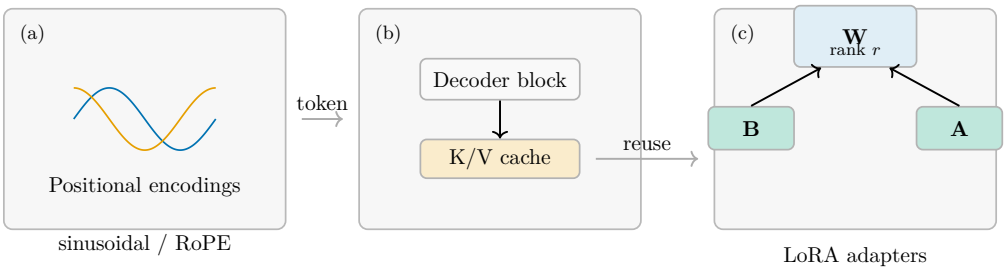


Figure 64: Transformer micro-views. Left: positional encodings (sinusoidal/rotary) add order information. Center: KV cache stores past keys/values so decoding a new token reuses prior context. Right: LoRA freezes a base weight matrix $\mathbf{W}$ and trains a low-rank update $\Delta\mathbf{W} = \mathbf{BA}$ for parameter-efficient tuning.

14.5 Positional encodings (order without recurrence)

Transformers lack recurrence, so order has to be injected explicitly. A simple baseline (from Vaswani et al. (2017)) is a sinusoidal positional encoding: for position pos and feature index i,

$$\begin{aligned} \mathrm{PE}(\mathrm{pos},\, 2i) &= \sin\!\left(\frac{\mathrm{pos}}{10000^{2i/d_{\mathrm{model}}}}\right), \\ \mathrm{PE}(\mathrm{pos},\, 2i+1) &= \cos\!\left(\frac{\mathrm{pos}}{10000^{2i/d_{\mathrm{model}}}}\right). \end{aligned} \tag{14.6}$$

You add this vector (or a learned alternative) to the token embedding at each position. The engineering point is not the specific constant; it is that different dimensions oscillate at different frequencies, so nearby positions look similar in some coordinates and far positions look different in others.

A tempting baseline is to represent the position as an ID (or a one-hot vector) and treat it like any other lookup. That *can* work as a table, but it is a poor fit for how attention does its comparisons. One-hot positions are orthogonal: under dot products they only tell you "same position" versus "different position," not whether two positions are one step apart or fifty steps apart. A fixed table also has a hard maximum length: if you trained up to $n = 512$ and you now see position 513, you must extend (and usually re-train) the lookup.

Sinusoidal encodings bake in a gentle geometry. For one frequency ω, the

sin/cos pair satisfies

$$\begin{bmatrix} \sin(\omega(\text{pos}+\Delta)) \\ \cos(\omega(\text{pos}+\Delta)) \end{bmatrix} = \begin{bmatrix} \cos(\omega\Delta) & \sin(\omega\Delta) \\ -\sin(\omega\Delta) & \cos(\omega\Delta) \end{bmatrix} \begin{bmatrix} \sin(\omega\,\text{pos}) \\ \cos(\omega\,\text{pos}) \end{bmatrix},$$

so a fixed offset Δ acts like a rotation in that 2D subspace. Across many frequencies, this becomes a multi-scale position code: low-frequency terms vary slowly (coarse, long-range cues) while high-frequency terms vary quickly (fine, local cues). Because attention is built on dot products and similarities, that structure gives the model a usable notion of "near" and "far" without having to relearn distance from scratch, and it lets you evaluate $\text{PE}(\text{pos}, \cdot)$ beyond the training window.

You can view the input to the first block as

$$\mathbf{H}_0 = \text{Embed}(\text{tokens}) + \text{PosEnc}(\text{positions}),$$

where both terms produce vectors in $\mathbb{R}^{d_{\text{model}}}$ and the sum is taken position-wise.

In modern practice you will also see learned positional embeddings, relative-position schemes, and rotary position embeddings (RoPE). The chapter keeps sinusoidal PE as the clean reference, and we treat other variants as drop-in replacements that mainly change how well models extrapolate to longer contexts or shifting windows.

14.6 Objectives and masks

Two masks show up so often that it is worth naming them early. A *causal mask* forbids attention to future positions; this is what makes next-token generation well-defined. A *padding mask* forbids attention to padded positions inserted only to make batches rectangular. Both are easy to get wrong: a single missing mask can leak information from the future or silently change what the model is allowed to use. See Figure 65 for a concrete picture of both patterns (queries on rows, keys on columns).

In implementation, masks are usually applied additively to attention logits

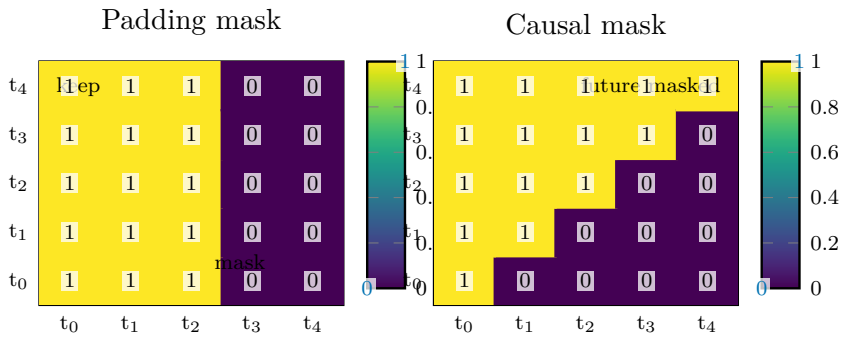


Figure 65: Attention masks as heatmaps (queries on rows, keys on columns). Left: padding mask zeroes out attention to padded *key columns* of a shorter sequence in a packed batch; padded-query rows are typically ignored downstream by output/loss masking. Right: causal mask enforces autoregressive flow by blocking attention to future tokens.

before softmax:

$$\mathbf{S}_{\text{mask}} = \frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} + \mathbf{M}, \qquad M_{ij} = \begin{cases} 0, & \text{if key } j \text{ is visible to query } i, \\ -\infty, & \text{if key } j \text{ is masked.} \end{cases} \tag{14.7}$$

Then $\mathbf{A} = \text{softmax}(\mathbf{S}_{\text{mask}})$ row-wise and $\mathbf{Z} = \mathbf{A}\mathbf{V}$, so masked positions contribute zero probability.

One subtle point is *why* the causal mask is needed during training. At inference time the future does not exist, so a decoder cannot look ahead even if it wanted to. But during training we often feed the full target sequence (teacher forcing: provide the ground-truth previous tokens) so we can compute a loss at every position. Without a causal mask, the decoder would "cheat" by attending to future target tokens. The causal mask enforces the same information constraint at training time that you will have at decoding time. This connects directly to KV caching in decoder inference: with one-token-at-a-time decoding, the cache only stores past keys/values, so there are no future columns to attend to; during training (full sequence in parallel), the explicit causal mask is still required.

The training objective is usually a cross-entropy (CE) loss, i.e., a negative log-likelihood (NLL) of the correct label under the model's predicted distribution. In sequence modeling, you typically sum (or average) that loss across time positions

and across batch elements.

We will return to the main Transformer families (encoder-only vs. decoder-only vs. encoder–decoder) after we write down the block structure in Section 14.7. The core idea is simple: families differ mostly by which masks they apply and which prediction problem they are trained on, not by a different attention mechanism.

14.7 Transformer blocks and training stabilizers

Transformer blocks are built from the same few ingredients repeated many times: attention, a small feedforward network, and stability glue (residual connections, normalization, dropout). Residual connections keep a direct path for information and gradients: the input to a sublayer is added back to its output so the model can refine a representation without destroying it. LayerNorm stabilizes scale by normalizing each token vector across its feature dimension; unlike BatchNorm it does not depend on batch statistics, which matters for variable-length sequences and small batches.

In the *pre-LN* layout (common in modern implementations), each sublayer sees a normalized input and then its output is added back to the residual stream:

$$\begin{aligned} \mathbf{H}_1 &= \mathbf{H} + \mathrm{MHA}(\mathrm{LayerNorm}(\mathbf{H})), \\ \mathbf{H}_{\mathrm{out}} &= \mathbf{H}_1 + \mathrm{FFN}(\mathrm{LayerNorm}(\mathbf{H}_1)). \end{aligned} \tag{14.8}$$

The original Transformer applied LayerNorm after each residual add (post-LN); the attention and FFN computations are the same, only the placement of normalization changes.

The feedforward sublayer (FFN) is applied independently to each position (row). In matrix form it is typically two linear layers with a nonlinearity:

$$\mathrm{FFN}(\mathbf{H}) = \sigma(\mathbf{H}\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2, \tag{14.9}$$

where σ is commonly ReLU (original paper) or GELU (many modern stacks), $\mathbf{W}_1 \in \mathbb{R}^{d_{\mathrm{model}} \times d_{\mathrm{ff}}}$, $\mathbf{W}_2 \in \mathbb{R}^{d_{\mathrm{ff}} \times d_{\mathrm{model}}}$, and the biases are broadcast across positions. Dropout and label smoothing are common.

Author's note: attention mixes; the FFN refines (two different jobs)

Self-attention is where tokens "talk" to each other. It is the part of the block that *mixes across positions*: each token forms a query, assigns weights over other tokens' keys, and pulls a weighted average of their values.
The feedforward sublayer is different. It is applied *independently* to each token: the same small MLP processes every position, taking whatever the attention step mixed in and turning it into a cleaner set of features for the next layer. If you like metaphors, I think of attention as the class discussion and the FFN as the private tutor: the discussion lets you hear from others; the tutor helps you rewrite your notes in a form you can use.

Decoder blocks: masked self-attention plus cross-attention. Encoder blocks use self-attention (typically with a padding mask) plus an FFN. Decoder blocks add a second attention step: first *masked* self-attention over the partially generated target (causal mask), then cross-attention into the encoder outputs. In that cross-attention, the query comes from the decoder stream while the keys and values come from the encoder stream. It is the same weighted-average mechanism, but now the memory lives on the source side and the questions live on the target side.

Implementation sketch: block pseudocode, training defaults, and one step

Block pseudocode (pre-LN).

```
function Block(H, mask=None):
    # Pre-normalize inputs (pre-LN stabilizes deep stacks)
    H_norm = LayerNorm(H)
    attn = MHA(H_norm, H_norm, H_norm, mask)
    H = H + Dropout(attn)
    H_norm = LayerNorm(H)
    ff = FFN(H_norm)
    return H + Dropout(ff)
```

Decoder blocks add a causal mask on self-attention, and encoder–decoder

variants also insert cross-attention (queries from decoder; keys/values from encoder outputs). Pre-LN (shown here) is now common because it keeps gradients well behaved for very deep stacks; post-LN (original Transformer) is still used in smaller models. **Training defaults (decoder-only, practical baseline).** AdamW with cosine decay and 1–3% warmup; LR $\sim 10^{-3}$ for small models, $1\text{–}2 \times 10^{-4}$ for mid-size. Weight decay ≈ 0.01 (exclude biases/LayerNorm gains). Attention/MLP dropout ≈ 0.1; clip global norm to 1.0. Mixed precision (FP16/BF16) plus gradient checkpointing for long contexts; tie input embeddings to the LM head; use causal masks for CLM and padding masks for packed batches.

One training step (decoder-only, causal mask).

```
x = tokenizer(batch_text)                # [B, T]
mask = causal_mask(x)                    # [B, 1, T, T]
h = embed(x) + pos(x)                    # [B, T, d_model]
for block in blocks:
    h = block(h, mask)                   # pre-LN MHA + FFN
logits = lm_head(h)                      # [B, T, vocab]
loss = CE(logits[:, :-1], x[:, 1:])      # next-token
loss.backward()
clip_grad_norm_(model.parameters(), 1.0)
opt.step(); opt.zero_grad()
```

At inference, reuse cached K/V states (see Figure 64 and Section 14.14).

KV cache in plain terms (decoder inference). At generation step t, the decoder computes one new query vector $\mathbf{q}_t$. The old keys/values from steps $1, \ldots, t-1$ are already stored in memory, so you do *not* recompute them. You only compute $\mathbf{k}_t, \mathbf{v}_t$, append them to the cache, and attend over the full cached set:

$$\mathbf{z}_t = \mathrm{softmax}\left(\frac{\mathbf{q}_t \mathbf{K}_{1:t}^\top}{\sqrt{d_k}}\right) \mathbf{V}_{1:t}, \tag{14.10}$$

where $\mathbf{K}_{1:t}$ and $\mathbf{V}_{1:t}$ are the matrices formed by stacking all cached keys/values up to step t. The engineering tradeoff is simple: bigger context means bigger

cache. So context size is not only a modeling choice; it is also a memory/latency budget choice at inference time.

Tokenization note from a GPT-2-style implementation (why students should care)

The nanochat reference implementation reinforces four practical points:

- GPT-style tokenizers are often *byte-level BPE*: this avoids unknown-token failures on odd characters and mixed text.
- A begin-of-sequence delimiter is treated as a real token (often named `<|bos|>` or historically `<|endoftext|>`).
- The same tokenizer must be used at train and inference time; changing it changes token IDs and breaks model behavior.
- Tokenization affects effective context usage: if text splits into fewer tokens, the same context window can carry more content.

For readers who are not programming-focused: think of tokenization as the model's "measurement scale." If you change the scale, all learned weights are calibrated to the wrong units. For a compact, end-to-end study repo (tokenizer through serving), nanochat is a good example to browse. (Karpathy, 2025)

Code–math dictionary. In code blocks, `x` is the token-index tensor (input IDs), `h` is the token-representation array $\mathbf{H}$, `mask` is the attention mask, and `embed(x)` denotes an embedding lookup into the learned table $\mathbf{W}$ (written algebraically as a row-selection, e.g., $\mathbf{W}[w_t]$, in Chapter 13).

14.8 BERT vs. GPT vs. encoder–decoder

Once you have attention blocks, most of the "model family" distinctions come from two choices: (i) which directions a position is allowed to attend to (masking), and (ii) what prediction problem you train on. Encoder-only stacks are built to produce strong *representations*; decoder-only stacks are built to *generate*; encoder–decoder stacks are built to condition one sequence on another (translation is the clean example).

BERT: encoder-only, bidirectional self-attention, masked-token prediction. BERT (Devlin et al., 2019) is a stack of Transformer *encoders*. There is no causal mask: each token can attend to both left and right context, so the representation at position i can use the whole sentence. In practice you still apply a padding mask when you batch variable-length sequences; "bidirectional" means "not restricted to past-only," not "no masking ever." Pre-training uses *masked language modeling* (MLM): choose a subset of token positions $\mathcal{M}$ and ask the model to reconstruct the original tokens from a corrupted input. If $w_{1:n}$ are the original token IDs and $\tilde{w}_{1:n}$ is the corrupted sequence (where some positions were replaced by `[MASK]` or other substitutions), a convenient loss view is

$$\mathcal{L}_{\text{MLM}} = -\sum_{i \in \mathcal{M}} \log p_\theta(w_i \mid \tilde{w}_{1:n}) \,, \tag{14.11}$$

where the model produces a vocabulary distribution at each position and we only score the masked positions. Because masked tokens are predicted using *both* left and right context, MLM does not train a left-to-right next-token factorization. This is why BERT is usually used as an encoder (representations + classifiers) rather than as a free-form decoder without additional machinery. The original BERT recipe masked about 15% of tokens and used a pragmatic corruption mix: most masked positions are replaced by `[MASK]`, some by a random token, and some are left unchanged so the model does not overfit to seeing `[MASK]` at fine-tuning time (Devlin et al., 2019).

Author's note: Word2Vec vs. BERT (the intuition that still helps)

Word2Vec (Mikolov et al., 2013) taught us a useful lesson: meaning leaks into geometry through co-occurrence. If you repeatedly see certain words together, a vector model can pick that up and cluster them.

I like to see BERT as a more powerful version of the same distributional idea. The training question is still "can you predict a missing piece from what surrounds it?" The difference is that the representation is no longer one vector *per vocabulary item*. BERT produces a *contextual* vector per token occurrence: the vector for "bank" in "river bank" is not forced to match the vector for "bank" in "bank account," because the surrounding tokens are part of the computation.

`[CLS]` as a learned pooling token. BERT-style models typically prepend a special `[CLS]` token and (for sentence pairs) separate segments with `[SEP]`. In the original setup, a small *token-type* (segment) embedding marks whether each token came from sentence A or sentence B (Devlin et al., 2019). Because `[CLS]` participates in the same self-attention as every other token, its representation vector is repeatedly updated as the model mixes information across the sequence. In practice, the final-layer `[CLS]` vector is treated as a learned, task-adaptable summary and is fed to a small classifier head for sentence-level tasks.

BERT inputs in one line (token + position + segment)

At position i, BERT forms an input vector by summing three components (Devlin et al., 2019):

$$\mathbf{x}_i = \mathbf{e}_{\text{tok}}(w_i) + \mathbf{e}_{\text{pos}}(i) + \mathbf{e}_{\text{seg}}(s_i),$$

where w_i is the token ID, i is the position index, and $s_i \in \{\text{A}, \text{B}\}$ indicates which sentence/segment the token belongs to (for single-sentence inputs, s_i is constant).

The original BERT paper also used an auxiliary "next sentence prediction" (NSP) objective for sentence pairs; later variants often drop it. For our purposes, the core encoder-only story is: bidirectional self-attention plus an objective that forces representations to be predictive of missing content.

GPT: decoder-only, causal self-attention, next-token prediction. GPT-style models are stacks of Transformer *decoders* with the cross-attention removed: they use masked (causal) self-attention plus an FFN in each block. The causal mask enforces the autoregressive factorization:

$$\begin{aligned} p_\theta(w_{1:n}) &= \prod_{t=1}^{n-1} p_\theta(w_{t+1} \mid w_{1:t}) \,, \\ \mathcal{L}_{\text{CLM}} &= -\sum_{t=1}^{n-1} \log p_\theta(w_{t+1} \mid w_{1:t}) \,. \end{aligned} \tag{14.12}$$

The training objective (causal language modeling, CLM) is the corresponding negative log-likelihood in (14.12). This is where it is easy to mix up training and

generation. During training you do *not* feed back the model's sampled token at each step; you feed the true prefix (teacher forcing) and compute the loss for all positions in parallel under the causal mask (as in the code block earlier). During inference you do sample/argmax and append one token at a time, and that is where decoding settings (temperature, top-k, top-p) matter; see Section 14.11.

Historically, GPT-1 used a 12-layer decoder-only stack with about 117M parameters and showed that simple next-token pre-training can transfer to many downstream tasks with surprisingly little architectural customization (Radford et al., 2018). GPT-2 and later work scaled the same basic recipe with larger models and broader data (Radford et al., 2019; Brown et al., 2020).

Encoder–decoder (seq2seq): conditional generation via cross-attention. Encoder–decoder Transformers keep the clean translation structure: the encoder reads the source; the decoder generates the target with causal self-attention plus cross-attention into the encoder outputs. This remains a natural fit whenever the output is explicitly conditioned on an input sequence (translation, summarization, structured prediction). Text-to-text models such as T5 make this conditioning view explicit by casting many tasks into a single seq2seq format (Raffel et al., 2020).

14.9 Long Contexts and Efficient Attention

Memory and compute scale quadratically with sequence length. Practical systems therefore mix several tricks:

- **Sparse or local attention** (e.g., Longformer, BigBird) to limit each query to a sliding or block-sparse neighborhood.
- **Low-rank/kernelized approximations** and recurrent chunking (Performer, Transformer-XL) so that computation/storage grows roughly linearly in context length.
- **I/O-aware kernels** such as FlashAttention that stream tiles through SRAM so the $O(n^2)$ attention computation remains exact while memory stays manageable.

Relative/rotary position schemes and KV caching are summarized in Section 14.14 and Figure 64.

14.10 Fine-Tuning and Parameter-Efficient Adaptation

Pre-training gives you a general-purpose language model; fine-tuning adapts that model to a task, domain, or interaction style. *Full fine-tuning* updates all weights, which can yield the best performance when you have enough high-quality data and a stable objective, but it is also the easiest to overfit or to accidentally "forget" useful general behavior.

Parameter-efficient fine-tuning (PEFT) methods (LoRA, adapters, prefix/prompt tuning, and related variants) inject small trainable modules while freezing most of the base model, enabling rapid adaptation with lower memory and more predictable changes. In LoRA, you freeze a weight matrix $\mathbf{W}$ and learn a low-rank update $\Delta\mathbf{W} = \mathbf{BA}$ (rank r), so only a small number of parameters move during tuning. These methods are attractive when you want many task-specific variants of a shared base model, or when you need to keep the base weights fixed for deployment and auditing.

Audit hooks for adaptation. Regardless of whether you update all weights or only a small adapter, treat fine-tuning like any other ERM pipeline: keep a held-out evaluation set that matches the deployment slice, monitor calibration and failure modes (not just loss), and log the exact base checkpoint, tokenizer, and data snapshot so results are reproducible. When fine-tuning for instruction-following or conversational behavior, add explicit tests for regressions (refusals, hallucinations on factual probes, and brittleness to prompt variants) rather than relying on a single aggregate score.

14.11 Decoding and Evaluation

Training produces a distribution over the next token; decoding turns that distribution into an actual sequence. For decoder-only models, decoding runs autoregressively: predict $p_\theta(x_{t+1} \mid x_{1:t})$, choose a token, append it, and repeat. This is also where the KV cache matters (see Figure 64): you reuse past keys/values so generating one more token does not require recomputing attention over the entire prefix from scratch.

From token representations to a next-token distribution. At position t, the final block produces a token representation $\mathbf{h}_t \in \mathbb{R}^{d_{\text{model}}}$. A language model

head is a linear map into the vocabulary $\mathcal{V}$:

$$\boldsymbol{\ell}_t = \mathbf{h}_t \mathbf{W}_{\text{out}} + \mathbf{b}_{\text{out}}, \qquad \boldsymbol{\ell}_t \in \mathbb{R}^{|\mathcal{V}|}, \tag{14.13}$$

where $\boldsymbol{\ell}_t$ are the logits (unnormalized scores). Softmax converts logits to probabilities, $p_\theta(x_{t+1} = v \mid x_{1:t}) = \text{softmax}(\boldsymbol{\ell}_t)_v$, and the cross-entropy (CE) loss trains the model to put mass on the observed next token.

Temperature and sampling (how you choose a token). Temperature rescales logits before softmax:

$$p_T(v \mid x_{1:t}) = \text{softmax}(\boldsymbol{\ell}_t/T)_v\,, \qquad T > 0. \tag{14.14}$$

Small T makes the distribution sharper (more deterministic); large T makes it flatter (more random). The limit $T \to 0$ approaches greedy argmax behavior; the limit $T \to \infty$ approaches a uniform distribution. Temperature $T = 0$ is not defined mathematically; in code it is simply implemented as "take arg max." A common sampling implementation is the roulette-wheel (inverse-CDF) rule: draw $u \sim \text{Uniform}(0, 1)$, accumulate the probabilities until the cumulative sum exceeds u, and pick that token. In practice you often combine this with top-k or nucleus (top-p) truncation: restrict to a shortlist of likely tokens, renormalize, then sample.

Greedy decoding takes the argmax at each step; it is fast and often strong for short, factual completions, but it can get stuck in repetitive loops. Beam search keeps multiple partial hypotheses; it can improve likelihood but sometimes harms perceived quality in open-ended generation. Sampling (often with top-k or top-p truncation) trades certainty for diversity; pick it deliberately and log the settings.

For evaluation, perplexity summarizes next-token performance for language modeling (see Chapter 13), but it does not tell you whether generations are useful, safe, or faithful. For downstream classification, prefer metrics that match the deployment slice (e.g., AUPRC for imbalanced problems) and keep decoding settings logged alongside checkpoints so results are reproducible.

14.12 Audit and failure modes (short list)

Audit and failure modes (engineering view)

- **Masking bugs:** missing/incorrect causal masks can leak future tokens; missing padding masks can let padding dominate attention in batched training.
- **Train/inference mismatch:** teacher forcing during training does not automatically tell you how errors compound during decoding; test the decoding strategy you plan to ship.
- **Long-context degradation:** attention enables long-range access, but quality can still decay with length; measure how performance changes as you increase context.
- **Calibration vs. correctness:** high probability is not a guarantee of correctness; audit reliability on slices and stress tests, not just average loss.
- **Reproducibility:** tokenizer choices, data filters, and decoding hyperparameters can swing results; log them as part of the experiment.

14.13 Alignment (Brief)

Post-training *alignment* shapes model behavior to match human preferences, safety constraints, and interaction norms. In broad terms, alignment objectives do not change the Transformer mechanics; they change what you reward during optimization (and therefore what behaviors are reinforced).

RLHF optimizes a policy against a learned reward model (with careful regularization to avoid drifting too far from the base model). Preference-based objectives such as DPO, KTO, or ORPO optimize directly from ranked pairs without a full reinforcement-learning loop.

Alignment is not a proof of correctness. Alignment can improve helpfulness and reduce obvious failures, but it can also introduce new ones (reward hacking, over-refusal, brittleness to prompt phrasing, or degraded performance off-distribution). Treat it as an engineering stage with explicit test suites and

logging: evaluate on held-out tasks, check calibration and refusal behavior, and track changes relative to the pre-alignment model.

14.14 Advanced notes (optional)

- **Relative/rotary positions.** RoPE (Su et al., 2021) encodes position by rotating query and key vectors rather than adding a fixed embedding to the token representation. Concretely, for a position m the d-dimensional query is partitioned into $d/2$ pairs of dimensions; each pair (q_{2i}, q_{2i+1}) is rotated by angle $m\theta_i$ where $\theta_i = 10000^{-2i/d}$. Because the dot product of a rotated query at position m with a rotated key at position n depends only on the *relative* displacement $m - n$ (by the rotation subtraction identity), the model can attend to relative distance without explicitly computing it. This property allows trained RoPE models to extrapolate better to sequence lengths beyond those seen during training, which is why nearly all modern large-language-model families (LLaMA, Mistral, Qwen) adopt RoPE as their default positional scheme. ALiBi (Press et al., 2022) takes a simpler approach: it subtracts a linear bias proportional to token distance from attention logits, requiring no change to embeddings; the trade-off is that absolute tables keep fixed anchors useful for classification tokens while rotary/relative schemes favour extrapolation and smoothly sliding context windows.

- **KV-cache management.** Decoder-only inference stores per-layer key/-value tensors; chunked caching, paged attention, and sliding windows keep memory linear in context length. Speculative decoding and assisted decoding reuse a lightweight draft model to propose tokens that the full model verifies before committing.

- **Efficient kernels.** FlashAttention (Dao et al., 2022) computes attention blocks in streaming tiles to keep activations in SRAM. Long-context variants mix windowed attention, recurrent memory, or low-rank adapters; state-space models such as Mamba (Gu et al., 2023) provide linear-time alternatives that back-propagate through implicitly defined kernels.

- **Mixture-of-experts and routing.** Sparse MoE layers (Shazeer et al., 2017) add conditional capacity; router z-losses, capacity factors, and load-

balancing losses are essential to avoid expert collapse.

- **Test-time scaling.** Curriculum-based sampling (nucleus, temperature annealing), classifier-free guidance, and beam-search variants all tune the accuracy/latency frontier; plan to log decoding hyperparameters alongside checkpoints so experiments are reproducible.

Parameter-efficient tuning methods are covered in Section 14.10.

14.15 RNNs vs. Transformers: When and Why

	RNN/LSTM/GRU	**Transformer**
Parallelism	Limited (sequential)	High (tokens in parallel)
Long context	Challenging (vanishing)	Natural; quadratic cost
Inductive bias	Order, recurrence	Content-based attention
Best for	Small data, streaming	Large data, global deps
Equivariance	N/A	Permutation-equivariant until positions (cf. conv translation equivariance in Chapter 11)

Key takeaways

Terminology. Masked-LM and next-token LM are *self-supervised* (targets derived from input), not unsupervised.

- Attention replaces recurrence with content-based mixing, enabling highly parallel training but introducing quadratic cost in sequence length.
- Practical stability depends on details (pre-LN vs. post-LN, optimizer choices, masking, and careful decoding/evaluation).
- Architecture choices (encoder/decoder, positions, caching) are not cosmetic: they determine what the model can reuse at inference time.

Minimum viable mastery.

- Compute masked attention for a short sequence and explain why the mask enforces causality.
- Explain pre-LN vs. post-LN and why residual paths influence optimization stability.
- Describe KV caching and how it changes inference-time cost compared to training-time cost.

Common pitfalls to watch for.

- Incorrect masking (future leakage) or inconsistent tokenization between training and evaluation.
- Reporting speed or memory without stating context length, batch size, and caching assumptions.
- Treating decoding strategy as an afterthought; greedy, beam, and sampling regimes change observed quality.

Exercises

Exercise 1 (Easy). Hand-compute one masked self-attention step for a 3-token prefix. Write the causal mask, compute the attention weights, and compute the output vectors.
Deliverable: the mask matrix + the attention weights (as numbers) + a short note explaining exactly which future tokens were blocked and why.

Exercise 2 (Medium). Build a tiny decoder-only model for next-token prediction on a small synthetic corpus (or a tiny character corpus) and run a controlled masking experiment: train once with a correct causal mask and once with an incorrect mask that leaks future tokens.
Deliverable: the training curves + one concrete "cheating" example that the leaky model can solve but the correctly masked model cannot.

Exercise 3 (Medium). Verify the representation claim: without positional information, a self-attention block is permutation-equivariant. Test this by permuting token order and checking that outputs permute the same way; then add positional encodings and show the equivariance breaks in the expected way.
Deliverable: one small numerical sanity check (before/after positions) + one paragraph linking the observation to why positional design matters.

If you are skipping ahead. After this chapter, the book pivots away from neural sequence models, so treat this chapter as the last stop for masking discipline and evaluation hygiene. If you need the embedding objectives and bias/deployment checklist, see Chapter 13.

Where we head next. Up to here, most updates were differentiable: loss → gradients → parameters. Chapter 15 pivots to a different design point under the same engineering loop: *explicit rules and graded concepts* (fuzzy systems), where the representation is linguistic membership and the update mechanism is rule composition and operator choices. Chapter 19 then closes the loop with *budgeted black-box search* (evolutionary optimization) for cases where gradients are unavailable, objectives are rugged, or constraints dominate.

Part II takeaways

- Representation and training are coupled: expressivity is only useful if gradients can reach the parameters.
- Depth and inductive bias trade parameters for structure (MLPs vs. CNNs vs. recurrence vs. attention).
- Stability tools recur across architectures: initialization, normalization, regularization, and validation-driven stopping.
- Sequence models turn memory into computation: unrolling, masking, and caching define what information can flow.
- **Unifying lens:** this Part fixes the representation and feedback signal, and the rest is choosing an update rule plus audits that prevent silent failure.

Part III: Soft computing and fuzzy reasoning

How to read Part III (rolling window)

- **A trilogy by design:** fuzzy sets → fuzzy relations → fuzzy inference; each chapter reuses the thermostat to add one layer of machinery.
- **Repetition is intentional:** concepts reappear as intuition, then as algebra, then as end-to-end behavior. Read repeated material as "preview vs. formalization vs. sanity-check."
- **Audit like an engineer:** keep universes/units explicit, pick operators explicitly, and verify edge cases numerically before tuning aesthetics.

Part III bridge: specification becomes the model

Parts I–II trained models by minimizing losses: we picked a hypothesis class, optimized it, and then audited generalization with careful validation. That approach is powerful when (i) you can collect enough data and (ii) the target behavior is well captured by a smooth objective.

Part III is about a different pain point: in many engineered systems the bottleneck is not data, but *specification*. The concepts you need are linguistic ("too warm", "slightly fast", "acceptable risk"), the boundaries are negotiable, and the decision logic often needs to remain auditable.

Fuzzy reasoning treats that interface explicitly: we write membership functions to represent concepts, pick operators to combine them, and assemble rules whose effects you can inspect. Everyday language is good at expressing graded categories; fuzzy logic turns that graded meaning into a mathematical object you can test, debug, and tune.

Read the repetition in this part as intentional: it is there to build familiarity with the machinery and to separate intuition from algebra. We revisit the same thermostat-style scenario as a preview, then as formal derivations, then as end-to-end behavior. Each pass adds one layer: fuzzy sets (building blocks), fuzzy relations (transfer and composition), and fuzzy inference (a complete controller).

When you get stuck, debug like an engineer: keep universes and units explicit, state your operator defaults, and test edge cases numerically before you tune aesthetics.

15 Introduction to Soft Computing

Parts I–II emphasized two complementary toolkits: probabilistic modeling and ERM (model–loss–optimize–audit), and biologically inspired learning (distributed representations trained by gradient descent). They shine when you can write down a smooth objective and collect enough data. Many engineered systems, however, are constrained less by data or compute than by specification: the relevant concepts are linguistic ("too warm," "slightly fast," "acceptable risk"), the boundaries are negotiable, and the decision logic must remain auditable.

Soft computing makes that vagueness explicit. Instead of insisting on exact equations everywhere, we represent human concepts with graded membership functions, combine them with transparent rules, and then tune those choices (and their trade-offs) with optimization rather than brittle hand tweaks. The shift is not away from rigor; it is toward placing approximations where they belong—at the interface between messy human predicates and systems that still have to act.

Learning Outcomes

- Articulate why soft computing (fuzzy logic, evolutionary search, neural hybrids) complements the statistical strand from earlier chapters.
- Define fuzzy sets, linguistic variables, and rule bases at a conceptual level before Chapters 16 to 18 formalize them.
- Work through a compact thermostat/autofocus controller example that grounds the design choices introduced throughout the fuzzy trilogy.

Design motif

When precision is costly or ill-defined, make the vagueness explicit and keep the system auditable: represent linguistic concepts with membership functions, reason with operator choices you can explain, and tune those choices with optimization rather than brittle rules.

Running example (thermostat). Throughout the fuzzy trilogy we use a single running example: a smart thermostat that regulates a room by mapping two linguistic inputs to one linguistic output.

- **Input 1 — temperature error** $e = T_{\text{room}} - T_{\text{set}}$ (°C), ranging roughly from -5 to $+5$. Labels: *Cold*, *Comfy*, *Hot*.
- **Input 2 — rate of change** $\dot{e}$ ($^\circ\text{C}\,\text{min}^{-1}$), ranging from -2 to $+2$. Labels: *Falling*, *Stable*, *Rising*.
- **Output — heater power** $u \in [0, 1]$. Labels: *Off*, *Low*, *Medium*, *High*.

The fuzzy rule *"IF error is Cold AND rate is Falling THEN power is High"* is the kind of statement the system must represent, evaluate, and defuzzify. Chapter 16 parameterizes the membership functions for each label, Chapter 17 shows how labels transfer across related universes, and Chapter 18 assembles the complete inference pipeline. Keep this loop in mind as each formalism is introduced.

15.1 Hard Computing: The Classical Paradigm

Hard computing refers to the classical approach to computation where the goal is to produce precise, unambiguous, and mathematically exact outputs. This paradigm assumes that the relationships between inputs and outputs can be modeled accurately using well-defined mathematical equations. For example, Einstein's mass-energy equivalence formula,

$$E = mc^2, \tag{15.1}$$

is a precise, unambiguous, and exact mathematical expression.

In hard computing, the process typically involves:

- Precise inputs,
- Deterministic models,
- Exact outputs.

However, this approach is often inadequate for many real-world problems because:

1. The real world is pervasively *imprecise* and *uncertain*.
2. Achieving precision and certainty is often *costly* and *difficult*.

These limitations motivate the need for alternative computational frameworks that can tolerate and exploit imprecision and uncertainty.

15.2 Soft Computing: Motivation and Definition

Soft computing, a label Zadeh popularized in the 1990s (Zadeh, 1994), is a computational paradigm designed to handle problems where precision and certainty are either impossible or prohibitively expensive to obtain. Unlike hard

computing, soft computing tolerates *imprecision*, *uncertainty*, and *approximate reasoning* to achieve solutions that are:

- **Tractable:** Computationally feasible to obtain,
- **Robust:** Insensitive to noise and variations,
- **Low-cost:** Economical in terms of computational resources.

Formally, soft computing is not a single homogeneous methodology but rather a *partnership of distinct methods* that conform to these guiding principles. In Zadeh's broad usage, the principal constituents include:

- **Fuzzy Logic:** Handling imprecision and approximate reasoning,
- **Neurocomputing (and neuro-fuzzy hybrids):** Learning from data through neural networks, sometimes combined with fuzzy rule bases (e.g., adaptive neuro-fuzzy inference system (ANFIS)),
- **Genetic Algorithms:** Evolutionary optimization inspired by natural selection.

These components often overlap and complement each other in practical applications. In this book, probabilistic modeling is treated in the statistical strand (Chapters 3 to 4); the soft-computing block focuses on fuzzy systems and evolutionary search, with neuro-fuzzy hybrids serving as the bridge back to the neural chapters.

15.3 Why Soft Computing?

The key insight behind soft computing is to exploit the *tolerance for imprecision and uncertainty* inherent in many real-world problems. Consider the example of handwritten digit recognition using a convolutional neural network (CNN):

- The input is a handwritten digit, say the digit "4".
- The network extracts features and produces a probability distribution over possible digits.
- The output might be:

$$P(\text{digit} = 4) = 0.60, \quad P(\text{digit} = 7) = 0.20, \quad P(\text{digit} = 1) = 0.20. \tag{15.2}$$

This output is *not precise* in the classical sense; it expresses uncertainty and partial belief. The system tolerates this imprecision and still makes a decision based on the highest probability, demonstrating robustness and flexibility.

This is a probability story (uncertainty about which class is true). Fuzzy logic addresses a different interface: degrees of truth for linguistic predicates such as "warm" or "slightly fast," even when measurements are exact. Chapter 16 makes that distinction concrete by writing membership functions and checking them at specific inputs.

15.4 Relationship Between Hard and Soft Computing

We can conceptualize the landscape of computing as follows:

- **Hard Computing:** Precise, deterministic, mathematically exact.
- **Soft Computing:** Approximate, tolerant of imprecision and uncertainty, heuristic.

Hard computing insists on crisp logic and exact arithmetic: precise thresholds, deterministic algorithms, and guarantees when assumptions hold. Soft computing is approximate by design, trading exactness for robustness when concepts are linguistic, measurements are noisy, or the specification itself is negotiable. Many real systems blend the two: hard constraints and safety checks wrapped around a soft inner loop (approximate reasoning or heuristic search), chosen based on which assumptions you can justify and which failure modes you can tolerate.

15.5 Overview of Soft Computing Constituents

Fuzzy Logic: Deals with *fuzziness* or vagueness, allowing partial membership in sets and approximate reasoning. It is particularly useful when information is incomplete or linguistic in nature.

Neurocomputing: Encompasses various neural network architectures (multilayer perceptrons, convolutional networks, recurrent models, Hopfield networks, and Radial Basis Function (RBF) networks) as well as neuromorphic hardware that learn from data and approximate complex nonlinear mappings.

Probabilistic Reasoning: Manages uncertainty using probability theory, be-

lief networks, and Bayesian inference. It assumes known or estimable probability distributions.

Genetic Algorithms: Inspired by biological evolution, these algorithms perform heuristic search and optimization by mimicking natural selection and genetic variation.

15.6 Distinguishing Imprecision, Uncertainty, and Fuzziness

It is important to clarify the subtle differences among these concepts:

- **Uncertainty** refers to situations where the outcome is unknown but can be described probabilistically. For example, a classifier might assign a 60% probability to a particular class.
- **Imprecision** refers to limited resolution or vagueness in the available descriptions or measurements. Saying that the outside temperature is "warm" rather than specifying 24.5°C is imprecise because we are unsure about the precise boundary that should separate "warm" from "hot."
- **Fuzziness** captures graded membership in a linguistic category; for instance, the extent to which a day is "warm." Membership values range continuously between 0 and 1 instead of forcing a binary decision.

In short, imprecision concerns our knowledge about a precise boundary, whereas fuzziness is a property of the concept itself: even with perfect measurements, "warm" transitions smoothly into "hot." For example, reading 24.5°C from a thermometer with ±1°C resolution is an *imprecise* observation, whereas deciding whether 24.5°C should be labelled "warm" or "hot" is a *fuzzy* membership question that remains even if the thermometer were infinitely precise.

Imprecision vs. Fuzziness

Imprecision concerns uncertainty about the exact value or boundary (e.g., measurement error or coarse resolution). **Fuzziness** concerns graded membership in a concept (e.g., the degree to which a day is "warm") even when measurements are exact. Probability quantifies uncertainty about events; fuzziness quantifies degree of truth of linguistic predicates.

Table 5: Fuzzy vs. probabilistic reasoning. Distinguishes randomness (probability) from graded concepts (fuzziness) when interpreting uncertainty.

	Fuzzy logic	**Probabilistic logic**
Semantics	Degree of membership (vagueness); e.g., "temperature is high to degree 0.7."	Degree of belief/uncertainty—probability that an event occurs.
Operators	t-norms / s-norms (min, product, max) model AND/OR; implication via fuzzy rules.	Sum / product rules, Bayes' theorem govern AND/OR/conditionals.
Outputs	Fuzzy sets defuzzified to crisp actions (e.g., heater power).	Numeric probabilities used for expectation, decision thresholds.
Typical use	Rule bases, approximate control, linguistic policies.	Stochastic modeling, hypothesis testing, Bayesian inference.

15.7 Soft Computing: Motivation and Overview

Soft computing is defined in Section 15.2; here we map its constituents to where they are developed in this book so you can follow the learning outcomes above without re-reading the same motivation.

- **Fuzzy logic:** this chapter and the fuzzy trilogy Chapters 16 to 18 build membership functions, rules, and inference pipelines.
- **Neurocomputing and hybrids:** neural models live in Chapters 3 to 14; neuro-fuzzy hybrids such as ANFIS (Jang, 1993) sit at the interface.
- **Probabilistic modeling:** the statistical strand in Chapters 3 to 4 remains complementary to fuzzy possibility views (Dubois and Prade, 1988).
- **Evolutionary computation:** Chapter 19 shows how evolutionary search tunes rule bases and membership parameters (Herrera and Lozano, 2008; Ishibuchi and Nakashima, 2007).

Table 6 maps crisp logic operators to their fuzzy counterparts.

Table 6: Boolean vs. fuzzy operators. Mapping from crisp logic rules to graded operators for fuzzy inference.

	Boolean logic	**Fuzzy logic**
AND	$\min(a, b)$	t-norm (e.g., min, product)
OR	$\max(a, b)$	s-norm (e.g., max, prob. sum)
NOT	$1 - a$	complement $1 - a$

15.8 Fuzzy Logic: Capturing Human Knowledge Linguistically

One of the most compelling aspects of fuzzy logic is its ability to represent human knowledge and experience in a linguistic form that machines can process. Consider the everyday reasoning:

If you wake up late and the traffic is congested, then you will be late.

This statement involves vague concepts such as "late," "congested," and "will be late," which are not crisply defined but are intuitively understood by humans. Fuzzy logic allows us to formalize such rules without requiring precise probabilistic models or extensive training data.

Fuzzy Rules and Approximate Reasoning A fuzzy rule typically has the form:

$$\text{IF } A \text{ AND } B \text{ THEN } C,$$

where A, B, and C are fuzzy propositions characterized by membership functions rather than crisp sets.

For example:

- A: "Wake up late" could be represented by a membership function $\mu_{\text{late}}(t)$ over the waking time t.
- B: "Traffic is congested" could be represented by a membership function $\mu_{\text{congested}}(x)$ over traffic density x.
- C: "You will be late" is the fuzzy output.

Each membership function maps from the relevant universe of discourse to $[0, 1]$, i.e., $\mu_{\text{late}} : \mathbb{R} \to [0, 1]$, so that linguistic labels become numeric degrees of

support. The fuzzy inference system combines these membership values using *t-norm* operators (e.g., min, product) to model logical conjunction and *s-norms* (e.g., max) to model disjunction, thereby inferring the degree to which the conclusion C holds. In practical systems the resulting fuzzy set is often *defuzzified* (e.g., via centroid or maximum-membership methods) to obtain a single crisp recommendation.

Why this is useful in practice Classical rule systems and purely statistical pipelines often assume either crisp thresholds or well-specified probability models. Fuzzy logic is useful when neither assumption is realistic: the data are coarse, the concepts are linguistic, and the control policy must still be auditable. In that setting, fuzzy sets let us encode expert knowledge directly, preserve graded boundaries, and still produce an interpretable decision path.

15.9 Comparison with Other Soft Computing Paradigms

Neural networks, evolutionary search, probabilistic modeling, and fuzzy logic solve different bottlenecks, even when they are applied to the same system. Neural networks are strongest when enough data exist and predictive accuracy is the main objective. Evolutionary methods are effective when the objective is rugged, discontinuous, simulator-in-the-loop, or otherwise unfriendly to gradients. Probabilistic models are strongest when uncertainty semantics matter (calibration, risk, and decision thresholds). Fuzzy logic is strongest when the bottleneck is *specification*: concepts are linguistic, boundaries are negotiable, and the decision logic must remain inspectable.

In practice, these paradigms are complementary building blocks. A fuzzy controller can sit on top of neural features; evolutionary search can tune membership functions, rule weights, or neural hyperparameters; and probabilistic tools can sit alongside either when you need the output to carry a defensible notion of uncertainty. The goal of this part is not to pick a winner, but to give you multiple ways to represent, update, and audit intelligent behavior depending on what the application can actually provide (data, expert knowledge, simulators, or constraints).

15.10 Zadeh's Insight and the Birth of Fuzzy Logic

Lotfi Zadeh, in the late 1960s, observed that classical statistics and probability theory demand precise knowledge of distributions and exact calculations, which is often unrealistic for human decision-making. Humans rely on approximate, linguistic knowledge rather than exact numerical data.

Zadeh's contribution was to formalize a practical middle ground: represent imprecise concepts with membership functions, reason with them compositionally, and keep the rules close to human language. The point was not to replace probability, but to model vagueness directly when boundaries are conceptual rather than random.

This approach revolutionized how we model uncertainty and reasoning in artificial intelligence and control systems.

15.11 Challenges in Fuzzy Logic Systems

Despite its advantages, fuzzy logic has two honest challenges. First, there is rarely a single systematic recipe that turns "expert knowledge" into a complete, consistent rule base; design is iterative and coverage-driven. Second, linguistic labels still need engineering choices: membership functions and ranges come from elicitation or data-driven tuning, and poor choices show up directly as brittle or non-monotone behavior. We treat this as a design-and-audit problem: make defaults explicit, test edge cases numerically, and tune only after the basic behavior is correct.

15.12 Mathematical Languages as Foundations for Fuzzy Logic

Recall that the motivation behind fuzzy logic was to develop a mathematical and linguistic framework capable of handling imprecision and uncertainty in a principled way. To achieve this, Lotfi Zadeh drew inspiration from several well-established mathematical languages, each with its own syntax, semantics, and rules of inference. Understanding these languages helps us appreciate how fuzzy logic extends and generalizes classical logic to accommodate vagueness.

15.12.1 Relational Algebra

Relational algebra is a formal language used primarily in database theory to manipulate sets and relations. It provides operators such as union ($\cup$), intersection ($\cap$), and set difference ($\setminus$) that operate on sets:

$$\begin{aligned} A \cup B &= \{x \mid x \in A \text{ or } x \in B\}, \\ A \cap B &= \{x \mid x \in A \text{ and } x \in B\}. \end{aligned} \tag{15.3}$$

The third canonical operator is the set difference

$$A \setminus B = \{x \mid x \in A \text{ and } x \notin B\}, \tag{15.4}$$

which removes from A any elements that also belong to B. For instance, if A is the set of all graduate students and B the set of research assistants, then $A \setminus B$ contains graduate students who are not currently research assistants.

These operators have well-defined meanings and predictable outputs, making relational algebra a precise language for reasoning about collections of elements. The vocabulary is limited but sufficient for set-theoretic operations.

15.12.2 Boolean Algebra

Boolean algebra is the algebraic structure underlying classical logic and digital circuits. It operates on binary variables taking values in $\{0, 1\}$, with logical operators such as `AND` ($\wedge$), `OR` ($\vee$), and `XOR` ($\oplus$):

$$\begin{aligned} A \vee B &= 1 \quad \text{if } A = 1 \text{ or } B = 1, \\ A \wedge B &= 1 \quad \text{if } A = 1 \text{ and } B = 1, \\ A \oplus B &= 1 \quad \text{if } A \neq B. \end{aligned}$$

Conversely, $A \vee B = 0$ only when both inputs are 0, and $A \wedge B = 0$ unless both inputs equal 1; the XOR operator returns 0 exactly when both operands share the same truth value.

Boolean algebra provides a crisp, binary framework where propositions are either true or false, with no intermediate values. This crispness is a limitation when modeling real-world phenomena involving gradations of truth.

15.12.3 Predicate Algebra

Predicate algebra extends Boolean algebra by incorporating quantifiers and variables, allowing statements about properties of elements in a domain. For example, a predicate statement might be:

$$\forall x \in \mathbb{R}, \quad x^2 \geq 0,$$

which reads: "For all real numbers x, x^2 is greater than or equal to zero." This language combines logical connectives with quantifiers such as $\forall$ (for all) and $\exists$ (there exists), enabling more expressive statements about sets and relations.

An example involving two domains could be:

$$\forall x \in \text{Rabbits}, \quad \forall y \in \text{Tortoises}, \quad \text{Faster}(x, y),$$

meaning "For any rabbit x and any tortoise y, x is faster than y."

Predicate algebra thus provides a linguistic and symbolic framework to express complex relationships and properties.

15.12.4 Propositional Calculus

Propositional calculus (or propositional logic) deals with propositions and their logical connectives. It focuses on the relationships between propositions without internal structure. The basic form involves premises and conclusions, such as:

$$P \implies Q, \quad P \quad \Rightarrow \quad Q.$$

where P and Q are propositions, and $\implies$ denotes implication.

Modus Ponens One fundamental rule of inference in propositional calculus is *modus ponens*:

If $P \implies Q$ and P is true, then Q must be true.

Symbolically,

$$P \implies Q, \quad P \quad \vdash \quad Q.$$

This rule affirms the consequent by affirming the antecedent.

Modus Tollens Another inference rule is *modus tollens*:

If $P \implies Q$ and Q is false, then P must be false.

Symbolically,

$$P \implies Q, \quad \neg Q \quad \vdash \quad \neg P.$$

This rule denies the antecedent by denying the consequent. However, as noted, this inference can sometimes be risky or invalid in practical scenarios due to exceptions or additional factors.

Hypothetical Syllogism A further inference pattern is the *hypothetical syllogism*:

If $P \implies Q$ and $Q \implies R$, then $P \implies R$.

Symbolically,

$$P \implies Q, \quad Q \implies R \quad \vdash \quad P \implies R.$$

This transitive property of implication allows chaining of logical statements.

Fuzzy logic as a mathematical language. Zadeh's insight was to synthesize these classical mathematical languages into a framework with graded truth values; the next section builds the intuition and contrasts crisp versus fuzzy reasoning.

15.13 Fuzzy Logic: Motivation and Intuition

Recall that classical (crisp) logic deals with binary truth values: a proposition is either true (1) or false (0). For example, the question "Was the exam easy?" can be answered crisply as "Yes" or "No." However, many real-world situations are not so black-and-white. Often, we want to express uncertainty or partial truth, such as "The exam was somewhat easy," or "The exam was easy to a certain degree."

Fuzzy truth values allow us to express such intermediate degrees of truth. Instead of restricting truth values to $\{0, 1\}$, fuzzy logic permits any value in the continuous interval $[0, 1]$. For instance, if the exam was moderately easy, we might assign a truth value of 0.6 or 0.7, indicating partial truth.

This flexibility captures the inherent vagueness in many human concepts and perceptions. For example, when asked "Did you enjoy your lunch?" one might respond "sort of," reflecting a fuzzy assessment rather than a crisp yes/no.

Why fuzzy logic?

- **Tolerance for imprecision:** Observations and measurements are often noisy or uncertain.
- **Expressiveness:** Allows linguistic hedging such as "somewhat," "maybe," or "approximately."
- **Robustness:** Systems can handle ambiguous or incomplete information gracefully.

Author's note: why study fuzzy systems now?

Fuzzy inference began in the 1960s, so it is reasonable to ask why this material still matters in a world shaped by large language models. The answer is practical and methodological. Practically, fuzzy systems remain useful when specifications are linguistic, data are limited, and the decision path must stay auditable (for example, policy-heavy control settings). Methodologically, they expose a core AI design pattern: represent vague concepts explicitly, compose rules, and inspect the resulting behavior under constraints.

This chapter is therefore not nostalgia. It is one lens for how intelligent behavior can be engineered. Modern systems can combine lenses: neural models for representation learning, fuzzy rules for explicit policy structure, and evolutionary search for tuning when gradients are unreliable.

15.14 From Crisp Sets to Fuzzy Sets

Crisp sets are classical sets where an element either belongs or does not belong to the set. Formally, for a universe X, a crisp set $A \subseteq X$ is characterized by its *characteristic function*:

$$\chi_A(x) = \begin{cases} 1 & \text{if } x \in A, \\ 0 & \text{if } x \notin A. \end{cases} \tag{15.5}$$

Example: Consider two classes:

$$\text{Class 1} = \{\text{Li}, \text{Rajnish}\}, \quad \text{Class 2} = \{\text{Hamid}, \text{John}, \text{Julia}, \text{Yet}\}.$$

These are crisp sets since no student belongs to both classes simultaneously.

Fuzzy sets generalize this notion by allowing partial membership. A fuzzy set $\tilde{A}$ on X is characterized by a *membership function*:

$$\mu_{\tilde{A}} : X \to [0, 1], \tag{15.6}$$

where $\mu_{\tilde{A}}(x)$ quantifies the degree to which x belongs to $\tilde{A}$.

Example: Sizes as fuzzy sets

Consider the linguistic labels `Small`, `Medium`, and `Large` for weights (in kilograms). A crisp partition such as $[0, 10], [11, 20], [21, 30]$ is disjoint; fuzzy sets allow these labels to overlap smoothly so a weight can belong to both `Medium` and `Large` to different degrees. See Chapter 16 (especially Section 16.8) for the explicit membership formulas and plots; here keep the intuition that fuzzy labels overlap and map a universe of discourse into $[0, 1]$.

Thermostat at a glance. Throughout Chapters 16 to 18 we reuse a fuzzy thermostat: inputs are temperature error and rate (linguistic labels such as `Cold`, `Warm`, `Hot`; `Falling`, `Stable`, `Rising`); rules map these to heater power; defuzzification turns the fuzzy action into a crisp control signal. Keep this loop in mind as membership functions and operators are introduced.

Lab prep: fuzzy thermostat starter

- Install `scikit-fuzzy` and `matplotlib`.
- Define triangular membership functions for `Cold`/`Warm`/`Hot` over a temperature universe; plot the overlap.
- Write one rule: IF error is `Cold` AND rate is `Rising` THEN power is `Low`; preview centroid defuzzification.

Exercises (Chapter 15)

- Classify three scenarios as imprecision vs. uncertainty vs. fuzziness; justify each.
- Write two fuzzy thermostat rules and reason qualitatively about the output for a borderline input.
- Compare min vs. product t-norm on the same antecedent degrees (e.g., 0.4 and 0.7); explain the impact.
- Sketch (or code) a triangular membership and a simple IF–THEN rule; describe how defuzzification would proceed.
- Identify where probability (Chapter 3) and fuzzy possibility (this chapter) would lead to different interpretations.

Forward pointer. Chapter 16 builds the membership functions and universes for the thermostat inputs/outputs; Chapter 17 provides the transfer operators (extension principle, projection, composition) that move fuzzy labels between related variables; and Chapter 18 assembles full inference and defuzzification. Chapter 19 later returns to the other soft-computing pillar: evolutionary search as a tuning tool for rule bases and memberships.

15.15 Wrapping up soft computing and previewing the fuzzy trilogy

This chapter set the motivation: when specifications are linguistic and boundaries are negotiable, it is often better to model that vagueness explicitly than to force brittle crisp thresholds. The next three chapters turn that motivation into a workflow you can execute and audit.

Fuzzy sets recap and why it matters in practice See Section 15.14 for the formal membership-function definition and Section 15.13 for the crisp-versus-fuzzy contrast. The practical payoff is that one observation can carry multiple linguistic truths at once. For example, a height value can be partly "short" and partly "medium," and that overlap is not ambiguity in the bad sense; it is deliberate tolerance around boundaries that would otherwise be brittle in a crisp model.

The same idea controls actuators more smoothly. In braking or thermostat control, we do not need to jump between “off” and “full.” We can combine fuzzy evidence and produce graded actions such as “slightly increase” or “strongly reduce,” which usually improves stability near operating thresholds.

Next steps: membership functions, transformations, and inference Chapters 16 to 18 now turn this motivation into a complete engineering pipeline. Chapter 16 builds the membership primitives. Chapter 17 moves those memberships between related universes through the extension principle. Chapter 18 composes full rule systems and defuzzification into a final control decision. Keep this map in view: the next chapters add algebraic detail, but they serve one continuous workflow.

Part transition thesis As we move from the fuzzy trilogy to evolutionary computing, the design contract stays the same while the update mechanism changes. In Part III, behavior is shaped by rule composition over linguistic memberships. In Part IV, behavior is improved by population search over candidate encodings. Two different lenses, one engineering goal: build systems that perform reliably under realistic constraints.

Key takeaways

- Soft computing embraces imprecision via fuzzy logic, evolutionary search, and neural networks.
- Fuzzy operators (t-norms, implications) enable approximate reasoning under uncertainty.
- Choosing operators and membership functions matches problem semantics to inference behavior.

Minimum viable mastery.

- Define what is being approximated (truth values, search steps, or function classes) in each soft-computing pillar.
- Explain why operator choices matter (they encode semantics and shape the resulting decision surfaces).
- Connect fuzzy-set primitives to the later inference pipeline (fuzzify, combine, imply, aggregate, defuzzify).

Common pitfalls.

- Mixing operator families inconsistently across a pipeline and then debugging symptoms at the end.
- Treating "soft" as a license to skip validation: soft methods still require measurable objectives and checks.
- Ignoring scaling and units when defining universes and membership functions.

Exercises

Exercise 1 (Easy). Rewrite one crisp threshold policy (braking, thermostat actuation, or an alarm trigger) as fuzzy labels and test at least three boundary cases where crisp logic flips abruptly.
Deliverable: the crisp policy + the fuzzy rewrite + a short table of boundary-case inputs and outputs.

Exercise 2 (Medium). On one small rule base, compare max–min and max–product choices end-to-end and explain the behavioral difference in terms of "how conjunction is interpreted."
Deliverable: one input→output curve under both operators + one paragraph describing which regime is more conservative/aggressive and why.

Exercise 3 (Medium). Build a reproducibility sheet for a fuzzy controller: universes (with units), memberships (with parameters), operator defaults, discretization step, and test points. Rerun the same controller later and verify you can reproduce the outputs exactly.
Deliverable: the one-page sheet + a short note explaining which missing detail would make reproduction impossible.

Fast-track note. As you move into Chapters 16 to 18, keep operator choices explicit and consistent. Many downstream mismatches come from hidden defaults set at the start.

Where we head next. Chapters 16 to 18 carry the thermostat running example end-to-end: Chapter 16 builds membership functions, Chapter 17 transports information across universes through relations, and Chapter 18 assembles full inference plus defuzzification.

16 Fuzzy Sets and Membership Functions: Foundations and Representations

Building on Chapter 15, we formalize the foundations of fuzzy logic: universes of discourse, fuzzy sets, membership functions, and the operators used throughout the fuzzy trilogy. You will see some ideas from Chapter 15 again, but

now in a form you can compute and sanity-check numerically (a deliberate preview→formalization pattern). These tools define the thermostat labels used later and prepare the transfer/inference steps developed in Chapters 17 to 18.

Learning Outcomes

After this chapter, you should be able to:

- Distinguish imprecision (uncertain value/boundary) from fuzziness (graded membership).
- Define and interpret membership functions in discrete and continuous domains.
- Apply fuzzy set operations and De Morgan's laws using max/min forms.
- Execute an end-to-end Mamdani inference and compute the centroid defuzzification.

Design motif

Treat vagueness as a first-class modeling choice: write the membership functions down, pick operators explicitly, and then audit how those choices shape the behavior of a rule base.

Running example checkpoint. For the thermostat scenario introduced in Chapter 15, the universe of discourse for the inputs is $[-5, 5]$°C temperature error and $[-2, 2]$°C/min rate-of-change. As you study triangular, trapezoidal, and Gaussian membership functions, imagine parameterizing the linguistic labels *Cold*, *Comfy*, and *Hot* for these inputs. We reuse those shapes when composing rules in Chapters 17 to 18.

16.1 Motivating example: designing a membership function from measurements

Consider the thermostat's temperature error $e = T_{\text{room}} - T_{\text{set}}$ in degrees Celsius (so $e < 0$ means "too cold"). We want a linguistic label *Cold* that is clearly true when the room is far below setpoint, clearly false when the room is at/above setpoint, and graded in between. A simple, auditable choice is a piecewise-linear

"shoulder" membership:

$$\mu_{\text{Cold}}(e) = \begin{cases} 1, & e \leq -4, \\ -\frac{e}{4}, & -4 < e < 0, \\ 0, & e \geq 0. \end{cases} \tag{16.1}$$

With this design, $e = -2$ yields $\mu_{\text{Cold}}(-2) = 0.5$: "cold" is half true (and in the linear region $e < 0$, so $-e > 0$). Later chapters reuse this shoulder *shape* but may shift its breakpoints to reflect different sensors, comfort bands, or operating assumptions. This number is not a probability; it is a degree of truth for a linguistic predicate. The rest of this chapter is about making these mappings explicit (shapes, overlaps, and operators) so they can be inspected and tuned rather than assumed.

Author's note: membership design is where judgment enters

Membership functions are an explicit way to encode boundary subjectivity. At the extremes, many labels are uncontroversial (very short vs. very tall, clearly cold vs. clearly hot). The modeling work happens in the overlap zone. We choose shapes and breakpoints so borderline cases are represented as graded truth rather than forced yes/no decisions. In practice, start by anchoring obvious extremes from measurements, then tune overlap widths to match observed behavior and control smoothness.

16.2 Fuzzy sets and the universe of discourse

A *fuzzy set* A in a universe of discourse X is characterized by a *membership function* $\mu_A : X \to [0, 1]$. This membership function assigns to each element $x \in X$ a degree of membership $\mu_A(x)$, which quantifies the extent to which x belongs to the fuzzy set A. Throughout the fuzzy trilogy, this notation stays fixed: μ_A always means membership of set A over the currently named universe X, so when we switch from temperature error to speed or rule consequents, we restate X rather than changing the meaning of μ_A.

- If $\mu_A(x) = 1$, then x fully belongs to A.
- If $\mu_A(x) = 0$, then x does not belong to A at all.
- If $0 < \mu_A(x) < 1$, then x partially belongs to A to the degree $\mu_A(x)$.

This contrasts with classical (crisp) sets, where membership is binary (either 0 or 1).

16.3 Membership Functions: Definition and Interpretation

A *membership function* $\mu_A(x)$ maps each element x in the universe X to a membership grade in the interval $[0, 1]$. The shape and parameters of μ_A encode the fuzziness or uncertainty associated with the concept represented by A.

Example: Consider the fuzzy set *Slow Speed* defined over the universe of speeds $X \subseteq \mathbb{R}$. The membership function $\mu_{\text{Slow}}(x)$ might assign high membership values to speeds near 20 km/h and gradually decrease as speed increases, reflecting the gradual transition from "slow" to "not slow."

Mathematical representation. For each $x \in X$, the membership grade satisfies $\mu_A(x) \in [0, 1]$.

The fuzzy set A can be represented as the collection of ordered pairs:

$$A = \{(x, \mu_A(x)) \mid x \in X\}. \tag{16.2}$$

16.4 Discrete vs. Continuous Universes of Discourse

The universe X can be either discrete or continuous, which affects how fuzzy sets and membership functions are represented.

16.4.1 Discrete Universe

When $X = \{x_1, x_2, \ldots, x_n\}$ is finite or countable, the fuzzy set A is represented as a finite collection of ordered pairs:

$$A = \{(x_1, \mu_A(x_1)), (x_2, \mu_A(x_2)), \ldots, (x_n, \mu_A(x_n))\}.$$

Typically, membership values equal to zero are omitted for brevity, since they indicate no membership.

Example: Suppose $X = \{1, 2, 3, 4, 5\}$ and the membership function values are:

$$\mu_A(1) = 0, \quad \mu_A(2) = 0.1, \quad \mu_A(3) = 0.3, \quad \mu_A(4) = 0.7, \quad \mu_A(5) = 0.$$

Then,

$$A = \{(2, 0.1), (3, 0.3), (4, 0.7)\}.$$

16.4.2 Continuous Universe

When $X \subseteq \mathbb{R}$ is continuous (e.g., an interval), the fuzzy set A is described by a membership function $\mu_A(x)$ defined for all $x \in X$. The representation is functional rather than enumerative:

$$A = \int_{x \in X} \mu_A(x)/x, \tag{16.3}$$

where the notation $\int \mu_A(x)/x$ denotes the continuous collection of pairs $(x, \mu_A(x))$.

Interpretation: The integral sign here is symbolic, indicating a continuous aggregation over X, not a numerical integral in the calculus sense.

Example: Consider a triangular membership function centered at c with base width w:

$$\mu_A(x) = \max\left(0, 1 - \frac{|x - c|}{w}\right). \tag{16.4}$$

This function assigns membership 1 at $x = c$, decreasing linearly to zero at $x = c \pm w$.

16.5 Crisp Sets versus Fuzzy Sets

Crisp (classical) sets assign membership values in the binary set $\{0, 1\}$, so each element either belongs to the set or it does not. In contrast, fuzzy sets allow intermediate membership values, enabling gradual transitions between full inclusion and full exclusion. Understanding this contrast highlights why membership functions are central to fuzzy logic.

Imprecision vs. Fuzziness (recap)

As discussed in Section 15.6, **imprecision** concerns uncertainty about the exact value or boundary (e.g., measurement noise or coarse resolution), whereas **fuzziness** concerns graded membership in a concept (e.g., the degree to which a speed is "slow") even when measurements are exact. Probability models uncertainty about events; fuzzy logic models degrees of truth of linguistic predicates.

16.6 Membership Functions in Fuzzy Sets

With the universe X fixed and the concept A named, the membership function $\mu_A : X \to [0, 1]$ assigns to each element $x \in X$ a degree of membership $\mu_A(x)$ indicating the extent to which x belongs to A.

Triangular Membership Function One of the simplest and most intuitive membership functions is the *triangular* membership function. It is defined by three parameters $a < b < c$ and given by

$$\mu_A(x) = \begin{cases} 0, & x \leq a \\ \frac{x-a}{b-a}, & a < x \leq b \\ \frac{c-x}{c-b}, & b < x < c \\ 0, & x \geq c \end{cases} \tag{16.5}$$

This function attains its maximum value 1 at $x = b$, representing the point of highest confidence that x belongs to the fuzzy set A. The membership decreases linearly on either side of b, reaching zero at a and c. This shape expresses a strong belief in membership near b and uncertainty elsewhere.

Trapezoidal Membership Function The trapezoidal membership function generalizes the triangular shape by allowing a flat top, representing a range of

values with full membership. It is defined by four parameters $a < b \leq c < d$:

$$\mu_A(x) = \begin{cases} 0, & x \leq a \\ \frac{x-a}{b-a}, & a < x \leq b \\ 1, & b < x \leq c \\ \frac{d-x}{d-c}, & c < x < d \\ 0, & x \geq d \end{cases} \tag{16.6}$$

This function models situations where there is full confidence that all values between b and c belong to the fuzzy set, with gradual transitions on the edges. The thermostat's *Cold* label in Equation (16.1) is a left-shoulder special case of this family: membership stays at 1 up to the cold threshold, then decreases linearly to 0 at the comfort boundary.

Gaussian Membership Function The Gaussian membership function is widely used due to its smoothness and differentiability, which are advantageous in optimization and learning algorithms. It is defined by parameters c (center) and $\sigma > 0$ (width):

$$\mu_A(x) = \exp\left(-\frac{(x-c)^2}{2\sigma^2}\right). \tag{16.7}$$

This bell-shaped curve smoothly assigns membership values, with the highest membership at $x = c$ and decreasing membership as x moves away from c. The parameter σ controls the spread or fuzziness of the set.

Generalized Bell Membership Function Another flexible membership function is the generalized bell function, defined by parameters a, b, c:

$$\mu_A(x) = \frac{1}{1 + \left|\frac{x-c}{a}\right|^{2b}}. \tag{16.8}$$

This function allows control over the width and slope of the membership curve, interpolating between shapes similar to triangular and Gaussian functions.

16.7 Comparison of Membership Functions

- **Triangular and Trapezoidal:** These are piecewise linear, computationally inexpensive, and easy to interpret. However, they are not differentiable at the vertices, which can be a limitation in gradient-based learning.
- **Gaussian and Bell:** These are smooth and differentiable, making them suitable for optimization and adaptive systems. They provide more modeling flexibility but are computationally more expensive.

Example: Grading System as Fuzzy Sets Consider a typical university grading scale as an example of fuzzy sets. Traditional crisp sets assign grades as follows:

$$\text{F} : [0, 59], \quad \text{D} : [60, 69], \quad \text{C} : [70, 79], \quad \text{B} : [80, 89], \quad \text{A} : [90, 100].$$

In a crisp set, membership is binary: a score of 75 is fully in C and not in B.

However, students and instructors may perceive these boundaries differently. For example, some may consider 75 to be a borderline B, or 68 to be a borderline C. This uncertainty can be modeled by fuzzy sets with overlapping membership functions.

For instance, the membership function for grade C could be trapezoidal:

$$\mu_C(x) = \begin{cases} 0, & x \leq 65, \\ \dfrac{x-65}{5}, & 65 < x \leq 70, \\ 1, & 70 < x \leq 75, \\ \dfrac{80-x}{5}, & 75 < x \leq 80, \\ 0, & x > 80. \end{cases}$$

Similarly, the membership for grade B could be written as

$$\mu_B(x) = \begin{cases} 0, & x \le 75, \\ \dfrac{x-75}{5}, & 75 < x \le 80, \\ 1, & 80 < x \le 85, \\ \dfrac{90-x}{5}, & 85 < x \le 90, \\ 0, & x > 90, \end{cases}$$

so a borderline score such as $x = 79$ yields $\mu_C(79) = (80-79)/5 = 0.2$ and $\mu_B(79) = (79-75$ th analogous expressions for the zoids capture the intuition that a elong to both C and B to differen

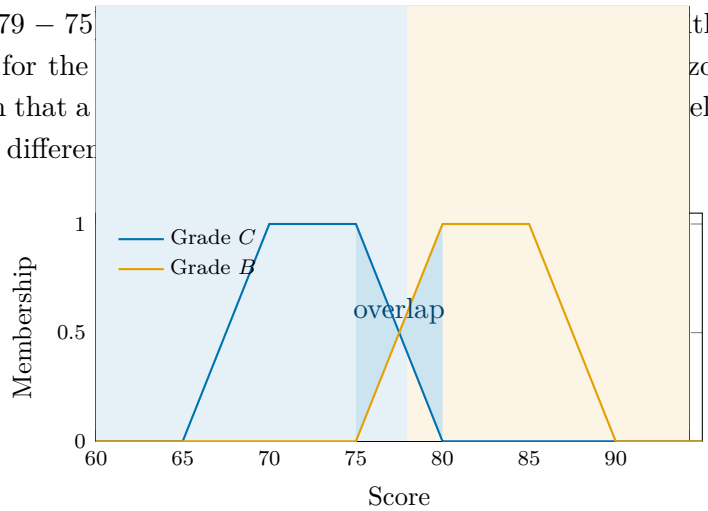


Figure 66: Trapezoidal membership functions for grades C and B with the overlapping region shaded. Scores near 78–82 partially satisfy both grade definitions. This overlap design keeps boundary cases smooth instead of forcing abrupt label switches.

Figure 67 is the overlap diagnostic used when tuning membership coverage.

16.8 Example: Overlapping weight labels

Fuzzy labels often overlap so that borderline values belong to multiple sets. For weights measured in kilograms, one crisp partition is $[0, 10], [11, 20], [21, 30]$ for

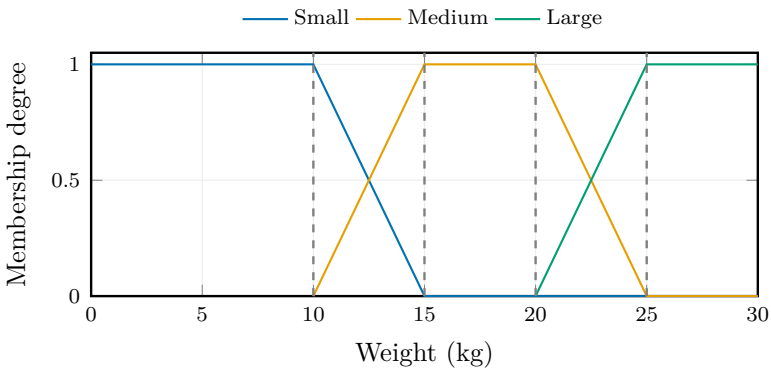


Figure 67: Overlapping membership functions for the "Small", "Medium", and "Large" weight labels. The shaded overlaps capture gradual transitions and reduce abrupt rule changes for small numeric perturbations.

`Small`, `Medium`, `Large`. A fuzzy partition smooths these transitions:

$$\mu_{\text{Small}}(x) = \begin{cases} 1, & x \le 10, \\ 1 - \dfrac{x-10}{5}, & 10 < x < 15, \\ 0, & x \ge 15, \end{cases}$$

$$\mu_{\text{Medium}}(x) = \begin{cases} 0, & x \le 10, \\ \dfrac{x-10}{5}, & 10 < x < 15, \\ 1, & 15 \le x \le 20, \\ \dfrac{25-x}{5}, & 20 < x < 25, \\ 0, & x \ge 25, \end{cases}$$

and

$$\mu_{\text{Large}}(x) = \begin{cases} 0, & x \le 20, \\ \dfrac{x-20}{5}, & 20 < x < 25, \\ 1, & x \ge 25. \end{cases}$$

The overlap reflects the vagueness of the labels: a weight near 22 kg partially satisfies both `Medium` and `Large`. For example, at $x = 22$ we have $\mu_{\text{Medium}}(22) = (25-22)/5 = 0.6$ and $\mu_{\text{Large}}(22) = (22-20)/5 = 0.4$.

Quick plotting snippet. With `scikit-fuzzy` or plain `matplotlib` you can visualize overlaps to debug label choices:

```
import numpy as np, matplotlib.pyplot as plt
x = np.linspace(0, 30, 400)
mu_small  = np.clip(1 - (x-10)/5, 0, 1) * (x <= 15)
mu_med    = np.clip((x-10)/5, 0, 1) * (x < 15)
mu_med   += ((x>=15) & (x<=20))
mu_med   += np.clip((25-x)/5, 0, 1) * (x > 20)
mu_large  = np.clip((x-20)/5, 0, 1)
plt.plot(x, mu_small, label="Small")
plt.plot(x, mu_med, label="Medium")
plt.plot(x, mu_large, label="Large")
plt.legend(); plt.show()
```

16.9 Fuzzy Sets: Core Concepts and Terminology

Recall that a *fuzzy set* A on a universe X is characterized by a membership function $\mu_A : X \to [0, 1]$, where $\mu_A(x)$ quantifies the degree to which element x belongs to A. Unlike crisp sets, where membership is binary (0 or 1), fuzzy sets allow partial membership.

Support Set The *support* of a fuzzy set A is the set of all elements with nonzero membership:

$$\text{supp}(A) = \{x \in X \mid \mu_A(x) > 0\}.$$

This set captures all elements that belong to A to some degree.

Core Set The *core* of A is the set of elements fully belonging to A:

$$\text{core}(A) = \{x \in X \mid \mu_A(x) = 1\}.$$

The core set generalizes the notion of crisp membership to fuzzy sets.

Normality A fuzzy set A is said to be *normal* if there exists at least one element $x \in X$ such that $\mu_A(x) = 1$. Otherwise, A is *subnormal*. Normality

ensures the fuzzy set has at least one element fully included.

Crossover Points For many membership functions, especially triangular or trapezoidal shapes, the *crossover points* c_A^- and c_A^+ are defined as the points where the membership function crosses the value 0.5:

$$\mu_A(c_A^-) = \mu_A(c_A^+) = 0.5.$$

These points are useful for interpreting the ”core region” and the ”fuzzy boundary” of the set.

Shoulder and Bounded-Core Fuzzy Sets Three shape families cover most linguistic labels:

- A *left-shoulder set* (sometimes called an open-left set in older literature) reaches membership 1 for all sufficiently small x and then decreases smoothly to 0; useful for labels such as “Very Cold” or “Very Low.”
- A *right-shoulder set* (open-right) mirrors this for large x; useful for labels such as “Very Hot” or “Very High.”
- A *bounded-core set* (sometimes called a closed set in older texts, though that usage is non-standard) attains membership 1 only on a bounded interval and tapers to 0 on both sides; triangular and trapezoidal shapes are the canonical examples (e.g., “Comfy” temperature, “Medium” speed).

These distinctions help in modeling asymmetric uncertainties or preferences. This shoulder/bounded-core terminology is standard in modern fuzzy-set writing; avoid the informal “open/closed” labels in formal definitions.

Type-2 Fuzzy Sets: handling uncertain membership

In a standard (type-1) fuzzy set the membership value $\mu_A(x)$ is a crisp number. When expert knowledge is itself imprecise or when membership functions are learned from noisy data, it is more realistic to let the membership value itself be a fuzzy set. This leads to *type-2 fuzzy sets* (Zadeh, 1975).

The **footprint of uncertainty (FOU)** is the bounded region in the (x, μ) plane swept out by all admissible membership grades:

$$\mathrm{FOU}(A) = \bigcup_{x \in X} [{}^{l}\mu_A(x),\ {}^{u}\mu_A(x)],$$

where ${}^{l}\mu_A$ and ${}^{u}\mu_A$ are the lower and upper membership functions bounding the uncertainty.

An **interval type-2 (IT2) fuzzy set** is the practically dominant special case: at each x the secondary membership is uniformly 1 over the FOU interval and 0 outside. IT2 sets are used in robust fuzzy controllers (thermostat, traffic signal, servo drives) when sensor noise or time-varying parameters make crisp breakpoints unreliable.

Reduction: before using an IT2 fuzzy set for inference, it must be *type-reduced* to a type-1 set (or a crisp value). The Karnik–Mendel iterative algorithm is the standard method; simpler closed-form approximations (Nie–Tan method) are often sufficient in practice.

16.10 Probability vs. Possibility

It is crucial to distinguish between *probability* and *possibility* when interpreting membership functions:

- **Probability** models uncertainty about events under a probabilistic model. In both frequentist and Bayesian interpretations, probabilities satisfy additivity over mutually exclusive events:

  $$\sum_i P(E_i) = 1.$$

  For example, the probabilities of mutually exclusive outcomes (red, blue,

black) in one draw sum to 1.

- **Possibility** measures compatibility/plausibility of values with a vague concept, and it is not additive in the probabilistic sense. A possibility distribution $\pi(x)$ is usually normalized by $\sup_x \pi(x) = 1$, not by summation.

Thus, membership functions in fuzzy sets represent *possibility* rather than *probability*. This distinction is fundamental in fuzzy logic and inference (cf. Table 5 in Chapter 15). When treating a membership as a possibility distribution $\pi(x)$, we normalize by $\sup_x \pi(x) = 1$, yielding $\Pi(A) = \sup_{x \in A} \pi(x)$ and $N(A) = 1 - \Pi(A^c)$.

A practical sanity check is to ask what "0.7" means in context. If it is probability, it speaks to event uncertainty under a probability model. If it is fuzzy membership, it means "compatible to degree 0.7" with a linguistic predicate. The same number can appear in both settings, but the semantics and valid operations are different.

Alpha-cuts, convexity, and fuzzy numbers

- **Alpha-cut:** $A_\alpha = \{x \in X \mid \mu_A(x) \geq \alpha\}$ for $\alpha \in (0, 1]$; A_0 is the support. Alpha-cuts turn fuzzy sets into nested crisp sets.
- **Convex fuzzy set:** A is convex if every alpha-cut A_α is convex. Normal + convex fuzzy sets with bounded support are called *fuzzy numbers*.
- **Why it matters:** Alpha-cuts commute with many continuous/monotone maps, making them a practical tool for the extension principle (Chapter 17) and for defuzzification (centroid in Chapter 18).

16.11 Fuzzy Set Operations

Operator defaults used in Chapters 16 to 18

Unless stated otherwise, the fuzzy trilogy uses the *standard* operators:

- Complement (negation): $C(\mu) = 1 - \mu$ (so $C(C(\mu)) = \mu$).
- Intersection and union: $T_{\min}(a, b) = \min(a, b)$, $S_{\max}(a, b) = \max(a, b)$.
- De Morgan's laws are interpreted with this standard complement.

Alternative t-/s-norms or complements are called out explicitly when they appear.

Notation handoff

In the fuzzy trilogy, $\mu_A(x)$ always denotes membership grade, T denotes a t-norm when generalized conjunction is needed, and S denotes an s-norm for generalized union. If these symbols clash with other parts, use the local chapter meaning and consult Appendix A.

Fuzzy logic introduces operations on fuzzy sets that generalize classical set operations but operate on membership functions. Let A and B be fuzzy sets on X with membership functions μ_A and μ_B.

Union The union $A \cup B$ is defined by the membership function:

$$\mu_{A \cup B}(x) = \max\big(\mu_A(x), \mu_B(x)\big). \tag{16.9}$$

This generalizes the classical union by taking the maximum membership degree at each element.

Intersection The intersection $A \cap B$ is defined by:

$$\mu_{A \cap B}(x) = \min\big(\mu_A(x), \mu_B(x)\big). \tag{16.10}$$

This corresponds to the minimum membership degree, reflecting the degree to which x belongs to both sets.

Complement The complement A^c is given by:

$$\mu_{A^c}(x) = 1 - \mu_A(x). \tag{16.11}$$

This generalizes the classical complement by inverting the membership degree. Parameterized complements C_λ (e.g., Yager, Sugeno classes) are sometimes used to alter the "steepness" of the negation; they rarely satisfy involution ($C(C(x)) = x$). A common Sugeno form is

$$C_p(\mu) = \frac{1-\mu}{1+p\,\mu}, \quad p \geq 0,$$

which preserves $C_p(0) = 1$ and $C_p(1) = 0$ but is involutive only when $p = 0$. Whenever strict involution is required (as in many De Morgan identities), we default to the standard complement $C(\mu) = 1 - \mu$.

Remarks These operations satisfy properties analogous to classical set theory but adapted to fuzzy membership values. For completeness, De Morgan's laws in fuzzy logic can be written either as equivalences between sets or explicitly in max/min form:

$$\mu_{(A\cap B)^c}(x) = \mu_{A^c\cup B^c}(x) = \max\big(1-\mu_A(x),\, 1-\mu_B(x)\big),$$
$$\mu_{(A\cup B)^c}(x) = \mu_{A^c\cap B^c}(x) = \min\big(1-\mu_A(x),\, 1-\mu_B(x)\big).$$

Throughout the book we adopt $\wedge = \min$ and $\vee = \max$ as the default t-/s-norm pair with the standard complement $1 - \mu$ (the De Morgan triple used again in Chapter 18 unless noted otherwise); alternative norms appear later in operator tables.

Reminder on basic operators Equations (16.9) to (16.11) already define the max/min/standard-complement pair that we use by default. Rather than restate them, we emphasise their practical role: unions aggregate rule consequents, intersections combine antecedents, and complements capture linguistic negations. The thermostat example later in the chapter uses these defaults unless stated otherwise.

Graphical interpretation For continuous universes, the union and intersection membership functions are the pointwise maximum and minimum of the two curves, respectively. A complement mirrors memberships around $\mu = 0.5$: every membership degree m maps to $1-m$. When you sketch these operations quickly, plot both curves on the same axes and take the upper (union) or lower (intersection) envelope. The same pointwise definitions apply to discrete universes; you simply compute the max/min values at each listed x_i.

16.12 Additional Fuzzy Set Operations

Beyond the basic operations, several other algebraic operations are defined on fuzzy sets:

Algebraic Product The algebraic product of fuzzy sets A and B is defined by the product of their membership values:

$$\mu_{A \cdot B}(x) = \mu_A(x) \cdot \mu_B(x), \quad \forall x \in X. \tag{16.12}$$

Scalar Multiplication Given a scalar $\alpha \in [0, 1]$, scalar multiplication of a fuzzy set A is:

$$\mu_{\alpha A}(x) = \alpha \cdot \mu_A(x), \quad \forall x \in X. \tag{16.13}$$

Algebraic Sum The algebraic sum of fuzzy sets A and B is given by:

$$\mu_{A+B}(x) = \mu_A(x) + \mu_B(x) - \mu_A(x) \cdot \mu_B(x), \quad \forall x \in X. \tag{16.14}$$

This operation ensures the resulting membership values remain within $[0, 1]$.

Difference The difference between fuzzy sets A and B, denoted $A - B$, can be defined as:

$$\mu_{A-B}(x) = \mu_A(x) \wedge \big(1 - \mu_B(x)\big) = \min\big(\mu_A(x), 1 - \mu_B(x)\big), \tag{16.15}$$

where $\wedge$ denotes the minimum operator.

Bounded Difference An alternative definition of difference is the bounded difference:

$$\mu_{A \ominus B}(x) = \max\big(0, \mu_A(x) - \mu_B(x)\big). \tag{16.16}$$

Remarks:

- The difference operation in (16.15) corresponds to the intersection of A with the complement of B.
- The bounded difference in (16.16) ensures membership values remain non-negative.
- These operations extend classical set difference to fuzzy sets, but their interpretations can vary depending on the application.

16.13 Example: Union and Intersection of Fuzzy Sets

Use the pointwise definitions in (16.9)–(16.10) to compute unions or intersections for any pair of fuzzy sets; the next subsection lifts these operations to relations via Cartesian products.

Example. Let $X = \{x_1, x_2, x_3\}$ and define memberships in the order (x_1, x_2, x_3):

$$\mu_A = \{0.2,\ 0.7,\ 0.4\}, \qquad \mu_B = \{0.6,\ 0.3,\ 0.9\}.$$

Then

$$\mu_{A \cup B} = \{\max(0.2, 0.6),\ \max(0.7, 0.3),\ \max(0.4, 0.9)\} = \{0.6,\ 0.7,\ 0.9\},$$
$$\mu_{A \cap B} = \{\min(0.2, 0.6),\ \min(0.7, 0.3),\ \min(0.4, 0.9)\} = \{0.2,\ 0.3,\ 0.4\}.$$

The union keeps the larger membership at each point, while the intersection keeps the smaller.

16.14 Cartesian Product of Fuzzy Sets

Using the membership-function definition from Section 16.9, the *Cartesian product* of two fuzzy sets A on X and B on Y is a fuzzy relation on the product space $X \times Y$.

Definition: The membership function of the Cartesian product $A \times B$ is defined as

$$\mu_{A\times B}(x, y) = \min\big(\mu_A(x), \mu_B(y)\big), \quad \forall x \in X, y \in Y. \tag{16.17}$$

This operation generalizes the classical Cartesian product of crisp sets to fuzzy sets by taking the minimum membership grade of the paired elements.

Example: Suppose

$$A = \{(x_1, 1.0), (x_2, 0.8), (x_3, 0.4)\}, \quad B = \{(y_1, 0.6), (y_2, 0.8), (y_3, 1.0)\}.$$

Then the Cartesian product $A \times B$ is represented by the matrix of membership values:

$\mu_{A\times B}(x, y)$	y_1	y_2	y_3
x_1	$\min(1.0, 0.6) = 0.6$	$\min(1.0, 0.8) = 0.8$	$\min(1.0, 1.0) = 1.0$
x_2	$\min(0.8, 0.6) = 0.6$	$\min(0.8, 0.8) = 0.8$	$\min(0.8, 1.0) = 0.8$
x_3	$\min(0.4, 0.6) = 0.4$	$\min(0.4, 0.8) = 0.4$	$\min(0.4, 1.0) = 0.4$

Note that the Cartesian product lifts the fuzzy sets from one-dimensional membership functions to a two-dimensional fuzzy relation.

16.15 Properties of Fuzzy Set Operations

The fuzzy set operations (union, intersection, complement) satisfy several important algebraic properties analogous to classical set theory, but defined in terms of membership functions.

Commutativity:

$$\mu_{A\cap B}(x) = \mu_{B\cap A}(x),$$
$$\mu_{A\cup B}(x) = \mu_{B\cup A}(x).$$

Associativity:

$$\mu_{(A\cap B)\cap C}(x) = \mu_{A\cap(B\cap C)}(x),$$
$$\mu_{(A\cup B)\cup C}(x) = \mu_{A\cup(B\cup C)}(x).$$

Distributivity:

$$\mu_{A\cup(B\cap C)}(x) = \mu_{(A\cup B)\cap(A\cup C)}(x),$$
$$\mu_{A\cap(B\cup C)}(x) = \mu_{(A\cap B)\cup(A\cap C)}(x).$$

Identity Elements:

$$\mu_{A\cup\emptyset}(x) = \mu_A(x),$$
$$\mu_{A\cap X}(x) = \mu_A(x),$$

where $\emptyset$ is the empty fuzzy set with membership zero everywhere, and X is the universal fuzzy set with membership one everywhere.

Involution:

$$\mu_{(A^c)^c}(x) = \mu_A(x).$$

In operator notation this reads $C(C(\mu_A(x))) = \mu_A(x)$: applying the complement twice recovers the original membership degree. For the standard fuzzy complement $C(\mu_A(x)) = 1 - \mu_A(x)$, involution is just the identity

$$1 - \big(1 - \mu_A(x)\big) = \mu_A(x),$$

so the membership "returns" to its original value after two applications.

De Morgan's Laws: With the standard complement A^c and the max/min operators in Equations (16.9) to (16.11), the classical De Morgan identities hold: $(A \cap B)^c = A^c \cup B^c$ and $(A \cup B)^c = A^c \cap B^c$.

These properties ensure that fuzzy set operations behave in a consistent and algebraically sound manner, enabling the extension of classical set theory to fuzzy logic.

16.16 Fuzzy Set Operators

While operations such as union, intersection, and complement define how to combine or modify fuzzy sets, *operators* formalize the rules behind those combinations. In practice, an operator is a mapping on membership grades that

encodes how conservative or permissive the model should be when evidence conflicts.

Three operator families appear repeatedly in the fuzzy trilogy:

- **Unary operators:** complement and linguistic hedges (for example, "not," "very," "more or less") that reshape a single membership function.
- **Binary operators:** t-norms and s-norms that implement fuzzy AND/OR and determine how multiple antecedents combine.
- **Relation operators:** extension, projection, and composition operators that transport fuzzy information between related universes.

The next subsection starts with unary negation operators, because complement choice directly affects De Morgan duality and downstream rule behavior.

16.17 Complement Operators in Fuzzy Logic

In classical logic, the complement of a proposition A is simply $1 - \mu_A(x)$, where $\mu_A(x)$ is the membership function of A. However, in fuzzy logic, this complement operation can be generalized to allow more flexible modeling of uncertainty and partial membership.

Standard complement We use the standard fuzzy negation $C(\mu) = 1 - \mu$, so $\mu_{A^c}(x) = 1 - \mu_A(x)$ (see (16.11)). It is simple, involutive, and usually the safest default unless a parameterized negation is explicitly required.

Parameterized Complement Operators To generalize the complement, choose a negation operator $C_p : [0,1] \to [0,1]$ and apply it pointwise: $\mu_{C_p(A)}(x) = C_p(\mu_A(x))$. One common (Sugeno-type) family is

$$C_p(\mu) = \frac{1-\mu}{1+p\mu}, \qquad p \geq 0,$$

which reduces to the standard complement when $p = 0$.

Another simple family is a power-law negation:

$$C_p(\mu) = (1-\mu)^p, \qquad p > 0,$$

which recovers the standard complement when $p = 1$ and adjusts the steepness for other p.

These operators allow for a nonlinear mapping of the complement, reflecting different degrees of confidence or hesitation in the membership values. Unlike the standard complement, most parameterized families *do not* preserve involution $C(C(\mu)) = \mu$ for arbitrary p; they are typically designed to satisfy boundary conditions and monotonicity instead. When strict involution is required, it is safest to use the standard complement.

Properties of Complement Operators A commonly desired set of properties for a complement operator C is:

- **Boundary conditions:** $C(0) = 1$ and $C(1) = 0$.
- **Monotonicity:** $\mu_A(x) \leq \mu_B(x) \implies C(\mu_A(x)) \geq C(\mu_B(x))$.
- **Involution (optional):** $C(C(\mu_A(x))) = \mu_A(x)$.

The standard complement satisfies all three. Parameterized complements typically satisfy the first two, while involution may be relaxed to gain extra modeling flexibility; one should check involution explicitly if it is required by a particular application.

16.18 Triangular norms (t-norms)

Motivation In fuzzy logic, the logical `AND` operation is generalized by *triangular norms* (t-norms). These are binary operators that combine membership values while preserving certain desirable properties analogous to intersection in classical set theory.

Definition A **t-norm** is a binary operator $T : [0,1]^2 \to [0,1]$ satisfying the following properties for all $x, y, z \in [0,1]$:

1. **Commutativity:**
$$T(x, y) = T(y, x).$$

2. **Associativity:**
$$T(x, T(y, z)) = T(T(x, y), z).$$

3. **Monotonicity:**

$$x \leq x', \quad y \leq y' \implies T(x, y) \leq T(x', y').$$

4. **Boundary condition (Identity):**

$$T(x, 1) = x, \quad T(x, 0) = 0.$$

These properties ensure that T behaves like a generalized intersection operator.

Examples of t-norms

- **Minimum t-norm:**

$$T_{\min}(x, y) = \min(x, y).$$

  This corresponds to the classical intersection in fuzzy sets.

- **Algebraic product t-norm:**

$$T_{\text{prod}}(x, y) = x \cdot y.$$

  This is a smooth, multiplicative generalization of intersection.

- **Łukasiewicz t-norm:**

$$T_{\text{Luk}}(x, y) = \max(0, x + y - 1).$$

For a geometric intuition, Figure 68 visualizes two common choices for fuzzy `AND`: the conservative minimum and the multiplicative product.

Each t-norm captures different semantics of conjunction in fuzzy logic.

Interpretation The t-norm generalizes the classical intersection operator to fuzzy sets by ensuring the output membership value remains within $[0, 1]$ and respects the ordering and boundary conditions expected of an intersection.

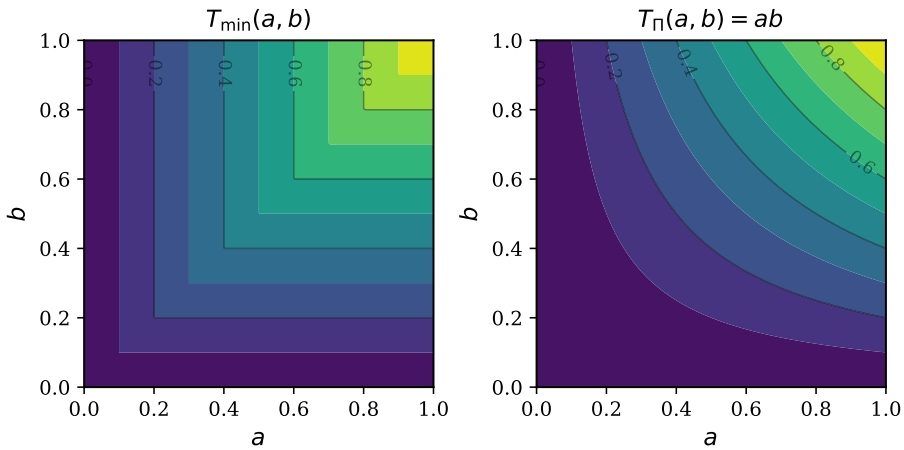


Figure 68: Fuzzy AND surfaces comparing minimum versus product t-norms; analogous OR surfaces show similar differences. Choices here influence rule aggregation in Chapter 18. The plot makes the design choice visible: conservative minimum versus multiplicative attenuation. *Source: author-generated.*

16.19 Triangular conorms (t-conorms / s-norms)

Definition The dual concept to t-norms is the **triangular conorm** (t-conorm), also called an *s-norm*, which generalizes the logical `OR` operation. A t-conorm $S : [0, 1]^2 \to [0, 1]$ satisfies:

1. **Commutativity:**
$$S(x, y) = S(y, x).$$

2. **Associativity:**
$$S(x, S(y, z)) = S(S(x, y), z).$$

3. **Monotonicity:** If $x \leq x'$ and $y \leq y'$, then
$$S(x, y) \leq S(x', y').$$

4. **Boundary conditions:**
$$S(x, 0) = x, \qquad S(x, 1) = 1.$$

These axioms mirror those of t-norms but with 1 as the neutral element instead of 0. Standard examples include the maximum s-norm $S_{\max}(x, y) = \max(x, y)$, the algebraic sum $S_{\text{sum}}(x, y) = x + y - xy$, and the bounded sum $S_{\text{bs}}(x, y) = \min(1, x + y)$; explicit formulas and their dual t-norms appear in the next subsection. Note that the algebraic sum explicitly enforces $S_{\text{sum}}(x, y) = x + y - xy \leq 1$ for all $x, y \in [0, 1]$.

16.20 T-Norms and S-Norms: Complementarity and Properties

We use the t-norm and s-norm definitions from the previous two subsections; here we focus on their complementarity via negation.

An important relationship between t-norms and s-norms is their complementarity via a negation operator. Throughout this section we use the *standard* fuzzy negation $N(x) = 1 - x$, so that the complement of μ_A and the duality between T and S can be written together as

$$\begin{aligned} \mu_{A^c}(x) &= 1 - \mu_A(x), \\ T(\mu_A(x), \mu_B(x)) &= 1 - S(1 - \mu_A(x), 1 - \mu_B(x)), \\ S(\mu_A(x), \mu_B(x)) &= 1 - T(1 - \mu_A(x), 1 - \mu_B(x)). \end{aligned} \tag{16.18}$$

This duality ensures that the fuzzy intersection and union are consistent with respect to complementation, generalizing classical De Morgan's laws.

16.21 Examples of common t-norm / s-norm pairs

Several standard t-norms and their corresponding s-norms are widely used:

- **Minimum t-norm and maximum s-norm:**

$$T_{\min}(x, y) = \min(x, y), \quad S_{\max}(x, y) = \max(x, y).$$

- **Algebraic product t-norm and algebraic sum s-norm:**

$$T_{\text{prod}}(x, y) = x \cdot y, \quad S_{\text{sum}}(x, y) = x + y - xy.$$

- **Bounded difference t-norm and bounded sum s-norm:**

$$T_{\text{bd}}(x, y) = \max(0, x + y - 1), \quad S_{\text{bs}}(x, y) = \min(1, x + y).$$

Each of these pairs satisfies the complementarity relation (16.18).

16.22 Fuzzy Set Inclusion and Subset Relations

In classical set theory, $A \subseteq B$ means every element of A is also in B. For fuzzy sets, the notion of subset is generalized via membership functions.

Definition (Fuzzy Subset). A fuzzy set A is a *subset* of fuzzy set B, denoted $A \subseteq B$, if and only if

$$\mu_A(x) \leq \mu_B(x), \quad \forall x \in X, \tag{16.19}$$

where X is the universe of discourse.

If the inequality is strict for at least one x, i.e., $\mu_A(x) < \mu_B(x)$ for some x, then A is a *proper fuzzy subset* of B.

Interpretation: Since membership functions represent degrees of belonging, the subset relation is graded rather than binary. This leads naturally to the concept of *degree of inclusion*.

16.23 Degree of Inclusion

Because fuzzy membership values lie in $[0, 1]$, the subset relation in (16.19) can also be quantified by a scalar measure indicating *how much* A is included in B.

For practical work we often use an *aggregate* measure:

$$\mathrm{incl}(A, B) = \frac{\sum_{x \in X} \min(\mu_A(x), \mu_B(x))}{\sum_{x \in X} \mu_A(x)} \tag{16.20}$$

for discrete universes (integrals for continuous, assuming finite mass). It summarizes how much of the mass of A lies inside B's support. A *pointwise* alternative relies on an implicator I and defines $\mathrm{Inc}(A, B) = \inf_x I(\mu_A(x), \mu_B(x))$ (see below); implicator-based grades avoid division by small μ_B and behave well when B has zeros. Both constructions satisfy $0 \leq \mathrm{incl}(A, B) \leq 1$, where 1 means A is fully included in B.

16.24 Set Operations and Inclusion Properties

Given fuzzy sets A, B, and C, the following properties hold for the standard t-norm and s-norm operations:

- If $A \subseteq B$, then $A \cap C \subseteq B \cap C$ and $A \cup C \subseteq B \cup C$. Explicitly,

$$\mu_{A\cap C}(x) = \min(\mu_A(x), \mu_C(x)) \leq \min(\mu_B(x), \mu_C(x)) = \mu_{B\cap C}(x),$$

  and analogously for the union/max operator.

- If $A \subseteq B$, applying any t-norm T and its dual s-norm S preserves inclusion: $T(A, C) \subseteq T(B, C)$ and $S(A, C) \subseteq S(B, C)$. In terms of memberships,

$$\mu_{T(A,C)}(x) \leq \mu_{T(B,C)}(x) \quad \text{and} \quad \mu_{S(A,C)}(x) \leq \mu_{S(B,C)}(x), \ \forall x.$$

- Complements reverse inclusion: $A \subseteq B \Rightarrow B^c \subseteq A^c$ because complements flip the ordering of memberships. $\mu_{B^c}(x) = 1 - \mu_B(x) \leq 1 - \mu_A(x) = \mu_{A^c}(x)$.

16.25 Grades of Inclusion and Equality in Fuzzy Sets

Recall that in classical set theory, the notion of subset and equality is crisp: a set A is a subset of B if every element of A is also in B, and $A = B$ if they contain exactly the same elements. In fuzzy set theory, these notions are generalized via *grades* of inclusion and equality, which quantify the degree to which one fuzzy set is included in or equal to another.

Grade of Inclusion Given two fuzzy sets A and B defined on the universe X, with membership functions $\mu_A(x)$ and $\mu_B(x)$, respectively, the *grade of inclusion* of A in B, denoted $\mathrm{Inc}(A, B)$, measures how much A is a subset of B.

One way to define this grade is:

$$\mathrm{Inc}(A, B) = \inf_{x\in X} I\big(\mu_A(x), \mu_B(x)\big), \tag{16.21}$$

where I is an *implicator* function, often derived from a chosen t-norm T. A common choice is the Gödel implicator:

$$I(a, b) = \begin{cases} 1, & \text{if } a \leq b, \\ b, & \text{otherwise.} \end{cases} \tag{16.22}$$

Alternatively, if T is part of a residuated pair (T, I), one sometimes writes

$$\mathrm{Inc}(A, B) = \inf_{x \in X} T\big(\mu_A(x), \mu_B(x)\big),$$

which should be interpreted as computing the tightest lower bound obtainable from the chosen T; this coincides with the implicator-based definition when I is the residuum of T.

Example Suppose A and B are fuzzy sets with membership functions such that for some x we have $\mu_A(x) \leq \mu_B(x)$, and for others $\mu_A(x) > \mu_B(x)$. Using the Gödel implicator,

$$I_G(\mu_A(x), \mu_B(x)) = \begin{cases} 1, & \mu_A(x) \leq \mu_B(x), \\ \mu_B(x), & \mu_A(x) > \mu_B(x), \end{cases}$$

so the overall grade of inclusion is $\inf_{x \in X} I_G(\mu_A(x), \mu_B(x))$. This explicitly shows how the implicator returns the smaller membership where A exceeds B.

Grade of Equality Similarly, the *grade of equality* between fuzzy sets A and B, denoted $\mathrm{Eq}(A, B)$, measures how close the two sets are to being equal. It can be defined as:

$$\mathrm{Eq}(A, B) = \inf_{x \in X} J\big(\mu_A(x), \mu_B(x)\big), \tag{16.23}$$

where J is an equality function. One convenient choice is

$$J(a, b) = \begin{cases} 1, & \text{if } a = b, \\ T(a, b), & \text{otherwise}, \end{cases} \tag{16.24}$$

with T a t-norm, so that exact agreement receives unit credit while disagreements are down-weighted via T. Other smooth symmetry measures (e.g., $J(a, b) = 1 - |a - b|$) can also be used; the key requirement is that J be symmetric, bounded in $[0, 1]$, and reach 1 only when $a = b$.

This definition allows for a graded notion of equality, reflecting the fuzzy nature of the sets.

16.26 Dilation and Contraction of Fuzzy Sets

Motivation Constructing fuzzy sets with appropriate membership functions is a challenging task. Often, one starts with an initial fuzzy set A and wishes to generate related fuzzy sets that represent concepts such as "more or less A" or "somewhat A". This leads to the operations of *dilation* and *contraction* of fuzzy sets, which modify the membership function to reflect these linguistic hedges.

Definitions Given a fuzzy set A with membership function $\mu_A(x)$, we introduce two non-negative shape parameters constrained to $\alpha \geq 1$ (dilation gain) and $\beta \geq 1$ (contraction gain) so that the resulting hedges behave monotonically:

$$\mu_{A^{(d)}}(x) = \big(\mu_A(x)\big)^{1/\alpha}, \qquad \alpha \geq 1 \quad \text{(dilation)}. \tag{16.25}$$

$$\mu_{A^{(c)}}(x) = \big(\mu_A(x)\big)^{\beta}, \qquad \beta \geq 1 \quad \text{(contraction)}. \tag{16.26}$$

Using separate symbols α and β avoids the notational clash that occurs when a single parameter k is forced to satisfy both $0 < k \leq 1$ (for dilation) and $k \geq 1$ (for contraction). In some references these two operations are also called *expansion* and *narrowing*; we treat the terms as synonyms.

Note that:

- For dilation, $0 < \mu_A(x) < 1$ implies $\mu_A(x)^{1/\alpha} \geq \mu_A(x)$ when $\alpha \geq 1$, so every membership value moves closer to 1, making the fuzzy set "larger" or more inclusive. Setting $\alpha = 1$ leaves the set unchanged.
- For contraction, $0 < \mu_A(x) < 1$ implies $\mu_A(x)^{\beta} \leq \mu_A(x)$ when $\beta \geq 1$, so the membership values move toward 0, making the fuzzy set "smaller" or more restrictive. Again, $\beta = 1$ recovers the original set.

Properties

- The *core* of the fuzzy set, i.e., the elements with membership 1, remains unchanged under dilation or contraction because $1^{1/\alpha} = 1^{\beta} = 1$ for all positive α, β:

$$\mu_A(x) = 1 \implies \mu_{A^{(d)}}(x) = 1 \text{ and } \mu_{A^{(c)}}(x) = 1.$$

16.27 Closure of Membership Function Derivations

This chapter completes the toolkit for generating new membership functions from existing ones via fuzzy-set operations. Membership functions encode graded belonging over a universe of discourse, and algebraic manipulation of those functions is central to fuzzy logic and fuzzy inference.

Two transfer cues matter for what comes next. First, treat membership functions as *objects you can manipulate*: once the shapes are written down, you can reason about overlaps, negations, and edge cases without guessing. Second, the same max/min mindset will reappear when we move from sets to *relations*: we will build fuzzy relations over Cartesian products and then compose them (a matrix-like operation) to propagate graded information between universes.

As you generate new membership functions below, keep a short audit checklist: verify the output stays in $[0, 1]$, check endpoints and monotonicity where intended, and confirm that the universe/units match the linguistic meaning. Those checks are invariants you will want to preserve later when you tune breakpoints, operator defaults, or rule bases.

16.27.1 Generating New Membership Functions via Set Operations

Given two membership functions, for example, $\mu_{\text{young}}(x)$ and $\mu_{\text{old}}(x)$, defined over the same universe X, we can construct new membership functions by applying the following operations:

Dilation (Expansion) Dilation increases the support of a fuzzy set, effectively "loosening" the membership criteria. For instance, dilating the old membership function yields a new fuzzy set more or less old:

$$\mu_{\text{more or less old}}(x) = \text{dilate}(\mu_{\text{old}}(x))$$

This operation broadens the range of x values considered "old" to some degree, reflecting linguistic vagueness.

Contraction (Narrowing) Contraction tightens the membership function, focusing on a core subset. For example, contracting $\mu_{\text{old}}(x)$ produces $\mu_{\text{too old}}(x)$:

$$\mu_{\text{too old}}(x) = \text{contract}(\mu_{\text{old}}(x))$$

This captures a stricter notion of "old."

Complement The complement of a fuzzy set reverses membership degrees:

$$\mu_{\text{not } A}(x) = 1 - \mu_A(x)$$

For example, $\mu_{\text{not young}}(x) = 1 - \mu_{\text{young}}(x)$.

Intersection The intersection of two fuzzy sets corresponds to the minimum of their membership functions:

$$\mu_{A \cap B}(x) = \min\{\mu_A(x), \mu_B(x)\}$$

This operation models the logical AND.

Union The union corresponds to the maximum:

$$\mu_{A \cup B}(x) = \max\{\mu_A(x), \mu_B(x)\}$$

16.27.2 Examples of Constructed Membership Functions

Using these operations, we can create nuanced fuzzy sets:

- **Not young and not old:**

  $$\mu_{\text{not young and not old}}(x) = \min\big(1 - \mu_{\text{young}}(x),\, 1 - \mu_{\text{old}}(x)\big)$$

  This set captures individuals who are neither young nor old, representing a middle-aged group.

- **Young but not too old:** First, contract $\mu_{\text{old}}(x)$ to get $\mu_{\text{too old}}(x)$, then take its complement, and intersect with $\mu_{\text{young}}(x)$:

  $$\mu_{\text{young but not too old}}(x) = \min\big(\mu_{\text{young}}(x),\, 1 - \mu_{\text{too old}}(x)\big)$$

  This set isolates those who are young but excludes those considered "too old," refining the concept of youthfulness.

- **More or less old:** Applying dilation to $\mu_{\text{old}}(x)$ expands the fuzzy set:

$$\mu_{\text{more or less old}}(x) = \text{dilate}(\mu_{\text{old}}(x))$$

Remark on Normality Note that some constructed membership functions may not be *normal*, i.e., their maximum membership degree may be less than 1. This reflects the inherent fuzziness and partial membership in linguistic concepts.

16.28 Implications for Fuzzy Inference Systems

The ability to generate new membership functions from a small set of base functions (e.g., μ_{young} and μ_{old}) is powerful. It allows us to encode complex human knowledge and linguistic nuances into fuzzy sets, which can then be used in fuzzy inference systems.

For example, consider an inference system with inputs:

$$\text{Age} \quad (x), \quad \text{Exercise Level} \quad (e)$$

and output:

$$\text{Health Status} \quad (h)$$

We can define membership functions for *age* (e.g., young, old) and *exercise level* (e.g., low, high), then use fuzzy operators (intersection, union, complement) to combine these inputs according to rules such as:

IF Age is old AND Exercise is high
THEN Health is good

In a Mamdani-style controller the conjunction "AND" is typically modeled by the minimum operator and the implication uses the same t-norm (i.e., the consequent is clipped at the firing strength). Other choices include using the product t-norm for conjunction, Larsen-style scaling for implication, and max for rule aggregation. Any alternative should be stated explicitly.

The next step is to formalize the *implication* and *aggregation* operators that map these fuzzy inputs to fuzzy outputs, and then perform *defuzzification* to obtain crisp outputs.

Table 7 serves as the operator-choice checklist in the Mamdani design dis-

Table 7: Typical operator choices in fuzzy inference and their qualitative effects. Here the t-norm implements fuzzy AND, the s-norm implements fuzzy OR, and the implication shapes consequents. The table supports default-operator selection for a Mamdani-style pipeline and anticipates qualitative output behavior.

t-norm $T(a,b)$	s-norm $S(a,b)$	Implication $\Rightarrow$	Qualitative behavior
$\min(a,b)$	$\max(a,b)$	Mamdani (clipping: $\min(\alpha,\mu_B)$)	Sharp, piecewise-linear surfaces; conservative.
$a \cdot b$	$a+b-ab$	Larsen (scaling: $\alpha\,\mu_B$)	Smoother transitions; products damp small activations.
$\max(0,a+b-1)$	$\min(1,a+b)$	Bounded (e.g., Łukasiewicz)	Bounded sums; useful when saturation is desired.

cussion.

16.29 Worked Example: Mamdani Fuzzy Inference (End-to-End)

We illustrate a complete Mamdani pipeline with one antecedent (temperature) and one consequent (fan speed).

Universes and membership functions

- Temperature $T \in [0,40]\,^\circ\mathrm{C}$ with fuzzy sets

$$\mu_{\mathrm{Cold}}(t) = \max\Big(0,\ 1-\tfrac{t-0}{15-0}\Big) \qquad (0,0,15),$$
$$\mu_{\mathrm{Warm}}(t) = \max\Big(0,\ 1-\tfrac{|t-20|}{10}\Big) \qquad (10,20,30),$$
$$\mu_{\mathrm{Hot}}(t) = \max\Big(0,\ \tfrac{t-25}{40-25}\Big) \qquad (25,40,40).$$

- Fan speed $S \in [0, 1]$ with fuzzy sets

$$\begin{aligned}
\mu_{\text{Low}}(s) &= \max\Big(0,\ 1 - \tfrac{s-0}{0.5-0}\Big) && (0, 0, 0.5),\\
\mu_{\text{Medium}}(s) &= \max\Big(0,\ 1 - \tfrac{|s-0.5|}{0.25}\Big) && (0.25, 0.5, 0.75),\\
\mu_{\text{High}}(s) &= \max\Big(0,\ \tfrac{s-0.5}{1-0.5}\Big) && (0.5, 1, 1).
\end{aligned}$$

Rule base

1. IF T is Cold THEN S is Low.
2. IF T is Warm THEN S is Medium.
3. IF T is Hot THEN S is High.

Fuzzify input and compute firing strengths For an input temperature $T = 27\,℃$,

$$\begin{aligned}
\mu_{\text{Cold}}(27) &= 0,\\
\mu_{\text{Warm}}(27) &= \frac{30-27}{10} = 0.3,\\
\mu_{\text{Hot}}(27) &= \frac{27-25}{15} = \frac{2}{15} \approx 0.133.
\end{aligned}$$

Using min-implication (clipping), the consequents become

$$\begin{aligned}
\mu'_{\text{Low}}(s) &= \min\big(0,\ \mu_{\text{Low}}(s)\big) = 0,\\
\mu'_{\text{Medium}}(s) &= \min\big(0.3,\ \mu_{\text{Medium}}(s)\big),\\
\mu'_{\text{High}}(s) &= \min\big(0.133,\ \mu_{\text{High}}(s)\big).
\end{aligned}$$

Aggregating by max yields the overall output fuzzy set

$$\mu_{\text{out}}(s) = \max\big(\mu'_{\text{Low}}(s),\ \mu'_{\text{Medium}}(s),\ \mu'_{\text{High}}(s)\big).$$

Defuzzification (centroid) The crisp fan speed is the centroid

$$s^\star = \frac{\int_0^1 s\,\mu_{\text{out}}(s)\,ds}{\int_0^1 \mu_{\text{out}}(s)\,ds}.$$

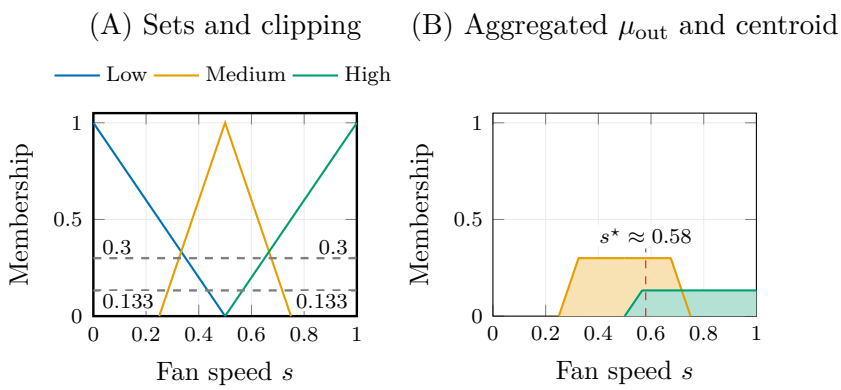


Figure 69: End-to-end fuzzy inference example. (A) Consequent membership functions with clipping levels from firing strengths at T = 27 deg C. (B) Aggregated output set (max of truncated consequents) and a centroid marker near s* approx 0.58. This trace provides an end-to-end sanity check for clipping, aggregation, and centroid defuzzification.

For symmetric triangles, the centroid of a truncated Medium set remains at 0.5, and the centroid of High is at ≈ 0.833. Approximating the centroid of the max-aggregated set by a convex combination of these centroids weighted by their peak heights,

$$s^\star \approx \frac{0.3 \cdot 0.5 + 0.133 \cdot 0.833}{0.3 + 0.133} \approx 0.58.$$

An exact centroid can be computed analytically or numerically by integrating the clipped shapes; the approximation above matches a direct trapezoidal integration on a uniform grid (10k points), which yields $s^\star \approx 0.580$ to three decimals. See Figure 69 for the membership functions and clipping levels used in this example. Practical tip: libraries such as `scikit-fuzzy` provide a tested `defuzz` (centroid) routine; when in doubt, compute the centroid numerically rather than relying on heuristic convex combinations.

Key takeaways

- Fuzzy sets map elements to degrees in $[0, 1]$; membership shapes (triangular, trapezoidal, Gaussian) encode semantics.
- Support/core and set operations (intersection/union/complement) generalize crisp logic.
- Visualizing membership and operations clarifies design of fuzzy controllers.

Minimum viable mastery.

- Construct a universe of discourse, define overlapping memberships, and compute degrees for concrete inputs.
- Apply basic operations (min/max, product, complements) and predict how the choice changes shape.
- Translate a design intent ("smooth", "conservative", "aggressive") into membership overlap and operator choices.

Common pitfalls.

- Setting memberships without checking units/scales, producing labels that never activate or always saturate.
- Using too many overlapping labels without justification (hard to tune, hard to interpret).
- Tuning by aesthetics alone instead of checking downstream sensitivity and stability.

Exercises

Exercise 1 (Easy). Define 3–5 fuzzy labels for a new universe (e.g., vehicle speed, battery state of charge, or network latency). Compute membership degrees for three sample points and explain one design choice (coverage, overlap, or saturation).
Deliverable: the membership definitions + the three computed degree vectors + one sentence justifying your overlap.

Exercise 2 (Medium). Compare semantics, not just formulas: pick two t-norm/s-norm pairs (e.g., max–min vs. max–product) and compute union/intersection at several points. Interpret the numerical differences as different meanings of "AND" and "OR."
Deliverable: a small pointwise table + one paragraph explaining which pair is more strict/lenient and in what sense.

Exercise 3 (Medium). Run a units/scale audit: take one set of membership functions and rescale the universe (change units or normalize), then show what must be adjusted so that behavior remains consistent downstream.
Deliverable: before/after plots (or parameter tables) + one paragraph describing how scale mistakes create controllers that "never fire" or "always saturate."

Fast-track note. Later fuzzy chapters build directly on these primitives. If rule outputs become unstable, revisit universe scaling and membership overlap first.

Where we head next. Chapter 17 moves from set-level primitives to *transfer*: how to push fuzzy labels through mappings between related universes (extension principle, projection, and composition). Chapter 18 then uses those relation tools to assemble complete rule composition and defuzzification pipelines.

17 Fuzzy Relations: Composition, Projection, and the Extension Principle

Building on Chapter 16, we address a fundamental question: *How do we transfer fuzzy knowledge from one universe of discourse to another related universe?* This arises whenever the same linguistic label must be reused across units, sensors, or derived variables (Celsius vs. Fahrenheit; position vs. velocity), each with its own universe and membership functions. This chapter develops that transfer layer so Chapter 18 can assemble full inference pipelines.

Learning Outcomes

- Apply the extension principle (single and multi-variable) to transport fuzzy knowledge across domains.
- Select appropriate t-norms/t-conorms and understand how those choices affect projection, dilation, and composition.
- Tie these transformations to the running thermostat/autofocus example to anticipate how inference rules will behave in Chapter 18.

Design motif

Preserve meaning while changing representation: the extension principle is a disciplined way to push fuzzy concepts through transformations so downstream inference remains interpretable (see Chapter 18).

Running example checkpoint. We treat the thermostat's heater power as the target universe while the inputs remain error/rate. When mapping "Comfortable" from Celsius to Fahrenheit or translating error/rate pairs into control actions, the extension principle tells us how those fuzzy labels transfer; keep that single example in mind as you work through the upcoming dilation and projection formulas.

Rolling-window note (what changes vs. Chapter 16). In Chapter 16, the main design work was *within* a universe: choose membership shapes, overlaps, and operators. Here the memberships are assumed given; the new question is *between* universes: how does a fuzzy label move through a mapping without losing

meaning? Treat this chapter as the transfer layer that makes the later inference pipeline (Chapter 18) reusable across sensors, units, and derived variables.

17.1 Context and Motivation

Previously, we studied operations such as *dilation* and *contraction* on fuzzy sets within a single universe of discourse. For example, given a fuzzy set representing the concept *young*, we can generate related fuzzy sets like *less young* or *too old* by applying these operations. By combining these fuzzy sets, we can express nuanced concepts such as *not too young* or *not too old* within the same universe.

However, what if we want to extend this reasoning to a *different* universe of discourse that is related to the original one? For instance, consider the following scenarios:

- Mapping temperature from Celsius to Fahrenheit.
- Transforming a variable x to $y = x^2$.
- Relating speed and acceleration to derive new fuzzy sets.

In such cases, the new universe is a function of the original universe, and we want to *preserve* and *transfer* the fuzzy knowledge encoded in the original fuzzy sets to the new universe.

Operator defaults in this trilogy

Unless stated otherwise in Chapters 16 to 18, we use the standard De Morgan triple: $\wedge = \min$, $\vee = \max$, and complement $C(\mu) = 1 - \mu$. Parameterized complements (Yager/Sugeno) are noted when used, but they generally lose involution ($C(C(\mu)) \neq \mu$) unless the parameter is zero.

Notation. When we introduce a general t-norm T, it appears explicitly in the multi-variable extension principle and in the general relation-composition rule. We point to it where those formulas are introduced. For symbol overloads when reading across parts, see Appendix A.

17.2 Problem Statement

Let X and Y be two universes of discourse, with a known mapping function

$$y = f(x), \quad x \in X, \quad y \in Y.$$

Suppose we have a fuzzy set $A \subseteq X$ with membership function $\mu_A : X \to [0, 1]$. We want to define a fuzzy set $B \subseteq Y$ with membership function $\mu_B : Y \to [0, 1]$ that corresponds to A under the transformation f.

The key questions are:

- How do we compute $\mu_B(y)$ for each $y \in Y$?
- How do we handle the fact that multiple $x \in X$ may map to the same $y \in Y$?
- How do we combine membership values $\mu_A(x)$ for all x such that $f(x) = y$?

17.3 Intuition and Challenges

It is tempting to define $\mu_B(y) = \mu_A(x)$ where $y = f(x)$, but this is generally insufficient because:

- The mapping f may not be one-to-one; multiple x values can map to the same y.
- Membership values represent degrees of truth or compatibility, not numerical values to be transformed arithmetically.
- Simply applying f to membership values (e.g., squaring them) does not preserve the semantic meaning of membership.

Therefore, we need a principled method to aggregate membership values from all preimages of y under f.

17.4 Formal Definition of the Transformed Membership Function

Given the fuzzy set $A \subseteq X$ with membership function μ_A, and the mapping $y = f(x)$, the membership function μ_B of the fuzzy set $B \subseteq Y$ is defined by

$$\mu_B(y) = \sup_{x \in X : f(x) = y} \mu_A(x) \tag{17.1}$$

The strongest preimage membership determines the membership of y. When the mapping depends on multiple fuzzy variables (e.g., $f(x_1, x_2)$), the individual memberships are combined with a chosen t-norm before taking the supremum, as shown later in (17.3).

Remarks:

- The sup (supremum) operator generalizes the maximum operator, capturing the highest membership value among all x mapping to y; when X is finite the supremum collapses to an ordinary maximum.
- If no $x \in X$ maps to y, then $\mu_B(y) = 0$.
- For single-input transformations no additional t-norm is needed; the aggregation shows up only when several input memberships must be combined before mapping through f.
- In continuous settings we assume f is measurable so that the preimage sets $\{x \mid f(x) = y\}$ are well-defined and the supremum exists.

17.5 Interpretation

For single-input mappings, the output membership is the strongest preimage; when multiple fuzzy inputs are involved we first combine their memberships with a chosen t-norm (cf. (17.3)) and then take the supremum. It helps to read this as a two-stage logic:

Within one candidate explanation (one preimage tuple), a t-norm plays the role of "AND" by combining the memberships that must simultaneously hold. *Across all candidate explanations* that map to the same y, the supremum plays the role of "OR" by keeping the strongest support. In discrete examples the supremum collapses to a max, which makes the meaning especially transparent.

This matters because membership values are degrees of compatibility, not probabilities to be summed. If many different x values map to the same y, we do not want y to look "more true" simply because there are many redundant preimages; we want y to inherit the best support that the mapping can provide. This approach preserves the logical interpretation of membership values and respects the structure of the mapping f.

17.6 Transformation of Fuzzy Sets Between Universes

We continue our discussion on fuzzy set transformations, focusing on mapping fuzzy sets from one universe to another via a function $y = f(x)$. The extension principle itself was defined in (17.1); here we work through a concrete discrete example that instantiates that definition.

Example: Mapping via $y = x^2$ Consider a fuzzy set A defined on universe $X = \{-1, 0, 1, 2\}$ with membership values:

$$\mu_A(-1) = 0.340, \quad \mu_A(0) = 0.141, \quad \mu_A(1) = 0.242, \quad \mu_A(2) = 0.4.$$

Note that A is not *normal* because no element achieves membership 1; a fuzzy set is normal precisely when $\sup_{x \in X} \mu_A(x) = 1$.

Define the transformation $y = x^2$. The image universe Y consists of:

$$Y = \{0^2, (-1)^2, 1^2, 2^2\} = \{0, 1, 4\}.$$

Calculating explicitly:

$$\begin{aligned} \mu_B(0) &= \mu_A(0) = 0.141, \\ \mu_B(1) &= \max\{\mu_A(-1), \mu_A(1)\} = \max\{0.340, 0.242\} = 0.340, \\ \mu_B(4) &= \mu_A(2) = 0.4, \\ B &= \{(0, 0.141), (1, 0.340), (4, 0.4)\}. \end{aligned} \tag{17.2}$$

Even on this very small domain the mapping $f(x) = x^2$ is *non-injective* (many-to-one): $x = -1$ and $x = 1$ both map to $y = 1$. The supremum rule is precisely what handles this case — it selects the strongest preimage membership rather than double-counting or averaging. Notice also that no x maps to a negative y, so the mapping is *non-surjective* onto $\mathbb{R}$; for those output values the extension principle correctly returns $\mu_B(y) = 0$ (no supporting evidence). Together, these two properties — non-injectivity and non-surjectivity — are the general cases the definition (17.1) is designed to handle.

Worked example: monotone map (Celsius → Fahrenheit)

Let $A = \text{Comfortable}_C$ be triangular on Celsius with breakpoints $(21, 23, 25)$. For the affine map $f(x) = 1.8x + 32$, the image $B = f(A)$ is triangular with breakpoints $f(21) = 69.8$, $f(23) = 73.4$, $f(25) = 77.0$. Because f is strictly increasing, $\mu_B(y) = \mu_A(f^{-1}(y))$ and every α-cut maps directly: $B_\alpha = f(A_\alpha)$. This is the fastest way to reuse the same linguistic label across units without recomputing via (17.1).

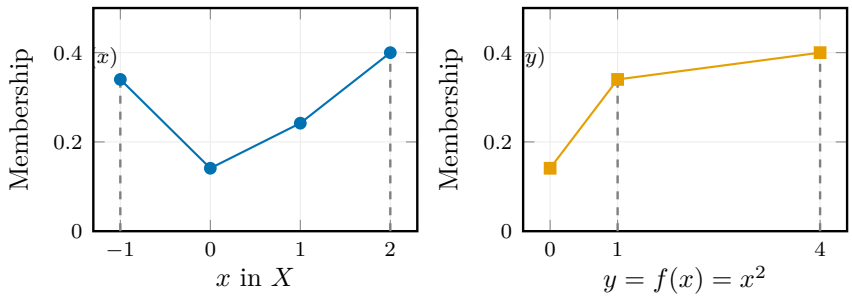


Figure 70: Mapping a fuzzy set through the function "y = x-squared". The membership at an output value y is the supremum over all preimages x that map to y; shared images such as $x = \pm 1$ map to y = 1 using the maximum membership. Helpful when applying the extension principle to a non-invertible function.

Visual intuition. Figure 70 walks through a simple mapping $y = x^2$, showing how memberships on X lift to memberships on Y via the supremum over all preimages that map to the same point.

Figure 71 makes the non-monotone alpha-cut mapping explicit for piecewise image transforms.

Extension to Multiple Fuzzy Sets Suppose now we have two fuzzy sets A_1 and A_2 defined on the same universe $X = \{-1, 0, 1, 2\}$, with membership functions listed in the order $(-1, 0, 1, 2)$:

$$\mu_{A_1} = \{0.4, 0.7, 0.5, 0.13\}, \quad \mu_{A_2} = \{0.5, 0.1, 0.4, 0.7\}.$$

Define a function $y = f(x_1, x_2) = x_1^2 + x_2^2$, where $x_1, x_2 \in X$ and their degrees of membership are taken from A_1 and A_2 respectively.

The universe Y is the set of all possible sums of squares:

$$Y = \{x_1^2 + x_2^2 \mid x_1, x_2 \in X\}.$$

For example, some values in Y include:

$$0^2 + 0^2 = 0, \quad (-1)^2 + 0^2 = 1, \quad 1^2 + 1^2 = 2, \quad 2^2 + 2^2 = 8, \quad \ldots$$

Computing Membership Values in Y For two variables, the extension principle becomes:

$$\mu_B(y) = \sup_{(x_1,x_2):f(x_1,x_2)=y} \min\{\mu_{A_1}(x_1), \mu_{A_2}(x_2)\}. \tag{17.3}$$

The minimum t-norm plays the role of the generic operator $\otimes$; any other t-norm could be substituted so long as the same choice is applied throughout the inference pipeline.

The numeric workflow is consistent: list all preimages (x_1, x_2) with $f(x_1, x_2) = y$, compute the minimum membership for each pair, then take the maximum.

Example: Compute $\mu_B(0)$.

The only pair with $x_1^2 + x_2^2 = 0$ is $(0, 0)$, so

$$\mu_B(0) = \min\{\mu_{A_1}(0), \mu_{A_2}(0)\} = \min\{0.7, 0.1\} = 0.1.$$

Example: Compute $\mu_B(1)$.

The pairs (x_1, x_2) such that $x_1^2 + x_2^2 = 1$ are:

$$(-1, 0), \quad (0, -1), \quad (1, 0), \quad (0, 1).$$

The minimum membership values for each pair are

$$\begin{aligned}
\min\{\mu_{A_1}(-1), \mu_{A_2}(0)\} &= \min\{0.4, 0.1\} = 0.1,\\
\min\{\mu_{A_1}(0), \mu_{A_2}(-1)\} &= \min\{0.7, 0.5\} = 0.5,\\
\min\{\mu_{A_1}(1), \mu_{A_2}(0)\} &= \min\{0.5, 0.1\} = 0.1,\\
\min\{\mu_{A_1}(0), \mu_{A_2}(1)\} &= \min\{0.7, 0.4\} = 0.4.
\end{aligned}$$

Taking the supremum gives $\mu_B(1) = \max\{0.1, 0.5, 0.1, 0.4\} = 0.5$.

$$\begin{aligned}
\mu_B(0) &= \min\{\mu_{A_1}(0), \mu_{A_2}(0)\} = 0.1,\\
\mu_B(1) &= \max\{0.1, 0.5, 0.1, 0.4\} = 0.5.
\end{aligned} \tag{17.4}$$

17.7 Extension Principle Recap and Projection Operations

We now focus on computational choices (discretization vs. alpha-cuts) before moving on to projection operations.

Computation and discretisation tips

For discrete universes the extension principle costs $O(|X|)$ per y (or $O(|X|^n)$ for n-ary maps) because we evaluate every preimage tuple. In discrete settings sup reduces to a max. Continuous universes require discretisation: sample each input axis on a uniform or adaptive grid (typical 200–500 points per dimension), apply the t-norm/aggregation on that mesh, and approximate the supremum via max. Sparse grids or Monte Carlo sampling reduce the curse of dimensionality; always report the resolution so readers understand numeric fidelity.

Alpha-cuts as an alternative

- **Unary monotone f:** $B_\alpha = f(A_\alpha)$ for every $\alpha \in (0, 1]$; computationally trivial for affine/monotone maps.
- **Non-monotone f:** split X into monotone pieces D_k; compute $B_\alpha = \bigcup_k f(A_\alpha \cap D_k)$. This is the standard route for fuzzy arithmetic on fuzzy numbers.
- **When to use:** alpha-cuts are numerically stable for continuous domains and avoid sampling artifacts when f is smooth; pointwise sup is more convenient on discrete grids.

Figure 72 visualizes how relation entries project into the corresponding marginals.

17.8 Projection of Fuzzy Relations

Now, consider the case where we have a fuzzy relation $R \subseteq X \times Y$, where X and Y are universes of discourse. The fuzzy relation R is characterized by a membership function

$$\mu_R : X \times Y \to [0, 1].$$

This relation can be viewed as a fuzzy set on the Cartesian product $X \times Y$. We define projection operators in this section and later reuse them in the dimensional

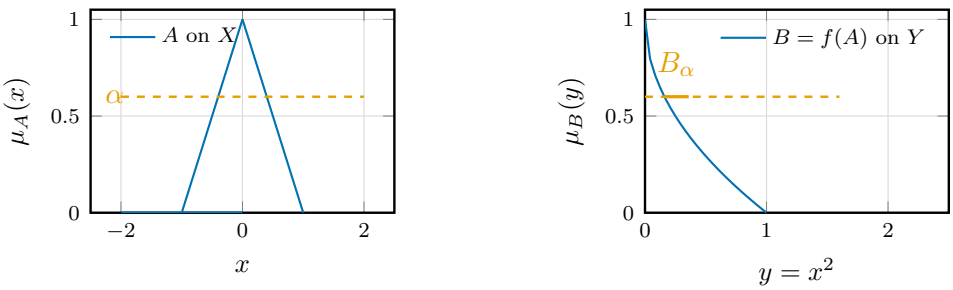


Figure 71: Alpha-cuts under the non-monotone map "y = x-squared". A symmetric triangular fuzzy set on X maps to a right-skewed fuzzy set on Y. Each alpha-cut on A splits into two intervals whose images union to the output alpha-cut. Use this to propagate a fuzzy set through a non-monotone map via alpha-cuts.

extension discussion; the goal here is to fix notation and walk a concrete table example.

Cartesian Product of Fuzzy Sets Given fuzzy sets $A \subseteq X$ and $B \subseteq Y$ with membership functions μ_A and μ_B, their Cartesian product $R = A \times B$ is defined by

$$\begin{aligned} \mu_R(x, y) &= T(\mu_A(x), \mu_B(y)), \\ T(a, b) &= \min(a, b). \end{aligned} \tag{17.5}$$

A *t-norm* is any binary operator $T : [0, 1]^2 \to [0, 1]$ that is commutative, associative, monotone in each argument, and has 1 as identity, so it faithfully generalizes set intersection to graded memberships. Popular choices include the minimum, the product ab, and the Łukasiewicz t-norm $\max(0, a + b - 1)$.

Example Suppose

$$\mu_A = \{0.5, 0.9\}, \quad \mu_B = \{0.8, 0.9\}.$$

Then the Cartesian product membership values are

$$\mu_R = \begin{bmatrix} \min(0.5, 0.8) & \min(0.5, 0.9) \\ \min(0.9, 0.8) & \min(0.9, 0.9) \end{bmatrix} = \begin{bmatrix} 0.5 & 0.5 \\ 0.8 & 0.9 \end{bmatrix}.$$

Table 8: Popular t-norms and their typical roles. Reference when choosing a default conjunction operator and understanding its qualitative behavior.

t-norm	**Dual t-conorm / identity**	**When to use**
Minimum $T_{\min}(a,b) = \min(a,b)$	Dual: $\max(a,b)$; idempotent	Linguistic rules mirroring classical AND; preserves interpretability.
Product $T_{\Pi}(a,b) = ab$	Dual: probabilistic sum $a+b-ab$	Smooth gradients, probabilistic semantics, differentiable control.
Łukasiewicz $T_{\text{Luk}}(a,b) = \max(0, a+b-1)$	Dual: bounded sum $\min(1, a+b)$	Allows partial satisfaction to accumulate; useful in preference aggregation and graded constraints; tolerates partial violations.

Here the first row corresponds to x_1, the second row to x_2, and the columns correspond to y_1 and y_2. Keeping that indexing explicit avoids ambiguity when reading off the projected membership values.

Often, we are interested in reducing the dimensionality of a fuzzy relation by projecting it onto one of its component universes. The projection operation extracts a fuzzy set on X or Y from the fuzzy relation R.

Definition (Projection onto X). The projection of R onto X, denoted $\pi_X(R)$, keeps each x-row and asks for the strongest compatible y-value in that row:

$$\begin{aligned} \mu_{\pi_X(R)}(x) &= \sup_{y \in Y} \mu_R(x,y), \\ \mu_{\pi_Y(R)}(y) &= \sup_{x \in X} \mu_R(x,y), \\ \mu_{\pi_{\text{total}}(R)} &= \sup_{x \in X, y \in Y} \mu_R(x,y). \end{aligned} \tag{17.6}$$

Definition (Projection onto Y). Similarly, projection onto Y, denoted $\pi_Y(R)$, keeps each y-column and takes the strongest supporting x-value in that column; the second line of (17.6) records that operator.

Total Projection The *total projection* of R is the maximum membership value over the entire relation; the third line of (17.6) gives this global summary.

Interpretation

- The projection onto X collapses the Y-dimension by taking the maximum membership value along each fixed x.
- The projection onto Y collapses the X-dimension similarly.
- The total projection gives the single highest membership value in the relation.

In implementation terms, these are row-wise and column-wise max operations on a relation matrix. That viewpoint becomes useful in Chapter 18, where rule relations are repeatedly projected back to the output universe before aggregation and defuzzification.

	y_1	y_2	y_3	$\pi_X(R)$	$\pi_Y(R)$
x_1	0.9	0.3	0.1	0.9	0.9
x_2	0.4	0.8	0.2	0.8	0.8
x_3	0.1	0.6	0.5	0.6	0.5

Figure 72: Illustrative fuzzy relation table (left) together with its projections onto the error universe (middle) and the rate-of-change universe (right). These are the exact quantities used in the running thermostat example before composing rules. Use it to build rule antecedents from a relation and verify which universe each projection inhabits.

Example (continued) Using the previous example matrix for μ_R:

$$\mu_R = \begin{bmatrix} 0.5 & 0.5 \\ 0.8 & 0.9 \end{bmatrix},$$

we compute

$$\mu_{\pi_X(R)} = \{\max(0.5, 0.5), \max(0.8, 0.9)\} = \{0.5, 0.9\},$$

$$\mu_{\pi_Y(R)} = \{\max(0.5, 0.8), \max(0.5, 0.9)\} = \{0.8, 0.9\},$$

and

$$\mu_{\pi_{\text{total}}(R)} = \max\{0.5, 0.8, 0.5, 0.9\} = 0.9.$$

17.9 Dimensional Extension and Projection in Fuzzy Set Operations

In practice, we frequently need to combine fuzzy sets and fuzzy relations that live on *different* universes of discourse. For example, one object may be defined on a one-dimensional universe X, while another lives on a product space $X \times Y$ (as relations do). Before we can take unions/intersections or compose rules, we must reconcile dimensions. Two operators do the heavy lifting: *cylindrical extension* lifts a set into a higher-dimensional space without changing its meaning, and *projection* collapses a relation back onto the universe we want to summarize.

Cylindrical Extension The *cylindrical extension* is a technique used to extend a fuzzy set defined on a lower-dimensional universe to a higher-dimensional universe by replicating membership values along the new dimension(s).

Suppose we have a fuzzy set $A \subseteq X$ with membership function $\mu_A : X \to [0, 1]$. To extend A to $X \times Y$, define the cylindrical extension A^* as:

$$\mu_{A^*}(x, y) = \mu_A(x), \quad \forall x \in X, y \in Y. \tag{17.7}$$

This operation "copies" the membership values of A uniformly along the Y-dimension, resulting in a fuzzy set over $X \times Y$.

Projection Projection collapses relations back onto a target universe; use the operators in (17.6) and apply them after cylindrical extension to recover the desired marginal.

Example Consider a fuzzy set A on $X = \{x_1, x_2\}$ with membership values $\mu_A(x_1) = 0.5$, $\mu_A(x_2) = 0.7$. Extending A cylindrically to $X \times Y$ where $Y = \{y_1, y_2, y_3\}$ yields:

$$\mu_{A^*}(x_i, y_j) = \mu_A(x_i), \quad i = 1, 2; \quad j = 1, 2, 3.$$

Thus, the membership values are replicated along the Y-axis, i.e.,

$$\mu_{A^*} = \begin{bmatrix} 0.5 & 0.5 & 0.5 \\ 0.7 & 0.7 & 0.7 \end{bmatrix},$$

with rows corresponding to x_1, x_2 and columns to y_1, y_2, y_3. In practice this extension step is often paired with projections to reconcile relation dimensions before composing rules and, later, to marginalize the inferred relation back onto the universe of interest.

17.10 Fuzzy Inference via Composition of Relations

The ultimate goal of building fuzzy logic systems is to perform *inference*, i.e., to compose fuzzy rules to generate predictions or decisions. This involves combining fuzzy relations that represent knowledge or rules.

Setup Suppose we have three universes of discourse X, Y, Z, and two fuzzy relations:

$$R_1 \subseteq X \times Y, \quad R_2 \subseteq Y \times Z,$$

with membership functions $\mu_{R_1}(x, y)$ and $\mu_{R_2}(y, z)$, respectively.

We seek the composed relation $R \subseteq X \times Z$ that relates X directly to Z by composing R_1 and R_2.

Composition of Fuzzy Relations The composition $R = R_1 \circ R_2$ is defined by:

$$\mu_R(x, z) = \sup_{y \in Y} T\big(\mu_{R_1}(x, y), \mu_{R_2}(y, z)\big). \tag{17.8}$$

We default to the *sup–min* (max–min) composition by taking $T(a, b) = \min(a, b)$; other t-norms are listed in Table 8.

Interpretation

- The min operator captures the degree to which x is related to y and y is related to z simultaneously.
- The sup (maximum) over all intermediate y aggregates all possible "paths" from x to z through y.

Dimensional Considerations Note that R_1 is defined on $X \times Y$, and R_2 on $Y \times Z$. The composition yields R on $X \times Z$. If the dimensions of the relations differ or if the universes are not aligned, cylindrical extension or projection can be applied to make the dimensions compatible before composition.

Example Let $X = \{x_1, x_2\}$, $Y = \{y_1, y_2\}$, and $Z = \{z_1, z_2\}$. Consider

$$\mu_{R_1} = \begin{bmatrix} 0.2 & 0.9 \\ 0.5 & 0.1 \end{bmatrix}, \qquad \mu_{R_2} = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.8 \end{bmatrix}.$$

Using the max–min composition,

$$\begin{aligned}
\mu_R(x_1, z_1) &= \max\{\min(0.2, 0.7), \min(0.9, 0.4)\} = \max\{0.2, 0.4\} = 0.4, \\
\mu_R(x_1, z_2) &= \max\{\min(0.2, 0.3), \min(0.9, 0.8)\} = \max\{0.2, 0.8\} = 0.8, \\
\mu_R(x_2, z_1) &= \max\{\min(0.5, 0.7), \min(0.1, 0.4)\} = \max\{0.5, 0.1\} = 0.5, \\
\mu_R(x_2, z_2) &= \max\{\min(0.5, 0.3), \min(0.1, 0.8)\} = \max\{0.3, 0.1\} = 0.3.
\end{aligned}$$

Therefore

$$\mu_R = \begin{bmatrix} 0.4 & 0.8 \\ 0.5 & 0.3 \end{bmatrix}.$$

Max–min composition as "fuzzy matrix multiply"

Given $R_1 \in [0,1]^{|X| \times |Y|}$ and $R_2 \in [0,1]^{|Y| \times |Z|}$,

```
for i in range(|X|):
    for k in range(|Z|):
        acc = 0
        for j in range(|Y|):
            acc = max(acc, min(R1[i, j], R2[j, k]))
        R[i, k] = acc
return R  # the composition R1 o R2
```

Swap min for another T (product, Łukasiewicz) and max for the corresponding t-conorm to instantiate other composition families.

17.11 Properties of Fuzzy Relation Composition

From max–min to sup–T Max–min composition is one instance of the broader sup–T family in (17.8). Any valid t-norm T (minimum, product, Łukasiewicz) gives a consistent composition rule; max–min remains the default because it is easiest to audit. Keep the worked example from Section 17.10 as the reference trace while reading the properties below.

The composition operation inherits several important algebraic properties, analogous to classical relations:

- **Associativity:** For fuzzy relations R_1, R_2, R_3,

$$(R_1 \circ R_2) \circ R_3 = R_1 \circ (R_2 \circ R_3).$$

  This allows chaining multiple relations without ambiguity.

- **Non-commutativity:** Generally,

$$R_1 \circ R_2 \neq R_2 \circ R_1,$$

  reflecting the directional nature of relations.

- **Distributivity:** Composition distributes over union:

$$R_1 \circ (R_2 \cup R_3) = (R_1 \circ R_2) \cup (R_1 \circ R_3).$$

- **De Morgan's Laws and Inclusion:** These extend naturally to fuzzy relations and their complements, intersections, and unions.

17.12 Alternative Composition Operators

While max–min is standard, other t-norms and aggregation rules can be useful when you need smoother transitions or stronger attenuation of partial matches. The key requirement is consistency: pick one family and carry it through the full inference pipeline.

- **Max-Product Composition:**

$$\mu_R(x, z) = \max_y \left(\mu_{R_1}(x, y) \cdot \mu_{R_2}(y, z)\right). \tag{17.9}$$

Compared with max–min, this penalizes weak intermediate links more strongly and often yields smoother control surfaces.

- **Averaging-style aggregations:** In some applications, the final $\max_y$ is replaced by weighted averaging over y-paths to preserve contributions from multiple moderate paths rather than only the strongest one. This can improve smoothness, but it also weakens the clear "best support path" interpretation of max-based composition.

Author's note: choosing an operator family

Start with max–min when safety and monotonicity matter; its outputs stay within the tightest support and preserve ordering. Swap to max–product or algebraic t-norms when you need smoother surfaces or when small disagreements should be penalized multiplicatively (e.g., sensor fusion). If the resulting surfaces are too flat, tighten the t-norm; if they are too brittle, loosen it. Operator choice is an engineering dial, not an article of faith.

The choice of composition operator therefore follows a practical rule: begin with max–min for its interpretability and stability, and reach for the alternatives catalogued in Table 8 only when the application demands smoother or more aggressive aggregation.

Key takeaways

Minimum viable mastery.

- Apply the extension principle to move a fuzzy set through a mapping (and state the domain assumptions explicitly).
- Compute one sup–T composition end-to-end and explain where sup (aggregation) and T (conjunction strength) enter.
- Choose an operator family once (t-norm/s-norm/complement) and carry it consistently through composition, projection, and aggregation.

Common pitfalls.

- Mixing operator defaults across steps and then trying to "debug" the surface after the fact.
- Losing track of universes/units so the transfer is computed on the wrong domain.
- Treating smoother alternatives as "better" by default: they change semantics, not just aesthetics.

Exercises

Exercise 1 (Easy). Build a tiny fuzzy relation $R(X, Y)$ on a discrete grid and compute both projections: "what Y is plausible given X" and "what X is plausible given Y."
Deliverable: the relation table + both projected fuzzy sets + one sentence interpreting each projection.

Exercise 2 (Medium). Compute one sup–T composition end-to-end for two relations under max–min and under max–product. Treat the comparison as a semantics question: what changes when conjunction is "bottleneck" vs. "multiplicative"?
Deliverable: the composed relation under both choices + a short explanation of one entry that differs and why.

Exercise 3 (Medium). Apply the extension principle to a many-to-one mapping (e.g., $y = x^2$) under a constrained domain and show how the constraint changes the transferred membership.
Deliverable: a small numerical check (a few membership values) + one paragraph naming the implicit assumption you are making about admissible pre-images.

Where we head next. Chapter 18 operationalizes these relation tools: each rule induces a fuzzy relation on input-output space, then sup-T composition and projection produce the implied output set before aggregation and defuzzification. The same defaults (max–min with standard complement) carry over, as summarized in (18.1)–(18.3).

18 Fuzzy Inference Systems: Rule Composition and Output Calculation

Building on Chapter 17, where transfer operators (projection, composition, and the extension principle) move fuzzy information between related universes, we now assemble complete fuzzy inference systems (FIS): rule composition and output calculation. This closes the fuzzy trilogy by turning set/relation machinery into end-to-end decisions.

Learning Outcomes

- Execute full Mamdani/Larsen style inference: antecedent aggregation, implication, aggregation, and defuzzification.
- Compare implication/aggregation choices (product vs. min, max vs. sum) and their impact on the running thermostat/autofocus example.
- Contrast Mamdani systems with Sugeno/Takagi–Sugeno systems to know when weighted-average consequents are preferable.

Design motif

Local linguistic rules become a global behavior only after you commit to concrete operators (t-norm, implication, aggregation, defuzzifier) and then sanity-check the resulting surface.

Running example checkpoint. For the thermostat, each rule combines the temperature error (*Cold*, *Slightly Warm*, ...) and rate-of-change to set heater power. Here "AND" is a t-norm (default: min), not Boolean conjunction; details appear in the operator subsection below. As you work through antecedent aggregation, implication, and defuzzification, keep one concrete rule base (e.g., "IF error is Cold AND rate is Stable THEN heater power is High") in mind; the formulas below map directly onto that setup.

18.1 Context and Motivation

Notation handoff

Throughout this chapter we keep the trilogy defaults from Chapters 16 to 17: $\wedge = \min$, $\vee = \max$, and the standard complement $1 - \mu$. Aggregation over rule consequents defaults to the max s-norm unless stated otherwise; alternatives live in Table 8. If symbols conflict with earlier chapters, use local definitions and refer to Appendix A.

Recall that a fuzzy inference system maps crisp inputs to fuzzy outputs by applying a set of fuzzy rules. Each rule typically has the form:

$$\textit{If } x_1 \text{ is } A_1 \textit{ and } x_2 \text{ is } A_2 \textit{ and } \cdots \textit{ then } y \text{ is } B,$$

where A_i and B are fuzzy sets defined on the respective universes of discourse. The antecedent (premise) combines multiple fuzzy conditions on inputs, and the consequent (conclusion) specifies the fuzzy output.

Author's note: rules as lived experience

These rules are not immutable physical laws; they are codified experience. We record facts such as "if it is morning then the sun is in the east," yet real observations may arrive at noon. Fuzzy inference exists to bridge that gap: observed memberships are composed with stored rules so that slight deviations in the antecedent produce softened consequents instead of brittle yes/no responses. When you carry out the algebra below, keep that picture of "experience vs. observation" in mind.

The key challenge is to systematically combine the antecedent fuzzy sets and then infer the output fuzzy set for each rule, before aggregating all rules to produce a final output.

Author's note: what makes a rule base "good"?

There is no single scalar score that certifies a rule base as "good" in all contexts. In practice, we look for four properties together: (i) **coverage** across normal and extreme operating regions, (ii) **controlled overlap** so neighboring rules blend instead of fighting, (iii) **monotonic trends** where the application expects them, and (iv) **conflict handling** when multiple rules fire strongly with different consequents. If one of these is missing, the controller may look reasonable in nominal conditions but fail abruptly at boundaries.

18.2 Rule Antecedent Composition

Given a rule with n antecedents, each associated with a fuzzy set A_i and an input value x_i, the degree to which the rule is activated (also called the *firing strength*) is computed by combining the membership values of each antecedent condition.

Membership values of antecedents. For each input x_i, evaluate its degree of compatibility with label A_i:

$$\mu_{A_i}(x_i) \in [0, 1].$$

These values are the local evidence carried by the rule antecedent. The antecedent operator (min, product, or another T) then compresses that evidence vector into one firing strength that drives implication.

Aggregation operator: The combined antecedent membership is obtained by applying a fuzzy logical operator, typically the *minimum* (intersection) or the *product* operator:

$$\begin{aligned} \mu_{\text{antecedent}}(x_1, \ldots, x_n) &= \min_{i=1}^{n} \mu_{A_i}(x_i), \qquad \text{(min operator)}, \\ \mu_{\text{antecedent}}(x_1, \ldots, x_n) &= \prod_{i=1}^{n} \mu_{A_i}(x_i), \qquad \text{(product operator)}. \end{aligned} \tag{18.1}$$

This value quantifies the degree to which the entire antecedent condition is satisfied by the input vector $\mathbf{x} = (x_1, \ldots, x_n)$. More generally, any t-norm T can be used in place of the min or product, provided it satisfies the standard properties (commutativity, associativity, monotonicity, and $T(a, 1) = a$); the chosen t-norm shapes how strictly the rule demands simultaneous satisfaction of all antecedents.

18.3 Rule Consequent and Output Fuzzy Set

Once the antecedent firing strength α is computed, it is used to modify the consequent fuzzy set B. The consequent fuzzy set is typically defined by its membership function $\mu_B(y)$ over the output universe.

Implication operator: The implication step adjusts the consequent membership function based on the firing strength α. Commonly used implication methods include:

- **Minimum implication:** Truncate the consequent membership function at level α,
- **Product implication:** Scale the consequent membership function by α,

$$\begin{aligned} \mu_{B'}(y) &= \min\big(\alpha, \mu_B(y)\big), \qquad \text{(minimum implication)}, \\ \mu_{B'}(y) &= \alpha \cdot \mu_B(y), \qquad \text{(product implication)}. \end{aligned} \tag{18.2}$$

The resulting fuzzy set B' represents the *output fuzzy set* contributed by this particular rule.

Implication choices

Mamdani uses min-implication (clipping), Larsen uses product-implication (scaling). Other options include residuated and axiomatic implicators paired with their t-norms (e.g., Gödel, Product/Goguen, Łukasiewicz; see Klement et al., 2000; Dubois and Prade, 1988). Pick by desired smoothness: clipping preserves shape and interpretability; scaling yields smoother surfaces and is friendlier to gradient-based tuning.

18.4 Aggregation of Multiple Rules

When multiple rules are present, each produces an output fuzzy set B'_j with membership function $\mu_{B'_j}(y)$, where j indexes the rules. These are aggregated to form a combined output fuzzy set:

$$\mu_{B_{\text{agg}}}(y) = \max_j \mu_{B'_j}(y). \tag{18.3}$$

We denote this aggregated set by $\mu_{B_{\text{agg}}}$; in the worked thermostat example we write $\mu_{\text{out}}(u)$ for the same quantity to emphasize the output variable.

The *max* operator corresponds to the fuzzy union of the individual rule outputs, capturing the overall inference result.

Other aggregations Algebraic sum or bounded sum (Table 8) are useful when max is too brittle and you want multiple moderate rules to contribute jointly. The trade-off is saturation: if many rules fire at once, sum-style aggregation can flatten distinctions between candidate outputs. In controller-style designs, max is usually the safer default unless smoother blending is an explicit requirement.

18.5 Summary of the Fuzzy Inference Process

For a compact algorithmic view, see the pipeline box after the worked example; it instantiates these same steps with explicit operators.

Failure modes to monitor

Most deployment failures come from five sources: rule explosion (opaque behavior), operator mismatch (surface shape changes), membership miscalibration (systematic over/under firing), defuzzifier sensitivity near boundaries, and sensor-scale drift after deployment. Keep these as explicit audit checks whenever you modify membership sets, operators, or normalization.

In sup-T form this is the same compositional rule of inference used in Chapter 17; here T defaults to min (or product) and sup reduces to a max on discrete grids.

The final step, defuzzification, converts B_{agg} into a crisp output value. One widely used approach is the centroid (center-of-gravity) method, which computes

$$y^* = \frac{\int_Y y\, \mu_{B_{\text{agg}}}(y)\, dy}{\int_Y \mu_{B_{\text{agg}}}(y)\, dy}. \tag{18.4}$$

This expression balances all candidate output values y by weighting them according to their membership grade in the aggregated fuzzy set. In discrete implementations, the integral is replaced with a sum over sampled output points.

Other defuzzifiers Common alternatives are mean/center of maxima (robust to multi-modal sets), smallest/largest of maxima (conservative tie-breaks), and center of sums (less sensitive to overlap than max aggregation). Choose centroid for smoothness, a max-based rule for fast or safety-critical switches, and always handle the zero-mass case explicitly.

Zero-mass fallback If the denominator in (18.4) is zero (e.g., all consequents clipped to zero), fall back to a max-membership or rule-based tie-breaker to avoid NaNs; log the condition for debugging.

Computation note With uniform sampling over m output points, centroid costs $O(mR)$ per evaluation for R rules. Non-singleton inputs add a convolution step but reuse the same aggregation/defuzz pipeline; refine the grid near peaks to reduce bias.

Centroid stability and tie-breaking

- **Multi-modal sets.** When B_{agg} has multiple peaks, the centroid may fall between modes. Log numerator and denominator separately and check that $\int \mu_{B_{\text{agg}}}(y)\,dy$ is non-zero; otherwise fall back to max-membership or a rule-based tie-break.
- **Discretisation.** Sampling the universe with too few points biases the centroid. Use uniform grids for smooth consequents and adaptive refinement near peaks for multi-modal sets. Report the step size (e.g., 0.5°C) to show numeric fidelity.

18.6 Worked example: thermostat inference with numbers

We now run one complete inference pass for a compact thermostat rule base with two inputs and one output. The goal is not to defend a particular membership design, but to show how the algebra produces a concrete output that you can compute, audit, and debug.

Universes and memberships. Let temperature error $e \in [-5, 5]\,°\text{C}$ and rate $r \in [-2, 2]\,°\text{C/min}$. Let the controller output $u \in [0, 1]$ denote heater power as a fraction of maximum. We use simple piecewise-linear labels:

$$\mu_{\text{Cold}}(e) = \begin{cases} 1, & e \le -5, \\ \frac{0-e}{5}, & -5 < e < 0, \\ 0, & e \ge 0, \end{cases} \qquad \mu_{\text{Comfy}}(e) = \max\left(0,\ 1 - \frac{|e|}{2.5}\right),$$

$$\mu_{\text{Stable}}(r) = \max(0,\ 1 - |r|)\,, \qquad \mu_{\text{Rising}}(r) = \begin{cases} 0, & r \le 0, \\ \frac{r}{2}, & 0 < r < 2, \\ 1, & r \ge 2. \end{cases}$$

For u, use the standard three-label output family (Low/Medium/High) from Chapter 16:

$$\mu_{\text{Low}}(u) = \max\Big(0,\ 1 - \tfrac{u}{0.5}\Big),$$
$$\mu_{\text{Medium}}(u) = \max\Big(0,\ 1 - \tfrac{|u-0.5|}{0.25}\Big),$$
$$\mu_{\text{High}}(u) = \max\Big(0,\ \tfrac{u-0.5}{0.5}\Big).$$

Rule base.

1. IF e is Cold AND r is Stable THEN u is High.
2. IF e is Cold AND r is Rising THEN u is Medium.
3. IF e is Comfy THEN u is Low.

Fuzzify inputs and compute firing strengths. For inputs $e = -2$ and $r = 0.5$,

$$\mu_{\text{Cold}}(-2) = 0.4, \quad \mu_{\text{Comfy}}(-2) = 0.2, \quad \mu_{\text{Stable}}(0.5) = 0.5, \quad \mu_{\text{Rising}}(0.5) = 0.25,$$
$$\alpha_1 = \min(0.4, 0.5) = 0.4, \quad \alpha_2 = \min(0.4, 0.25) = 0.25, \quad \alpha_3 = 0.2. \tag{18.5}$$

These come directly from the definitions: $\mu_{\text{Cold}}(-2) = (0 - (-2))/5 = 0.4$ and $\mu_{\text{Rising}}(0.5) = 0.5/2 = 0.25$. The second line of (18.5) uses $T = \min$ from (18.1).

Implication, aggregation, and defuzzification. Using Mamdani min-implication (the first line of (18.2)), each consequent is clipped at its firing strength (e.g., $\mu'_{\text{High}}(u) = \min(\alpha_1, \mu_{\text{High}}(u))$). Aggregating consequents by max ((18.3)) yields $\mu_{\text{out}}(u)$. Finally, centroid defuzzification ((18.4)) produces a crisp heater-power command. Using a uniform grid on $u \in [0, 1]$ (step 10^{-4}) gives

$$u^\star \approx 0.577. \tag{18.6}$$

This value is above the midpoint because the *Cold* rules (High/Medium consequents) fire more strongly than the *Comfy* rule, while the positive rate term (*Rising*) tempers full-*High* output. In implementations, compute the centroid numerically (and log the grid/step size) rather than relying on heuristic convex

combinations when overlap patterns become complex.

Pipeline at a glance (Mamdani/Larsen)

```
for each rule j:
    alpha_j = T( mu_A1(x1), ..., mu_An(xn) )
        # firing strength
    mu_Bprime_j(y) = implication(alpha_j, mu_Bj(y))
        # clip or scale
mu_out(y) = S( mu_Bprime_1(y), ..., mu_Bprime_R(y) )
    # aggregate (mu_out == mu_Bagg)
y_star = centroid(mu_out(y))
    # or another defuzzifier
```

Defaults: $T = \min$ or product; $S = \max$; centroid defuzzification. Non-singleton fuzzification (convolving input uncertainty with μ_{A_i}) uses the same pipeline once the input blend is computed.

Design checklist

- Define universes/labels; ensure coverage and reasonable overlap.
- Pick $T/S/\Rightarrow$ via Table 8 to get the smoothness/interpretability you need.
- Verify rule-base coverage; avoid contradictions/holes.
- Choose defuzzifier and sampling resolution; set a minimum-mass fallback.
- Test monotonicity/saturation; refine membership widths or rule weights.

Common pitfalls

- Max aggregation can mask contributions from several moderate rules; algebraic sum can over-saturate.
- Memberships that are too narrow yield sparse firing; too wide produce mushy outputs.
- Coarse grids bias centroids; inconsistent units across labels break interpretability.
- Neglecting non-singleton inputs: if sensor noise matters, blur inputs before fuzzifying.

18.7 Mamdani vs. Sugeno/Takagi–Sugeno systems

Mamdani-style inference (clipped fuzzy consequents, centroid defuzzification) excels when linguistic interpretability is a priority and when rule consequents must remain human-readable. Sugeno/Takagi–Sugeno (TSK) systems replace fuzzy consequents with crisp functions such as affine models. Let r denote the rate-of-change ($r = \dot{e}$):

$$\text{IF } e \text{ is } A_i \text{ AND } r \text{ is } B_i \text{ THEN } u_i = p_i e + q_i r + r_i.$$

Each rule still produces a firing strength λ_i via a t-norm, but the final output becomes the weighted average

$$u^* = \frac{\sum_i \lambda_i u_i}{\sum_i \lambda_i},$$

eliminating the defuzzification integral. The trade-offs are:

- **Mamdani:** transparent consequents, straightforward incorporation of expert knowledge, but higher computational cost due to aggregation and centroid evaluation.
- **Sugeno/TSK:** faster evaluation (weighted averages), amenable to gradient-based tuning of the consequent parameters, yet less interpretable because consequents are numerical functions rather than linguistic labels.

For the thermostat example, Mamdani rules ("IF error is Cold AND rate is Sta-

ble THEN heater power is High") are ideal when operators must audit decisions, whereas a Sugeno/TSK variant is preferable when embedding the controller into a high-speed or automatically tuned loop. [3]

18.8 Neuro-Fuzzy Hybrid: ANFIS

A natural limitation of handcrafted fuzzy systems is that membership breakpoints and rule weights must be tuned by domain experts, which is time-consuming and prone to subjectivity. *Adaptive Neuro-Fuzzy Inference System* (ANFIS) (Jang, 1993) resolves this by embedding a Sugeno-type fuzzy rule base into a five-layer feedforward network that is trained end-to-end using gradient descent.

Architecture (five layers, two-input example).

1. **Fuzzification layer.** Each node computes the membership degree $\mu_{A_i}(x_1)$ or $\mu_{B_j}(x_2)$ using trainable membership function parameters (e.g., Gaussian centres and widths).
2. **Rule strength layer.** Each node i computes the product (or minimum) of its antecedent memberships: $w_i = \mu_{A_i}(x_1) \cdot \mu_{B_i}(x_2)$.
3. **Normalisation layer.** Each firing strength is normalised: $\bar{w}_i = w_i / \sum_j w_j$.
4. **Consequent layer.** Each node computes $\bar{w}_i f_i$ where $f_i = p_i x_1 + q_i x_2 + r_i$ is a linear (Sugeno) consequent with trainable parameters (p_i, q_i, r_i).
5. **Output layer.** Sums the weighted consequents: $\hat{y} = \sum_i \bar{w}_i f_i$.

Hybrid learning. ANFIS uses a hybrid of gradient-descent (backpropagation, for the membership-function parameters in layer 1) and least-squares estimation (LSE, for the linear consequent parameters in layer 4). The LSE pass is exact and fast; the gradient pass tunes the shape of the antecedents. Iterating these two passes converges reliably for moderate-sized problems.

[3] Classic sources: Mamdani and Assilian (1975) for clipping implication; Takagi and Sugeno (1985) for TSK; Jang (1993) for ANFIS; see also (Klement et al., 2000; Dubois and Prade, 1988) for operator/implicator theory.

Connection to the fuzzy trilogy. ANFIS closes the loop between the three Part III chapters: the membership functions of Chapter 16 become trainable nodes, the extension/composition ideas of Chapter 17 govern how rule strengths propagate, and the Sugeno inference of this chapter provides the differentiable output. The parameters that ANFIS tunes automatically (breakpoints, widths, rule weights) can also be optimised with a genetic algorithm — see Chapter 19 for that population-based alternative when the search space is non-smooth or multi-modal.

Key takeaways

- Fuzzy inference composes rule antecedents (via a t-norm) and modifies consequents by implication.
- Aggregation and defuzzification (e.g., centroid) produce crisp outputs from fuzzy rule bases.
- Design choices (operators, shapes) trade interpretability and control smoothness.
- ANFIS embeds a Sugeno rule base into a trainable network, enabling data-driven tuning of membership functions via gradient descent and LSE.

Minimum viable mastery.

- For a rule base, compute firing strengths, apply implication, aggregate consequents, and compute a centroid output.
- Explain the Mamdani vs. Sugeno/TSK tradeoff (interpretability vs. efficiency/tunability).
- Treat operator choice as part of the specification and keep it consistent across examples and implementations.

Common pitfalls.

- Defining memberships on incompatible scales (inputs never trigger, or everything triggers).
- Writing many redundant rules and then tuning symptoms rather than consolidating the rule base.
- Comparing controllers without stating operators, defuzzification, and evaluation protocol.

Exercises

Exercise 1 (Easy). Reproduce the thermostat worked example in code end-to-end: fuzzify, compute firing strengths, apply implication, aggregate, and centroid-defuzzify.
Deliverable: a short table showing the computed μ's and α_j's for $e = -2$, $r = 0.5$, plus the final $u^\star$ computed by your centroid routine.

Exercise 2 (Medium). Treat operator choice as a semantic design decision: run the same rule base under max–min vs. max–product and show how the output surface changes over (e, r). Audit at least two properties (e.g., monotonicity in $|e|$ at fixed r, saturation at large $|e|$, or sensitivity to discretization step size).
Deliverable: two control-surface plots + one paragraph explaining one difference that matters operationally.

Exercise 3 (Medium). Build a Sugeno/TSK variant of the thermostat controller and compare it against the Mamdani version under identical test trajectories. Focus on interpretability and debugging: explain which variant makes it easier to localize a misbehavior to one rule/parameter.
Deliverable: one trajectory plot (Mamdani vs. TSK) + a short debugging note naming the first place you would look when outputs drift.

Fast-track note. Before moving to Chapter 19, keep the same audit discipline: log operator choices, memberships, and evaluation settings so results remain interpretable and reproducible.

Where we head next. This trilogy treated fuzzy inference as a *behavioral* design tool: we copied the *form* of human reasoning explicitly (linguistic predicates, auditable operators, and rules you can inspect). Chapter 19 shifts to a different nature-inspired paradigm: instead of copying reasoning logic, we copy an *adaptation process* we associate with natural evolution (selection + variation under a fitness budget) and hope useful structure emerges from the search—think Karl Sims-style evolved behaviors as the intuition. The goal is not faithfulness to evolutionary theory; the goal is an algorithmic optimizer that can tune memberships, rule weights, and other discrete/continuous knobs when hand-tuning is brittle and evaluations are noisy, expensive, or constrained.

Part III takeaways

- Soft computing makes uncertainty explicit: degrees of membership and rule-based aggregation.
- Interpretability is a design variable: rules, operators, and defuzzification encode assumptions.
- Debugging is empirical: test edge cases, inspect rule firing, and validate monotonicity and scale.
- The same audit mindset applies: define failure modes up front and track them as you tune operators and rules.
- **Unifying lens:** this Part fixes the representation and feedback signal, and the rest is choosing an update rule plus audits that prevent silent failure.

Part IV: Evolutionary optimization

How to read Part IV (rolling window)

- **Optimization lens:** treat evolutionary algorithms as a budgeted optimizer for black-box, noisy, or constrained objectives (not as biology).
- **Repetition is intentional:** the GA loop appears as a big-picture story, then as operator details, then as a numeric trace you can verify in code.
- **Reporting is part of correctness:** match evaluation budgets, run multiple seeds, and report distributions and constraint handling explicitly.

19 Evolutionary Computing: Search, Selection, and Adaptation

Parts I–III built three complementary toolkits: optimize models against data (ERM), learn representations with gradients, and encode auditable heuristics with fuzzy rules. Each is powerful when you can write a smooth objective or commit to a compact rule base. Many engineering choices, however, live outside that comfort zone: the knobs are discrete or tightly constrained, evaluations are noisy or expensive, and the landscape is rugged enough that local improvement is unreliable.

Evolutionary computing treats that situation as search. Instead of following gradients or hand-tuning rules, we maintain a population of candidate designs, score them with a fitness function, and use selection plus variation to explore and refine solutions within a fixed budget. After the fuzzy trilogy (Chapters 15 to 18), this chapter develops the evolutionary strand introduced in Chapter 1 and focuses on population-based optimization under constraints, budgets, and stochastic noise.

A distinct paradigm (and what "nature-inspired" means here). Fuzzy inference systems are a behavioral modeling tool: we write down the reasoning logic directly so a human can audit why a particular decision was made. Evolutionary computing aims at a different target. It encodes *candidate designs* and uses a simplified selection–variation loop—an algorithmic echo of natural evolution—to search for designs that perform well under a fixed evaluation budget (the Karl Sims-style intuition: emergence via selection and variation, not hand-coded reasoning; see Sims, 1994). The point is not to be faithful to evolutionary biology; the point is that this process can produce designs that respond well to situations under constraints and noise even when gradients are unavailable and the knobs are mixed discrete/continuous.

Learning Outcomes

- Explain the evolutionary-computation toolbox (GAs, GP, CMA-ES, DE) and when each is well-suited.
- Implement population-based optimization loops with selection, crossover/mutation, and constraint handling.
- Diagnose convergence, premature stagnation, and feasibility trade-offs on examples such as controller tuning.

Design motif

When gradients are unavailable or the landscape is rugged, treat design as search: maintain a diverse population, score candidates with a fitness function, and let selection plus variation drive improvement under constraints.

19.1 Context and Motivation

Evolutionary computing is the strand you reach for when the objective is a black box (simulation, experiment, or expensive audit) or when the design space is a mix of continuous and discrete choices. The modeling contract stays simple: pick an encoding, define a fitness function, state the constraints, and commit to an evaluation budget. The algorithm then trades exploration (diversity) against exploitation (selecting what works) as it searches a rugged landscape.

Connection to the Ch. 1 case study. The autonomous-vehicle example sketched in Chapter 1 highlights a familiar outer-loop problem: some design choices are continuous, some are discrete, and the only reliable score comes from closed-loop evaluation under realistic constraints. Evolutionary search is a practical tool in exactly that regime, where threshold policies, architecture choices, and trade-offs must be tuned under a fixed evaluation budget.

In the fuzzy trilogy, once you commit to membership shapes and a rule base, a practical question remains: how do you tune those choices (and their trade-offs) when hand tweaks do not scale? Evolutionary search provides a disciplined answer: score candidate controllers, select the better ones, perturb/recombine them, and keep iterating until the improvements flatten or the budget runs out.

Author's note: why GAs remain relevant in modern practice

Genetic algorithms remain relevant because many high-impact optimization loops are still black-box, noisy, and mixed discrete/continuous. Hyperparameter and pipeline tuning are canonical examples: one evaluation may be expensive, stochastic, and constraint-laden, while gradients are unavailable or misleading at the system level. In that setting, population search is not a historical artifact; it is an engineering tool for budgeted improvement.

This also explains the bridge from Part III. Fuzzy systems expose structured, interpretable policy components (membership shapes and rules), and evolutionary search provides a practical mechanism for tuning those components when manual adjustment does not scale.

Application note: budgeted search in the wild (locomotion, ML/LLMs, and physical experiments)

Robot gait / locomotion. Suppose we want a legged robot to move forward robustly across slightly different terrains. The design has mixed knobs: continuous parameters (gains, trajectory coefficients, penalty weights) and discrete choices (controller mode, safety logic, constraint-handling policy). A single trial is expensive and noisy (contact dynamics, initial conditions, sensor noise), so it is natural to treat performance as a black-box fitness, average over multiple seeds, and search under a strict evaluation budget.

Hyperparameter and pipeline tuning (including LLM systems). Many modern learning pipelines have the same structure: discrete design choices (model family, system configuration variant, data filters) plus continuous knobs (learning rates, thresholds, temperatures), measured by an empirical score with variance across runs. Evolutionary search is attractive when gradients are unavailable, the evaluation surface is rugged, or the objective invites silent failure modes (metric gaming, constraint violations). Read this chapter as the non-differentiable sibling of ERM: same objective-and-audit discipline, different update mechanism.

Physical-world analogy (jet nozzle). Classic evolution-strategy examples describe optimizing a jet-nozzle shape via random mutations plus selection: you do not differentiate the wind tunnel; you spend a limited trial budget to evolve better designs. The point of the analogy is not biology, but budgeted optimization under constraints.

Application example: GA tuning of a previously built fuzzy controller

Revisit the Mamdani-style fuzzy controller from the fuzzy trilogy (Chapters 15 to 18). Inputs such as tracking error $e_t = r_t - y_t$ and change-of-error Δe_t are fuzzified, rules fire, consequents are aggregated, and a centroid defuzzifier produces a crisp action u_t.

A GA enters only as a tuning layer *around* that fixed pipeline. A chromosome encodes quantities already introduced in Part III: membership-function parameters (e.g., breakpoints of triangular/trapezoidal sets), optional rule weights, and (if you allow it) a discrete choice of operator defaults from a small admissible set (t-norm, s-norm, implication, defuzzifier).

If you want the membership functions to reflect graded human judgments (not only closed-loop performance), you can tune those same parameters against expert-provided pairs $\{(x_i, \hat{\mu}_i)\}$ while enforcing ordering and overlap constraints. The GA does not change what a membership function *means*; it only gives you a disciplined search procedure for fitting its shape under constraints.

Each chromosome is evaluated by running the closed loop and scoring a fitness/objective such as

$$J = w_1 \,\mathrm{MAE}(e) + w_2 \,\mathrm{Overshoot} + w_3 \,\mathrm{ControlEffort} + \lambda \,\mathrm{Penalty}_{\mathrm{constraints}},$$

where the penalty term covers hard limits (actuator saturation, safety constraints, monotonicity constraints on memberships). This is the same fuzzy controller you can audit by inspection, with a different answer to the tuning question: instead of hand tweaking, you search under a fixed evaluation budget.

Application example: GA over neural networks (outer-loop design search)

Neural networks are trained by gradients, but many *design* decisions around them are discrete, constrained, and evaluation-limited. Consider designing a CNN (Chapter 11), an RNN (Chapter 12), or a Transformer (Chapter 14) for a fixed task and dataset. The inner loop is familiar: train by backpropagation and score generalization on a validation split. The outer loop is the hard part: choose architecture and training knobs under a compute budget.

A GA makes that outer loop explicit. Encode a candidate design as a chromosome $\mathbf{g}$ containing discrete choices (depth, width, kernel sizes, pooling schedule, sequence length/window, optimizer family) and continuous knobs (learning rate, weight decay, dropout rate). Define an objective that matches the book's audit discipline:

$$J(\mathbf{g}) = \widehat{\mathcal{L}}_{\text{val}}(\mathbf{g}) + \lambda \, \text{Penalty}_{\text{latency/memory}}(\mathbf{g}),$$

where $\widehat{\mathcal{L}}_{\text{val}}(\mathbf{g})$ is measured after a fixed training budget (epochs, steps, or wall time) using the same monitoring habits from earlier chapters (learning curves, calibration, slice checks).

The point is not to replace backpropagation. The point is that evolutionary search can sit *above* backpropagation when the knobs are mixed and the evaluation surface is noisy: it proposes designs, you train them, you score them, and selection plus variation keeps improving what you try next under a fixed budget.

Application example: jet-nozzle shape optimization (experiment in the loop)

In a wind-tunnel setting, the "model" is a physical experiment and the design variables are geometric. Let $\mathbf{x}$ parameterize a nozzle profile (for example, a small set of spline control points subject to thickness and smoothness constraints). A natural objective is "maximize thrust" (or efficiency) while penalizing constraint violations:

$$\min_{\mathbf{x}\in\mathcal{D}} J(\mathbf{x}) = -T(\mathbf{x}) + \rho\,\mathrm{Penalty}_{\mathrm{manufacturing/safety}}(\mathbf{x}).$$

There is no gradient through the tunnel. Each evaluation consumes time and budget, and measurements are noisy. Evolutionary search fits this regime: maintain a population of candidate shapes, test them, keep the better ones, perturb/recombine, and iterate until the improvement stalls or the trial budget is exhausted.

When GA is not the first choice

If the objective is smooth, gradients are trustworthy, and constraints are easy to encode, gradient-based or convex optimization methods are usually faster and easier to tune. When the evaluation budget is tiny (tens to hundreds of trials), Bayesian optimization can be a strong alternative (Frazier, 2018). Use GA when those assumptions fail: discontinuities, mixed-variable encodings, simulator-in-the-loop objectives, or hard constraints that make local gradients uninformative.

19.2 Philosophical and Historical Background

Evolutionary computing traces its roots back to the 1950s and 1960s, contemporaneous with early developments in neural networks. It is important to recognize that evolutionary algorithms are not direct scientific models of biological evolution; rather, they are inspired by a simplified, abstracted view of evolutionary principles such as selection, mutation, and reproduction.

Key Insight: These algorithms are *heuristics*; they provide practical methods to find *good enough* solutions to problems that are otherwise computationally

intractable, rather than guaranteed optimal solutions. Consequently, convergence proofs typically ensure improvement in expectation or under restrictive assumptions, but not attainment of the true global optimum.

Author's note: a pragmatic take on evolution

Evolutionary algorithms borrow the language of biology to provide a disciplined way to search rugged landscapes, not to recreate population genetics. The design mandate is pragmatic: deliver a respectable solution within the computational budget, even if it is only approximately optimal. Keep that lens in mind when evaluating selection, mutation, or recombination operators; they are tuned because they help optimization, not because they are biologically faithful.

19.3 Problem Setting: Optimization

Notation handoff

Vectors of candidate solutions are written as $\mathbf{x}$, objective values as $J(\mathbf{x})$, and population members as individuals/chromosomes depending on encoding. If symbol reuse from earlier chapters becomes ambiguous, default to local definitions and consult Appendix A.

Consider an optimization problem where the goal is to find an input vector $\mathbf{x} \in \mathbb{R}^n$ that minimizes (or maximizes) a given objective function $J : \mathbb{R}^n \to \mathbb{R}$. Formally, we want to solve

$$\mathbf{x}^* = \arg\min_{\mathbf{x} \in \mathcal{D}} J(\mathbf{x}), \tag{19.1}$$

where $\mathcal{D} \subseteq \mathbb{R}^n$ is the feasible domain incorporating any bound, equality, or inequality constraints required by the application.

Challenges:

- The function J may be *non-convex*, exhibiting multiple local minima and maxima.
- The objective or constraints may be undefined or discontinuous in parts of the domain, breaking gradient assumptions.

- The feasible set may include combinatorial or integer constraints that resist continuous methods.
- There may be no closed-form or deterministic method to find the global optimum, especially for NP-hard variants.
- The search space $\mathcal{D}$ can be large or complex, making exhaustive search (brute force) computationally prohibitive.
- Real-time or practical constraints often require solutions within limited time frames.

19.4 Illustrative Example

Imagine a function J with multiple peaks and valleys (local maxima and minima). The global minimum is the lowest valley, but many local minima exist that can trap naive optimization methods. A local method (think: "follow the slope downhill") commits to one trajectory: once it falls into a basin, it has no built-in mechanism to "jump" to a different valley unless you restart it or inject noise.

Evolutionary computing approaches the same picture by keeping *many* candidates alive at once. A population gives you coverage: some candidates sit near promising regions, others explore alternatives. Selection then biases the next round toward candidates that look better under J, while variation operators propose new points that can move across basins.

Goal: Instead of guaranteeing the global optimum, evolutionary computing aims to find a *good enough* solution—one that is sufficiently close to optimal and found within a reasonable computational budget. This goal is practical: when one objective evaluation is expensive (simulation, hardware, or human checks), the right question is often "what is the best design I can get before I run out of budget?"

19.5 Why Not Brute Force?

While brute force search guarantees finding the global optimum by evaluating all possible candidates, it is often infeasible due to:

- **Computational complexity:** The number of candidate solutions can be astronomically large.

- **Time constraints:** Real-world applications often require timely decisions, making exhaustive search impractical.

For example, in control systems, one might want to tune parameters to regulate temperature or pressure optimally. Waiting for a brute force search to complete could be unacceptable, whereas a near-optimal solution found quickly is valuable. In realistic workflows, one evaluation may involve simulation runtime, hardware experiments, or human-in-the-loop checks. Under those costs, exhaustive enumeration is not just slow but operationally infeasible, which is why evolutionary methods are framed as budgeted search.

19.6 Summary

Checkpoint: you now have the *why* for evolutionary search—rugged landscapes, constraints, discontinuities, and budgets that make local improvement unreliable. The rest of this chapter develops one canonical workflow in three passes: (i) a big-picture story (population + evaluation + variation), (ii) operator-level definitions (selection/crossover/mutation and constraint handling), and (iii) an implementation-facing sanity-check (flowchart, pseudocode, and a one-generation numeric trace). Read repeated mentions of "the GA loop" as *preview* → *formalization* → *checklist*, not as duplication.

Keep the "rugged landscape" picture in mind when you later interpret convergence traces, premature stagnation, and diversity metrics in population-based search.

19.7 Challenges in Continuous Optimization and Motivation for Evolutionary Computing

In many continuous optimization problems, the objective function may be undefined or discontinuous in certain regions of the domain. For example, consider a function with singularities or points where the function value is not defined (akin to division by zero). Such characteristics pose significant challenges for classical optimization methods such as gradient descent or hill climbing, which rely on smoothness and continuity to navigate the search space effectively.

These issues compound the nonconvexity and constraint challenges above: discontinuities and combinatorial feasibility break gradient assumptions, and global guarantees are out of reach for many NP-hard variants. Given these

challenges, deterministic approaches may be infeasible or computationally expensive. Instead, we can tolerate approximate solutions and employ heuristic or metaheuristic methods that explore the search space more flexibly. This motivates the use of *evolutionary computing* methods.

19.8 Evolutionary Computing at a Glance

Evolutionary computing treats optimization as budgeted population search: evaluate candidates, bias reproduction toward better ones, generate variation (crossover/mutation), replace the population, and repeat until budget or convergence criteria trigger. Genetic algorithms are the canonical instance of this loop.

The core tradeoff is always the same: exploration for coverage versus exploitation for immediate gains. In engineering terms, you want enough selection pressure to make progress, but not so much that the population collapses around one mediocre basin too early. This is why GA work is rarely just "run the loop": you also watch traces (best/mean fitness) and at least one diversity proxy, and you adjust operators when the behavior stops matching your intent.

19.9 Biological Inspiration: Evolutionary Concepts

This section sets terminology and motivation. If you already think of a GA as "selection + variation + replacement under a fitness budget," use it as a notation pass; the algorithmic details begin in Section 19.12.

To understand GAs, we briefly review relevant biological concepts:

Chromosomes and Genes In biology, an organism's genetic information is encoded in chromosomes, which are long sequences of DNA. Each chromosome contains many genes, which determine specific traits. In GA terms, this maps to representation: one chromosome encodes one candidate solution, and each gene encodes one design decision (binary switch, integer slot, real-valued parameter, or subtree element).

Cell Division: Mitosis vs. Meiosis

- **Mitosis:** A process where a cell divides to produce two genetically identical daughter cells, each containing the full chromosome set (e.g., 46 chro-

mosomes, i.e., 23 pairs in humans). This process is responsible for growth and tissue repair.

- **Meiosis:** A specialized form of cell division that produces gametes (sperm or egg cells) with half the number of chromosomes (haploid). When two gametes combine during fertilization, they form a new cell with a full set of chromosomes (diploid), mixing genetic material from both parents.

Genetic Recombination and Variation During meiosis, chromosomes undergo *crossover* events. Segments of genetic material are exchanged between paired chromosomes. Recombination increases genetic diversity.

This intuition is carried directly into algorithm design: crossover recombines useful partial structures from different parents, while mutation injects fresh variation so search does not collapse prematurely around one basin.

Inheritance and Heredity The offspring's chromosomes are a mixture of the parents' genetic material, but not a simple half-and-half split. Instead, genes from multiple previous generations contribute to the genetic makeup, introducing variability and enabling adaptation over time.

19.10 Implications for Genetic Algorithms

In GA terms, the biology analogy reduces to an engineering cycle: keep a population (coverage), apply fitness-based selection (pressure), recombine parents (structure transfer), mutate offspring (novelty), and replace survivors (iteration). Each operator controls a specific failure mode: too much pressure collapses diversity, too little pressure stalls progress, and weak mutation traps the search in local basins.

Two clarifications keep the analogy honest. First, "genes" are not physical objects in a GA; they are coordinates in an *encoding*. That means operator design must respect the representation: swapping substrings is natural for binary strings, but it can break permutation encodings unless you use permutation-safe crossover/mutation. Second, the point of stochasticity is not randomness for its own sake; it is controlled variability that lets the algorithm explore complex, multimodal landscapes when deterministic methods struggle or when gradients are not trustworthy.

19.11 Summary of Biological Mechanisms Modeled in GAs

Biological Process	GA Analog
Chromosomes and genes	Encoding of candidate solutions (chromosomes)
Meiosis and fertilization	Crossover of parent chromosomes to produce offspring
Genetic recombination	Mixing of solution components
Mutation	Random perturbations in offspring
Selection	Fitness-based selection of parents and survivors
Generations	Iterative improvement over time

The remainder of this section formalizes how candidate solutions are encoded and how genetic operators manipulate those encodings during evolution.

GA hyperparameters at a glance

As a starting point for binary encodings of length L, choose population sizes between $5L$ and $10L$, tournament selection with small tournaments (2–4 individuals), one-point or uniform crossover, and mutation probability near $1/L$ per bit. For real-coded GAs, replace bit-flips with Gaussian mutations on parameters and use simulated binary crossover (SBX) or blend-style crossover to mix parents smoothly (Deb and Agrawal, 1995); then tune population size and mutation scale empirically based on convergence speed and population diversity.

Author's note: population and mutation heuristics

Budget dictates population size and diversity mechanisms. If evaluations are cheap, spend them on larger populations to cover the search space; if evaluations are expensive, keep populations modest but invest in diversity-preserving steps (mutation, niching, restarts) to avoid premature convergence. Use the defaults in the box above as a first pass, then tune based on convergence traces and diversity diagnostics.

19.12 Genetic Algorithms: Modeling Chromosomes

In the previous discussion, we introduced the concept of diversity in genetic algorithms (GAs) and the probabilistic nature of evolutionary processes. We now delve deeper into modeling chromosomes and the mechanisms of genetic inheritance, crossover, and mutation, drawing parallels to optimization problems.

Genetic algorithm at a glance

Objective: Optimize an objective $J(\mathbf{x})$ over a discrete or continuous search space by evolving a population of encoded candidate solutions according to selection, crossover, and mutation.
Key hyperparameters: Population size, selection pressure (tournament size or selection temperature), crossover and mutation rates, encoding scheme, and stopping criteria (max generations, no-improvement window, target fitness).
Common pitfalls: Premature convergence due to excessive selection pressure or low mutation, deceptive fitness landscapes, constraint violations when mutation or crossover produce infeasible solutions, and overinterpreting stochastic runs without multiple seeds.

Chromosomes as Information Carriers Recall that chromosomes in GAs represent candidate solutions encoded as strings of data. For modeling purposes, we consider each chromosome as a sequence of bits or symbols, each encoding a piece of information relevant to the problem domain. Formally, let a chromosome be represented as

$$\mathbf{c} = (c_1, c_2, \ldots, c_L), \tag{19.2}$$

where each gene c_i encodes a particular trait or parameter, and L is the chromosome length.

Inheritance and Crossover During reproduction, offspring inherit genes from parents via crossover and mutation (with some genes passing through unchanged). The next figure gives an intuitive operator-level picture; full selection and crossover details follow in Section 19.17 and Section 19.18.

Figure 73 summarizes selection, crossover, and constraint handling at the operator level.

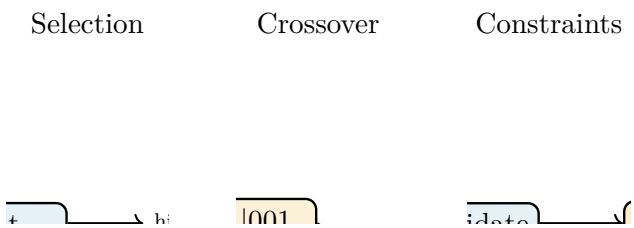


Figure 73: Evolutionary micro-operators. Left: fitter individuals get sampled more often (roulette/tournament). Middle: crossover splices parents by a mask (one-point shown). Right: constraint handling routes offspring through repair/penalty/feasibility before evaluation. Use this to map an implementation to the canonical operator loop.

Modeling the Genetic Operations Let $\mathbf{p}_1$ and $\mathbf{p}_2$ be parent chromosomes. The offspring chromosome $\mathbf{o}$ is formed by combining segments from $\mathbf{p}_1$ and $\mathbf{p}_2$ according to a crossover pattern, and then applying mutation:

$$\mathbf{o} = \text{Mutate}\big(\text{Crossover}(\mathbf{p}_1, \mathbf{p}_2, \mathbf{u})\big). \tag{19.3}$$

The crossover operator selects which genes come from which parent, often modeled by a binary mask $\mathbf{u} \in \{0,1\}^L$, where

$$o_i = \begin{cases} (p_1)_i, & \text{if } u_i = 0, \\ (p_2)_i, & \text{if } u_i = 1. \end{cases} \tag{19.4}$$

Mutation (see Section 19.19) perturbs genes with a small probability p_m.

Fitness and selection (preview) Chromosomes encode candidate solutions (phenotypes), and fitness scores quantify how well each candidate meets the objective. For example, consider chromosomes representing facial feature variants with fitness values

$$f = \{80, 75, 60, 65, 40, 20\}.$$

Higher-fitness chromosomes should be sampled more often, but not deterministically: occasional selection of weaker candidates preserves diversity and reduces premature convergence. A simple baseline is roulette selection: normalize non-negative fitness values into probabilities and sample with replacement. The point is the *bias* toward what works (not biological fidelity); we formalize selection and

work a roulette example in Section 19.17.

19.13 Mapping Genetic Algorithms to Optimization Problems

Genetic algorithms can be viewed as heuristic optimization methods. To formalize this analogy, consider the components of an optimization problem:

- **Objective function:** $J(\mathbf{x})$, which we seek to maximize or minimize.
- **Constraints:** Conditions restricting the feasible set of solutions.
- **Input parameters:** Decision variables $\mathbf{x}$.

In GAs, the chromosome encodes the input parameters $\mathbf{x}$, and the fitness function corresponds to the objective function $J(\mathbf{x})$.

Key GA Components in Optimization Terms

- **Encoding:** The method of representing $\mathbf{x}$ as chromosomes.
- **Initial population:** The starting set of candidate solutions.
- **Fitness evaluation:** Computing $f(\mathbf{c}) = J(\mathbf{x})$ for each chromosome.
- **Selection:** Choosing chromosomes for reproduction based on fitness.
- **Crossover and mutation:** Generating new candidate solutions by recombining and perturbing chromosomes.
- **Convergence criteria:** Determining when the algorithm has sufficiently optimized the objective.

Constraint handling in GAs

Realistic optimization problems often involve constraints $g_j(\mathbf{x}) \leq 0$ or $h_k(\mathbf{x}) = 0$. Common strategies include (i) *repair*, which projects infeasible offspring back into the feasible set; (ii) *penalties*, which modify the fitness as $\tilde{f}(\mathbf{x}) = f(\mathbf{x}) - \rho \sum_j \max(0, g_j(\mathbf{x}))$ for some $\rho > 0$; and (iii) *feasibility rules*, which prefer feasible individuals over infeasible ones at equal objective value. Penalty methods are simple and mesh well with existing GA code, while repair and feasibility rules are preferable when constraints encode hard physical or safety limits.

Fitness as Objective Function Proxy The fitness function guides the search towards optimal solutions. The closer a chromosome's phenotype is to the desired optimum, the higher its fitness:

$$f(\mathbf{c}) \propto \text{closeness to optimum.}$$

This relationship allows GAs to explore the solution space stochastically, balancing exploitation of high-fitness regions and exploration via mutation and recombination.

19.14 Encoding in Genetic Algorithms

We use the chromosome representation from Section 19.12; the focus here is how different encodings affect operators, constraints, and search efficiency.

Genotype and Phenotype

- **Genotype:** The encoded representation of a solution, e.g., a binary string.
- **Phenotype:** The decoded solution in the problem domain, e.g., real-valued parameters.

The goal is to design an encoding scheme that allows efficient exploration of the search space while respecting constraints and enabling effective genetic operations.

19.14.1 Common Encoding Schemes

1. Binary Encoding Each parameter is represented as a binary string of fixed length. For example, if a parameter x_i is to be represented with precision p, the length of the binary string is chosen accordingly.

- Advantages: Simple, well-studied, easy to implement crossover and mutation.
- Disadvantages: May suffer from Hamming cliffs (large phenotypic changes from small genotypic changes).

2. Floating-Point Encoding Parameters are represented directly as floating-point numbers.

- Advantages: No decoding needed, natural representation for real-valued parameters.
- Genetic operators can be adapted, e.g., crossover by averaging.
- Disadvantages: More complex mutation and crossover operators; may require specialized operators to maintain diversity.

3. Gray Coding A binary encoding where consecutive numbers differ by only one bit, reducing the Hamming distance between adjacent values.

- Useful to reduce large jumps in phenotype space due to small genotypic changes.
- Decoding involves mapping Gray code to decimal values.

19.14.2 Example: Binary Encoding of Parameters

Suppose we want to encode four parameters x_1, x_2, x_3, x_4 each represented by a binary string of length l_i. The genotype is the concatenation of these binary strings:

$$\underbrace{b_{1,1}b_{1,2}\cdots b_{1,l_1}}_{x_1} \quad \underbrace{b_{2,1}b_{2,2}\cdots b_{2,l_2}}_{x_2} \quad \underbrace{b_{3,1}b_{3,2}\cdots b_{3,l_3}}_{x_3} \quad \underbrace{b_{4,1}b_{4,2}\cdots b_{4,l_4}}_{x_4}$$

For example, a genotype might look like:

$$011 \quad 00100 \quad 0101 \quad 011110$$

Each substring is decoded to a decimal or real value according to the encoding scheme.

19.14.3 Example Problem: Minimization with Constraints

Consider the problem:

$$\min_x \quad f(x) = \frac{x}{2} + \frac{125}{x}$$

subject to

$$0 < x \leq 15$$

For example, $x = 5$ gives $f(x) = 5/2 + 125/5 = 27.5$.

Encoding Strategy

- Since x is bounded between 0 and 15, we can encode x as a binary string representing integers in $[1, 15]$.
- For example, 4 bits can represent integers from 0 to 15, so we can use 4 bits and exclude zero.
- Each genotype corresponds to a candidate solution x.

Decoding

$$x = \text{decimal value of binary string}$$

If the decoded value is zero, it is invalid due to division by zero, so such genotypes are discarded or penalized.

Fitness Evaluation For minimization problems, selection still expects "higher is better," so define fitness as a monotone transform of the cost (e.g., $F(x) = -f(x)$ or $F(x) = 1/(1 + f(x))$), and then incorporate penalties or feasibility rules for constraint violations.

19.15 Population Initialization and Size

Once encoding is decided, the initial population is generated by sampling genotypes within the search space. Purely random initialization is the default because it makes minimal assumptions and gives broad coverage early. If you have domain knowledge, you can also seed a fraction of the population with plausible candidates (and keep the remainder random) so the search starts with both coverage and a few "good guesses."

For constrained problems, initialization is also where you decide how you will treat feasibility: sample only feasible candidates (possibly with a repair step), or allow infeasible ones but attach penalties/feasibility rules during fitness evaluation. Either way, the initialization policy quietly shapes what the algorithm can discover in its first few generations.

Population Size

- Larger populations provide better coverage of the search space but increase computational cost.

- Smaller populations may converge prematurely.
- Typical sizes range from 20 to several hundreds depending on problem complexity.

A useful engineering view is to think in terms of evaluation budget. If you can afford B objective evaluations total, then (up to bookkeeping overhead) you are choosing how to split that budget between population size M and number of generations G: roughly $B \approx M \times G$. Large M buys diversity per generation; large G buys more refinement steps. The right balance depends on evaluation cost and on how quickly selection pressure collapses diversity for your encoding.

Example For the problem above, a population of 50 individuals with 4-bit genotypes representing $x \in [1, 15]$ can be initialized by randomly generating 50 binary strings of length 4.

19.16 Genetic Operators

After initialization, genetic operators are applied to evolve the population. In practice, these operators are the *control knobs* that decide how aggressively the search exploits what looks good versus how much it explores new regions of the design space.

Where operators sit in the GA loop. At a high level, each generation repeats the same pipeline:

1. Evaluate fitness (and apply penalties/feasibility rules if constrained).
2. Select parents (controls *selection pressure*).
3. Generate variation via crossover and mutation (controls *mixing* and *novelty*).
4. Form the next population (often with elitism or replacement rules).

The next subsections unpack these pieces: selection in Section 19.17, crossover in Section 19.18, and mutation in Section 19.19.

19.17 Selection in Genetic Algorithms

Selection answers the question: *who gets to reproduce?* It is the main mechanism that biases sampling toward better candidates, and it therefore determines the

algorithm's *selection pressure*. Too much pressure collapses diversity and risks premature convergence; too little reduces the GA to near-random search. In this section we formalize how fitness values translate into parent choice (roulette-style probabilities), then compare alternatives (ranking and tournament selection) that are often more stable when raw fitness values are poorly scaled. The goal is to connect the big-picture GA loop to implementable parent selection rules you can audit and tune.

19.17.1 Fitness and Selection Probability

Given a population of N chromosomes, each chromosome i has an associated fitness value f_i. The fitness function quantifies the quality of the solution represented by the chromosome.

A common approach to selection is to assign each chromosome a probability of being chosen proportional to its fitness. This can be expressed as:

$$p_i = \frac{f_i}{\sum_{j=1}^{N} f_j}, \quad i = 1, 2, \ldots, N, \tag{19.5}$$

where p_i is the probability that chromosome i is selected.

Practical note (fitness scaling). Proportional (roulette) selection assumes *nonnegative* fitness values. If your objective can be negative (as in Table 9) or you are minimizing a cost, define a shifted/scaled fitness, e.g., $\tilde{f}_i = f_i - \min_j f_j + \varepsilon$ with $\varepsilon > 0$, or use rank/tournament selection, which only relies on order comparisons.

Roulette Wheel Selection This proportional selection method is often called *roulette wheel selection*. Imagine a wheel divided into N slices, each slice corresponding to a chromosome and sized proportionally to p_i. To select a chromosome, a random number is generated to "spin" the wheel, and the chromosome corresponding to the slice where the wheel stops is chosen.

Key properties:

- Chromosomes with higher fitness have a larger slice and thus a higher chance of being selected.
- The same chromosome can be selected multiple times, reflecting its relative

superiority.

- This stochastic process maintains diversity but can be sensitive to fitness scaling.

Example Suppose we have 5 chromosomes with fitness values:

$$f = [10, 20, 5, 15, 50].$$

The total fitness is 100, so the selection probabilities are:

$$p = [0.10, 0.20, 0.05, 0.15, 0.50].$$

For instance, $p_2 = 20/100 = 0.20$ and $p_5 = 50/100 = 0.50$. Chromosome 5 has a 50% chance of selection, making it likely to be chosen multiple times.

19.17.2 Ranking Selection

When fitness values are close or vary widely, roulette wheel selection may not perform well. For example, if fitness values are very close, selection probabilities become nearly uniform, reducing selection pressure. Conversely, if one chromosome dominates, diversity may be lost prematurely.

Ranking selection addresses this by assigning selection probabilities based on the rank of chromosomes rather than raw fitness values.

Procedure

1. Sort chromosomes by fitness in descending order.
2. Assign ranks r_i such that the best chromosome has rank 1, the second best rank 2, and so forth.
3. Define a selection probability function $p(r_i)$ decreasing with rank.

A simple linear ranking scheme is:

$$p(r_i) = \frac{2-s}{N} + \frac{2(r_i - 1)(s-1)}{N(N-1)}, \tag{19.6}$$

where $s \in [1, 2]$ controls selection pressure. When $s = 1$, all chromosomes have equal probability; when $s = 2$, the best chromosome has twice the average

probability.

Elitism Ranking selection can be combined with *elitism*, where the best chromosome(s) are guaranteed to survive to the next generation. This ensures that the highest-quality solutions are preserved.

Advantages

- Controls selection pressure explicitly.
- Prevents premature convergence by maintaining diversity.
- Avoids issues with scaling fitness values.

Selection pressure and exploration/exploitation

- **Knobs:** Tournament size, rank-selection parameter s, crossover/mutation rates, and elitism all modulate pressure.
- **High pressure** (large tournaments, strong elitism, low mutation) speeds exploitation but risks premature convergence.
- **Low pressure** (small tournaments, higher mutation) preserves diversity but slows progress.
- **Practical default:** start with tournament size 2–3, modest elitism (top 1–5%), $p_c \approx 0.8$–0.9, and mutation tuned so 1–5% of bits/genes change per generation.

19.18 Crossover Operator

After selection, the *crossover* operator generates new offspring chromosomes by recombining parts of two parent chromosomes; this formalizes the intuition previewed above and supports controlled exploration.

19.18.1 One-Point Crossover

Consider two parent chromosomes represented as binary strings of length L:

$$\text{Parent 1: } \mathbf{c}^{(1)} = (c_1^{(1)}, c_2^{(1)}, \ldots, c_L^{(1)})$$

$$\text{Parent 2: } \mathbf{c}^{(2)} = (c_1^{(2)}, c_2^{(2)}, \ldots, c_L^{(2)})$$

One-point crossover proceeds as follows:

1. Choose a crossover point k uniformly at random from $\{1, 2, \ldots, L-1\}$.
2. Create two offspring by exchanging the tails of the parents at point k:

$$\begin{aligned} \text{Offspring 1} &= (c_1^{(1)}, \ldots, c_k^{(1)}, c_{k+1}^{(2)}, \ldots, c_L^{(2)}), \\ \text{Offspring 2} &= (c_1^{(2)}, \ldots, c_k^{(2)}, c_{k+1}^{(1)}, \ldots, c_L^{(1)}). \end{aligned} \tag{19.7}$$

This operator allows mixing of genetic material between parents to create novel combinations.

Multi-point and uniform crossover. Two-point or multi-point crossover swaps multiple segments between parents; uniform crossover swaps each gene independently with a fixed probability. These variants increase mixing but can disrupt building blocks if overused.

Crossover probability. Apply crossover with probability p_c (typically 0.6–0.9). When crossover is skipped, offspring are usually copied forward and then mutated, which helps preserve good structures while maintaining exploration.

19.19 Mutation Operator

Mutation introduces random alterations to individual chromosomes, mimicking biological mutations. It serves to maintain genetic diversity within the population and helps the algorithm escape local optima.

Biological motivation Mutation is a rare event in nature but crucial for evolution. For example, the white coloration of polar bears is a mutation that provided an adaptive advantage in snowy environments. Similarly, environmental pressures can select for mutations, such as female elephants in Africa evolving to lack ivory tusks to avoid poaching.

Role in optimization Mutation allows the algorithm to explore new regions of the search space that are not reachable by crossover alone. Consider a fitness landscape with multiple local maxima and minima. Mutation can randomly

perturb a solution, potentially moving it from a local minimum to a region near a global maximum.

Implementation of mutation In binary-encoded chromosomes, mutation typically involves flipping a bit:

$$0 \to 1, \quad 1 \to 0$$

The mutation is applied with a small *mutation probability* p_m, often on the order of 10^{-3} to 10^{-1}.

Mutation operator formalization Given a binary chromosome $\mathbf{c} \in \{0,1\}^L$, mutation produces $\mathbf{c}'$ by mutating each bit c_i independently with probability p_m:

$$c_i' = \begin{cases} 1 - c_i, & \text{with probability } p_m, \\ c_i, & \text{with probability } 1 - p_m. \end{cases} \tag{19.8}$$

19.20 Summary of Genetic Operators and Their Probabilities

You have now seen each operator in isolation. When you implement a GA, you typically configure the loop with a small set of *operator knobs*:

- **Selection scheme + pressure:** roulette/ranking/tournament selection, plus an explicit pressure parameter (e.g., fitness scaling, rank parameter s, tournament size) and often an elitism rate (top k or top $e\%$ copied forward).
- **Crossover probability p_c:** with probability p_c, recombine two parents; otherwise copy parents forward (and still allow mutation).
- **Mutation probability p_m:** for binary encodings, mutate each bit independently with probability p_m (a common default is $p_m \approx 1/L$ for length L); for real encodings, p_m usually corresponds to a mutation *scale* (e.g., Gaussian noise standard deviation) rather than bit flips.

To keep notation compact later, we write these scheme-dependent settings

abstractly as

$$
\begin{aligned}
p_s &\equiv \text{selection pressure parameter(s) (scheme-dependent)},\\
p_c &= \text{probability of applying crossover},\\
p_m &= \text{mutation probability (per gene/bit, unless stated otherwise)}.
\end{aligned}
$$

Tuning these settings controls exploration versus exploitation and strongly affects whether the algorithm stagnates early or continues to make progress.

19.21 Known Issues in Genetic Algorithms

Risk & audit

- **Premature convergence:** aggressive selection or weak mutation collapses diversity before good regions are explored.
- **Budget mismatch:** comparisons are invalid unless algorithms share evaluation budgets and stopping rules.
- **Constraint leakage:** hidden infeasibility can inflate scores; use explicit feasibility checks in logs and plots.
- **Single-run overclaim:** stochastic algorithms require multi-seed reporting with spread, not one best trajectory.
- **Objective gaming:** fitness proxies can drift from deployment goals; audit secondary metrics and failure cases.

While genetic algorithms (GAs) provide a powerful heuristic framework for optimization, several well-known issues can affect their performance and reliability:

Premature Convergence Because GAs rely on heuristic search without a global optimality guarantee, they often converge prematurely to local minima rather than the global minimum. This is especially common if the initial population is not diverse or if the selection pressure is too high, causing loss of genetic diversity early on.

Diversity maintenance

- **Crowding/sharing:** penalize overly similar individuals to keep multiple niches in multi-modal landscapes.
- **Restarts/islands:** run multiple subpopulations (often in parallel) with occasional migration; robust against stagnation.
- **Adaptive mutation:** increase mutation or inject random individuals when diversity drops.

Mutation Interference Mutation is intended to introduce diversity and help escape local minima by randomly altering genes. However, excessive or poorly controlled mutation can cause oscillations, where beneficial mutations are undone by subsequent mutations. This back-and-forth effect can prevent convergence and degrade solution quality.

Deception Deception refers to situations where the encoding or representation of solutions misleads the GA's fitness evaluation. Low-order schemata with high observed fitness may actually guide the search away from the global optimum, so that combining "good" building blocks produces worse offspring. There is no single formal definition, but a deceptive fitness landscape is one in which local improvements inferred from schemata systematically lead the GA to suboptimal basins of attraction.

Fitness Misinterpretation Since selection is driven by fitness values, any inaccuracies or misleading fitness evaluations can cause the GA to make poor decisions about which individuals to propagate. This can arise from noisy fitness functions, poorly designed objective functions, or deceptive encodings.

19.22 Convergence Criteria

Determining when to stop the GA is a critical practical consideration. Common convergence criteria include:

- **Fixed number of generations:** Run the GA for a predetermined number of iterations.

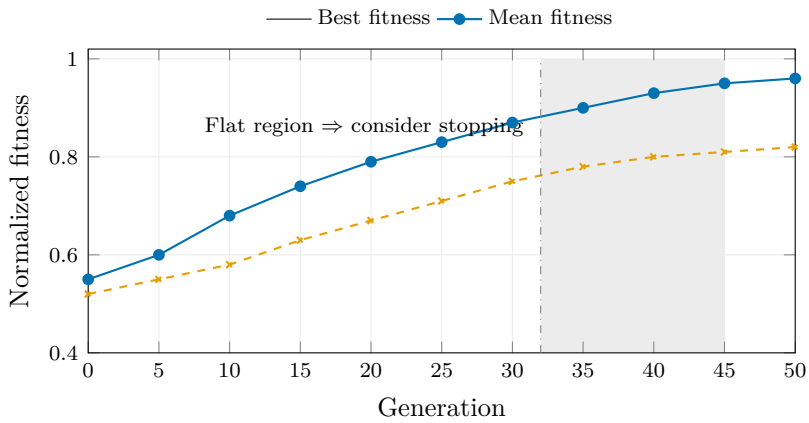


Figure 74: Illustrative GA run showing the best and mean normalized fitness over 50 generations. Flat regions motivate "no improvement" stopping rules, while steady separation between best and mean indicates ongoing selection pressure. Helpful for diagnosing premature convergence versus ongoing exploration.

- **Time limit:** Stop after a fixed amount of computational time.
- **No improvement:** Terminate if the best fitness value has not improved over a specified number of generations.
- **Manual inspection:** Periodically inspect the population to decide if the solutions are satisfactory.

In practice, a combination of these criteria is often used. For example, one might stop if *either* (a) no improvement in the best fitness is observed for 10 consecutive generations, *or* (b) the run reaches 100 generations in total, whichever condition is met first. Figure 74 visualizes such a run, making it easy to spot plateaus and the persistent gap between best and mean fitness.

19.23 Summary of Genetic Algorithm Workflow

This is the implementation-facing assembly of the GA loop. Earlier sections motivated the need for population-based search and defined each operator; here we put them back together and give three complementary sanity-check views: Figure 75 for control flow, Table 9 for a one-generation numeric trace, and Section 19.24 for a minimal implementation recipe. All three describe the same loop at different levels of detail.

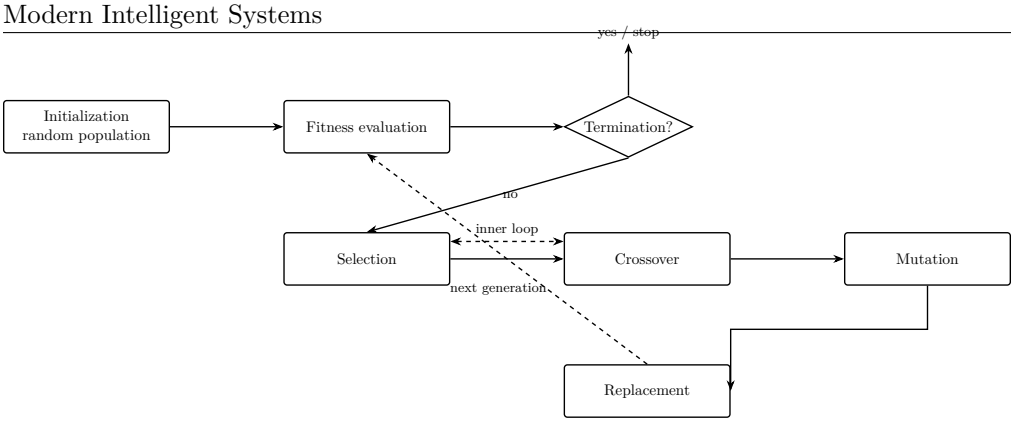


Figure 75: GA flowchart showing the iterative process: initialization leads to fitness evaluation and a termination check. If not terminated, the algorithm proceeds through selection, crossover, mutation, and replacement, which then feeds the next generation's fitness evaluation. Reference when verifying that your implementation preserves the intended control flow.

For the toy generation in Table 9, fitness values are computed using $f(x) = \cos(5\pi x)\exp\left(-x^2\right)$ from Section 19.25.

Step	Example bitstrings	Decoded x	Fitness $f(x)$
Initial (subset)	0001_2, 0010_2, 0101_2, 0111_2	0.0625, 0.125, 0.3125, 0.4375	0.553, -0.377, 0.177, 0.687
Select parents (example)	0010_2, 0101_2	0.125, 0.3125	-0.377, 0.177
Crossover (one-point)	Parents: 00|10, 01|01 $\rightarrow$ offspring: 00|01, 01|10	0.0625, 0.3750	0.553, 0.803
Mutation (flip one bit)	$0110_2 \rightarrow 0111_2$	0.4375	0.687

Table 9: Toy GA generation on a bounded interval. One crossover and mutation illustrate how the fitness function guides selection before the next generation. Use this to explain how variation operators interact with selection pressure.

Table 9 is the step-by-step toy generation trace used to ground the operator sequence.

Defaults that work (DE/CMA-ES)

Differential Evolution (DE): start with population $10-20 \times D$ (dimension D), mutation scale $F \in [0.5, 0.8]$, crossover $C_r \in [0.7, 0.9]$. **CMA-ES:** population $\lambda \approx 4 + \lfloor 3 \ln D \rfloor$, initial step-size $\sigma_0 \approx 0.3$ of the variable range. These defaults are robust first tries before tuning.

19.24 Pseudocode Representation

The GA can be expressed in pseudocode as follows. Treat this as the "minimum viable" skeleton; a production implementation additionally records random seeds, evaluation budgets, constraint handling decisions, and per-generation diagnostics (best/mean fitness, diversity) so results are reproducible and interpretable.

```
Initialize population P with N chromosomes
Evaluate fitness of each chromosome in P

while termination criteria not met do
    Select parents from P based on fitness
    Apply crossover to parents to create offspring
    Apply mutation to offspring
    Evaluate fitness of offspring
    Replace some or all of P with offspring
end while

Return best chromosome found
```

19.25 Example: GA for a Constrained Optimization Problem

Consider the problem of *maximizing* the function:

$$f(x) = \cos(5\pi x) \cdot \exp\left(-x^2\right)$$

subject to the constraint:

$$0 \leq x \leq 0.5$$

with a precision of three decimal places.

GA Parameters:

- Population size: 10 chromosomes
- Encoding: Fixed-point with resolution 0.001: store an integer $n \in \{0, 1, \ldots, 500\}$ and decode via $x = n/1000$ (a 9-bit representation covers 0–511)

- Crossover probability: 25%
- Mutation probability: 10%
- Selection: Truncation selection (example): select the top five chromosomes by fitness as parents each generation

Initialization: Generate 10 random values of x uniformly distributed in $[0, 0.5]$, each rounded to three decimal places. When prior designs or surrogate models exist, *warm start* a few chromosomes with those known-good solutions before filling the rest randomly; seeding accelerates convergence without losing diversity if you keep most of the population stochastic.

Fitness Evaluation: Calculate $f(x)$ for each chromosome and treat it as a fitness score (higher is better). If instead you are minimizing a cost, convert it to fitness via a monotone transform (e.g., $-J(x)$, a shifted score, or a rank-based scheme) so that selection still prefers better candidates.

Evolutionary Cycle: Apply selection, crossover, and mutation to produce new offspring, then evaluate their fitness. Repeat this process for multiple generations until convergence criteria are met.

Remarks: In practice, some initial chromosomes may fall outside the constraint bounds due to rounding or mutation; these should be clipped or repaired to maintain feasibility.

As an illustration, if the initial population contains

$$x \in \{0.04,\ 0.09,\ 0.13,\ 0.18,\ 0.22,\ 0.27,\\ 0.31,\ 0.36,\ 0.42,\ 0.48\},$$

then the corresponding objective values are

$$f(x) \in \{0.808,\ 0.155,\ -0.446,\ -0.921,\ -0.906,\ -0.422,\\ 0.142,\ 0.711,\ 0.797,\ 0.245\},$$

rounded to three decimals. Each chromosome uses a 9-bit fixed-point code (3 fractional digits), decoded by interpreting the bits as an integer n and scaling

via $x = n/1000$. Any decoded values outside [0, 0.5] are repaired (e.g., clipped to bounds); note that $x = 0$ is allowed in this example.

A single generation could proceed as follows:

- Select the top five chromosomes by fitness.
- Apply one-point crossover on the 9-bit fixed-point codes (MSB-first). For example, $0.203 \mapsto 011001011_2$ and $0.359 \mapsto 101100111_2$; cutting after 5 bits and swapping tails yields offspring $011000111_2 \mapsto 0.199$ and $101101011_2 \mapsto 0.363$.
- Mutate each bit with probability 0.1, ensuring all decoded values remain within [0, 0.5].
- Re-evaluate fitness and retain the best ten individuals for the next generation.

Constraint handling playbook

- **Penalty methods** soften constraints by augmenting the fitness with a violation term, e.g., $F(\mathbf{x}) = f(\mathbf{x}) + \lambda \sum_k \max\{0, g_k(\mathbf{x})\}^2$; increase λ when infeasible individuals survive too often.
- **Repair operators** project infeasible chromosomes back into the feasible region (clip bound violations, renormalize equality constraints, or rerun a problem-specific solver) before evaluation.
- **Feasibility-first selection** ranks feasible candidates ahead of infeasible ones, then compares raw fitness only within each group; among infeasible solutions, select those with the smallest violation.
- **Deb's feasibility rules** (Deb, 2001): (i) if one solution is feasible and the other is not, pick the feasible one; (ii) if both are feasible, pick the better objective; (iii) if both are infeasible, pick the one with smaller total constraint violation. Adaptive penalties can be layered on top when violations persist.

Reproducibility and fair comparison Fix random seeds when debugging, run many seeds (e.g., 20+) for reporting, match evaluation budgets across algorithms, and report mean/median best-so-far with variability bands. Log all hyperparameters and share code/configs to make comparisons fair, following Appendix B as the default reporting protocol. Penalty terms are easy to implement, repair operators exploit domain knowledge, and feasibility-first policies are useful in safety-critical controllers where violating constraints is unacceptable even temporarily.

19.26 Genetic Algorithms: Iterative Process and Convergence

This section focuses on convergence behavior; the GA cycle itself is summarized in Section 19.23 and the control flow in Figure 75. Over generations, populations tend to cluster around better regions of the fitness landscape; flat best-fitness curves signal stagnation, while a persistent gap between best and mean indicates ongoing selection pressure (see Figure 74). Convergence typically occurs after a problem-dependent number of generations, and the best-so-far solution should

be treated as a high-quality approximation rather than a guaranteed global optimum.

19.27 Beyond canonical GAs: real-coded strategies

Bit-string encodings are ideal for combinatorial search, yet most engineering problems have continuous decision variables. Two mature real-coded families are now standard tools:

- **Covariance Matrix Adaptation Evolution Strategy (CMA-ES)** (Hansen and Ostermeier, 2001) maintains a multivariate Gaussian search distribution. Successful steps update the mean, adapt the global step size, and rotate the covariance to align with the landscape's principal directions. CMA-ES shines on smooth, ill-conditioned black-box functions where gradients are unavailable but the objective rewards second-order adaptation.
- **Differential Evolution (DE)** (Storn and Price, 1997) perturbs a target vector with scaled differences of two other individuals, $\mathbf{v} = \mathbf{x}_r + F(\mathbf{x}_p - \mathbf{x}_q)$, then mixes $\mathbf{v}$ with the original via binomial or exponential crossover. This simple mechanism balances exploration/exploitation with only three hyperparameters (F, C_r, N) and handles noisy, non-smooth objectives well.

Evolution strategies (ES): step-size adaptation and self-adaptation

Evolution strategies are a real-coded evolutionary family that makes one engineering idea explicit: *mutation step size is part of the algorithm's state, not a fixed constant.* In a simple isotropic form,

$$\mathbf{x}' = \mathbf{x} + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I}),$$

where σ controls exploration (large σ) versus local refinement (small σ). A classic adaptive heuristic is the "1/5 success rule": if more than 20% of recent mutations improve fitness, increase σ; if fewer than 20% succeed, decrease σ. Self-adaptive variants go one step further by embedding σ (and sometimes covariance structure) into the chromosome itself so selection pressure co-evolves the search behavior alongside the solution parameters.

Both algorithms plug into the same evaluation loop shown earlier and can reuse the constraint-handling policies in the preceding box. In practice many

teams prototype with DE (fast, few knobs) and switch to CMA-ES when the problem demands higher precision or adaptive covariance modeling.

19.28 Particle Swarm Optimisation (PSO)

Particle Swarm Optimisation is a population-based metaheuristic inspired by collective behaviour in bird flocks and fish schools. Each candidate solution ("particle") moves through the search space by blending its own best-known position with the swarm's best-known position.

Update equations. Each particle i has a position $\mathbf{x}_i$ and a velocity $\mathbf{v}_i$. At each iteration:

$$\begin{aligned}\mathbf{v}_i^{(t+1)} &= \omega\,\mathbf{v}_i^{(t)} + c_1 r_1\big(\mathbf{p}_i - \mathbf{x}_i^{(t)}\big) + c_2 r_2\big(\mathbf{g} - \mathbf{x}_i^{(t)}\big),\\ \mathbf{x}_i^{(t+1)} &= \mathbf{x}_i^{(t)} + \mathbf{v}_i^{(t+1)},\end{aligned}$$

where $\mathbf{p}_i$ is particle i's personal best position, $\mathbf{g}$ is the global best position across the entire swarm, ω is the *inertia weight* (controls exploration vs. exploitation), c_1 and c_2 are the *cognitive* and *social* coefficients, and $r_1, r_2 \sim \mathcal{U}(0,1)$ are fresh random draws each iteration.

Hyperparameter guidance. Common defaults are $\omega \approx 0.7$, $c_1 \approx c_2 \approx 1.5$, and a swarm size around 20–50 particles. A linearly decreasing inertia schedule (ω from roughly 0.9 to 0.4) shifts the search from broad exploration early to local refinement late. For constrained problems, apply Deb's feasibility rules or velocity clamping to keep particles within bounds.

Comparison with GAs and DE. PSO has no crossover or mutation operators; all information sharing happens through the velocity update. This makes PSO very easy to implement and well-suited to continuous, unconstrained (or lightly constrained) problems. GAs and DE generally outperform PSO when the search space has many discrete components or when diversity preservation is critical. PSO can converge prematurely on multimodal landscapes; restart heuristics or topology variants (ring, von Neumann) mitigate this.

Pointer: neuro-evolution and NEAT. Both GAs and PSO can optimise neural network weights and topologies simultaneously—an approach called *neuro-evolution*. The landmark algorithm in this space is NEAT (NeuroEvolution of Augmenting Topologies), which grows network structure incrementally using historical markings to align crossover across networks with different topologies. NEAT demonstrated that topology-and-weight evolution could solve problems that fixed-topology neuro-evolution could not, and it has experienced a resurgence in reinforcement learning settings where the loss landscape is non-differentiable or highly non-stationary. Modern descendants (HyperNEAT, POET, Open-Ended Learning) extend the idea to indirect encodings and competitive co-evolution.

19.29 Genetic Programming (GP)

Genetic programming extends the principles of genetic algorithms to the evolution of computer programs or symbolic expressions rather than fixed-length parameter vectors.

Problem Setup Consider a problem where the relationship between input variables $x_1, x_2, \ldots, x_n$ and output y is unknown. Unlike traditional parameter estimation, we do not assume a fixed functional form. Instead, we want to discover the function f such that

$$y = f(x_1, x_2, \ldots, x_n).$$

Representation of Programs In GP, candidate solutions are represented as tree-like structures encoding mathematical expressions or programs composed of:

- **Terminals:** Input variables $(x_1, x_2, \ldots)$ and constants.
- **Functions:** Arithmetic operations (addition, subtraction, multiplication, division), logical operations, or other domain-specific functions.

For example, a candidate program might represent the expression

$$(x_1 \times x_3) + (x_1 + x_4).$$

Genetic Operators in GP

- **Crossover:** Subtrees from two parent programs are exchanged to create offspring programs.
- **Mutation:** Random modifications are made to nodes in the program tree, such as changing an operator or replacing a subtree.

These operations allow the evolution of increasingly complex and effective programs.

Fitness Evaluation A candidate program is evaluated by executing it on a training set and comparing its outputs with the desired targets. Fitness functions often measure mean squared error, classification accuracy, or accumulated reward, and penalize programs that raise runtime exceptions or exceed resource limits. Individuals with higher fitness are more likely to be selected for reproduction.

Example Suppose we have the following initial program trees:

$$\begin{aligned} \text{Parent 1:} \quad & f_1 = (x_1 \times x_3) + (x_1 + x_4) \\ \text{Parent 2:} \quad & f_2 = (x_2 - 5) \times (x_4 + 1) \end{aligned}$$

Suppose we exchange the right subtree of f_1 (the addition node $x_1 + x_4$) with the left subtree of f_2 (the subtraction node $x_2 - 5$). The resulting offspring are

$$f_1' = (x_1 \times x_3) + (x_2 - 5), \qquad f_2' = (x_1 + x_4) \times (x_4 + 1).$$

Mutation might then replace the terminal x_4 in f_1' with a constant (e.g., 5) or switch the addition operator to multiplication, thereby exploring nearby program structures while keeping the tree depth bounded.

Recursive and Modular Programs GP can evolve recursive functions and modular code blocks (subroutines), enabling the discovery of complex behaviors and algorithms. In practice this is achieved by allowing trees to reference automatically defined functions (ADFs) or macros that are evolved alongside the main program. The depth of the program trees and the number of reusable modules are usually constrained to prevent uncontrolled growth and to keep execution cost manageable.

Applications Genetic programming is particularly useful for:

- Symbolic regression: discovering analytical expressions fitting data.
- Automated program synthesis: generating code for control, decision-making, or data processing.
- Robotics: evolving control programs for navigation, obstacle avoidance, or manipulation.

Example: Robot Obstacle Avoidance Consider evolving a program that controls a robot's movement based on sensor inputs indicating obstacles. The function set might include commands like `move_forward`, `turn_left`, `turn_right`, and conditional statements. The GP evolves sequences and combinations of these commands to maximize the robot's ability to navigate without collisions.

Summary Genetic programming generalizes genetic algorithms by evolving program structures (often trees) rather than fixed-length chromosomes.

19.30 Wrapping Up Genetic Algorithms and Genetic Programming

In this final segment of the chapter, we conclude our discussion on genetic algorithms (GAs) and genetic programming (GP), emphasizing their conceptual foundations, practical implications, and the distinctions between them.

Recap of Genetic Algorithms Genetic algorithms are population-based metaheuristics; the earlier sections already define selection, crossover, mutation, and the loop in detail. For recap, keep the design levers in view and revisit the worked traces when needed:

- **Representation:** How solutions are encoded (bit strings, real vectors, trees) and what constraints must be preserved.
- **Fitness and evaluation:** What you reward, how you handle noisy measurements, and how constraints enter the score.
- **Operator/loop settings:** Selection pressure, crossover/mutation rates, and stopping criteria.

See Section 19.23 for control flow, Table 9 for a numeric generation trace, and Sections 19.17 to 19.19 for operator details.

Genetic Programming: Structure over Parameters Genetic programming extends the GA paradigm by evolving computer programs or symbolic expressions rather than fixed-length parameter vectors. The fundamental difference is that GP searches over the space of program structures (trees of functions and terminals) instead of numeric parameter values.

Key points about GP include:

- **Representation:** Programs are represented as hierarchical trees, where internal nodes are functions (e.g., arithmetic operators, logical functions) and leaves are terminals (input variables, constants).
- **Evolution of Programs:** Genetic operators manipulate program trees:
  - *Crossover* exchanges subtrees between parent programs.
  - *Mutation* randomly modifies nodes or subtrees.
- **Fitness Evaluation:** Programs are executed on input data, and their outputs are compared against desired outputs to compute fitness.
- **Emergent Solutions:** GP can discover novel program structures that model complex phenomena without explicit programming, often yielding surprising and insightful results.

Applications and Insights Genetic programming is particularly powerful for modeling complex systems where the underlying relationships are unknown or difficult to specify explicitly. For example, given inputs such as wind speed, humidity, and temperature, GP can evolve models that predict environmental phenomena without prior assumptions about the functional form.

This capability highlights the strength of GP as a tool for automated model discovery and symbolic regression.

Further Topics and Extensions While this chapter provided a concise overview, the field of evolutionary computation encompasses many advanced topics, including:

- **Multi-objective Genetic Algorithms:** Handling optimization problems with multiple conflicting objectives.
- **Constraint Handling:** Incorporating problem-specific constraints into the evolutionary process.
- **Hybrid Methods:** Combining GAs/GP with other optimization or machine learning techniques.
- **Scalability and Parallelization:** Efficiently implementing evolutionary algorithms for large-scale problems.

Readers are encouraged to explore these topics through further reading and research; the short primer below highlights the most widely used multi-objective GA.

19.31 Multi-objective search and NSGA-II

When two or more objectives conflict, we seek a set of Pareto-optimal solutions rather than a single best point. The Non-dominated Sorting Genetic Algorithm II (NSGA-II) sorts each generation into Pareto fronts: rank-1 individuals are non-dominated, rank-2 are dominated only by rank-1, etc. Replacement preserves all members of the best fronts and uses crowding distance to maintain diversity along the trade-off curve. NSGA-II's combination of elitist survival and $O(N \log N)$ non-dominated sorting makes it the default baseline for multi-objective evolutionary optimization (Deb et al., 2002).

Metrics and variants Hypervolume (area/volume dominated by the front with respect to a reference point) is a common scalar indicator; report it alongside the spread of solutions (Zitzler et al., 2002). MOEA/D decomposes objectives into weighted subproblems; SPEA2/IBEA are popular alternatives. Always plot the Pareto set and budget-matched hypervolume traces when comparing algorithms.

Figure 76 is the trade-off view used to interpret multi-objective runs.

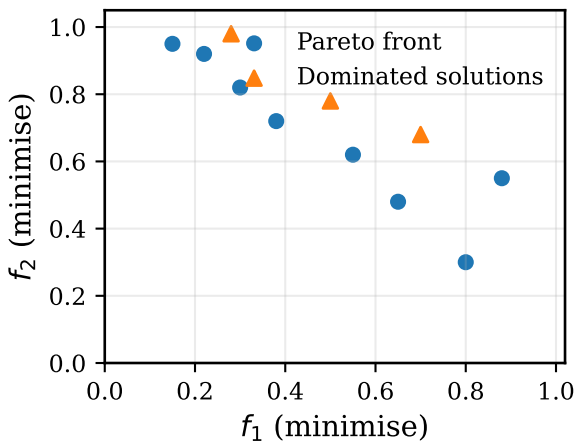


Figure 76: Sample Pareto front for two objectives. NSGA-II keeps all non-dominated points (blue) while pushing dominated solutions (orange) toward the front via selection, yielding a spread of trade-offs in one run. The point is to interpret multi-objective results as a set of trade-offs rather than a single optimum. *Source: author-generated.*

Key takeaways

- Evolutionary algorithms optimize without gradients by iterating selection, variation, and replacement under a fitness evaluation loop.
- Constraint handling is part of the design: penalties, repair, and feasibility-first selection each encode a different notion of "acceptable" search.
- Multi-objective search replaces a single optimum with a Pareto front; NSGA-II is a standard baseline for producing diverse trade-offs.

Minimum viable mastery.

- Specify genotype, variation operators, selection scheme, and termination criteria in a way that is reproducible.
- Distinguish constraint handling strategies and justify the one used (penalty vs. repair vs. feasibility rules).
- Report multi-objective results as trade-offs (Pareto fronts) rather than as a single scalar score.

Common pitfalls.

- Over-interpreting a single stochastic run (report multiple seeds and dispersion).
- Using aggressive selection and low mutation, collapsing diversity and getting stuck in deceptive basins.
- Comparing algorithms without matching evaluation budget (fitness calls), constraint handling, and stopping rules.

Ethical considerations. Evolutionary search will exploit whatever you write down as fitness. If the objective is a proxy for the real goal, the optimizer can "win" in ways you did not intend (metric gaming, unsafe corner solutions, hidden constraint violations). Treat objective design and constraint auditing as part of system safety, not as afterthoughts.

Exercises

Exercise 1 (Easy). Turn the chapter's one-generation trace into a reproducible unit test: implement the exact selection/crossover/mutation steps used in Table 9 and verify that your code reproduces the same intermediate population (parents chosen, offspring produced, and replacement).
Deliverable: a step-by-step table matching Table 9 + one sentence explaining what would make such a trace *non-reproducible*.

Exercise 2 (Medium). Study selection pressure vs. diversity on a multi-modal 1D objective (any bounded interval): run tournament selection with several tournament sizes (or roulette vs. tournament) under a fixed evaluation budget. Track both best fitness and a diversity proxy (e.g., genotype variance or number of unique individuals).
Deliverable: best-fitness curves (median over seeds) + one diversity plot + a short conclusion about when the GA "wins" by collapsing diversity too early.

Exercise 3 (Medium). Compare constraint handling strategies under the same fitness-call budget: penalty vs. feasibility-first selection (or repair). Use a constrained objective where the unconstrained optimum is infeasible so the difference is visible.
Deliverable: a plot of feasibility rate over generations + the best feasible fitness over time + one paragraph explaining what your constraint strategy is implicitly prioritizing.

Fast-track note. This chapter is largely self-contained. If you revisit earlier model chapters, keep the common thread explicit: each method still needs a clear objective, a diagnostic that exposes failure, and a reporting protocol that survives replication.

Where we head next. This chapter closes the soft-computing thread. For targeted refreshers, Appendix D summarizes linear-systems prerequisites and Appendix C consolidates kernel-method context used earlier in the book.

Part IV takeaways

- Evolutionary search is an optimization tool when gradients are unavailable or unreliable.
- Operators (selection, crossover, mutation) encode exploration vs. exploitation; tune them against variance across runs.
- Constraints and multi-objectives are first-class: define feasibility and trade-offs before optimizing.
- Report reproducibly: seeds, multiple runs, and distributions matter more than a single best trajectory.
- **Unifying lens:** this Part removes the gradient and asks what discipline is needed when feedback is noisy, sparse, or available only as a fitness rank.

Back matter

Key Takeaways

Chapter 1 *About This Book* sets a reading strategy, summarizes notation/figure conventions, and motivates the four-level taxonomy for systems thinking across chapters.

Chapter 2 *Symbolic Integration and Problem-Solving Strategies* frames safe substitutions, heuristic branches, and numeric fallbacks inside a transformation tree for algebraic problem solving.

Chapter 3 *Supervised Learning Foundations* develops ERM/MLE/MAP, contrasts loss families, and grounds diagnostics such as learning curves, calibration, and proper scoring rules.

Chapter 4 *Classification and Logistic Regression* builds probabilistic classifiers using the Chapter 3 toolkit, emphasizes ROC/PR analysis, and treats class imbalance, calibration, and optimization choices (Newton vs. first-order).

Chapter 5 *Introduction to Neural Networks* casts the perceptron as a thresholded linear classifier (vs. logistic as the smooth probabilistic counterpart), proves convergence guarantees, and catalogs pitfalls such as poor feature scaling or non-separable data.

Chapter 6 *MLP Foundations* formalizes forward/backward passes with matrix-calculus identities, highlights caching/normalization for numerical stability, and frames bias–variance behavior for deep linear stacks.

Chapter 7 *Backpropagation in Practice* turns the derivatives into SGD/mini-batch pseudocode, adds early-stopping heuristics, and compares optimization tweaks (momentum, adaptive schedules) against the diagnostics from Chapter 3.

Chapter 8 *Radial Basis Function Networks* interprets RBFs as local

"bubbles," covers center/width selection (including practical σ rules), contrasts primal vs. dual training formulations, and connects the finite-basis view to kernel methods (e.g., kernel ridge regression and SVMs with RBF kernels).

Chapter 9 *Self-Organizing Maps* explains neighborhood competition/cooperation phases, quality measures (quantization/topographic error), and visualization tricks for prototype-based embedding.

Chapter 10 *Hopfield and Energy-Based Memories* derives discrete/continuous dynamics, capacity bounds, and asynchronous vs. synchronous update strategies for associative recall.

Chapter 11 *Convolutional Neural Networks and Deep Training Tools* details convolution/cross-correlation, pooling, receptive-field growth, and the engineering defaults behind modern CNN blocks and training loops.

Chapter 12 *Recurrent Neural Networks* develops BPTT, gating strategies, and conditioning tricks (teacher forcing, scheduled sampling) for sequential modeling while connecting to earlier diagnostics.

Chapter 13 *NLP Pipelines and Responsible Deployment* links static/contextual embeddings to downstream tasks, adds bias/calibration checklists, and closes with a deployment-readiness assessment.

Chapter 14 *Transformers and Attention* consolidates scaled dot-product attention, multi-head blocks, encoder/decoder stacks, long-context strategies (RoPE/ALiBi, FlashAttention, KV caches), and PEFT techniques.

Chapter 15 *Soft Computing Orientation* positions fuzzy logic, neurocomputing, probabilistic reasoning, and

evolutionary search as complementary tools and introduces the running thermostat example used in Chapters 16 to 18.

Chapter 16 *Fuzzy Sets and Membership Functions* defines linguistic variables, membership design patterns, set operations, and inclusion metrics that quantify vagueness and overlap.

Chapter 17 *Fuzzy Relations and the Extension Principle* covers Cartesian products, projections, and composition operators (max–min, algebraic, Łukasiewicz) that transfer fuzzy information across universes.

Chapter 18 *Fuzzy Inference Systems* assembles Mamdani and Sugeno pipelines (aggregation, implication, defuzzification) and studies operator choices, scaling, and thermostat/autofocus examples.

Chapter 19 *Evolutionary and Population-Based Search* surveys canonical GAs, GP, CMA-ES, and Differential Evolution, emphasizing constraint handling, budget-aware population sizing, and integration with the rest of the toolkit.

Four-level taxonomy in practice

Level 1 (reactive systems): Feedback loops and associative memories that respond instantly, e.g., Hopfield dynamics in Chapter 10 or the thermostat-style fuzzy controllers introduced across Chapters 16 to 18.
Level 2 (deliberative planners): Rule-based systems that reason over an internal linguistic state before acting; see the fuzzy relation and inference machinery of Chapters 16 to 18, where conditions aggregate before a crisp recommendation is issued.
Level 3 (adaptive learners): Data-driven models that update parameters from data, spanning the ERM toolkit (Chapters 3 to 4), perceptrons/MLPs/RBFs/CNNs (Chapters 5 to 8 and Chapter 11), sequence models and Transformers (Chapters 12 to 14), SOMs (Chapter 9), and population heuristics (Chapter 19).
Level 4 (meta-cognitive agents): Algorithms that reason about their own learning loops: calibration and uncertainty estimation (Chapters 3 to 4), training diagnostics and early-stop policies (Chapter 7), alignment/PEFT tooling for Transformers (Chapter 14), and self-adaptive evolutionary strategies (Chapter 19). These examples illustrate early steps toward systems that refine their own policies.

Figure 77 and Table 10 are the book's synthesis: they show that the families in Parts I–IV are not disconnected topics but design points under one contract. To read the map, ask four questions: (1) what is the representation, (2) what is the update/search operator, (3) what feedback signal trains it, and (4) what audits prevent silent failure. Use the map to move laterally: if you know one design point (e.g., ERM with gradients), you can locate its siblings (energy descent, rule composition, population search) and predict what will break (credit assignment, calibration, constraint handling).

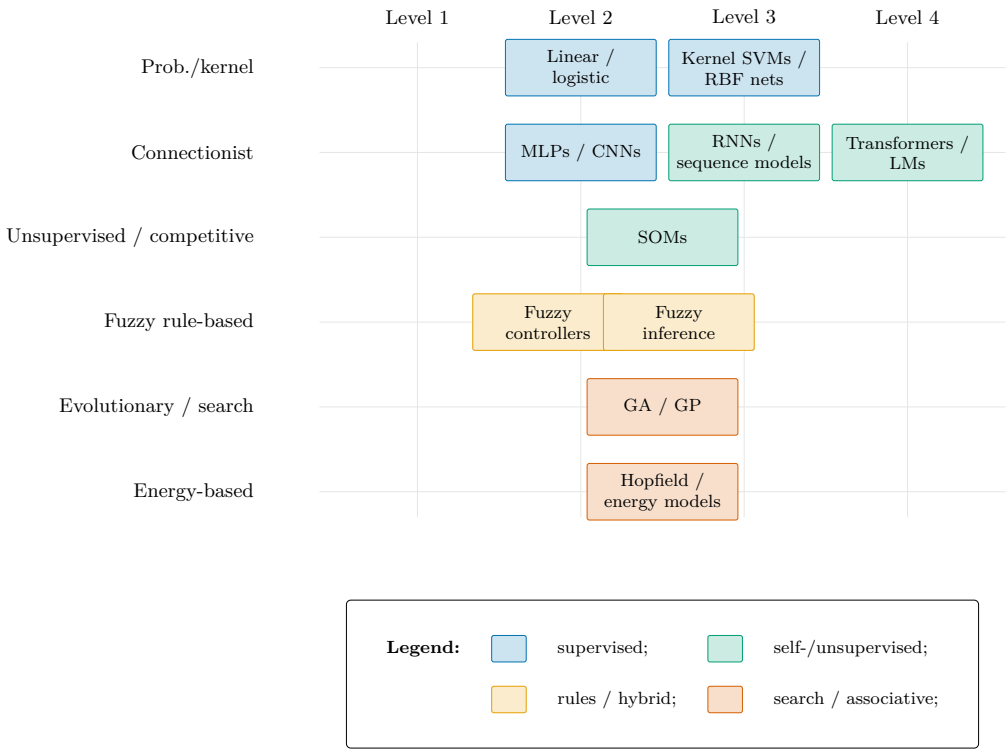


Figure 77: Unified design-space map of the book. Each family is a design point defined by (representation, update/search operator, feedback signal, and audit obligations), plotted across agent level, model nature, and learning signal. Use this to translate intuitions across paradigms (e.g., ERM ↔ energy descent ↔ rule composition ↔ fitness-driven search).

Table 10: Big-picture view of model families across the taxonomy and learning paradigms. Each entry represents a family introduced in the book; supervision labels indicate the dominant training signal rather than strict exclusivity. The table serves as a compact lookup for model-family placement across levels and learning signals.

Model family	Level	Nature	Learning signal
Linear / logistic regression	2–3	probabilistic	supervised
Kernel SVMs and RBF networks	2–3	probabilistic / kernel	supervised
MLPs / CNNs	3	connectionist	supervised
RNNs / sequence models	3	connectionist	supervised / self-supervised
Transformers / attention LMs	3–4	connectionist	self-supervised
Self-organizing maps (SOMs)	2–3	unsupervised / competitive	unsupervised
Fuzzy controllers and inference systems	1–2	fuzzy rule-based	supervised / expert rules
Genetic algorithms and GP	1–3	evolutionary	search / fitness-driven
Hopfield and modern Hopfield variants	1–3	energy-based	associative / unsupervised

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A Notation collision index

This appendix lists the most common *notation collisions* in this book: symbols that appear in multiple domains (probability, optimization, deep learning) with different meanings. The goal is not to eliminate reuse—that is unrealistic—but to make the disambiguation rule explicit so reading remains fast and consistent.

Disambiguation rule (used throughout the book)

- **Function argument wins:** $\sigma(x)$ is the sigmoid; $f(\cdot)$ is an activation; $p(\cdot)$ is a density/pmf.
- **Plain scalars default to the local domain:** σ without an argument is a width/scale unless the paragraph is explicitly about the sigmoid.
- **Typography is a hint, not a guarantee:** bold symbols are vectors/matrices; subscripts usually indicate time, layer, or an index set.

Symbol	Common meanings	Where to look / how to disambiguate
σ	Sigmoid nonlinearity; standard deviation / scale; RBF width	$\sigma(x)$ is always sigmoid; plain σ is a width/scale (e.g., RBFNs) unless the section is explicitly about activations.
λ	Regularization strength; eigenvalue; Lagrange multiplier	Regularization uses λ alongside an objective; spectral topics use λ with matrices/operators; constraints use λ as a multiplier.
p	Probability / density; padding (CNNs); momentum/parameter name in code	$p(y \mid x)$ is probability; convolution padding is stated as p in the output-size formula with n, k, s.
L	Number of layers; sequence length; loss	Layer count uses L with indices $l = 1, \ldots, L$; loss is $\mathcal{L}$; sequence length is usually T.
n	Feature dimension; sequence length (tokens)	Supervised chapters use $\mathbf{x} \in \mathbb{R}^n$ for feature dimension; sequence chapters use T for length, but n may appear in shape summaries. Disambiguate by nearby symbols: $X \in \mathbb{R}^{N \times n}$, $\mathbf{w} \in \mathbb{R}^n$ (features) versus indices $t = 1, \ldots, T$ (tokens/time).
d	Data/feature dimension; embedding width	Most chapters: d is an input-feature dimension. In Transformers/NLP we write the embedding width as d_{model} (and per-head sizes d_k, d_v). When multiple "dimensions" appear together, prefer the subscripted forms.
t	Time index; target vector/component	Time is t with sequences x_t, h_t; targets use $\mathbf{t}$ (bold) or t_k in loss definitions.
h	Hypothesis function h_θ; hidden state h_t / hidden units h	Hypotheses appear as $h_\theta(\cdot)$ in supervised chapters; hidden state uses a time index h_t; hidden width uses a dimension symbol (e.g., h or d_h) in network-shape context.
ϕ, φ, Φ	Feature map; RBF basis function; design matrix	Kernel methods use $\phi(\cdot)$ for feature maps and Φ for the resulting design matrix; RBF chapters reserve $\varphi(\cdot)$ for radial basis functions. If you see $\Phi\mathbf{w}$ it is a design matrix; if you see $\varphi(\lVert\mathbf{x} - \mathbf{c}\rVert)$ it is an RBF.

Reading note. When a symbol is reused with a different meaning, chapters typically flag the local meaning near the first use. When reading out of order, use this index to disambiguate quickly and keep the local convention consistent.

B Reproducibility and Reporting Standards

This appendix defines the minimum reporting standard used throughout this book when experiments include stochastic training, model selection, or constrained optimization. The objective is simple: results should be auditable and repeatable by another reader with the same code and data access.

Core reporting template (required fields)

- **Data protocol:** split policy, leakage controls, and preprocessing/tokenization version.
- **Model protocol:** architecture configuration, parameter count, initialization policy, and regularization settings.
- **Optimization protocol:** optimizer, schedule, batch size, stopping rule, and checkpoint selection criterion.
- **Evaluation protocol:** primary metric, calibration/slice checks, and whether thresholds were tuned on validation only.
- **Variance protocol:** random seeds, number of runs, and summary statistics (median/mean plus spread).

Minimum acceptable evidence

- Report at least 5 seeds for low-cost experiments and 10+ seeds when claims depend on small metric differences.
- Show both central tendency and spread (e.g., median + interquartile range, or mean + standard deviation).
- Match evaluation budgets when comparing methods (same number of epochs/fitness calls and same stopping policy).
- Preserve a machine-readable experiment log (configuration + metrics + seed) for each run.

Common failure modes to avoid

- **Best-run reporting:** publishing only the strongest seed and omitting dispersion.
- **Budget mismatch:** giving one method more training/evaluation budget than another.
- **Selection leakage:** tuning hyperparameters on test data (explicitly or implicitly).
- **Unversioned preprocessing:** changing tokenization or normalization without recording the revision.

Chapter-specific notes

- **Supervised/Logistic chapters:** pair accuracy-like metrics with calibration and slice checks.
- **Deep-learning chapters:** track gradient health, early stopping criteria, and checkpoint restore policy.
- **Evolutionary chapter:** report evaluation-budget-normalized performance and multi-seed front dispersion.

Practical rule. If a claim changes a design decision, it must include enough protocol detail for a third party to reproduce the comparison.

C Kernel Methods and Support Vector Machines

This appendix is a concise reference for the "kernel/SVM" thread that appears throughout the book (hinge losses in Chapter 3, RBF feature maps in Chapter 8, and the geometry of margins). The goal is not to be exhaustive, but to make the notation and the relationship between explicit (finite) feature maps and implicit (kernel) feature maps unambiguous.

Kernel trick and Gram matrices

Let $\phi : \mathbb{R}^d \to \mathcal{H}$ be a (possibly high-dimensional) feature map into an inner-product space $\mathcal{H}$. A *kernel* is a symmetric, positive semidefinite function

$$k(\mathbf{x}, \mathbf{z}) \triangleq \langle \phi(\mathbf{x}), \phi(\mathbf{z}) \rangle_{\mathcal{H}}.$$

Given training points $\{\mathbf{x}_i\}_{i=1}^N$, the *Gram matrix* $K \in \mathbb{R}^{N \times N}$ has entries $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$; by construction, K is symmetric and positive semidefinite.

Kernel ridge regression (KRR)

Kernel ridge regression fits a function of the form

$$f(\mathbf{x}) = \sum_{i=1}^{N} \alpha_i \, k(\mathbf{x}_i, \mathbf{x}),$$

where the coefficient vector $\boldsymbol{\alpha} \in \mathbb{R}^N$ solves

$$(K + \lambda I)\boldsymbol{\alpha} = \mathbf{y}. \tag{C.1}$$

Here $\lambda > 0$ regularizes the solution and stabilizes the linear system when K is ill-conditioned. This is the fully kernelized analogue of ridge regression in a finite design matrix Φ. Chapter 8's dual viewpoint shows that a primal RBF network with centers at all data points ($M = N$) recovers this same predictor under an RBF kernel.

Soft-margin SVMs (primal and kernelized form)

For binary labels $y_i \in \{-1, +1\}$, the (linear) soft-margin SVM solves (Cortes and Vapnik, 1995)

$$\begin{aligned} \min_{\mathbf{w}, b, \boldsymbol{\xi}} \quad & \frac{1}{2}\|\mathbf{w}\|_2^2 + C \sum_{i=1}^{N} \xi_i \\ \text{s.t.} \quad & y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0. \end{aligned}$$

The parameter $C > 0$ trades margin size against slack violations. The decision function is $f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b$, with classification by $\operatorname{sign}(f(\mathbf{x}))$. In the kernelized

form, $\mathbf{w}$ is never formed explicitly; instead,

$$f(\mathbf{x}) = \sum_{i=1}^{N} \alpha_i y_i \, k(\mathbf{x}_i, \mathbf{x}) + b, \tag{C.2}$$

where many coefficients α_i become zero (only *support vectors* remain).

How kernels relate to RBFNs (and when to use which)

- **RBFN (explicit features):** choose $M \ll N$ centers and widths, build $\Phi \in \mathbb{R}^{N \times M}$, and solve a regularized linear system in M unknowns. This is efficient when you can afford explicit features and want direct control over locality and model size.
- **Kernel method (implicit features):** work directly with $K \in \mathbb{R}^{N \times N}$, which corresponds to an implicit feature map of potentially very high dimension. This is attractive for small-to-medium N when the kernel encodes a useful inductive bias, but training and storage scale poorly with N unless approximations are used.
- **Nyström / low-rank approximations:** choose M landmark points and project into a rank-M space, recovering an explicit finite basis that connects the kernel view back to the RBFN picture in Chapter 8.

D Linear Systems Primer

This appendix collects basic material on signals, linear time-invariant (LTI) systems, and state-space models. It serves as a reference for the dynamical viewpoints used in later chapters (e.g., Hopfield networks, RNNs, and control-oriented fuzzy systems).

Signals and Systems

A *signal* is a mapping from an index set (typically time or space) into a set of values that encode a physical or abstract quantity. Formally, a continuous-time *scalar* signal is a function $x : \mathbb{R} \to \mathbb{R}$ (or $\mathbb{C}$), while a continuous-time *vector* signal is $x : \mathbb{R} \to \mathbb{R}^n$ (or $\mathbb{C}^n$). Discrete-time signals are defined analogously on $\mathbb{Z}$. Signals may be deterministic, stochastic, scalar, or vector-valued depending on the context.

A *system* is an operator $\mathcal{T}$ that maps an input signal space $\mathcal{X}$ to an output signal space $\mathcal{Y}$, i.e., $y = \mathcal{T}\{x\}$. Systems are characterized by properties such as linearity, time-invariance, causality, and stability; determining which of these properties hold tells us which analytical tools (Fourier analysis, state-space models, etc.) are applicable.

Linear Time-Invariant Systems

LTI systems are a central class of models. They satisfy:

- **Linearity:** For any inputs $x_1(t)$, $x_2(t)$ and scalars a_1, a_2,

$$\mathcal{S}[a_1 x_1(t) + a_2 x_2(t)] = a_1 \mathcal{S}[x_1(t)] + a_2 \mathcal{S}[x_2(t)].$$

- **Time-invariance:** If the input is shifted in time by τ, the output is shifted by the same amount:

$$\mathcal{S}[x(t-\tau)] = y(t-\tau),$$

  where $y(t) = \mathcal{S}[x(t)]$.

Impulse Response and Convolution

The behavior of an LTI system is completely characterized by its *impulse response* $h(t)$, defined as the output when the input is a Dirac delta function $\delta(t)$:

$$h(t) = \mathcal{S}[\delta(t)].$$

For any input $x(t)$, the output $y(t)$ is given by the convolution integral

$$y(t) = (x * h)(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau.$$

In discrete time the integral is replaced by a sum over integer indices.

Frequency-Domain Representation

The Fourier transform is a standard tool for analysing signals and LTI systems in the frequency domain. For a signal $x(t)$, the Fourier transform $X(f)$ is

$$X(f) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi ft}\, dt.$$

Under suitable regularity conditions, convolution in time corresponds to multiplication in frequency:

$$Y(f) = H(f)X(f),$$

where $H(f)$ is the transform of $h(t)$.

State-Space Models and Transfer Functions

Many dynamical systems in this book are expressed in continuous-time state-space form:

$$\begin{aligned} \frac{d\mathbf{x}(t)}{dt} &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t), \\ \mathbf{y}(t) &= \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t), \end{aligned} \tag{D.1}$$

where $\mathbf{x}(t) \in \mathbb{R}^n$ is the state, $\mathbf{u}(t) \in \mathbb{R}^m$ the input, and $\mathbf{y}(t) \in \mathbb{R}^p$ the output.

Homogeneous solution. For the zero-input system, the state evolves as

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0),$$

where $e^{\mathbf{A}t}$ is the matrix exponential

$$e^{\mathbf{A}t} = \sum_{k=0}^{\infty} \frac{(\mathbf{A}t)^k}{k!}.$$

Key properties include $e^{\mathbf{A}0} = \mathbf{I}$ and $\frac{d}{dt}e^{\mathbf{A}t} = \mathbf{A}e^{\mathbf{A}t} = e^{\mathbf{A}t}\mathbf{A}$. If $\mathbf{A}$ is diagonalizable, $e^{\mathbf{A}t}$ can be computed efficiently via eigen-decomposition.

Forced response. With input $\mathbf{u}(t)$, the solution is

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)\, d\tau, \tag{D.2}$$

which mirrors the convolution expression for LTI systems: the kernel $e^{\mathbf{A}(t-\tau)}$ plays the role of a matrix-valued impulse response.

Transfer function. Taking the Laplace transform of (D.1) with zero initial conditions yields

$$\begin{aligned} s\mathbf{X}(s) &= \mathbf{A}\mathbf{X}(s) + \mathbf{B}\mathbf{U}(s), \\ \mathbf{Y}(s) &= \mathbf{C}\mathbf{X}(s) + \mathbf{D}\mathbf{U}(s). \end{aligned}$$

Solving for $\mathbf{X}(s)$ gives

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{U}(s),$$

and substituting into the output equation produces the transfer function matrix

$$\mathbf{G}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}. \tag{D.3}$$

This compactly describes the input–output behavior of the LTI system in the Laplace domain.

Further Reading

- Kailath, T. (1980). *Linear Systems.* Prentice Hall.
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Index

activation function, 106, 126
Adaline, 106
ANFIS (adaptive neuro-fuzzy inference system), 415
associative memory, 221
attention, 316
  cross-attention, 316
  self-attention, 316
attractor, 221
audit, 31

backpropagation, 140
backpropagation through time (BPTT), 263
backtracking, 51
batch normalization (BN), 242
BERT, 316
bias-variance tradeoff, 68

calibration, 68
centroid (defuzzification), 361
chain rule, 126
composition (fuzzy relations), 398
convolution, 242
convolutional neural network (CNN), 242
cross-entropy, 89, 140
cross-validation, 68
crossover (GA), 430

data augmentation, 242
decision boundary, 106
defuzzification, 361, 415
differential evolution (DE), 430
dropout, 242

empirical risk minimization (ERM), 68
energy function, 221
evolutionary computing, 430
extension principle, 398

feature map, 166
fine-tuning, 316
fitness function, 430
FlashAttention, 316
footprint of uncertainty (FOU), 361
fuzzy inference system (FIS), 415
fuzzy logic, 343
fuzzy relation, 398
fuzzy set, 361
  interval type-2 (IT2), 361
  type-2 fuzzy set, 361

genetic algorithm (GA), 430
genetic programming (GP), 430
GPT, 316
gradient, 140
gradient check, 140
gradient clipping, 242
gradient descent, 68
  stochastic gradient descent (SGD), 68
GRU, 263

Hebb rule, 106
heuristic, 51
Hopfield network, 221

hybrid learning (ANFIS), 415

intelligent system, 31

Karnik–Mendel algorithm, 361
kernel trick, 166
KV cache, 316

learning curve, 68
least squares, 166
linguistic variable, 343
logistic regression, 89
LoRA, 316
loss function, 68
LSTM, 263

Mamdani inference, 415
masked language modeling, 316
membership function, 343, 361
    Gaussian, 361
    trapezoidal, 361
    triangular, 361
meta-cognition, 31
MixUp, 242
multi-head attention, 316
multi-layer perceptron (MLP), 126
multi-objective optimisation, 430
mutation (GA), 430

natural language processing (NLP), 289
NEAT (NeuroEvolution of Augmenting Topologies), 430
negative log-likelihood (NLL), 89
negative sampling, 289
neural network, 106
neuro-evolution, 430
neuro-fuzzy system, 415
NSGA-II, 430

overfitting, 68

particle swarm optimisation (PSO), 430
perceptron, 106
perceptron convergence theorem, 106
positional encoding, 316
    ALiBi, 316
    RoPE (rotary position embedding), 316
pre-training, 316

quantization error (QE), 186

radial basis function network (RBFN), 166
receptive field, 242
recurrent neural network (RNN), 263
regularization, 68
representation, 31
ResNet, 242
ROC curve, 68
rule base, 415

scaled dot-product attention, 316
search, 51
selection (GA), 430
self-organizing map (SOM), 186
skip connection, 242
soft computing, 343
softmax, 89
spurious state, 221
Sugeno inference, 415
supervised learning, 68

symbolic integration, 51

t-conorm, 398
t-norm, 398
tokenization, 289
topographic error (TE), 186
train–test split, 68
transformer, 316

U-matrix, 186
underfitting, 68
universal approximation theorem (UAT), 106
unsupervised learning, 186

validation set, 68
vanishing gradient, 263

weight initialization, 242
    He initialization, 242
    Xavier initialization, 242
word embedding, 289
Word2Vec, 289

XOR, 126